- $L_1:\ x = 2s - 1,\ y = s - 1,\ z = s - 4$
- $L_2:\ x = t - 3,\ y = 3t + 8,\ z = 5 - 2t$

b.

- $L_1:\ x = -y = z$
- $L_2:\ \dfrac{x-3}{2} = y = z - 2$

c.

- $L_1:\ x = 6s - 1,\ y = -2s,\ z = 3s + 1$
- $L_2:\ \dfrac{x-4}{6} = \dfrac{y+3}{-2} = \dfrac{z-1}{3}$

Solution

a. Line L_1 has direction vector $\vec{\mathbf{v}}_1 = \langle 2, 1, 1\rangle$; line L_2 has direction vector $\vec{\mathbf{v}}_2 = \langle 1, 3, -2\rangle$. Because the direction vectors are not parallel vectors, the lines are either intersecting or skew. To determine whether the lines intersect, we see if there is a point, (x, y, z), that lies on both lines. To find this point, we use the parametric equations to create a system of equalities:

$$2s - 1 = t - 3;$$

$$s - 1 = 3t + 8;$$

$$s - 4 = 5 - 2t.$$

By the first equation, $t = 2s + 2$. Substituting into the second equation yields

$$s - 1 = 3(2s + 2) + 8$$

$$s - 1 = 6s + 6 + 8$$

$$5s = -15$$

$$s = -3.$$

Substitution into the third equation, however, yields a contradiction:

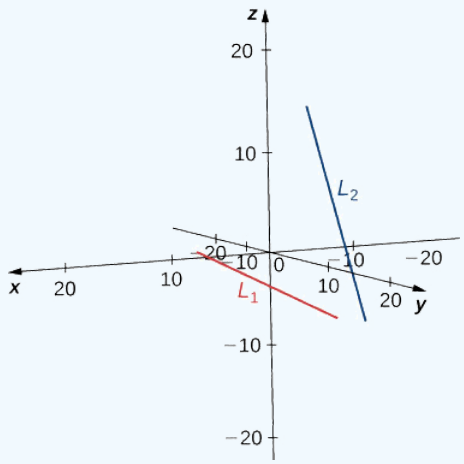
$$s - 4 = 5 - 2(2s + 2)$$

$$s - 4 = 5 - 4s - 4$$

$$5s = 5$$

$$s = 1.$$

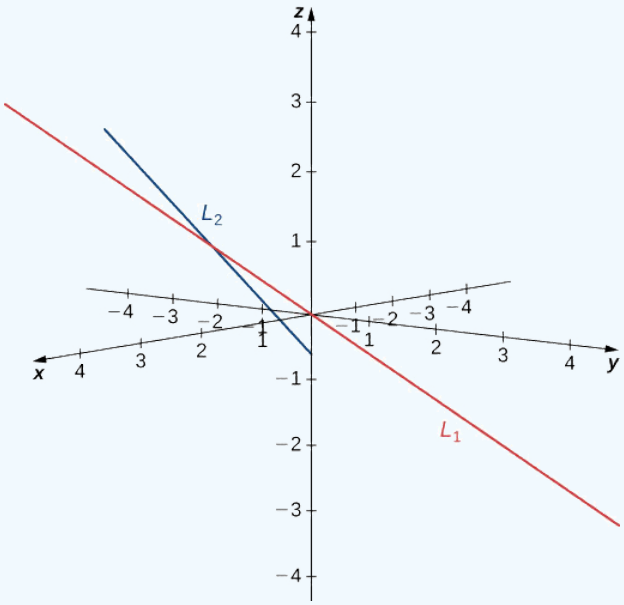
There is no single point that satisfies the parametric equations for L_1 and L_2 simultaneously. These lines do not intersect, so they are skew (see the following figure).



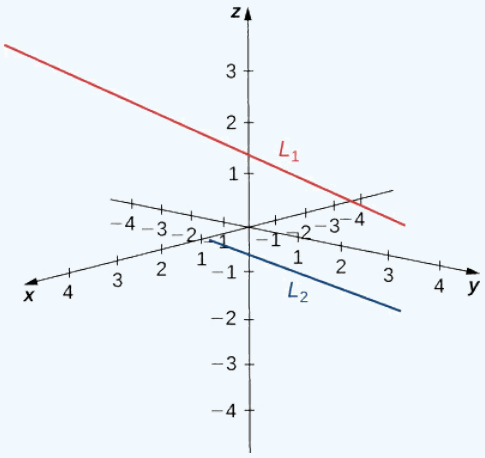
b. Line L_1 has direction vector $\vec{\mathbf{v}}_1 = \langle 1, -1, 1 \rangle$ and passes through the origin, $(0, 0, 0)$. Line L_2 has a different direction vector, $\vec{\mathbf{v}}_2 = \langle 2, 1, 1 \rangle$, so these lines are not parallel or equal. Let r represent the parameter for line L_1 and let s represent the parameter for L_2:

$$\begin{array}{ll} \text{Line } L_1: & \text{Line } L_2: \\ x = r & x = 2s + 3 \\ y = -r & y = s \\ z = r & z = s + 2 \end{array}$$

Solve the system of equations to find $r = 1$ and $s = -1$. If we need to find the point of intersection, we can substitute these parameters into the original equations to get $(1, -1, 1)$ (see the following figure).



c. Lines L_1 and L_2 have equivalent direction vectors: $\vec{\mathbf{v}} = \langle 6, -2, 3 \rangle$. These two lines are parallel (see the following figure).



Exercise 12.5.4

Describe the relationship between the lines with the following parametric equations:

$$x = 1 - 4t,\ y = 3 + t,\ z = 8 - 6t$$

$$x = 2 + 3s,\ y = 2s,\ z = -1 - 3s.$$

Hint

Start by identifying direction vectors for each line. Is one a multiple of the other?

Answer

These lines are skew because their direction vectors are not parallel and there is no point (x, y, z) that lies on both lines.

Equations for a Plane

We know that a line is determined by two points. In other words, for any two distinct points, there is exactly one line that passes through those points, whether in two dimensions or three. Similarly, given any three points that do not all lie on the same line, there is a unique plane that passes through these points. Just as a line is determined by two points, a plane is determined by three.

This may be the simplest way to characterize a plane, but we can use other descriptions as well. For example, given two distinct, intersecting lines, there is exactly one plane containing both lines. A plane is also determined by a line and any point that does not lie on the line. These characterizations arise naturally from the idea that a plane is determined by three points. Perhaps the most surprising characterization of a plane is actually the most useful.

Imagine a pair of orthogonal vectors that share an initial point. Visualize grabbing one of the vectors and twisting it. As you twist, the other vector spins around and sweeps out a plane. Here, we describe that concept mathematically. Let $\vec{\mathbf{n}} = \langle a, b, c\rangle$ be a vector and $P = (x_0, y_0, z_0)$ be a point. Then the set of all points $Q = (x, y, z)$ such that $\overrightarrow{PQ}$ is orthogonal to $\vec{\mathbf{n}}$ forms a plane (Figure 12.5.7). We say that $\vec{\mathbf{n}}$ is a **normal vector**, or perpendicular to the plane. Remember, the dot product of orthogonal vectors is zero. This fact generates the **vector equation of a plane**:

$$\vec{\mathbf{n}} \cdot \overrightarrow{PQ} = 0.$$

Rewriting this equation provides additional ways to describe the plane:

$$\begin{aligned} \vec{\mathbf{n}} \cdot \overrightarrow{PQ} &= 0 \\ \langle a, b, c\rangle \cdot \langle x - x_0, y - y_0, z - z_0\rangle &= 0 \\ a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0. \end{aligned}$$

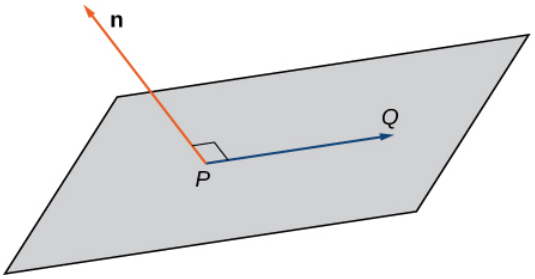


Figure 12.5.7: Given a point P and vector $\vec{\mathbf{n}}$, the set of all points Q with $\overrightarrow{PQ}$ orthogonal to $\vec{\mathbf{n}}$ forms a plane.

Definition: Scalar Equation of a Plane

Given a point P and vector $\vec{\mathbf{n}}$, the set of all points Q satisfying the equation $\vec{\mathbf{n}} \cdot \overrightarrow{PQ} = 0$ forms a plane. The equation

$$\vec{\mathbf{n}} \cdot \overrightarrow{PQ} = 0$$

is known as the *vector equation of a plane*.

The *scalar equation of a plane* (sometimes also called the *standard equation of a plane*) containing point $P = (x_0, y_0, z_0)$ with normal vector $\vec{n} = \langle a, b, c\rangle$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

This equation can be expressed as $ax + by + cz + d = 0$, where $d = -ax_0 - by_0 - cz_0$. This form of the equation is sometimes called the *general form of the equation of a plane.*

As described earlier in this section, any three points that do not all lie on the same line determine a plane. Given three such points, we can find an equation for the plane containing these points.

Example 12.5.5: Writing an Equation of a Plane Given Three Points in the Plane

Write an equation for the plane containing points $P=(1,1,-2), Q=(0,2,1)$, and $R=(-1,-1,0)$ in both standard and general forms.

Solution

To write an equation for a plane, we must find a normal vector for the plane. We start by identifying two vectors in the plane:

$$\begin{aligned}\overrightarrow{PQ} &= \langle 0-1, 2-1, 1-(-2)\rangle \\ &= \langle -1, 1, 3\rangle \\ \overrightarrow{QR} &= \langle -1-0, -1-2, 0-1\rangle \\ &= \langle -1, -3, -1\rangle.\end{aligned}$$

The cross product $\overrightarrow{PQ}\times\overrightarrow{QR}$ is orthogonal to both $\overrightarrow{PQ}$ and $\overrightarrow{QR}$, so it is normal to the plane that contains these two vectors:

$$\begin{aligned}\vec{\mathbf{n}} &= \overrightarrow{PQ}\times\overrightarrow{QR} \\ &= \begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -1 & 1 & 3 \\ -1 & -3 & -1\end{vmatrix} \\ &= (-1+9)\hat{\mathbf{i}} - (1+3)\hat{\mathbf{j}} + (3+1)\hat{\mathbf{k}} \\ &= 8\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}.\end{aligned}$$

Thus, $\vec{\mathbf{n}} = \langle 8, -4, 4\rangle$, and we can choose any of the three given points to write an equation of the plane:

$$\begin{aligned}8(x-1) - 4(y-1) + 4(z+2) &= 0 \\ 8x - 4y + 4z + 4 &= 0.\end{aligned}$$

The scalar equations of a plane vary depending on the normal vector and point chosen.

Example 12.5.6: Writing an Equation for a Plane Given a Point and a Line

Find an equation of the plane that passes through point $(1,4,3)$ and contains the line given by $x=\dfrac{y-1}{2}=z+1$.

Solution

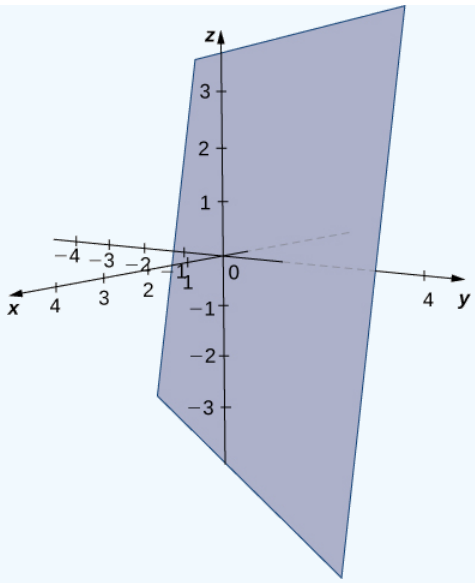
Symmetric equations describe the line that passes through point $(0,1,-1)$ parallel to vector $\vec{\mathbf{v}}_1=\langle 1,2,1\rangle$ (see the following figure). Use this point and the given point, $(1,4,3)$, to identify a second vector parallel to the plane:

$$\vec{\mathbf{v}}_2 = \langle 1-0, 4-1, 3-(-1)\rangle = \langle 1,3,4\rangle.$$

Use the cross product of these vectors to identify a normal vector for the plane:

$$\begin{aligned}\vec{\mathbf{n}} &= \vec{\mathbf{v}}_1\times\vec{\mathbf{v}}_2 \\ &= \begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2 & 1 \\ 1 & 3 & 4\end{vmatrix} \\ &= (8-3)\hat{\mathbf{i}} - (4-1)\hat{\mathbf{j}} + (3-2)\hat{\mathbf{k}} \\ &= 5\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}.\end{aligned}$$

The scalar equations for the plane are $5x-3(y-1)+(z+1)=0$ and $5x-3y+z+4=0$.



Exercise 12.5.6

Find an equation of the plane containing the lines L_1 and L_2:

$$L_1 : \ x = -y = z$$

$$L_2 : \ \frac{x-3}{2} = y = z - 2.$$

Hint

Hint: The cross product of the lines' direction vectors gives a normal vector for the plane.

Answer

$$-2(x-1) + (y+1) + 3(z-1) = 0$$

or

$$-2x + y + 3z = 0$$

Now that we can write an equation for a plane, we can use the equation to find the distance d between a point P and the plane. It is defined as the shortest possible distance from P to a point on the plane.

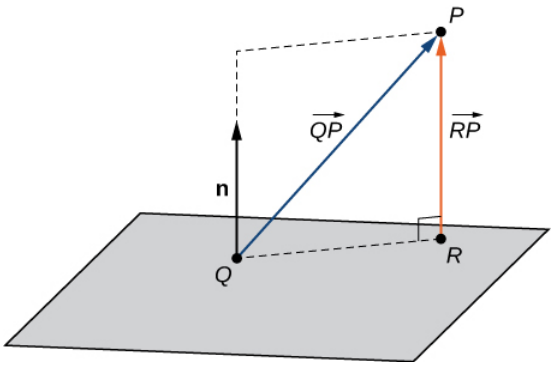


Figure 12.5.8: We want to find the shortest distance from point P to the plane. Let point R be the point in the plane such that, for any other point in the plane Q, $\|\overrightarrow{RP}\| < \|\overrightarrow{QP}\|$.

Just as we find the two-dimensional distance between a point and a line by calculating the length of a line segment perpendicular to the line, we find the three-dimensional distance between a point and a plane by calculating the length of a line segment

perpendicular to the plane. Let R be the point in the plane such that $\overrightarrow{RP}$ is orthogonal to the plane, and let Q be an arbitrary point in the plane. Then the projection of vector $\overrightarrow{QP}$ onto the normal vector describes vector $\overrightarrow{RP}$, as shown in Figure 12.5.8.

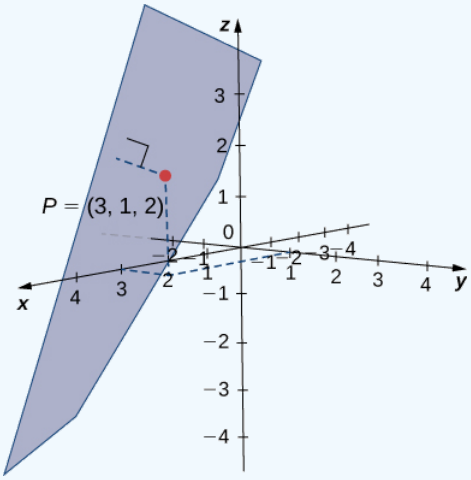
The Distance between a Plane and a Point

Suppose a plane with normal vector $\vec{\mathbf{n}}$ passes through point Q. The distance d from the plane to a point P not in the plane is given by

$$d = \|\text{proj}_{\vec{\mathbf{n}}} \overrightarrow{QP}\| = |\text{ comp}_{\vec{\mathbf{n}}} \overrightarrow{QP}| = \frac{|\overrightarrow{QP} \cdot \vec{\mathbf{n}}|}{\|\vec{\mathbf{n}}\|}. \tag{12.5.5}$$

Example 12.5.7: Distance between a Point and a Plane

Find the distance between point $P = (3, 1, 2)$ and the plane given by $x - 2y + z = 5$ (see the following figure).



Solution

The coefficients of the plane's equation provide a normal vector for the plane: $\vec{\mathbf{n}} = \langle 1, -2, 1\rangle$. To find vector $\overrightarrow{QP}$, we need a point in the plane. Any point will work, so set $y = z = 0$ to see that point $Q = (5, 0, 0)$ lies in the plane. Find the component form of the vector from Q to P:

$$\overrightarrow{QP} = \langle 3 - 5, 1 - 0, 2 - 0\rangle = \langle -2, 1, 2\rangle.$$

Apply the distance formula from Equation 12.5.5:

$$\begin{aligned} d &= \frac{\left|\overrightarrow{QP} \cdot \vec{\mathbf{n}}\right|}{\|\vec{\mathbf{n}}\|} \\ &= \frac{|\langle -2, 1, 2\rangle \cdot \langle 1, -2, 1\rangle|}{\sqrt{1^2 + (-2)^2 + 1^2}} \\ &= \frac{|-2 - 2 + 2|}{\sqrt{6}} \\ &= \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3} \text{ units.} \end{aligned}$$

Exercise 12.5.7

Find the distance between point $P = (5, -1, 0)$ and the plane given by $4x + 2y - z = 3$.

Hint

Point $(0, 0, -3)$ lies on the plane.

Answer

$$\frac{15}{\sqrt{21}} = \frac{5\sqrt{21}}{7} \text{ units}$$

Parallel and Intersecting Planes

We have discussed the various possible relationships between two lines in two dimensions and three dimensions. When we describe the relationship between two planes in space, we have only two possibilities: the two distinct planes are parallel or they intersect. When two planes are parallel, their normal vectors are parallel. When two planes intersect, the intersection is a line (Figure 12.5.9).

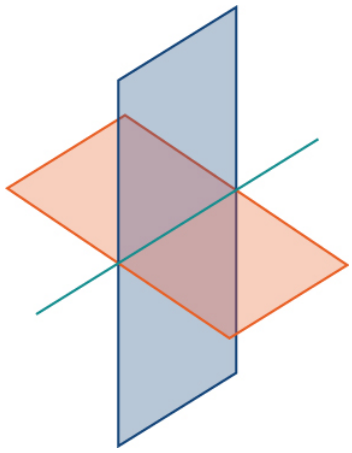
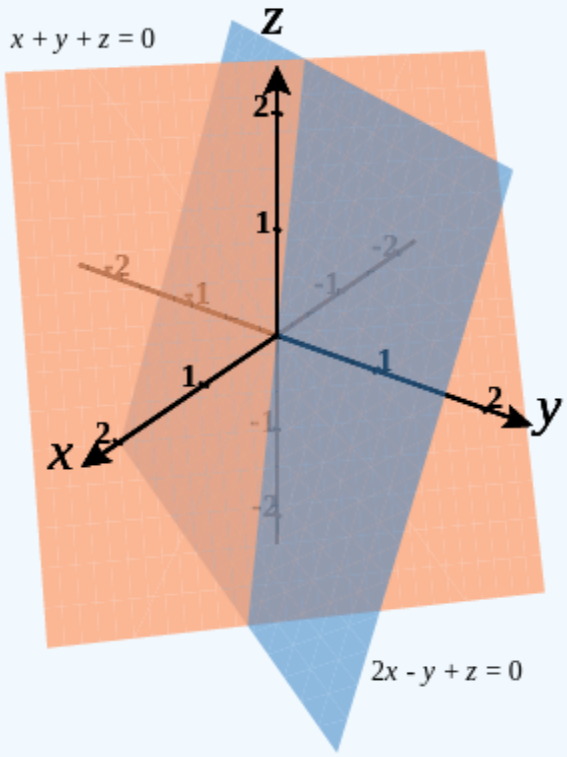


Figure 12.5.9: The intersection of two nonparallel planes is always a line.

We can use the equations of the two planes to find parametric equations for the line of intersection.

Example 12.5.8: Finding the Line of Intersection for Two Planes

Find parametric and symmetric equations for the line formed by the intersection of the planes given by $x + y + z = 0$ and $2x - y + z = 0$ (see the following figure).



Solution

Note that the two planes have nonparallel normals, so the planes intersect. Further, the origin satisfies each equation, so we know the line of intersection passes through the origin. Add the plane equations so we can eliminate one of the variables, in this case, y:

$$x + y + z = 0$$

$$2x - y + z = 0$$

$$_____$$

$$3x + 2z = 0.$$

This gives us $x = -\frac{2}{3}z$. We substitute this value into the first equation to express y in terms of z:

$$x + y + z = 0$$

$$-\frac{2}{3}z + y + z = 0$$

$$y + \frac{1}{3}z = 0$$

$$y = -\frac{1}{3}z$$

We now have the first two variables, x and y, in terms of the third variable, z. Now we define z in terms of t. To eliminate the need for fractions, we choose to define the parameter t as $t = -\frac{1}{3}z$. Then, $z = -3t$. Substituting the parametric representation of

z back into the other two equations, we see that the parametric equations for the line of intersection are $x = 2t,\ y = t,\ z = -3t$. The symmetric equations for the line are $\frac{x}{2} = y = \frac{z}{-3}$.

Exercise 12.5.8

Find parametric equations for the line formed by the intersection of planes $x + y - z = 3$ and $3x - y + 3z = 5$.

Hint

Add the two equations, then express z in terms of x. Then, express y in terms of x.

Answer

$x = t,\ y = 7 - 3t,\ z = 4 - 2t$

In addition to finding the equation of the line of intersection between two planes, we may need to find the angle formed by the intersection of two planes. For example, builders constructing a house need to know the angle where different sections of the roof meet to know whether the roof will look good and drain properly. We can use normal vectors to calculate the angle between the two planes. We can do this because the angle between the normal vectors is the same as the angle between the planes. Figure 12.5.10 shows why this is true.

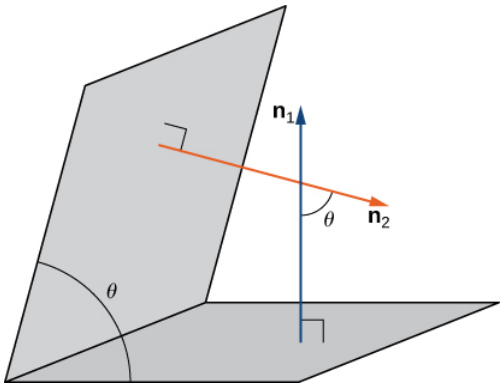


Figure 12.5.10: The angle between two planes has the same measure as the angle between the normal vectors for the planes.

We can find the measure of the angle θ between two intersecting planes by first finding the cosine of the angle, using the following equation:

$$\cos\theta = \frac{|\vec{\mathbf{n}}_1 \cdot \vec{\mathbf{n}}_2|}{\|\vec{\mathbf{n}}_1\|\|\vec{\mathbf{n}}_2\|}.$$

We can then use the angle to determine whether two planes are parallel or orthogonal or if they intersect at some other angle.

Example 12.5.9: Finding the Angle between Two Planes

Determine whether each pair of planes is parallel, orthogonal, or neither. If the planes are intersecting, but not orthogonal, find the measure of the angle between them. Give the answer in radians and round to two decimal places.

a. $x + 2y - z = 8$ and $2x + 4y - 2z = 10$
b. $2x - 3y + 2z = 3$ and $6x + 2y - 3z = 1$
c. $x + y + z = 4$ and $x - 3y + 5z = 1$

Solution:

a. The normal vectors for these planes are $\vec{\mathbf{n}}_1 = \langle 1, 2, -1\rangle$ and $\vec{\mathbf{n}}_2 = \langle 2, 4, -2\rangle$. These two vectors are scalar multiples of each other. The normal vectors are parallel, so the planes are parallel.
b. The normal vectors for these planes are $\vec{\mathbf{n}}_1 = \langle 2, -3, 2\rangle$ and $\vec{\mathbf{n}}_2 = \langle 6, 2, -3\rangle$. Taking the dot product of these vectors, we have

$$\begin{aligned}\vec{\mathbf{n}}_1 \cdot \vec{\mathbf{n}}_2 &= \langle 2, -3, 2\rangle \cdot \langle 6, 2, -3\rangle \\ &= 2(6) - 3(2) + 2(-3) = 0.\end{aligned}$$

The normal vectors are orthogonal, so the corresponding planes are orthogonal as well.

c. The normal vectors for these planes are $\vec{\mathbf{n}}_1 = \langle 1, 1, 1\rangle$ and $\vec{\mathbf{n}}_2 = \langle 1, -3, 5\rangle$:

$$\begin{aligned}\cos\theta &= \frac{|\vec{\mathbf{n}}_1 \cdot \vec{\mathbf{n}}_2|}{\|\vec{\mathbf{n}}_1\|\|\vec{\mathbf{n}}_2\|} \\ &= \frac{|\langle 1, 1, 1\rangle \cdot \langle 1, -3, 5\rangle|}{\sqrt{1^2 + 1^2 + 1^2}\sqrt{1^2 + (-3)^2 + 5^2}} \\ &= \frac{3}{\sqrt{105}}\end{aligned}$$

Then $\theta = \arccos \frac{3}{\sqrt{105}} \approx 1.27$ rad.

Thus the angle between the two planes is about 1.27 rad, or approximately $73°$.

? Exercise 12.5.9

Find the measure of the angle between planes $x + y - z = 3$ and $3x - y + 3z = 5$. Give the answer in radians and round to two decimal places.

Hint

Use the coefficients of the variables in each equation to find a normal vector for each plane.

Answer

1.44 rad

When we find that two planes are parallel, we may need to find the distance between them. To find this distance, we simply select a point in one of the planes. The distance from this point to the other plane is the distance between the planes.

Previously, we introduced the formula for calculating this distance in Equation 12.5.5:

$$d = \frac{\overrightarrow{QP} \cdot \vec{\mathbf{n}}}{\|\vec{\mathbf{n}}\|},$$

where Q is a point on the plane, P is a point not on the plane, and $\vec{n}$ is the normal vector that passes through point Q. Consider the distance from point (x_0, y_0, z_0) to plane $ax + by + cz + k = 0$. Let (x_1, y_1, z_1) be any point in the plane. Substituting into the formula yields

$$\begin{aligned}d &= \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|ax_0 + by_0 + cz_0 + k|}{\sqrt{a^2 + b^2 + c^2}}.\end{aligned}$$

We state this result formally in the following theorem.

Distance from a Point to a Plane

Let $P(x_0, y_0, z_0)$ be a point. The distance from P to plane $ax + by + cz + k = 0$ is given by

$$d = \frac{|ax_0 + by_0 + cz_0 + k|}{\sqrt{a^2 + b^2 + c^2}}.$$

Example 12.5.10: Finding the Distance between Parallel Planes

Find the distance between the two parallel planes given by $2x + y - z = 2$ and $2x + y - z = 8$.

Solution

Point $(1, 0, 0)$ lies in the first plane. The desired distance, then, is

$$\begin{aligned} d &= \frac{|ax_0 + by_0 + cz_0 + k|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|2(1) + 1(0) + (-1)(0) + (-8)|}{\sqrt{2^2 + 1^2 + (-1)^2}} \\ &= \frac{6}{\sqrt{6}} = \sqrt{6} \text{ units} \end{aligned}$$

Exercise 12.5.10:

Find the distance between parallel planes $5x - 2y + z = 6$ and $5x - 2y + z = -3$.

Hint

Set $x = y = 0$ to find a point on the first plane.

Answer

$$\frac{9}{\sqrt{30}} = \frac{3\sqrt{30}}{10} \text{ units}$$

Distance between Two Skew Lines

Finding the distance from a point to a line or from a line to a plane seems like a pretty abstract procedure. But, if the lines represent pipes in a chemical plant or tubes in an oil refinery or roads at an intersection of highways, confirming that the distance between them meets specifications can be both important and awkward to measure. One way is to model the two pipes as lines, using the techniques in this chapter, and then calculate the distance between them. The calculation involves forming vectors along the directions of the lines and using both the cross product and the dot product.

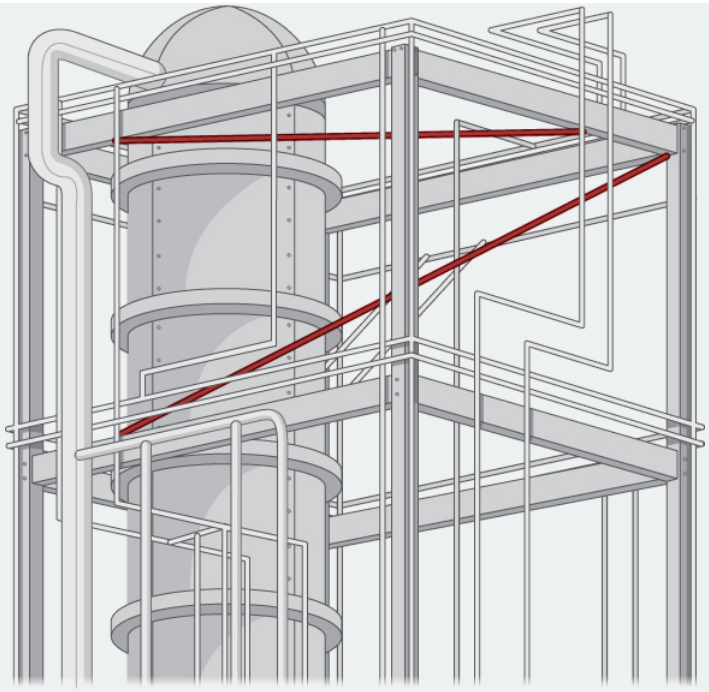


Figure 12.5.11: Industrial pipe installations often feature pipes running in different directions. How can we find the distance between two skew pipes?

The symmetric forms of two lines, L_1 and L_2, are

$$L_1 : \frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

$$L_2 : \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}.$$

You are to develop a formula for the distance d between these two lines, in terms of the values $a_1, b_1, c_1; a_2, b_2, c_2; x_1, y_1, z_1;$ and x_2, y_2, z_2. The distance between two lines is usually taken to mean the minimum distance, so this is the length of a line segment or the length of a vector that is perpendicular to both lines and intersects both lines.

1. First, write down two vectors, $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$, that lie along L_1 and L_2, respectively.

2. Find the cross product of these two vectors and call it $\vec{\mathbf{N}}$. This vector is perpendicular to $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$, and hence is perpendicular to both lines.

3. From vector $\vec{\mathbf{N}}$, form a unit vector $\vec{\mathbf{n}}$ in the same direction.

4. Use symmetric equations to find a convenient vector $\vec{\mathbf{v}}_{12}$ that lies between any two points, one on each line. Again, this can be done directly from the symmetric equations.

5. The dot product of two vectors is the magnitude of the projection of one vector onto the other—that is, $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = \|\vec{\mathbf{A}}\|\|\vec{\mathbf{B}}\| \cos\theta$, where θ is the angle between the vectors. Using the dot product, find the projection of vector $\vec{\mathbf{v}}_{12}$ found in step 4 onto unit vector $\vec{\mathbf{n}}$ found in step 3. This projection is perpendicular to both lines, and hence its length must be the perpendicular distance d between them. Note that the value of d may be negative, depending on your choice of vector $\vec{\mathbf{v}}_{12}$ or the order of the cross product, so use absolute value signs around the numerator.

6. Check that your formula gives the correct distance of $|-25|/\sqrt{198} \approx 1.78$ between the following two lines:

$$L_1 : \frac{x-5}{2} = \frac{y-3}{4} = \frac{z-1}{3}$$

$$L_2 : \frac{x-6}{3} = \frac{y-1}{5} = \frac{z}{7}.$$

7. Is your general expression valid when the lines are parallel? If not, why not? (Hint: What do you know about the value of the cross product of two parallel vectors? Where would that result show up in your expression for d?)

8. Demonstrate that your expression for the distance is zero when the lines intersect. Recall that two lines intersect if they are not parallel and they are in the same plane. Hence, consider the direction of $\vec{\mathbf{n}}$ and $\vec{\mathbf{v}}_{12}$. What is the result of their dot product?

9. Consider the following application. Engineers at a refinery have determined they need to install support struts between many of the gas pipes to reduce damaging vibrations. To minimize cost, they plan to install these struts at the closest points between adjacent skewed pipes. Because they have detailed schematics of the structure, they are able to determine the correct lengths of the struts needed, and hence manufacture and distribute them to the installation crews without spending valuable time making measurements.

The rectangular frame structure has the dimensions $4.0 \times 15.0 \times 10.0\,\text{m}$ (height, width, and depth). One sector has a pipe entering the lower corner of the standard frame unit and exiting at the diametrically opposed corner (the one farthest away at the top); call this L_1. A second pipe enters and exits at the two different opposite lower corners; call this L_2 (Figure 12.5.12).

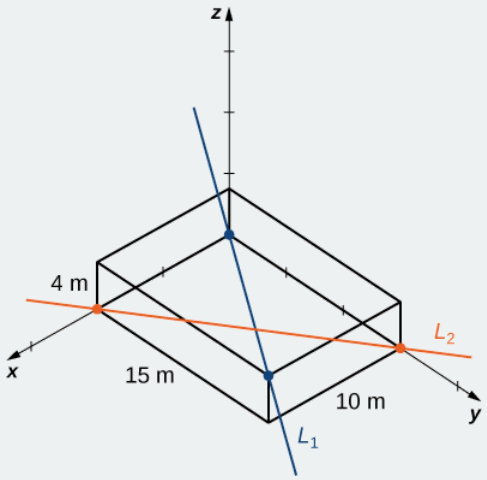


Figure 12.5.12: Two pipes cross through a standard frame unit.

Write down the vectors along the lines representing those pipes, find the cross product between them from which to create the unit vector $\vec{\mathbf{n}}$, define a vector that spans two points on each line, and finally determine the minimum distance between the lines. (Take the origin to be at the lower corner of the first pipe.) Similarly, you may also develop the symmetric equations for each line and substitute directly into your formula.

Key Concepts

- In three dimensions, the direction of a line is described by a direction vector. The vector equation of a line with direction vector $\vec{\mathbf{v}} = \langle a, b, c \rangle$ passing through point $P = (x_0, y_0, z_0)$ is $\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{v}}$, where $\vec{\mathbf{r}}_0 = \langle x_0, y_0, z_0 \rangle$ is the position vector of point P. This equation can be rewritten to form the parametric equations of the line: $x = x_0 + ta, y = y_0 + tb$, and $z = z_0 + tc$. The line can also be described with the symmetric equations $\dfrac{x - x_0}{a} = \dfrac{y - y_0}{b} = \dfrac{z - z_0}{c}$.
- Let L be a line in space passing through point P with direction vector $\vec{\mathbf{v}}$. If Q is any point not on L, then the distance from Q to L is $d = \dfrac{\|\overrightarrow{PQ} \times \vec{\mathbf{v}}\|}{\|\vec{\mathbf{v}}\|}$.
- In three dimensions, two lines may be parallel but not equal, equal, intersecting, or skew.
- Given a point P and vector $\vec{\mathbf{n}}$, the set of all points Q satisfying equation $\vec{\mathbf{n}} \cdot \overrightarrow{PQ} = 0$ forms a plane. Equation $\vec{\mathbf{n}} \cdot \overrightarrow{PQ} = 0$ is known as the *vector equation of a plane*.
- The scalar equation of a plane containing point $P = (x_0, y_0, z_0)$ with normal vector $\vec{\mathbf{n}} = \langle a, b, c \rangle$ is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$. This equation can be expressed as $ax + by + cz + d = 0$, where $d = -ax_0 - by_0 - cz_0$. This form of the equation is sometimes called the general *form of the equation of a plane*.

- Suppose a plane with normal vector n passes through point Q. The distance D from the plane to point P not in the plane is given by

$$D = \|\text{proj}_{\vec{\mathbf{n}}}\overrightarrow{QP}\| = |\text{ comp}_{\vec{\mathbf{n}}}\vec{QP}| = \frac{|\vec{QP}\cdot\vec{\mathbf{n}}|}{\|\vec{\mathbf{n}}\|}.$$

- The normal vectors of parallel planes are parallel. When two planes intersect, they form a line.
- The measure of the angle θ between two intersecting planes can be found using the equation: $\cos\theta = \frac{|\vec{\mathbf{n}}_1\cdot\vec{\mathbf{n}}_2|}{\|\vec{\mathbf{n}}_1\|\|\vec{\mathbf{n}}_2\|}$, where $\vec{\mathbf{n}}_1$ and $\vec{\mathbf{n}}_2$ are normal vectors to the planes.
- The distance D from point (x_0, y_0, z_0) to plane $ax + by + cz + d = 0$ is given by

$$D = \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

.

Key Equations

- **Vector Equation of a Line**

  $\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{v}}$

- **Parametric Equations of a Line**

  $x = x_0 + ta,\ y = y_0 + tb$, and $z = z_0 + tc$

- **Symmetric Equations of a Line**

  $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$

- **Vector Equation of a Plane**

  $\vec{\mathbf{n}}\cdot\overrightarrow{PQ} = 0$

- **Scalar Equation of a Plane**

  $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

- **Distance between a Plane and a Point**

  $d = \|\text{proj}_{\vec{\mathbf{n}}}\overrightarrow{QP}\| = |\text{ comp}_{\vec{\mathbf{n}}}\overrightarrow{QP}| = \frac{|\overrightarrow{QP}\cdot\vec{\mathbf{n}}|}{\|\vec{\mathbf{n}}\|}$

Glossary

direction vector

a vector parallel to a line that is used to describe the direction, or orientation, of the line in space

general form of the equation of a plane

an equation in the form $ax + by + cz + d = 0$, where $\vec{\mathbf{n}} = \langle a, b, c\rangle$ is a normal vector of the plane, $P = (x_0, y_0, z_0)$ is a point on the plane, and $d = -ax_0 - by_0 - cz_0$

normal vector

a vector perpendicular to a plane

parametric equations of a line

the set of equations $x = x_0 + ta, y = y_0 + tb$, and $z = z_0 + tc$ describing the line with direction vector $v = \langle a, b, c\rangle$ passing through point (x_0, y_0, z_0)

scalar equation of a plane

the equation $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$ used to describe a plane containing point $P=(x_0,y_0,z_0)$ with normal vector $n=\langle a,b,c\rangle$ or its alternate form $ax+by+cz+d=0$, where $d=-ax_0-by_0-cz_0$

skew lines

two lines that are not parallel but do not intersect

symmetric equations of a line

the equations $\frac{x-x_0}{a}=\frac{y-y_0}{b}=\frac{z-z_0}{c}$ describing the line with direction vector $v=\langle a,b,c\rangle$ passing through point (x_0,y_0,z_0)

vector equation of a line

the equation $\vec{\mathbf{r}}=\vec{\mathbf{r}}_0+t\vec{\mathbf{v}}$ used to describe a line with direction vector $\vec{\mathbf{v}}=\langle a,b,c\rangle$ passing through point $P=(x_0,y_0,z_0)$, where $\vec{\mathbf{r}}_0=\langle x_0,y_0,z_0\rangle$, is the position vector of point P

vector equation of a plane

the equation $\vec{\mathbf{n}}\cdot\overrightarrow{PQ}=0$, where P is a given point in the plane, Q is any point in the plane, and $\vec{\mathbf{n}}$ is a normal vector of the plane

---

This page titled 12.5: Equations of Lines and Planes in Space is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.5E: Exercises for Section 12.5

In exercises 1 - 4, points P and Q are given. Let L be the line passing through points P and Q.

a. Find the vector equation of line L.

b. Find parametric equations of line L.

c. Find symmetric equations of line L.

d. Find parametric equations of the line segment determined by P and Q.

1) $P(-3,5,9), \quad Q(4,-7,2)$

Answer

a. $\vec{\mathbf{r}} = \langle -3,5,9\rangle + t\langle 7,-12,-7\rangle, \quad t \in R;$
b. $x = -3+7t, \quad y = 5-12t, \quad z = 9-7t, \quad t \in R;$
c. $\dfrac{x+3}{7} = \dfrac{y-5}{-12} = \dfrac{z-9}{-7};$
d. $x = -3+7t, \quad y = 5-12t, \quad z = 9-7t, \quad 0 \le t \le 1$

2) $P(4,0,5), \quad Q(2,3,1)$

3) $P(-1,0,5), \quad Q(4,0,3)$

Answer

a. $\vec{\mathbf{r}} = \langle -1,0,5\rangle + t\langle 5,0,-2\rangle, \quad t \in R;$
b. $x = -1+5t, y = 0, z = 5-2t, \quad t \in R;$
c. $\dfrac{x+1}{5} = \dfrac{z-5}{-2}, \quad y = 0;$
d. $x = -1+5t, \quad y = 0, \quad z = 5-2t, \quad t \in [0,1]$

4) $P(7,-2,6), \quad Q(-3,0,6)$

For exercises 5 - 8, point P and vector $\vec{\mathbf{v}}$ are given. Let L be the line passing through point P with direction $\vec{\mathbf{v}}$.

a. Find parametric equations of line L.

b. Find symmetric equations of line L.

c. Find the intersection of the line with the xy-plane.

5) $P(1,-2,3), \ \vec{\mathbf{v}} = \langle 1,2,3\rangle$

Answer

a. $x = 1+t, \quad y = -2+2t, \quad z = 3+3t, \quad t \in R;$
b. $\dfrac{x-1}{1} = \dfrac{y+2}{2} = \dfrac{z-3}{3};$
c. $(0,-4,0)$

6) $P(3,1,5), \ \vec{\mathbf{v}} = \langle 1,1,1\rangle$

7) $P(3,1,5), \ \vec{\mathbf{v}} = \overrightarrow{QR}$, where $Q(2,2,3)$ and $R(3,2,3)$

Answer

a. $x = 3+t, \quad y = 1, \quad z = 5, \quad t \in R;$
b. $y = 1, \quad z = 5;$
c. The line does not intersect the xy-plane.

8) $P(2,3,0), \ \vec{\mathbf{v}} = \overrightarrow{QR}$, where $Q(0,4,5)$ and $R(0,4,6)$

For exercises 9 and 10, line L is given.

a. Find a point P that belongs to the line and a direction vector $\vec{v}$ of the line. Express $\vec{v}$ in component form.

b. Find the distance from the origin to line L.

9) $x = 1 + t, \quad y = 3 + t, \quad z = 5 + 4t, \quad t \in R$

Answer

a. A possible point and direction vector are $P(1, 3, 5)$ and $\vec{v} = \langle 1, 1, 4 \rangle$, but these answers are not unique.
b. $\sqrt{3}$ units

10) $-x = y + 1, \quad z = 2$

11) Find the distance between point $A(-3, 1, 1)$ and the line of symmetric equations

$x = -y = -z.$

Answer

$\frac{2\sqrt{2}}{\sqrt{3}} = \frac{2\sqrt{6}}{3}$ units

12) Find the distance between point $A(4, 2, 5)$ and the line of parametric equations

$x = -1 - t, \ y = -t, \ z = 2, \ t \in R.$

For exercises 13 - 14, lines L_1 and L_2 are given.

a. Verify whether lines L_1 and L_2 are parallel.

b. If the lines L_1 and L_2 are parallel, then find the distance between them.

13) $L_1 : x = 1 + t, \quad y = t, \quad z = 2 + t, \quad t \in R$ and $L_2 : x - 3 = y - 1 = z - 3$

Answer

a. Parallel;
b. $\frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$ units

14) $L_1 : x = 2, \quad y = 1, \quad z = t, \quad t \in R$ and $L_2 : x = 1, \quad y = 1, \quad z = 2 - 3t, \quad t \in R$

15) Show that the line passing through points $P(3, 1, 0)$ and $Q(1, 4, -3)$ is perpendicular to the line with equation $x = 3t, \quad y = -32 + 8t, \quad z = -9 + 6t, \quad t \in R.$

Answer

$\overrightarrow{PQ} = \langle -2, 3, -3 \rangle$ is the direction vector of the line through points P and Q, and the direction vector of the line defined by the parametric equations above is $\vec{v} = \langle 3, 8, 6 \rangle$.
Since $\vec{v} \cdot \overrightarrow{PQ} = -6 + 24 - 18 = 0$, the two direction vectors are orthogonal.
Now all we need to show is that the two lines intersect.
The line through points $P(3, 1, 0)$ and $Q(1, 4, -3)$ has parametric equations: $x = 3 - 2u$, $y = 1 + 3u$, and $z = -3u$.
Setting the x- and z-coordinates of the two lines equal, we obtain the system of equations:

$$3t = 3 - 2u \quad \text{and} \quad -9 + 6t = -3u$$

Solving this system using substitution gives us, $u = -3$ and $t = 3$. Plugging these values of t and u back into the parametric equations of these two lines gives us the intersection point with coordinates $(9, -8, 9)$ on both lines.
Therefore the lines intersect and the line through points P and Q with direction vector $\overrightarrow{PQ}$ is perpendicular to the other line.

16) Are the lines of equations $x = -2 + 2t, \quad y = -6, \quad z = 2 + 6t, \quad t \in R$ and $x = -1 + t, \quad y = 1 + t, \quad z = t, \quad t \in R$, perpendicular to each other?

17) Find the point of intersection of the lines of equations $x = -2y = 3z$ and $x = -5 - t, \quad y = -1 + t, \quad z = t - 11, \quad t \in R.$

Answer

$(-12, 6, -4)$

18) Find the intersection point of the x-axis with the line of parametric equations $x = 10 + t, \quad y = 2 - 2t, \quad z = -3 + 3t, \quad t \in R.$

For exercises 19 - 22, lines L_1 and L_2 are given. Determine whether the lines are *equal, parallel but not equal, skew,* or *intersecting*.

19) $L_1 : x = y - 1 = -z$ and $L_2 : x - 2 = -y = \dfrac{z}{2}$

Answer

The lines are skew.

20) $L_1 : x = 2t, \quad y = 0, \quad z = 3, \quad t \in R$ and $L_2 : x = 0, \quad y = 8 + s, \quad z = 7 + s, \quad s \in R$

21) $L_1 : x = -1 + 2t, \quad y = 1 + 3t, \quad z = 7t, \quad t \in R$ and $L_2 : x - 1 = \frac{2}{3}(y - 4) = \frac{2}{7}z - 2$

Answer

The lines are equal.

22) $L_1 : 3x = y + 1 = 2z$ and $L_2 : x = 6 + 2t, \quad y = 17 + 6t, \quad z = 9 + 3t, \quad t \in R$

23) Consider line L of symmetric equations $x - 2 = -y = \dfrac{z}{2}$ and point $A(1, 1, 1)$.

a. Find parametric equations for a line parallel to L that passes through point A.

b. Find symmetric equations of a line skew to L and that passes through point A.

c. Find symmetric equations of a line that intersects L and passes through point A.

Answer

a. $x = 1 + t, \quad y = 1 - t, \quad z = 1 + 2t, \quad t \in R$

b. For instance, the line passing through A with direction vector $j : x = 1, z = 1$

c. For instance, the line passing through A and point $(2, 0, 0)$ that belongs to L is a line that intersects; $L : \frac{x-1}{-1} = y - 1 = z - 1$

24) Consider line L of parametric equations $x = t, \quad y = 2t, \quad z = 3, \quad t \in R.$

a. Find parametric equations for a line parallel to L that passes through the origin.

b. Find parametric equations of a line skew to L that passes through the origin.

c. Find symmetric equations of a line that intersects L and passes through the origin.

For exercises 25 - 28, point P and vector $\vec{\mathrm{n}}$ are given.

a. Find the scalar equation of the plane that passes through P and has normal vector $\vec{\mathrm{n}}$.

b. Find the general form of the equation of the plane that passes through P and has normal vector $\vec{\mathrm{n}}$.

25) $P(0, 0, 0), \quad \vec{\mathrm{n}} = 3\hat{\mathrm{i}} - 2\hat{\mathrm{j}} + 4\hat{\mathrm{k}}$

Answer

a. $3x - 2y + 4z = 0$

b. $3x - 2y + 4z = 0$

26) $P(3, 2, 2), \quad \vec{\mathrm{n}} = 2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - \hat{\mathrm{k}}$

27) $P(1, 2, 3), \quad \vec{\mathrm{n}} = \langle 1, 2, 3 \rangle$

Answer

a. $(x-1)+2(y-2)+3(z-3)=0$

b. $x+2y+3z-14=0$

28) $P(0,0,0), \quad \vec{\mathbf{n}}=\langle -3,2,-1\rangle$

For exercises 29 - 32, the equation of a plane is given.

a. Find normal vector $\vec{\mathbf{n}}$ to the plane. Express $\vec{\mathbf{n}}$ using standard unit vectors.

b. Find the intersections of the plane with each of the coordinate axes (its intercepts).

c. Sketch the plane.

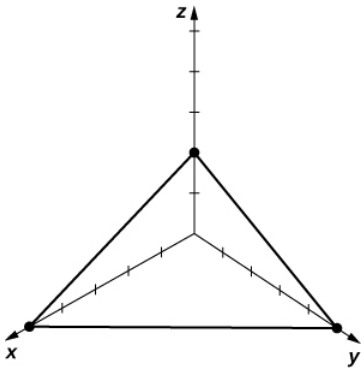
29) [T] $4x+5y+10z-20=0$

Answer

a. $\vec{\mathbf{n}}=4\hat{\mathbf{i}}+5\hat{\mathbf{j}}+10\hat{\mathbf{k}}$

b. $(5,0,0)$, $(0,4,0)$, and $(0,0,2)$

c.



30) $3x+4y-12=0$

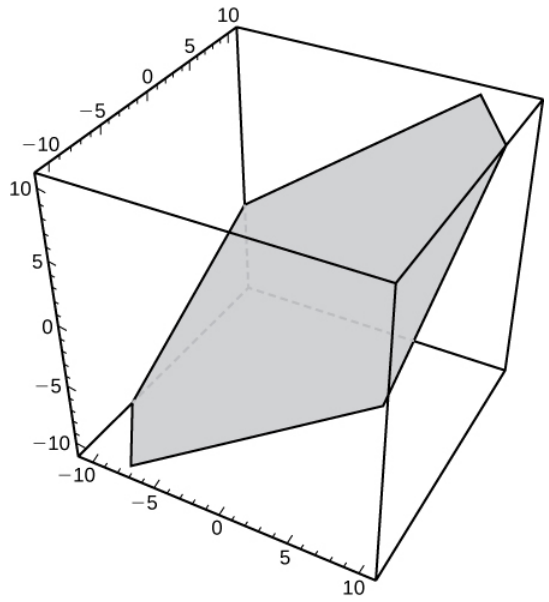
31) $3x-2y+4z=0$

Answer

a. $\vec{\mathbf{n}}=3\hat{\mathbf{i}}-2\hat{\mathbf{j}}+4\hat{\mathbf{k}}$

b. $(0,0,0)$

c.



32) $x + z = 0$

33) Given point $P(1, 2, 3)$ and vector $\vec{\mathbf{n}} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$, find point Q on the x-axis such that $\overrightarrow{PQ}$ and $\vec{\mathbf{n}}$ are orthogonal.

Answer

$(3, 0, 0)$

34) Show there is no plane perpendicular to $\vec{\mathbf{n}} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ that passes through points $P(1, 2, 3)$ and $Q(2, 3, 4)$.

35) Find parametric equations of the line passing through point $P(-2, 1, 3)$ that is perpendicular to the plane of equation $2x - 3y + z = 7$.

Answer

$x = -2 + 2t, \quad y = 1 - 3t, \quad z = 3 + t, \quad t \in R$

36) Find symmetric equations of the line passing through point $P(2, 5, 4)$ that is perpendicular to the plane of equation $2x + 3y - 5z = 0$.

37) Show that line $\dfrac{x-1}{2} = \dfrac{y+1}{3} = \dfrac{z-2}{4}$ is parallel to plane $x - 2y + z = 6$.

38) Find the real number α such that the line of parametric equations $x = t, \quad y = 2 - t, \quad z = 3 + t, \quad t \in R$ is parallel to the plane of equation $\alpha x + 5y + z - 10 = 0$.

For exercises 39 - 42, the equations of two planes are given.

a. Determine whether the planes are *parallel*, *orthogonal*, or *neither*.

b. If the planes are neither parallel nor orthogonal, then find the measure of the angle between the planes. Express the answer in degrees rounded to the nearest integer.

c. If the planes intersect, find the line of intersection of the planes, providing the parametric equations of this line.

39) [T] $x + y + z = 0, \quad 2x - y + z - 7 = 0$

Answer

a. The planes are neither parallel nor orthogonal.

b. $62°$

c. $x = -1 + 2t$

$y = -4 + t$
$z = 5 - 3t$

40) $5x - 3y + z = 4, \quad x + 4y + 7z = 1$

41) $x - 5y - z = 1, \quad 5x - 25y - 5z = -3$

Answer

a. The planes are parallel.

42) [T] $x - 3y + 6z = 4, \quad 5x + y - z = 4$

For exercises 43 - 46, determine whether the given line intersects with the given plane. If they do intersect, state the point of intersection.

43) Plane: $2x + y - z = 11$ Line: $x = 1 + t, \quad y = 3 - 2t, \quad z = 2 + 4t$

Answer

They intersect at point $(-1, 7, -6)$.

44) Plane: $-x + 2y + z = 2$ Line: $x = 1 + 2t, \quad y = -2 + t, \quad z = 5 - 3t$

Answer

They intersect at point $\left(-\frac{1}{3}, -\frac{8}{3}, 7\right)$.

45) Plane: $x - 3y + 2z = 4$ Line: $x = 2 - t, \quad y = t, \quad z = 4 + 2t$

Answer

The line does not intersect with this plane.

46) Plane: $x - 3y + 2z = 10$ Line: $x = 2 - t, \quad y = t, \quad z = 4 + 2t$

Answer

The line is actually fully contained in this plane, so every point on the line is on the plane. For example, when $t = 0$ we have the point, $(2, 0, 4)$.

47) Show that the lines of equations $x = t, \quad y = 1 + t, \quad z = 2 + t, \quad t \in R$, and $\frac{x}{2} = \frac{y - 1}{3} = z - 3$ are skew, and find the distance between them.

Answer

$\frac{1}{\sqrt{6}} = \frac{\sqrt{6}}{6}$ units

48) Show that the lines of equations $x = -1 + t, \quad y = -2 + t, \quad z = 3t, \quad t \in R$, and $x = 5 + s, \quad y = -8 + 2s, \quad z = 7s, \quad s \in R$ are skew, and find the distance between them.

49) Consider point $C(-3, 2, 4)$ and the plane of equation $2x + 4y - 3z = 8$.

a. Find the radius of the sphere with center C tangent to the given plane.

b. Find point P of tangency.

Answer

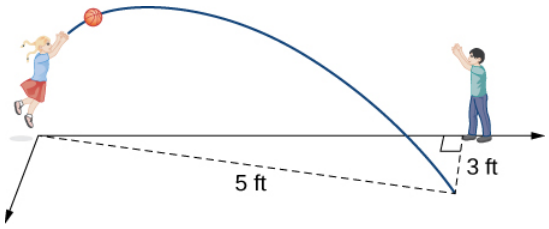
a. $r = \frac{18}{\sqrt{29}} = \frac{18\sqrt{29}}{29}$
b. $P\left(-\frac{51}{29}, \frac{130}{29}, \frac{62}{29}\right)$

50) Consider the plane of equation $x - y - z - 8 = 0$.

a. Find the equation of the sphere with center C at the origin that is tangent to the given plane.

b. Find parametric equations of the line passing through the origin and the point of tangency.

51) Two children are playing with a ball. The girl throws the ball to the boy. The ball travels in the air, curves 3 ft to the right, and falls 5 ft away from the girl (see the following figure). If the plane that contains the trajectory of the ball is perpendicular to the ground, find its equation.



Answer

$4x - 3y = 0$

52) [T] John allocates d dollars to consume monthly three goods of prices a, b, and c. In this context, the budget equation is defined as $ax + by + cz = d$, where $x \geq 0,\ y \geq 0$, and $z \geq 0$ represent the number of items bought from each of the goods. The budget set is given by $\{(x, y, z) \mid ax + by + cz \leq d,\ x \geq 0,\ y \geq 0,\ z \geq 0\}$, and the budget plane is the part of the plane of equation $ax + by + cz = d$ for which $x \geq 0,\ y \geq 0$, and $z \geq 0$. Consider $a = \$8,\ b = \$5,\ c = \$10$, and $d = \$500$.

a. Use a CAS to graph the budget set and budget plane.

b. For $z = 25$, find the new budget equation and graph the budget set in the same system of coordinates.

53) [T] Consider $\vec{\mathbf{r}}(t) = \langle \sin t, \cos t, 2t \rangle$ the position vector of a particle at time $t \in [0, 3]$, where the components of $\vec{\mathbf{r}}$ are expressed in centimeters and time is measured in seconds. Let $\overrightarrow{OP}$ be the position vector of the particle after 1 sec.

a. Determine the velocity vector $\vec{\mathbf{v}}(1)$ of the particle after 1 sec.

b. Find the scalar equation of the plane that is perpendicular to $v(1)$ and passes through point P. This plane is called the normal plane to the path of the particle at point P.

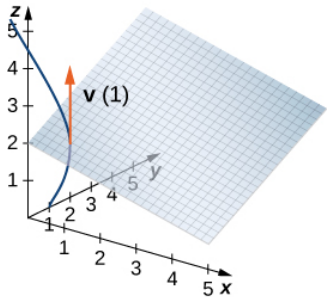
c. Use a CAS to visualize the path of the particle along with the velocity vector and normal plane at point P.

Answer

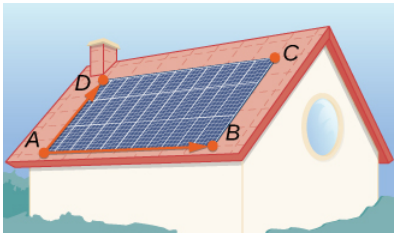
a. $\vec{\mathbf{v}}(1) = \langle \cos 1, -\sin 1, 2 \rangle$

b. $(\cos 1)(x - \sin 1) - (\sin 1)(y - \cos 1) + 2(z - 2) = 0$

c.



54) [T] A solar panel is mounted on the roof of a house. The panel may be regarded as positioned at the points of coordinates (in meters) $A(8, 0, 0),\ B(8, 18, 0),\ C(0, 18, 8)$, and $D(0, 0, 8)$ (see the following figure).



a. Find the general form of the equation of the plane that contains the solar panel by using points A, B, and C, and show that its normal vector is equivalent to $\overrightarrow{AB} \times \overrightarrow{AD}$.

b. Find parametric equations of line L_1 that passes through the center of the solar panel and has direction vector $\vec{s} = \frac{1}{\sqrt{3}}\hat{\mathbf{i}} + \frac{1}{\sqrt{3}}\hat{\mathbf{j}} + \frac{1}{\sqrt{3}}\hat{\mathbf{k}}$, which points toward the position of the Sun at a particular time of day.

c. Find symmetric equations of line L_2 that passes through the center of the solar panel and is perpendicular to it.

d. Determine the angle of elevation of the Sun above the solar panel by using the angle between lines L_1 and L_2.

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

Problems 15 and 43 - 46 created by Paul Seeburger

This page titled 12.5E: Exercises for Section 12.5 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.6: Quadric Surfaces

Learning Objectives

- Identify a cylinder as a type of three-dimensional surface.
- Recognize the main features of ellipsoids, paraboloids, and hyperboloids.
- Use traces to draw the intersections of quadric surfaces with the coordinate planes.

We have been exploring vectors and vector operations in three-dimensional space, and we have developed equations to describe lines, planes, and spheres. In this section, we use our knowledge of planes and spheres, which are examples of three-dimensional figures called *surfaces*, to explore a variety of other surfaces that can be graphed in a three-dimensional coordinate system.

Identifying Cylinders

The first surface we'll examine is the cylinder. Although most people immediately think of a hollow pipe or a soda straw when they hear the word cylinder, here we use the broad mathematical meaning of the term. As we have seen, cylindrical surfaces don't have to be circular. A rectangular heating duct is a cylinder, as is a rolled-up yoga mat, the cross-section of which is a spiral shape.

In the two-dimensional coordinate plane, the equation $x^2 + y^2 = 9$ describes a circle centered at the origin with radius 3. In three-dimensional space, this same equation represents a surface. Imagine copies of a circle stacked on top of each other centered on the z-axis (Figure $12.6.1$), forming a hollow tube. We can then construct a cylinder from the set of lines parallel to the z-axis passing through the circle $x^2 + y^2 = 9$ in the xy-plane, as shown in the figure. In this way, any curve in one of the coordinate planes can be extended to become a surface.

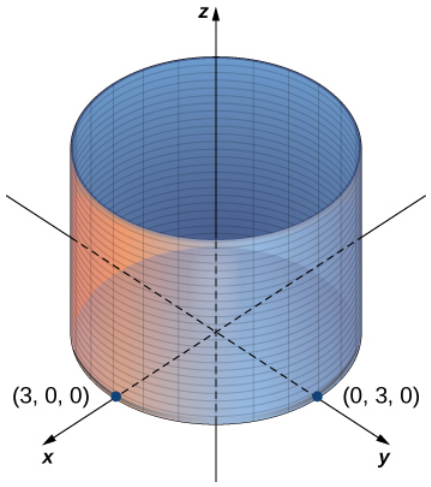


Figure $12.6.1$: In three-dimensional space, the graph of equation $x^2 + y^2 = 9$ is a cylinder with radius 3 centered on the z-axis. It continues indefinitely in the positive and negative directions.

Definition: cylinders and rulings

A set of lines parallel to a given line passing through a given curve is known as a cylindrical surface, or *cylinder***.** The parallel lines are called rulings**.**

From this definition, we can see that we still have a cylinder in three-dimensional space, even if the curve is not a circle. Any curve can form a cylinder, and the rulings that compose the cylinder may be parallel to any given line (Figure $12.6.2$).

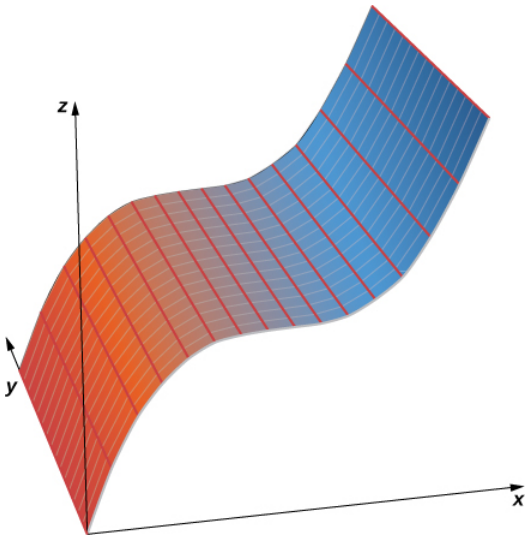


Figure 12.6.2: In three-dimensional space, the graph of equation $z = x^3$ is a cylinder, or a cylindrical surface with rulings parallel to the y-axis.

Example 12.6.1: Graphing Cylindrical Surfaces

Sketch the graphs of the following cylindrical surfaces.

a. $x^2 + z^2 = 25$
b. $z = 2x^2 - y$
c. $y = \sin x$

Solution

a. The variable y can take on any value without limit. Therefore, the lines ruling this surface are parallel to the y-axis. The intersection of this surface with the xz-plane forms a circle centered at the origin with radius 5 (see Figure 12.6.3).

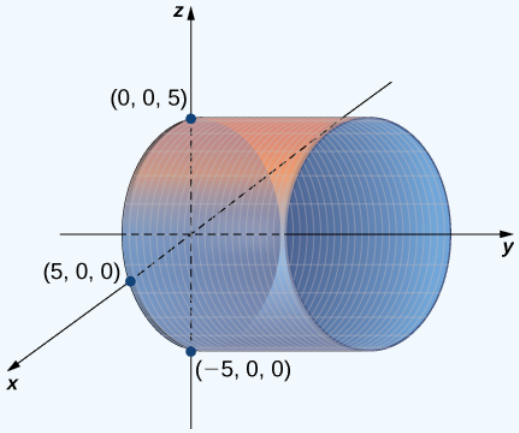


Figure 12.6.3: The graph of equation $x^2 + z^2 = 25$ is a cylinder with radius 5 centered on the y-axis.

b. In this case, the equation contains all three variables —x, y, and z— so none of the variables can vary arbitrarily. The easiest way to visualize this surface is to use a computer graphing utility (Figure 12.6.4).

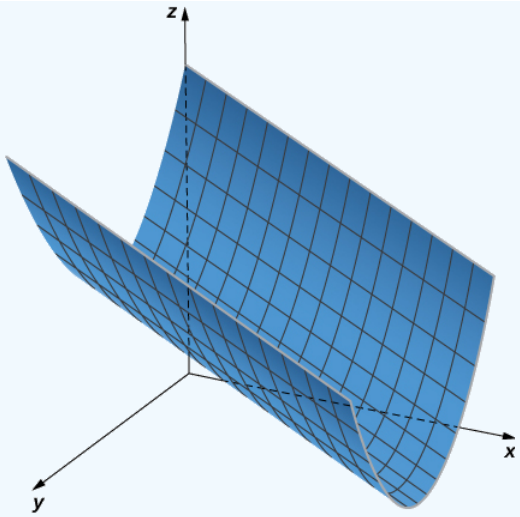


Figure 12.6.4

c. In this equation, the variable z can take on any value without limit. Therefore, the lines composing this surface are parallel to the z-axis. The intersection of this surface with the xy-plane outlines curve $y = \sin x$ (Figure 12.6.5).

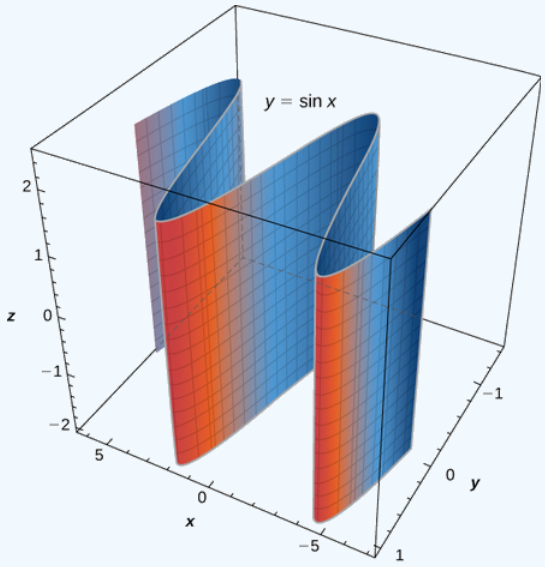


Figure 12.6.5: The graph of equation $y = \sin x$ is formed by a set of lines parallel to the z-axis passing through curve $y = \sin x$ in the xy-plane.

Exercise 12.6.1:

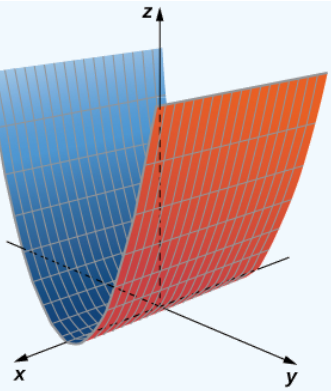
Sketch or use a graphing tool to view the graph of the cylindrical surface defined by equation $z = y^2$.

Hint

The variable x can take on any value without limit.

Answer

https://math.libretexts.org/@go/page/2591



When sketching surfaces, we have seen that it is useful to sketch the intersection of the surface with a plane parallel to one of the coordinate planes. These curves are called traces. We can see them in the plot of the cylinder in Figure 12.6.6.

Definition: traces

The *traces* of a surface are the cross-sections created when the surface intersects a plane parallel to one of the coordinate planes.

Traces are useful in sketching cylindrical surfaces. For a cylinder in three dimensions, though, only one set of traces is useful. Notice, in Figure 12.6.6, that the trace of the graph of $z = \sin x$ in the xz-plane is useful in constructing the graph. The trace in the xy-plane, though, is just a series of parallel lines, and the trace in the yz-plane is simply one line.

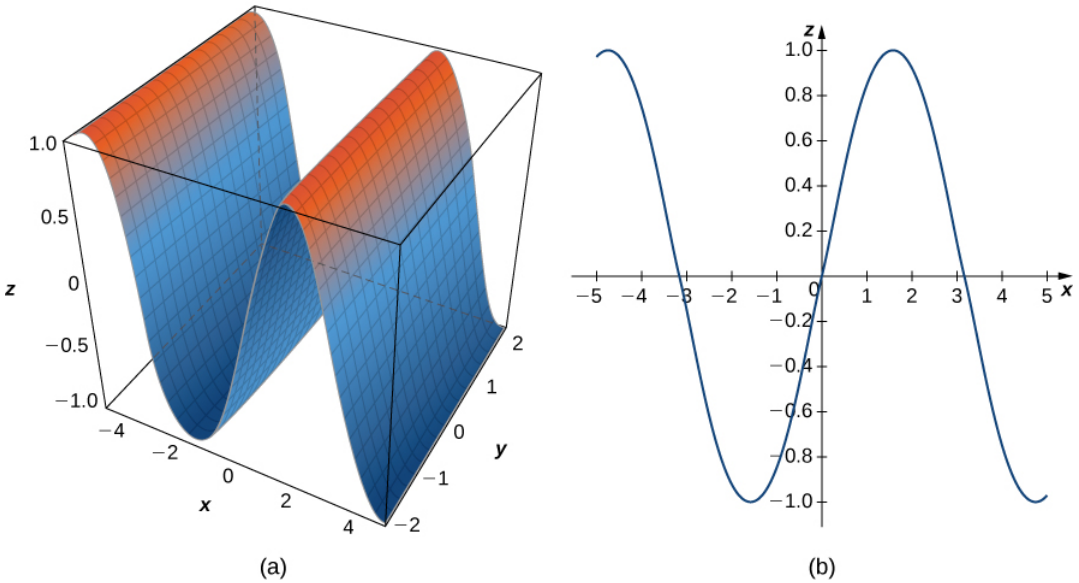


Figure 12.6.6: (a) This is one view of the graph of equation $z = \sin x$. (b) To find the trace of the graph in the xz-plane, set $y = 0$. The trace is simply a two-dimensional sine wave.

Cylindrical surfaces are formed by a set of parallel lines. Not all surfaces in three dimensions are constructed so simply, however. We now explore more complex surfaces, and traces are an important tool in this investigation.

Quadric Surfaces

We have learned about surfaces in three dimensions described by first-order equations; these are planes. Some other common types of surfaces can be described by second-order equations. We can view these surfaces as three-dimensional extensions of the conic sections we discussed earlier: the ellipse, the parabola, and the hyperbola. We call these graphs quadric surfaces

Definition: Quadric surfaces and conic sections

Quadric surfaces are the graphs of equations that can be expressed in the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Jz + K = 0.$$

When a quadric surface intersects a coordinate plane, the trace is a conic section.

An *ellipsoid* is a surface described by an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. Set $x = 0$ to see the trace of the ellipsoid in the yz -plane. To see the traces in the xy- and xz-planes, set $z = 0$ and $y = 0$, respectively. Notice that, if $a = b$, the trace in the xy-plane is a circle. Similarly, if $a = c$, the trace in the xz-plane is a circle and, if $b = c$, then the trace in the yz-plane is a circle. A sphere, then, is an ellipsoid with $a = b = c$.

Example 12.6.2: Sketching an Ellipsoid

Sketch the ellipsoid

$$\frac{x^2}{2^2} + \frac{y^2}{3^2} + \frac{z^2}{5^2} = 1.$$

Solution

Start by sketching the traces. To find the trace in the xy-plane, set $z = 0$: $\frac{x^2}{2^2} + \frac{y^2}{3^2} = 1$ (Figure 12.6.7). To find the other traces, first set $y = 0$ and then set $x = 0$.

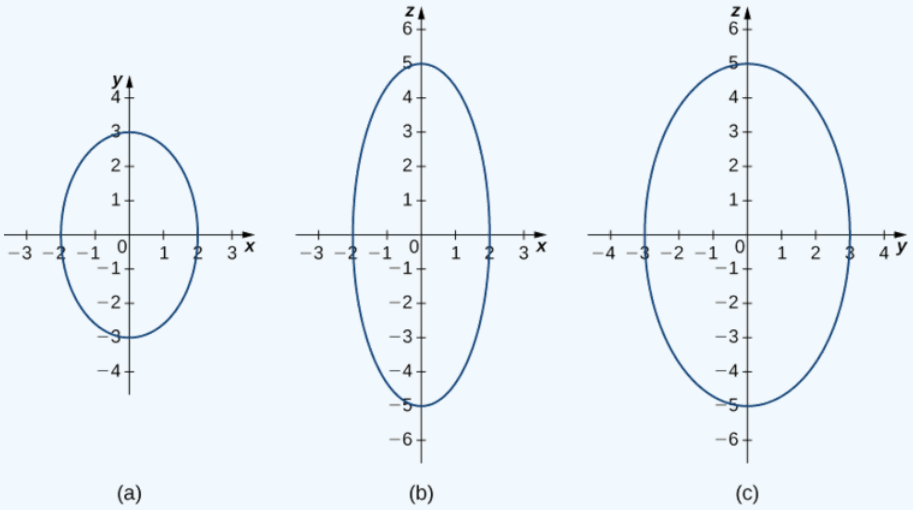


Figure 12.6.7: (a) This graph represents the trace of equation $\frac{x^2}{2^2} + \frac{y^2}{3^2} + \frac{z^2}{5^2} = 1$ in the xy-plane, when we set $z = 0$. (b) When we set $y = 0$, we get the trace of the ellipsoid in the xz-plane, which is an ellipse. (c) When we set $x = 0$, we get the trace of the ellipsoid in the yz-plane, which is also an ellipse.

Now that we know what traces of this solid look like, we can sketch the surface in three dimensions (Figure 12.6.8).

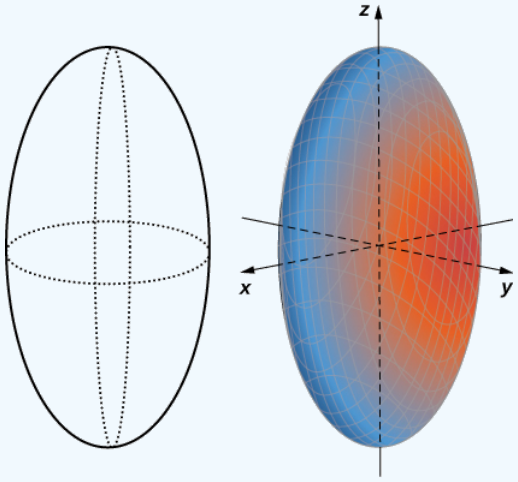


Figure 12.6.8: (a) The traces provide a framework for the surface. (b) The center of this ellipsoid is the origin.

The trace of an ellipsoid is an ellipse in each of the coordinate planes. However, this does not have to be the case for all quadric surfaces. Many quadric surfaces have traces that are different kinds of conic sections, and this is usually indicated by the name of the surface. For example, if a surface can be described by an equation of the form

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{z}{c}$$

then we call that surface an **elliptic paraboloid**. The trace in the xy-plane is an ellipse, but the traces in the xz-plane and yz-plane are parabolas (Figure 12.6.9). Other elliptic paraboloids can have other orientations simply by interchanging the variables to give us a different variable in the linear term of the equation $\frac{x^2}{a^2}+\frac{z^2}{c^2}=\frac{y}{b}$ or $\frac{y^2}{b^2}+\frac{z^2}{c^2}=\frac{x}{a}$.

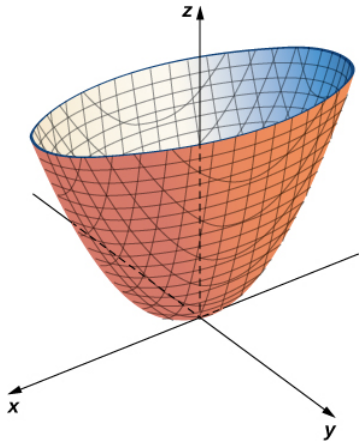


Figure 12.6.9: This quadric surface is called an elliptic paraboloid.

Example 12.6.3: Identifying Traces of Quadric Surfaces

Describe the traces of the elliptic paraboloid $x^2+\frac{y^2}{2^2}=\frac{z}{5}$.

Solution

To find the trace in the xy-plane, set $z=0: x^2+\frac{y^2}{2^2}=0$. The trace in the plane $z=0$ is simply one point, the origin. Since a single point does not tell us what the shape is, we can move up the z-axis to an arbitrary plane to find the shape of other traces of the figure.

The trace in plane $z = 5$ is the graph of equation $x^2 + \frac{y^2}{2^2} = 1$, which is an ellipse. In the xz-plane, the equation becomes $z = 5x^2$. The trace is a parabola in this plane and in any plane with the equation $y = b$.

In planes parallel to the yz-plane, the traces are also parabolas, as we can see in Figure 12.6.10.

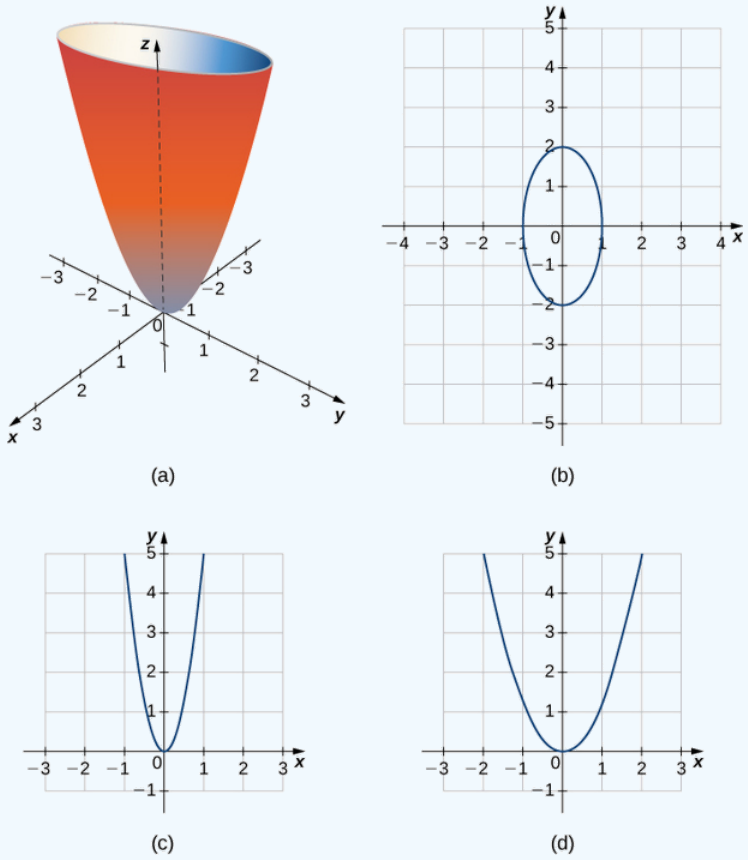


Figure 12.6.10: (a) The paraboloid $x^2 + \frac{y^2}{2^2} = \frac{z}{5}$. (b) The trace in plane $z = 5$. (c) The trace in the xz-plane. (d) The trace in the yz-plane.

Exercise 12.6.2:

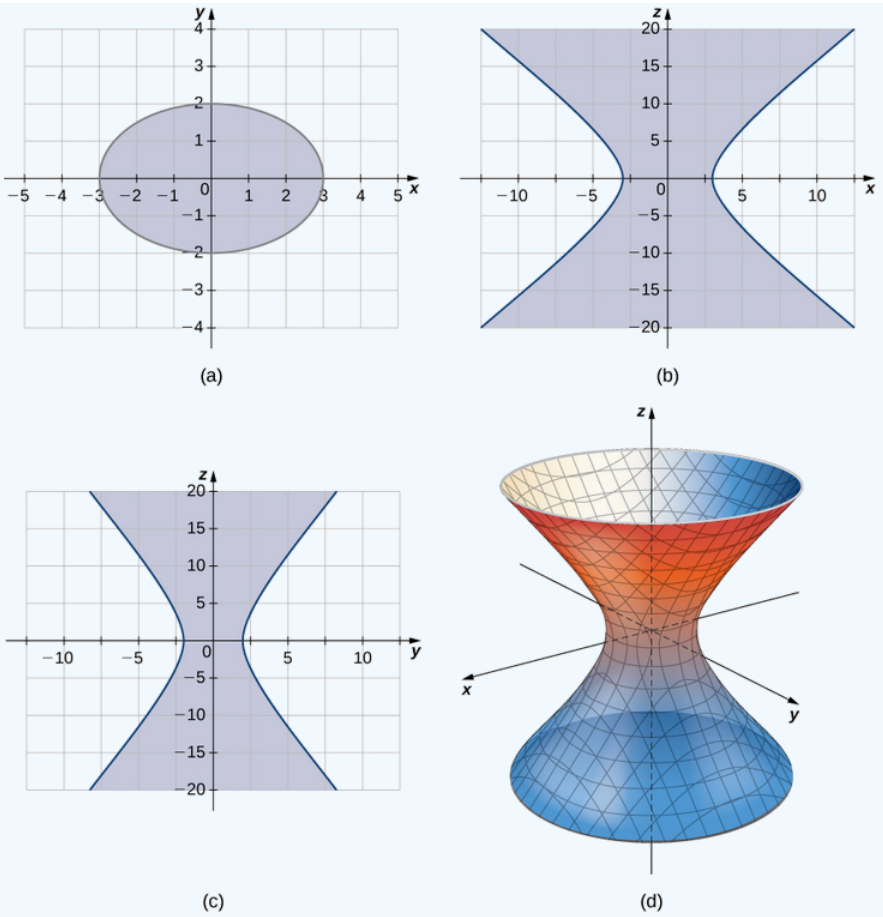
A hyperboloid of one sheet is any surface that can be described with an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$. Describe the traces of the hyperboloid of one sheet given by equation $\frac{x^2}{3^2} + \frac{y^2}{2^2} - \frac{z^2}{5^2} = 1$.

Hint

To find the traces in the coordinate planes, set each variable to zero individually.

Answer

The traces parallel to the xy-plane are ellipses and the traces parallel to the xz- and yz-planes are hyperbolas. Specifically, the trace in the xy-plane is ellipse $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$, the trace in the xz-plane is hyperbola $\frac{x^2}{3^2} - \frac{z^2}{5^2} = 1$, and the trace in the yz-plane is hyperbola $\frac{y^2}{2^2} - \frac{z^2}{5^2} = 1$ (see the following figure).



Hyperboloids of one sheet have some fascinating properties. For example, they can be constructed using straight lines, such as in the sculpture in Figure 12.6.11a. In fact, cooling towers for nuclear power plants are often constructed in the shape of a hyperboloid. The builders are able to use straight steel beams in the construction, which makes the towers very strong while using relatively little material (Figure 12.6.11b).

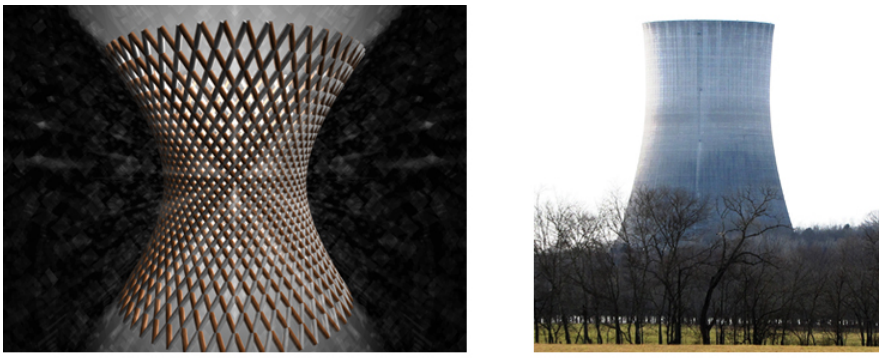


Figure 12.6.11: (a) A sculpture in the shape of a hyperboloid can be constructed of straight lines. (b) Cooling towers for nuclear power plants are often built in the shape of a hyperboloid.

Example 12.6.4: Chapter Opener: Finding the Focus of a Parabolic Reflector

Energy hitting the surface of a parabolic reflector is concentrated at the focal point of the reflector (Figure 12.6.12). If the surface of a parabolic reflector is described by equation $\frac{x^2}{100}+\frac{y^2}{100}=\frac{z}{4}$, where is the focal point of the reflector?

https://math.libretexts.org/@go/page/2591



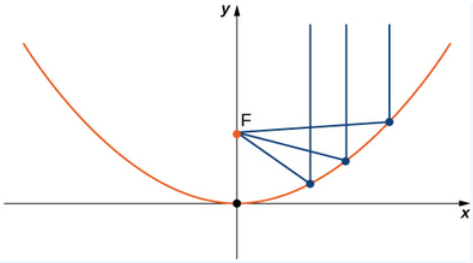


Figure 12.6.12: Energy reflects off of the parabolic reflector and is collected at the focal point. (credit: modification of CGP Grey, Wikimedia Commons)

Solution

Since z is the first-power variable, the axis of the reflector corresponds to the z-axis. The coefficients of x^2 and y^2 are equal, so the cross-section of the paraboloid perpendicular to the z-axis is a circle. We can consider a trace in the xz-plane or the yz-plane; the result is the same. Setting $y = 0$, the trace is a parabola opening up along the z-axis, with standard equation $x^2 = 4pz$, where p is the focal length of the parabola. In this case, this equation becomes $x^2 = 100 \cdot \frac{z}{4} = 4pz$ or $25 = 4p$. So p is 6.25 m, which tells us that the focus of the paraboloid is 6.25 m up the axis from the vertex. Because the vertex of this surface is the origin, the focal point is $(0, 0, 6.25)$.

Seventeen standard quadric surfaces can be derived from the general equation

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Jz + K = 0.$$

The following figures summarize the most important ones.

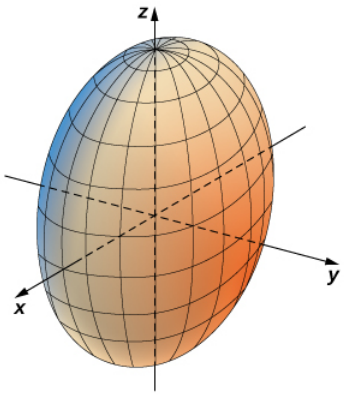
Characteristics of Common Quadric Surfaces

Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Traces
In plane $z = p$: an ellipse
In plane $y = q$: an ellipse
In plane $x = r$: an ellipse

If $a = b = c$, then this surface is a sphere.

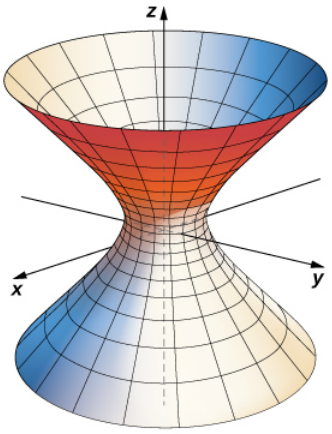


Hyperboloid of One Sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Traces
In plane $z = p$: an ellipse
In plane $y = q$: a hyperbola
In plane $x = r$: a hyperbola

In the equation for this surface, two of the variables have positive coefficients and one has a negative coefficient. The axis of the surface corresponds to the variable with the negative coefficient.



Hyperboloid of Two Sheets

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Traces
In plane $z = p$: an ellipse or the empty set (no trace)
In plane $y = q$: a hyperbola
In plane $x = r$: a hyperbola

In the equation for this surface, two of the variables have negative coefficients and one has a positive coefficient. The axis of the surface corresponds to the variable with a positive coefficient. The surface does not intersect the coordinate plane perpendicular to the axis.

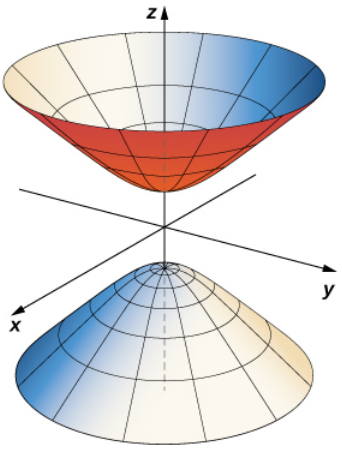


Figure 12.6.13: Characteristics of Common Quadric Surfaces: Ellipsoid, Hyperboloid of One Sheet, Hyperboloid of Two Sheets.

https://math.libretexts.org/@go/page/2591

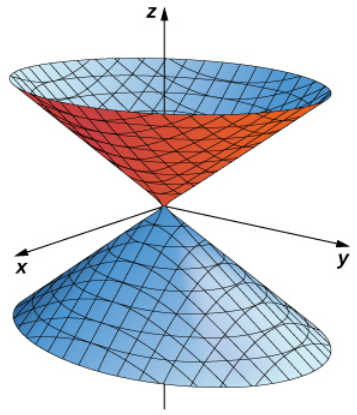
Characteristics of Common Quadric Surfaces

Elliptic Cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

Traces
In plane $z = p$: an ellipse
In plane $y = q$: a hyperbola
In plane $x = r$: a hyperbola
In the xz – plane: a pair of lines that intersect at the origin
In the yz – plane: a pair of lines that intersect at the origin

The axis of the surface corresponds to the variable with a negative coefficient. The traces in the coordinate planes parallel to the axis are intersecting lines.

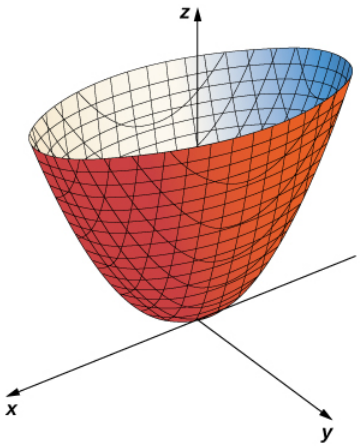


Elliptic Paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Traces
In plane $z = p$: an ellipse
In plane $y = q$: a parabola
In plane $x = r$: a parabola

The axis of the surface corresponds to the linear variable.



Hyperbolic Paraboloid

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

Traces
In plane $z = p$: a hyperbola
In plane $y = q$: a parabola
In plane $x = r$: a parabola

The axis of the surface corresponds to the linear variable.

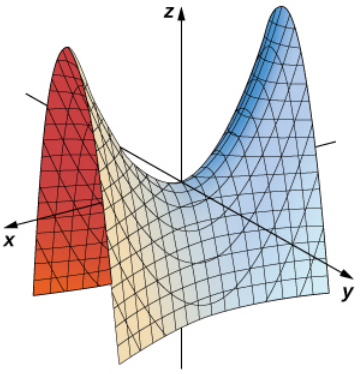


Figure 12.6.14: Characteristics of Common Quadric Surfaces: Elliptic Cone, Elliptic Paraboloid, Hyperbolic Paraboloid.

Example 12.6.5: Identifying Equations of Quadric Surfaces

Identify the surfaces represented by the given equations.

a. $16x^2 + 9y^2 + 16z^2 = 144$
b. $9x^2 - 18x + 4y^2 + 16y - 36z + 25 = 0$

Solution

a. The $x, y,$ and z terms are all squared, and are all positive, so this is probably an ellipsoid. However, let's put the equation into the standard form for an ellipsoid just to be sure. We have

$$16x^2 + 9y^2 + 16z^2 = 144.$$

Dividing through by 144 gives

$$\frac{x^2}{9} + \frac{y^2}{16} + \frac{z^2}{9} = 1.$$

So, this is, in fact, an ellipsoid, centered at the origin.

b. We first notice that the z term is raised only to the first power, so this is either an elliptic paraboloid or a hyperbolic paraboloid. We also note there are x terms and y terms that are not squared, so this quadric surface is not centered at the origin. We need to complete the square to put this equation in one of the standard forms. We have

$$\begin{aligned} 9x^2 - 18x + 4y^2 + 16y - 36z + 25 &= 0 \\ 9x^2 - 18x + 4y^2 + 16y + 25 &= 36z \\ 9(x^2 - 2x) + 4(y^2 + 4y) + 25 &= 36z \\ 9(x^2 - 2x + 1 - 1) + 4(y^2 + 4y + 4 - 4) + 25 &= 36z \\ 9(x-1)^2 - 9 + 4(y+2)^2 - 16 + 25 &= 36z \\ 9(x-1)^2 + 4(y+2)^2 &= 36z \\ \frac{(x-1)^2}{4} + \frac{(y+2)^2}{9} &= z. \end{aligned}$$

This is an elliptic paraboloid centered at $(1, -2, 0)$.

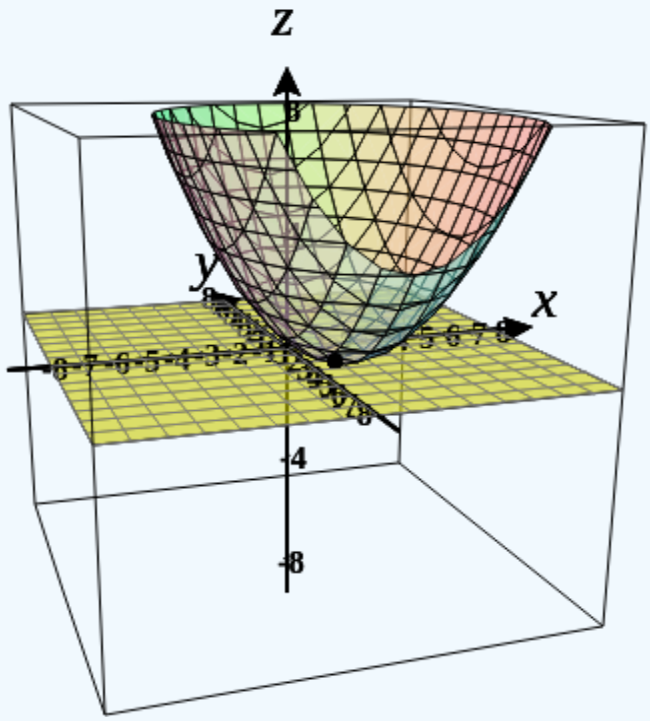


Figure 12.6.15: Plot of $z = \frac{(x-1)^2}{4} + \frac{(y+2)^2}{9}$ in CalcPlot3D.

Exercise 12.6.3

Identify the surface represented by equation $9x^2 + y^2 - z^2 + 2z - 10 = 0$.

Hint

Look at the signs and powers of the x, y, and z terms

Answer

Hyperboloid of one sheet, centered at $(0, 0, 1)$.

Key Concepts

- A set of lines parallel to a given line passing through a given curve is called a *cylinder*, or a cylindrical surface. The parallel lines are called *rulings*.
- The intersection of a three-dimensional surface and a plane is called a trace. To find the trace in the xy-, yz-, or xz-planes, set $z = 0$, $x = 0$, or $y = 0$, respectively.
- Quadric surfaces are three-dimensional surfaces with traces composed of conic sections. Every quadric surface can be expressed with an equation of the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Jz + K = 0.$$

- To sketch the graph of a quadric surface, start by sketching the traces to understand the framework of the surface.
- Important quadric surfaces are summarized in Figures 12.6.13 and 12.6.14.

Glossary

cylinder
a set of lines parallel to a given line passing through a given curve

ellipsoid
a three-dimensional surface described by an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$; all traces of this surface are ellipses

elliptic cone
a three-dimensional surface described by an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$; traces of this surface include ellipses and intersecting lines

elliptic paraboloid
a three-dimensional surface described by an equation of the form $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$; traces of this surface include ellipses and parabolas

hyperboloid of one sheet
a three-dimensional surface described by an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$; traces of this surface include ellipses and hyperbolas

hyperboloid of two sheets
a three-dimensional surface described by an equation of the form $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$; traces of this surface include ellipses and hyperbolas

quadric surfaces
surfaces in three dimensions having the property that the traces of the surface are conic sections (ellipses, hyperbolas, and parabolas)

rulings

parallel lines that make up a cylindrical surface

trace

the intersection of a three-dimensional surface with a coordinate plane

---

This page titled 12.6: Quadric Surfaces is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

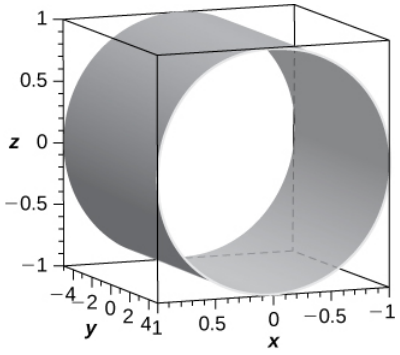
12.6E: Exercises for Section 12.6

For exercises 1 - 6, sketch and describe the cylindrical surface of the given equation.

1) [T] $x^2 + z^2 = 1$

Answer

The surface is a cylinder with the rulings parallel to the y-axis.

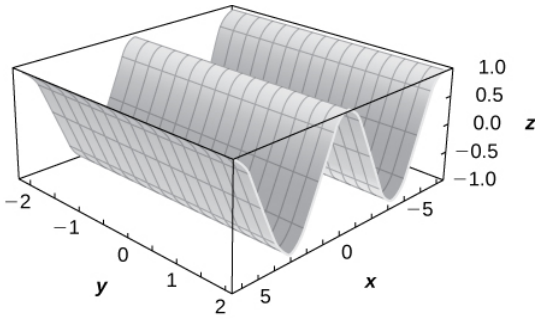


2) [T] $x^2 + y^2 = 9$

3) [T] $z = \cos\left(\frac{\pi}{2} + x\right)$

Answer

The surface is a cylinder with rulings parallel to the y-axis.



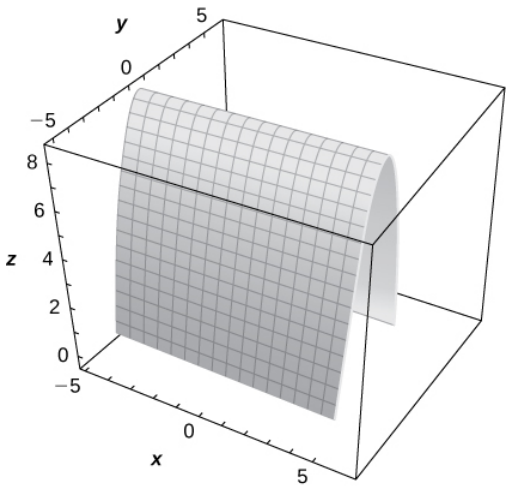
4) [T] $z = e^x$

5) [T] $z = 9 - y^2$

Answer

The surface is a cylinder with rulings parallel to the x-axis.

https://math.libretexts.org/@go/page/67610



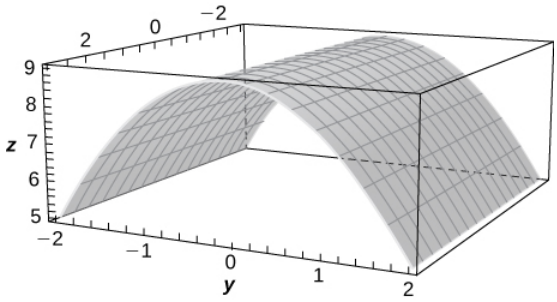
6) [T] $z = \ln x$

For exercises 7 - 10, the graph of a quadric surface is given.

a. Specify the name of the quadric surface.

b. Determine the axis of symmetry of the quadric surface.

7)

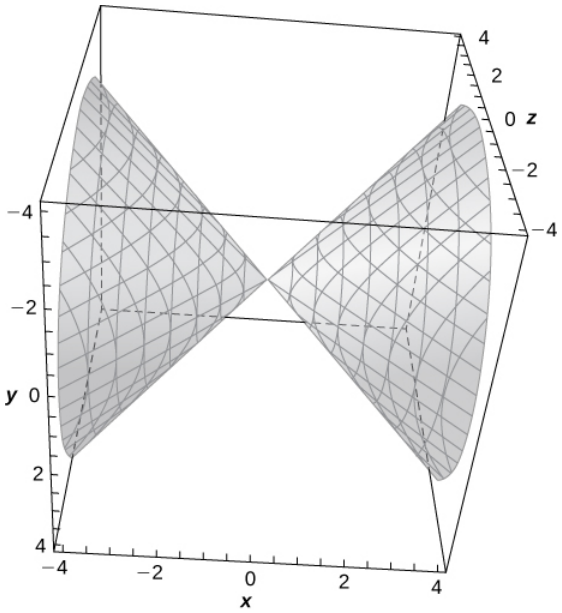


Answer

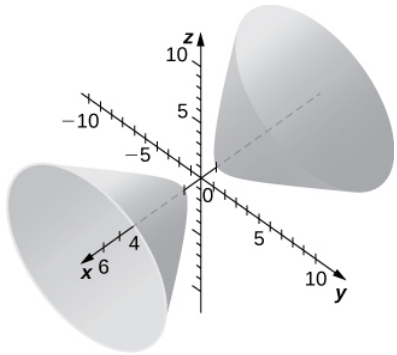
a. Cylinder; b. The x-axis

8)

https://math.libretexts.org/@go/page/67610



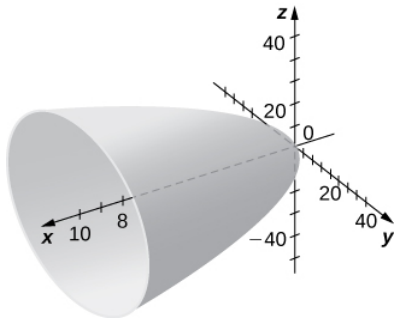
9)



Answer

a. Hyperboloid of two sheets; b. The x-axis

10)



For exercises 11 - 16, match the given quadric surface with its corresponding equation in standard form.

a. $\frac{x^2}{4}+\frac{y^2}{9}-\frac{z^2}{12}=1$

b. $\frac{x^2}{4}-\frac{y^2}{9}-\frac{z^2}{12}=1$

c. $\frac{x^2}{4}+\frac{y^2}{9}+\frac{z^2}{12}=1$

d. $z = 4x^2 + 3y^2$

e. $z = 4x^2 - y^2$

f. $4x^2 + y^2 - z^2 = 0$

11) Hyperboloid of two sheets

Answer

b.

12) Ellipsoid

13) Elliptic paraboloid

Answer

d.

14) Hyperbolic paraboloid

15) Hyperboloid of one sheet

Answer

a.

16) Elliptic cone

For exercises 17 - 28, rewrite the given equation of the quadric surface in standard form. Identify the surface.

17) $-x^2 + 36y^2 + 36z^2 = 9$

Answer

$-\frac{x^2}{9} + \frac{y^2}{\frac{1}{4}} + \frac{z^2}{\frac{1}{4}} = 1$, hyperboloid of one sheet with the x-axis as its axis of symmetry

18) $-4x^2 + 25y^2 + z^2 = 100$

19) $-3x^2 + 5y^2 - z^2 = 10$

Answer

$-\frac{x^2}{\frac{10}{3}} + \frac{y^2}{2} - \frac{z^2}{10} = 1$, hyperboloid of two sheets with the y-axis as its axis of symmetry

20) $3x^2 - y^2 - 6z^2 = 18$

21) $5y = x^2 - z^2$

Answer

$y = -\frac{z^2}{5} + \frac{x^2}{5}$, hyperbolic paraboloid with the y-axis as its axis of symmetry

22) $8x^2 - 5y^2 - 10z = 0$

23) $x^2 + 5y^2 + 3z^2 - 15 = 0$

Answer

$\frac{x^2}{15} + \frac{y^2}{3} + \frac{z^2}{5} = 1$, ellipsoid

24) $63x^2 + 7y^2 + 9z^2 - 63 = 0$

25) $x^2 + 5y^2 - 8z^2 = 0$

Answer

$\frac{x^2}{40}+\frac{y^2}{8}-\frac{z^2}{5}=0$, elliptic cone with the z-axis as its axis of symmetry

26) $5x^2-4y^2+20z^2=0$

27) $6x=3y^2+2z^2$

Answer

$x=\frac{y^2}{2}+\frac{z^2}{3}$, elliptic paraboloid with the x-axis as its axis of symmetry

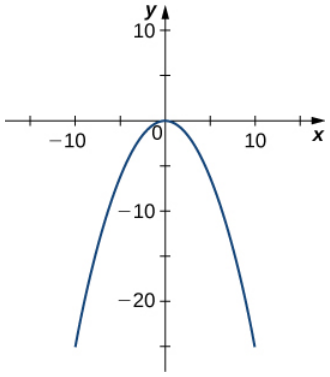
28) $49y=x^2+7z^2$

For exercises 29 - 34, find the trace of the given quadric surface in the specified plane of coordinates and sketch it.

29) [T] $x^2+z^2+4y=0, \quad z=0$

Answer

Parabola $y=-\frac{x^2}{4}$,

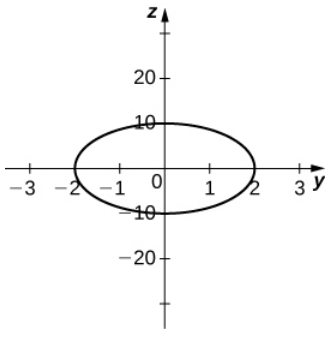


30) [T] $x^2+z^2+4y=0, \quad x=0$

31) [T] $-4x^2+25y^2+z^2=100, \quad x=0$

Answer

Ellipse $\frac{y^2}{4}+\frac{z^2}{100}=1$,

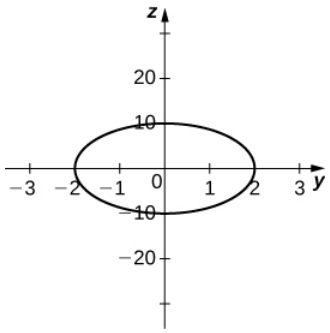


32) [T] $-4x^2+25y^2+z^2=100, \quad y=0$

33) [T] $x^2+\frac{y^2}{4}+\frac{z^2}{100}=1, \quad x=0$

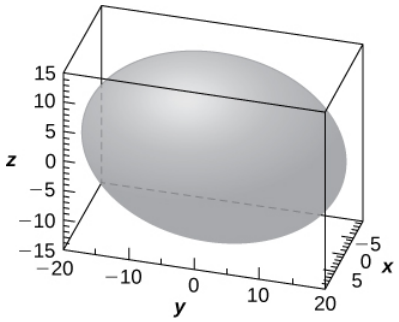
Answer

Ellipse $\frac{y^2}{4}+\frac{z^2}{100}=1$,



34) [T] $x^2 - y - z^2 = 1, \quad y = 0$

35) Use the graph of the given quadric surface to answer the questions.



a. Specify the name of the quadric surface.

b. Which of the equations—$16x^2 + 9y^2 + 36z^2 = 3600,\ 9x^2 + 36y^2 + 16z^2 = 3600$, or $36x^2 + 9y^2 + 16z^2 = 3600$—corresponds to the graph?

c. Use b. to write the equation of the quadric surface in standard form.

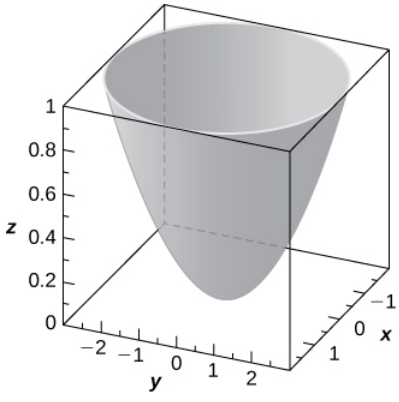
Answer

a. Ellipsoid

b. The third equation

c. $\frac{x^2}{100} + \frac{y^2}{400} + \frac{z^2}{225} = 1$

36) Use the graph of the given quadric surface to answer the questions.



a. Specify the name of the quadric surface.

b. Which of the equations—$36z = 9x^2 + y^2,\ 9x^2 + 4y^2 = 36z$, or $-36z = -81x^2 + 4y^2$—corresponds to the graph above?

c. Use b. to write the equation of the quadric surface in standard form.

For exercises 37 - 42, the equation of a quadric surface is given.

a. Use the method of completing the square to write the equation in standard form.

b. Identify the surface.

37) $x^2+2z^2+6x-8z+1=0$

Answer

a. $\dfrac{(x+3)^2}{16}+\dfrac{(z-2)^2}{8}=1$

b. Cylinder centered at $(-3,2)$ with rulings parallel to the y-axis

38) $4x^2-y^2+z^2-8x+2y+2z+3=0$

39) $x^2+4y^2-4z^2-6x-16y-16z+5=0$

Answer

a. $\dfrac{(x-3)^2}{4}+(y-2)^2-(z+2)^2=1$

b. Hyperboloid of one sheet centered at $(3,2,-2)$, with the z-axis as its axis of symmetry

40) $x^2+z^2-4y+4=0$

41) $x^2+\dfrac{y^2}{4}-\dfrac{z^2}{3}+6x+9=0$

Answer

a. $(x+3)^2+\dfrac{y^2}{4}-\dfrac{z^2}{3}=0$

b. Elliptic cone centered at $(-3,0,0)$, with the z-axis as its axis of symmetry

42) $x^2-y^2+z^2-12z+2x+37=0$

43) Write the standard form of the equation of the ellipsoid centered at the origin that passes through points $A(2,0,0),\ B(0,0,1)$, and $C\left(12,\sqrt{11},\frac{1}{2}\right)$.

Answer

$\dfrac{x^2}{4}+\dfrac{y^2}{16}+z^2=1$

44) Write the standard form of the equation of the ellipsoid centered at point $P(1,1,0)$ that passes through points $A(6,1,0),\ B(4,2,0)$ and $C(1,2,1)$.

45) Determine the intersection points of elliptic cone $x^2-y^2-z^2=0$ with the line of symmetric equations $\dfrac{x-1}{2}=\dfrac{y+1}{3}=z$.

Answer

$(1,-1,0)$ and $\left(\frac{13}{3},4,\frac{5}{3}\right)$

46) Determine the intersection points of parabolic hyperboloid $z=3x^2-2y^2$ with the line of parametric equations $x=3t,\ y=2t,\ z=19t$, where $t\in R$.

47) Find the equation of the quadric surface with points $P(x,y,z)$ that are equidistant from point $Q(0,-1,0)$ and plane of equation $y=1$. Identify the surface.

Answer

$x^2+z^2+4y=0$, elliptic paraboloid

48) Find the equation of the quadric surface with points $P(x,y,z)$ that are equidistant from point $Q(0,2,0)$ and plane of equation $y=-2$. Identify the surface.

49) If the surface of a parabolic reflector is described by equation $400z = x^2 + y^2$, find the focal point of the reflector.

Answer

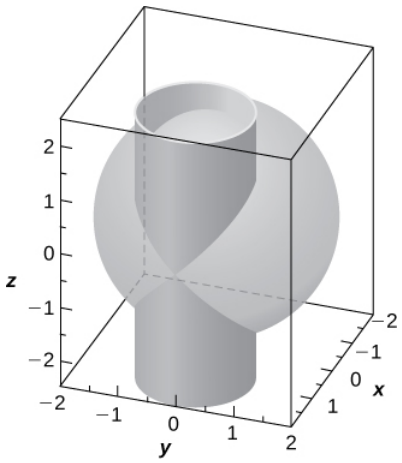
$(0, 0, 100)$

50) Consider the parabolic reflector described by equation $z = 20x^2 + 20y^2$. Find its focal point.

51) Show that quadric surface $x^2 + y^2 + z^2 + 2xy + 2xz + 2yz + x + y + z = 0$ reduces to two parallel planes.

52) Show that quadric surface $x^2 + y^2 + z^2 - 2xy - 2xz + 2yz - 1 = 0$ reduces to two parallel planes passing.

53) [T] The intersection between cylinder $(x - 1)^2 + y^2 = 1$ and sphere $x^2 + y^2 + z^2 = 4$ is called a *Viviani curve*.



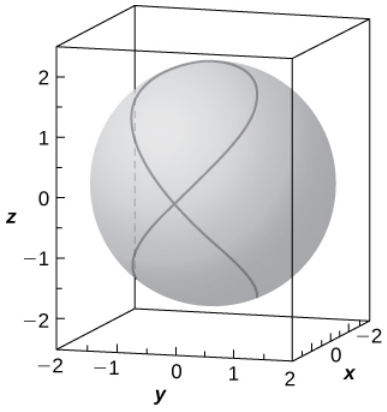
a. Solve the system consisting of the equations of the surfaces to find the equation of the intersection curve. (Hint: Find x and y in terms of z.)

b. Use a computer algebra system (CAS) or CalcPlot3D to visualize the intersection curve on sphere $x^2 + y^2 + z^2 = 4$.

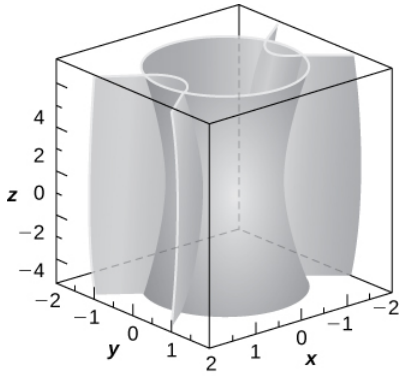
Answer

a. $x = 2 - \dfrac{z^2}{2}, \quad y = \pm\dfrac{z}{2}\sqrt{4 - z^2}$, where $z \in [-2, 2]$;

b.



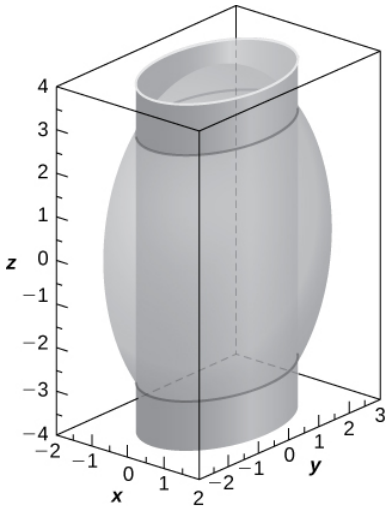
54) Hyperboloid of one sheet $25x^2 + 25y^2 - z^2 = 25$ and elliptic cone $-25x^2 + 75y^2 + z^2 = 0$ are represented in the following figure along with their intersection curves. Identify the intersection curves and find their equations (Hint: Find y from the system consisting of the equations of the surfaces.)



55) [T] Use a CAS or CalcPlot3D to create the intersection between cylinder $9x^2 + 4y^2 = 18$ and ellipsoid $36x^2 + 16y^2 + 9z^2 = 144$, and find the equations of the intersection curves.

Answer

two ellipses of equations $\frac{x^2}{2} + \frac{y^2}{\frac{9}{2}} = 1$ in planes $z = \pm 2\sqrt{2}$



56) [T] A spheroid is an ellipsoid with two equal semiaxes. For instance, the equation of a spheroid with the z-axis as its axis of symmetry is given by $\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{c^2} = 1$, where a and c are positive real numbers. The spheroid is called oblate if $c < a$, and prolate for $c > a$.

a. The eye cornea is approximated as a prolate spheroid with an axis that is the eye, where $a = 8.7mm$ and $c = 9.6mm$.Write the equation of the spheroid that models the cornea and sketch the surface.

b. Give two examples of objects with prolate spheroid shapes.

57) [T] In cartography, Earth is approximated by an oblate spheroid rather than a sphere. The radii at the equator and poles are approximately 3963mi and 3950mi, respectively.

a. Write the equation in standard form of the ellipsoid that represents the shape of Earth. Assume the center of Earth is at the origin and that the trace formed by plane $z = 0$ corresponds to the equator.

b. Sketch the graph.

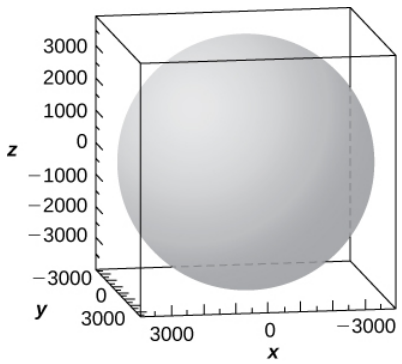
c. Find the equation of the intersection curve of the surface with plane $z = 1000$ that is parallel to the xy-plane. The intersection curve is called a *parallel.*

d. Find the equation of the intersection curve of the surface with plane $x + y = 0$ that passes through the z-axis. The intersection curve is called a meridian.

Answer

a. $\frac{x^2}{3963^2}+\frac{y^2}{3963^2}+\frac{z^2}{3950^2}=1$

b.



c. The intersection curve is the ellipse of equation $\frac{x^2}{3963^2}+\frac{y^2}{3963^2}=\frac{(2950)(4950)}{3950^2}$, and the intersection is an ellipse.

d. The intersection curve is the ellipse of equation $\frac{2y^2}{3963^2}+\frac{z^2}{3950^2}=1$.

58) [T] A set of buzzing stunt magnets (or “rattlesnake eggs”) includes two sparkling, polished, superstrong spheroid-shaped magnets well-known for children’s entertainment. Each magnet is 1.625 in. long and 0.5 in. wide at the middle. While tossing them into the air, they create a buzzing sound as they attract each other.

a. Write the equation of the prolate spheroid centered at the origin that describes the shape of one of the magnets.

b. Write the equations of the prolate spheroids that model the shape of the buzzing stunt magnets. Use a CAS or CalcPlot3D to create the graphs.

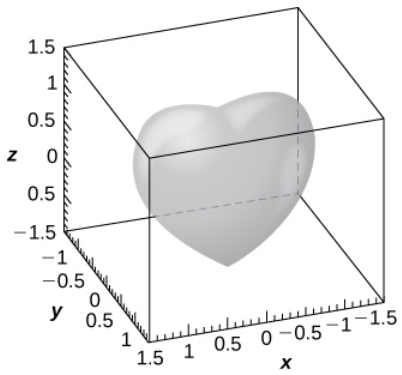
59) [T] A heart-shaped surface is given by equation $(x^2+\frac{9}{4}y^2+z^2-1)^3-x^2z^3-\frac{9}{80}y^2z^3=0.$

a. Use a CAS or CalcPlot3D to graph the surface that models this shape.

b. Determine and sketch the trace of the heart-shaped surface on the xz-plane.

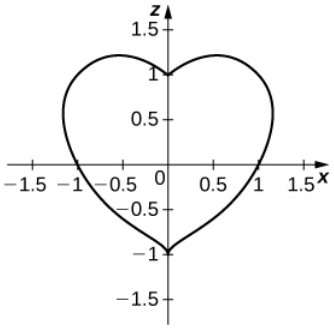
Answer

a.

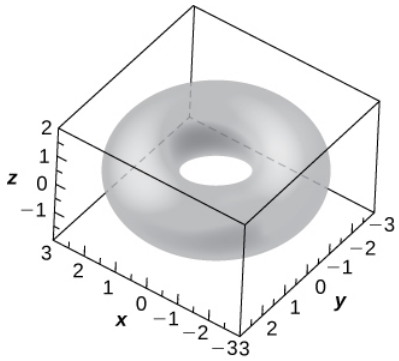


b. The intersection curve is $(x^2+z^2-1)^3-x^2z^3=0.$

https://math.libretexts.org/@go/page/67610



60) [T] The ring torus symmetric about the z-axis is a special type of surface in topology and its equation is given by $(x^2+y^2+z^2+R^2-r^2)^2 = 4R^2(x^2+y^2)$, where $R > r > 0$. The numbers R and r are called are the major and minor radii, respectively, of the surface. The following figure shows a ring torus for which $R = 2$ and $r = 1$.



a. Write the equation of the ring torus with $R = 2$ and $r = 1$, and use a CAS or CalcPlot3D to graph the surface. Compare the graph with the figure given.

b. Determine the equation and sketch the trace of the ring torus from a. on the xy-plane.

c. Give two examples of objects with ring torus shapes.

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 12.6E: Exercises for Section 12.6 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.7: Cylindrical and Spherical Coordinates

Learning Objectives

- Convert from cylindrical to rectangular coordinates.
- Convert from rectangular to cylindrical coordinates.
- Convert from spherical to rectangular coordinates.
- Convert from rectangular to spherical coordinates.

The Cartesian coordinate system provides a straightforward way to describe the location of points in space. Some surfaces, however, can be difficult to model with equations based on the Cartesian system. This is a familiar problem; recall that in two dimensions, polar coordinates often provide a useful alternative system for describing the location of a point in the plane, particularly in cases involving circles. In this section, we look at two different ways of describing the location of points in space, both of them based on extensions of polar coordinates. As the name suggests, cylindrical coordinates are useful for dealing with problems involving cylinders, such as calculating the volume of a round water tank or the amount of oil flowing through a pipe. Similarly, spherical coordinates are useful for dealing with problems involving spheres, such as finding the volume of domed structures.

Cylindrical Coordinates

When we expanded the traditional Cartesian coordinate system from two dimensions to three, we simply added a new axis to model the third dimension. Starting with polar coordinates, we can follow this same process to create a new three-dimensional coordinate system, called the cylindrical coordinate system. In this way, cylindrical coordinates provide a natural extension of polar coordinates to three dimensions.

Definition: The Cylindrical Coordinate System

In the cylindrical coordinate system**,** a point in space (Figure 12.7.1) is represented by the ordered triple (r, θ, z), where

- (r, θ) are the polar coordinates of the point's projection in the xy-plane
- z is the usual z-**coordinate** in the Cartesian coordinate system

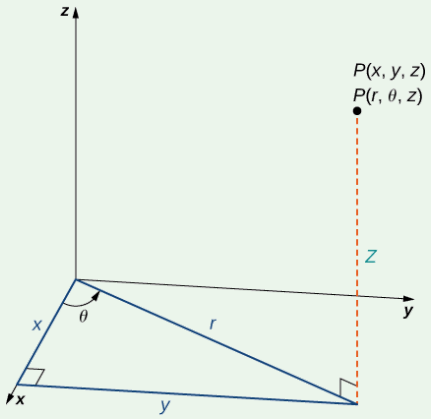


Figure 12.7.1: The right triangle lies in the xy-plane. The length of the hypotenuse is r and θ is the measure of the angle formed by the positive x-axis and the hypotenuse. The z-coordinate describes the location of the point above or below the xy-plane.

In the xy-plane, the right triangle shown in Figure 12.7.1 provides the key to transformation between cylindrical and Cartesian, or rectangular, coordinates.

Conversion between Cylindrical and Cartesian Coordinates

The rectangular coordinates (x, y, z) and the cylindrical coordinates (r, θ, z) of a point are related as follows:

These equations are used to convert from cylindrical coordinates to rectangular coordinates.

- $x = r\cos\theta$
- $y = r\sin\theta$
- $z = z$

These equations are used to convert from rectangular coordinates to cylindrical coordinates

1. $r^2 = x^2 + y^2$
2. $\tan\theta = \dfrac{y}{x}$
3. $z = z$

As when we discussed conversion from rectangular coordinates to polar coordinates in two dimensions, it should be noted that the equation $\tan\theta = \dfrac{y}{x}$ has an infinite number of solutions. However, if we restrict θ to values between 0 and 2π, then we can find a unique solution based on the quadrant of the xy-plane in which original point (x, y, z) is located. Note that if $x = 0$, then the value of θ is either $\dfrac{\pi}{2}, \dfrac{3\pi}{2}$, or 0, depending on the value of y.

Notice that these equations are derived from properties of right triangles. To make this easy to see, consider point P in the xy-plane with rectangular coordinates $(x, y, 0)$ and with cylindrical coordinates $(r, \theta, 0)$, as shown in Figure 12.7.2.

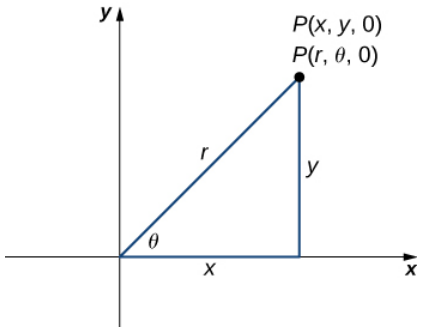


Figure 12.7.2: The Pythagorean theorem provides equation $r^2 = x^2 + y^2$. Right-triangle relationships tell us that $x = r\cos\theta, y = r\sin\theta$, and $\tan\theta = y/x$.

Let's consider the differences between rectangular and cylindrical coordinates by looking at the surfaces generated when each of the coordinates is held constant. If c is a constant, then in rectangular coordinates, surfaces of the form $x = c, y = c$, or $z = c$ are all planes. Planes of these forms are parallel to the yz-plane, the xz-plane, and the xy-plane, respectively. When we convert to cylindrical coordinates, the z-coordinate does not change. Therefore, in cylindrical coordinates, surfaces of the form $z = c$ are planes parallel to the xy-plane. Now, let's think about surfaces of the form $r = c$. The points on these surfaces are at a fixed distance from the z-axis. In other words, these surfaces are vertical circular cylinders. Last, what about $\theta = c$? The points on a surface of the form $\theta = c$ are at a fixed angle from the x-axis, which gives us a half-plane that starts at the z-axis (Figures 12.7.3 and 12.7.4).

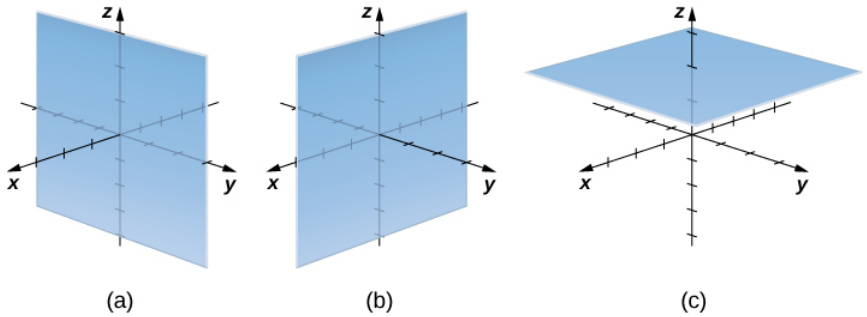


Figure 12.7.3: In rectangular coordinates, (a) surfaces of the form $x = c$ are planes parallel to the yz-plane, (b) surfaces of the form $y = c$ are planes parallel to the xz-plane, and (c) surfaces of the form $z = c$ are planes parallel to the xy-plane.

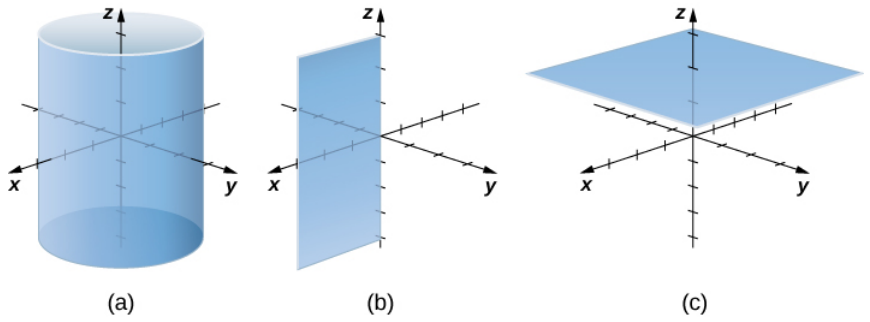


Figure 12.7.4: In cylindrical coordinates, (a) surfaces of the form $r = c$ are vertical cylinders of radius r, (b) surfaces of the form $\theta = c$ are half-planes at angle θ from the x-axis, and (c) surfaces of the form $z = c$ are planes parallel to the xy-plane.

Example 12.7.1: Converting from Cylindrical to Rectangular Coordinates

Plot the point with cylindrical coordinates $(4, \frac{2\pi}{3}, -2)$ and express its location in rectangular coordinates.

Solution

Conversion from cylindrical to rectangular coordinates requires a simple application of the equations listed in Conversion between Cylindrical and Cartesian Coordinates:

$$x = r\cos\theta = 4\cos\frac{2\pi}{3} = -2$$
$$y = r\sin\theta = 4\sin\frac{2\pi}{3} = 2\sqrt{3}.$$
$$z = -2$$

The point with cylindrical coordinates $(4, \frac{2\pi}{3}, -2)$ has rectangular coordinates $(-2, 2\sqrt{3}, -2)$ (Figure 12.7.5).

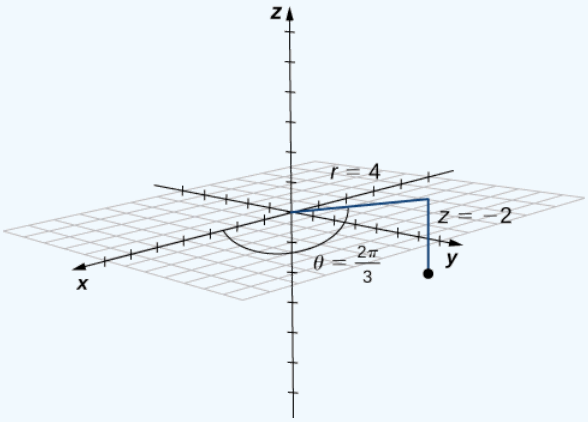


Figure 12.7.5: The projection of the point in the xy-plane is 4 units from the origin. The line from the origin to the point's projection forms an angle of $\frac{2\pi}{3}$ with the positive x-axis. The point lies 2 units below the xy-plane.

Exercise 12.7.1

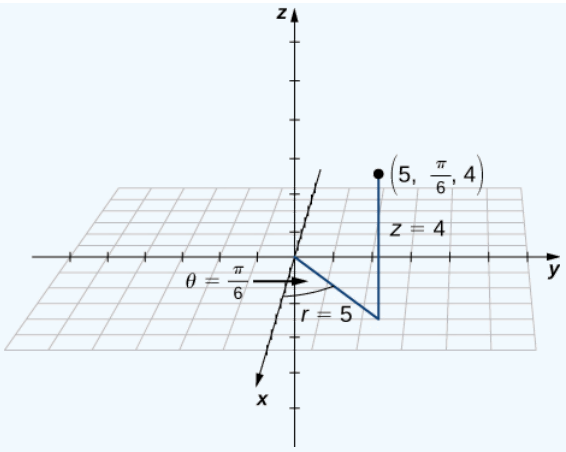
Point R has cylindrical coordinates $(5, \frac{\pi}{6}, 4)$. Plot R and describe its location in space using rectangular, or Cartesian, coordinates.

Hint

The first two components match the polar coordinates of the point in the xy-plane.

Answer

The rectangular coordinates of the point are $(\frac{5\sqrt{3}}{2}, \frac{5}{2}, 4)$.



If this process seems familiar, it is with good reason. This is exactly the same process that we followed in Introduction to Parametric Equations and Polar Coordinates to convert from polar coordinates to two-dimensional rectangular coordinates.

Example 12.7.2: Converting from Rectangular to Cylindrical Coordinates

Convert the rectangular coordinates $(1, -3, 5)$ to cylindrical coordinates.

Solution

Use the second set of equations from Conversion between Cylindrical and Cartesian Coordinates to translate from rectangular to cylindrical coordinates:

$$\begin{aligned} r^2 &= x^2 + y^2 \\ r &= \pm\sqrt{1^2 + (-3)^2} \\ &= \pm\sqrt{10}. \end{aligned}$$

We choose the positive square root, so $r = \sqrt{10}$.Now, we apply the formula to find θ. In this case, y is negative and x is positive, which means we must select the value of θ between $\frac{3\pi}{2}$ and 2π:

$$\begin{aligned} \tan\theta &= \frac{y}{x} & &= \frac{-3}{1} \\ \theta &= \arctan(-3) & &\approx 5.03\,\text{rad}. \end{aligned}$$

In this case, the z-coordinates are the same in both rectangular and cylindrical coordinates:

$$z = 5.$$

The point with rectangular coordinates $(1, -3, 5)$ has cylindrical coordinates approximately equal to $(\sqrt{10}, 5.03, 5)$.

Exercise 12.7.2

Convert point $(-8, 8, -7)$ from Cartesian coordinates to cylindrical coordinates.

Hint

$r^2 = x^2 + y^2$ and $\tan\theta = \frac{y}{x}$

Answer

$(8\sqrt{2}, \frac{3\pi}{4}, -7)$

The use of cylindrical coordinates is common in fields such as physics. Physicists studying electrical charges and the capacitors used to store these charges have discovered that these systems sometimes have a cylindrical symmetry. These systems have

complicated modeling equations in the Cartesian coordinate system, which make them difficult to describe and analyze. The equations can often be expressed in more simple terms using cylindrical coordinates. For example, the cylinder described by equation $x^2 + y^2 = 25$ in the Cartesian system can be represented by cylindrical equation $r = 5$.

Example 12.7.3: Identifying Surfaces in the Cylindrical Coordinate System

Describe the surfaces with the given cylindrical equations.

a. $\theta = \dfrac{\pi}{4}$
b. $r^2 + z^2 = 9$
c. $z = r$

Solution

a. When the angle θ is held constant while r and z are allowed to vary, the result is a half-plane (Figure 12.7.6).

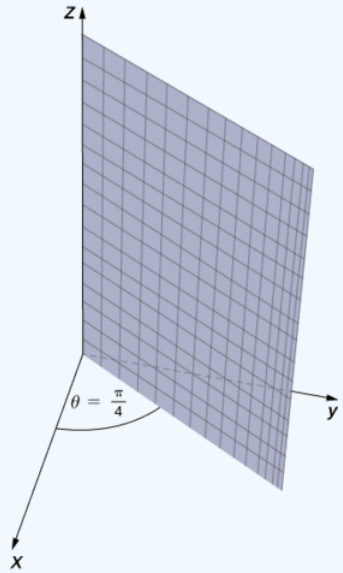


Figure 12.7.6: In polar coordinates, the equation $\theta = \pi/4$ describes the ray extending diagonally through the first quadrant. In three dimensions, this same equation describes a half-plane.

b. Substitute $r^2 = x^2 + y^2$ into equation $r^2 + z^2 = 9$ to express the rectangular form of the equation: $x^2 + y^2 + z^2 = 9$. This equation describes a sphere centered at the origin with radius 3 (Figure 12.7.7).

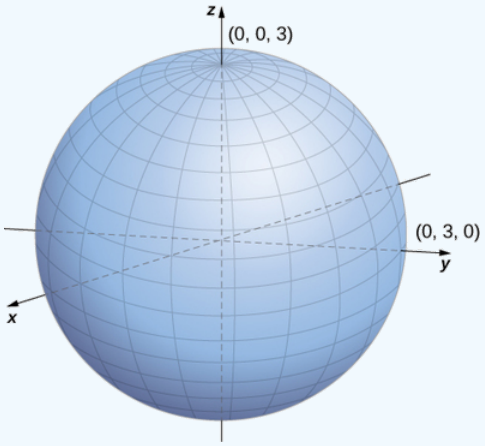


Figure 12.7.7: The sphere centered at the origin with radius 3 can be described by the cylindrical equation $r^2 + z^2 = 9$.

c. To describe the surface defined by equation $z = r$, is it useful to examine traces parallel to the xy-plane. For example, the trace in plane $z = 1$ is circle $r = 1$, the trace in plane $z = 3$ is circle $r = 3$, and so on. Each trace is a circle. As the value of z increases, the radius of the circle also increases. The resulting surface is a cone (Figure 12.7.8).

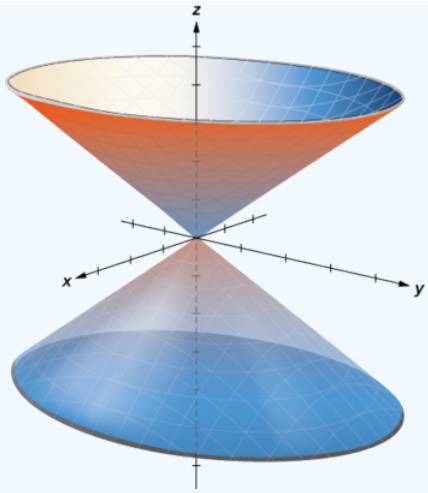


Figure 12.7.8: The traces in planes parallel to the xy-plane are circles. The radius of the circles increases as z increases.

? Exercise 12.7.3

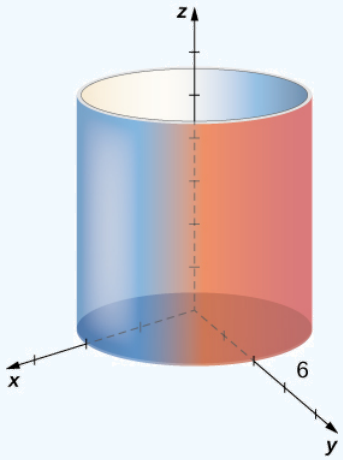
Describe the surface with cylindrical equation $r = 6$.

Hint

The θ and z components of points on the surface can take any value.

Answer

This surface is a cylinder with radius 6.



Spherical Coordinates

In the Cartesian coordinate system, the location of a point in space is described using an ordered triple in which each coordinate represents a distance. In the cylindrical coordinate system, the location of a point in space is described using two distances (r and z) and an angle measure (θ). In the spherical coordinate system, we again use an ordered triple to describe the location of a point in space. In this case, the triple describes one distance and two angles. Spherical coordinates make it simple to describe a sphere, just as cylindrical coordinates make it easy to describe a cylinder. Grid lines for spherical coordinates are based on angle measures, like those for polar coordinates.

Definition: spherical coordinate system

In the *spherical coordinate system*, a point P in space (Figure 12.7.9) is represented by the ordered triple (ρ, θ, φ) where

- ρ (the Greek letter rho) is the distance between P and the origin $(\rho \neq 0)$;
- θ is the same angle used to describe the location in cylindrical coordinates;
- φ (the Greek letter phi) is the angle formed by the positive z-axis and line segment $\overline{OP}$, where O is the origin and $0 \leq \varphi \leq \pi$.

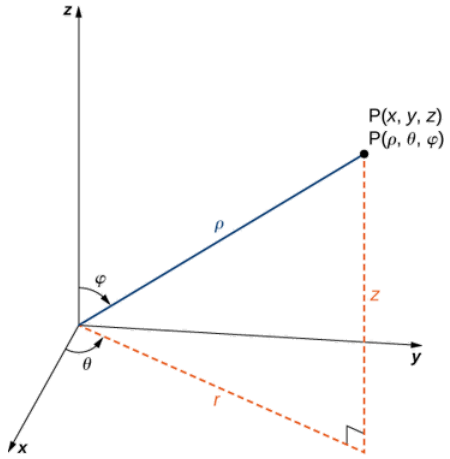


Figure 12.7.9: The relationship among spherical, rectangular, and cylindrical coordinates.

By convention, the origin is represented as $(0, 0, 0)$ in spherical coordinates.

HOW TO: Converting among Spherical, Cylindrical, and Rectangular Coordinates

Rectangular coordinates (x, y, z), cylindrical coordinates (r, θ, z), and spherical coordinates (ρ, θ, φ) of a point are related as follows:

Convert from spherical coordinates to rectangular coordinates

These equations are used to convert from spherical coordinates to rectangular coordinates.

- $x = \rho \sin\varphi \cos\theta$
- $y = \rho \sin\varphi \sin\theta$
- $z = \rho \cos\varphi$

Convert from rectangular coordinates to spherical coordinates

These equations are used to convert from rectangular coordinates to spherical coordinates.

- $\rho^2 = x^2 + y^2 + z^2$
- $\tan\theta = \dfrac{y}{x}$
- $\varphi = \arccos\left(\dfrac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$.

Convert from spherical coordinates to cylindrical coordinates

These equations are used to convert from spherical coordinates to cylindrical coordinates.

- $r = \rho \sin\varphi$
- $\theta = \theta$
- $z = \rho \cos\varphi$

Convert from cylindrical coordinates to spherical coordinates

These equations are used to convert from cylindrical coordinates to spherical coordinates.

- $\rho = \sqrt{r^2 + z^2}$

- $\theta = \theta$
- $\varphi = \arccos\left(\dfrac{z}{\sqrt{r^2+z^2}}\right)$

The formulas to convert from spherical coordinates to rectangular coordinates may seem complex, but they are straightforward applications of trigonometry. Looking at Figure 12.7.10, it is easy to see that $r = \rho \sin\varphi$. Then, looking at the triangle in the xy-plane with r as its hypotenuse, we have $x = r\cos\theta = \rho\sin\varphi\cos\theta$. The derivation of the formula for y is similar. Figure 12.7.10 also shows that $\rho^2 = r^2 + z^2 = x^2 + y^2 + z^2$ and $z = \rho\cos\varphi$. Solving this last equation for φ and then substituting $\rho = \sqrt{r^2+z^2}$ (from the first equation) yields $\varphi = \arccos\left(\dfrac{z}{\sqrt{r^2+z^2}}\right)$. Also, note that, as before, we must be careful when using the formula $\tan\theta = \dfrac{y}{x}$ to choose the correct value of θ.

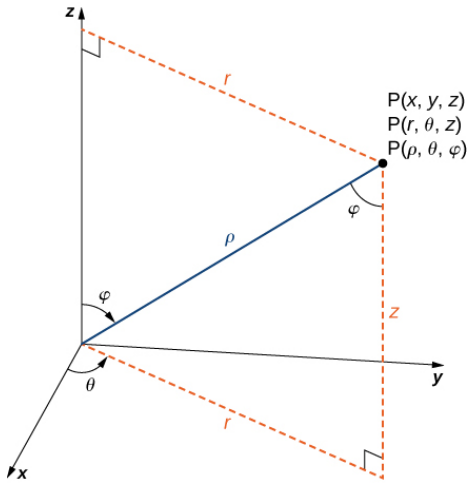


Figure 12.7.10: The equations that convert from one system to another are derived from right-triangle relationships.

As we did with cylindrical coordinates, let's consider the surfaces that are generated when each of the coordinates is held constant. Let c be a constant, and consider surfaces of the form $\rho = c$. Points on these surfaces are at a fixed distance from the origin and form a sphere. The coordinate θ in the spherical coordinate system is the same as in the cylindrical coordinate system, so surfaces of the form $\theta = c$ are half-planes, as before. Last, consider surfaces of the form $\varphi = c$. The points on these surfaces are at a fixed angle from the z-axis and form a half-cone (Figure 12.7.11).

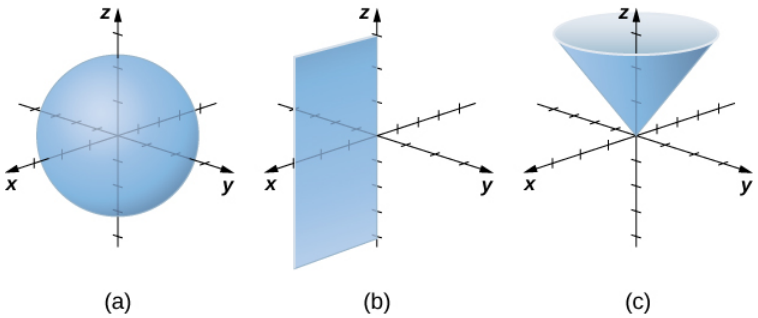


Figure 12.7.11: In spherical coordinates, surfaces of the form $\rho = c$ are spheres of radius ρ (a), surfaces of the form $\theta = c$ are half-planes at an angle θ from the x-axis (b), and surfaces of the form $\phi = c$ are half-cones at an angle ϕ from the z-axis (c).

Example 12.7.4: Converting from Spherical Coordinates

Plot the point with spherical coordinates $(8, \dfrac{\pi}{3}, \dfrac{\pi}{6})$ and express its location in both rectangular and cylindrical coordinates.

Solution

Use the equations in Converting among Spherical, Cylindrical, and Rectangular Coordinates to translate between spherical and cylindrical coordinates (Figure 12.7.12):

$$
\begin{aligned}
x &= \rho \sin\varphi \cos\theta \\
&= 8 \sin\left(\frac{\pi}{6}\right) \cos\left(\frac{\pi}{3}\right) \\
&= 8\left(\frac{1}{2}\right)\frac{1}{2} \\
&= 2 \\
y &= \rho \sin\varphi \sin\theta \\
&= 8 \sin\left(\frac{\pi}{6}\right) \sin\left(\frac{\pi}{3}\right) \\
&= 8\left(\frac{1}{2}\right)\frac{\sqrt{3}}{2} \\
&= 2\sqrt{3} \\
z &= \rho \cos\varphi \\
&= 8 \cos\left(\frac{\pi}{6}\right) \\
&= 8\left(\frac{\sqrt{3}}{2}\right) \\
&= 4\sqrt{3}
\end{aligned}
$$

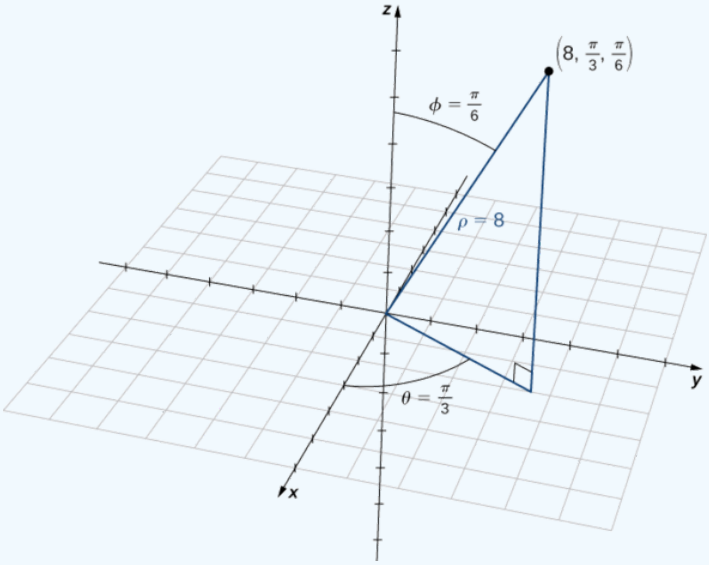


Figure 12.7.12: The projection of the point in the xy-plane is 4 units from the origin. The line from the origin to the point's projection forms an angle of $\pi/3$ with the positive x-axis. The point lies $4\sqrt{3}$ units above the xy-plane.

The point with spherical coordinates $(8, \frac{\pi}{3}, \frac{\pi}{6})$ has rectangular coordinates $(2, 2\sqrt{3}, 4\sqrt{3})$.

Finding the values in cylindrical coordinates is equally straightforward:

$$
\begin{aligned}
r &= \rho \sin\varphi \\
&= 8 \sin\frac{\pi}{6} \\
&= 4 \\
\theta &= \theta \\
z &= \rho \cos\varphi \\
&= 8 \cos\frac{\pi}{6} \\
&= 4\sqrt{3}.
\end{aligned}
$$

Thus, cylindrical coordinates for the point are $(4, \frac{\pi}{3}, 4\sqrt{3})$.

Exercise 12.7.4

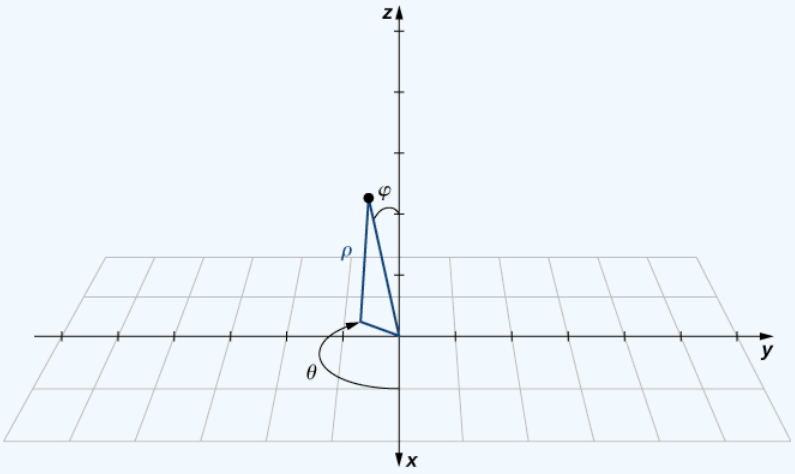
Plot the point with spherical coordinates $(2, -\frac{5\pi}{6}, \frac{\pi}{6})$ and describe its location in both rectangular and cylindrical coordinates.

Hint

Converting the coordinates first may help to find the location of the point in space more easily.

Answer

Cartesian: $(-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \sqrt{3})$, cylindrical: $(1, -\frac{5\pi}{6}, \sqrt{3})$



Example 12.7.5: Converting from Rectangular Coordinates

Convert the rectangular coordinates $(-1, 1, \sqrt{6})$ to both spherical and cylindrical coordinates.

Solution

Start by converting from rectangular to spherical coordinates:

$$\rho^2 = x^2 + y^2 + z^2 = (-1)^2 + 1^2 + (\sqrt{6})^2 = 8$$

$$\tan\theta = \frac{1}{-1}$$

$$\rho = 2\sqrt{2} \text{ and } \theta = \arctan(-1) = \frac{3\pi}{4}.$$

Because $(x, y) = (-1, 1)$, then the correct choice for θ is $\frac{3\pi}{4}$.

There are actually two ways to identify φ. We can use the equation $\varphi = \arccos\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$. A more simple approach, however, is to use equation $z = \rho\cos\varphi$. We know that $z = \sqrt{6}$ and $\rho = 2\sqrt{2}$, so

$$\sqrt{6} = 2\sqrt{2}\cos\varphi, \text{ so } \cos\varphi = \frac{\sqrt{6}}{2\sqrt{2}} = \frac{\sqrt{3}}{2}$$

and therefore $\varphi = \frac{\pi}{6}$. The spherical coordinates of the point are $(2\sqrt{2}, \frac{3\pi}{4}, \frac{\pi}{6})$.

To find the cylindrical coordinates for the point, we need only find r:

$$r = \rho\sin\varphi = 2\sqrt{2}\sin(\frac{\pi}{6}) = \sqrt{2}.$$

The cylindrical coordinates for the point are $(\sqrt{2}, \frac{3\pi}{4}, \sqrt{6})$.

Example 12.7.6: Identifying Surfaces in the Spherical Coordinate System

Describe the surfaces with the given spherical equations.

a. $\theta = \dfrac{\pi}{3}$
b. $\varphi = \dfrac{5\pi}{6}$
c. $\rho = 6$
d. $\rho = \sin\theta \sin\varphi$

Solution

a. The variable θ represents the measure of the same angle in both the cylindrical and spherical coordinate systems. Points with coordinates $(\rho, \dfrac{\pi}{3}, \varphi)$ lie on the plane that forms angle $\theta = \dfrac{\pi}{3}$ with the positive x-axis. Because $\rho > 0$, the surface described by equation $\theta = \dfrac{\pi}{3}$ is the half-plane shown in Figure 12.7.13.

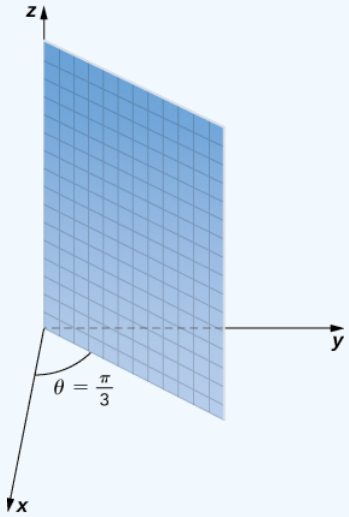


Figure 12.7.13: The surface described by equation $\theta = \dfrac{\pi}{3}$ is a half-plane.

b. Equation $\varphi = \dfrac{5\pi}{6}$ describes all points in the spherical coordinate system that lie on a line from the origin forming an angle measuring $\dfrac{5\pi}{6}$ rad with the positive z-axis. These points form a half-cone (Figure 12.7.14). Because there is only one value for φ that is measured from the positive z-axis, we do not get the full cone (with two pieces).

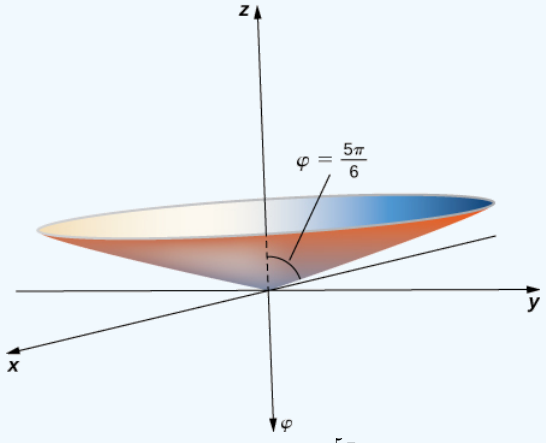


Figure 12.7.14: The equation $\varphi = \dfrac{5\pi}{6}$ describes a cone.

To find the equation in rectangular coordinates, use equation $\varphi = \arccos\left(\dfrac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$.

$$\frac{5\pi}{6} = \arccos\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$$

$$\cos\frac{5\pi}{6} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$-\frac{\sqrt{3}}{2} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\frac{3}{4} = \frac{z^2}{x^2+y^2+z^2}$$

$$\frac{3x^2}{4} + \frac{3y^2}{4} + \frac{3z^2}{4} = z^2$$

$$\frac{3x^2}{4} + \frac{3y^2}{4} - \frac{z^2}{4} = 0.$$

This is the equation of a cone centered on the z-axis.

c. Equation $\rho = 6$ describes the set of all points 6 units away from the origin—a sphere with radius 6 (Figure 12.7.15).

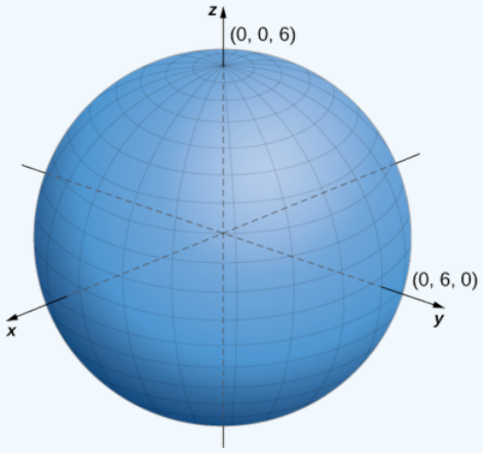


Figure 12.7.15: Equation $\rho = 6$ describes a sphere with radius 6.

d. To identify this surface, convert the equation from spherical to rectangular coordinates, using equations $y = \rho\sin\varphi\sin\theta$ and $\rho^2 = x^2+y^2+z^2$:

$\rho = \sin\theta\sin\varphi$

$\rho^2 = \rho\sin\theta\sin\varphi$ Multiply both sides of the equation by ρ.

$x^2+y^2+z^2 = y$ Substitute rectangular variables using the equations above.

$x^2+y^2-y+z^2 = 0$ Subtract y from both sides of the equation.

$x^2+y^2-y+\frac{1}{4}+z^2 = \frac{1}{4}$ Complete the square.

$x^2+\left(y-\frac{1}{2}\right)^2+z^2 = \frac{1}{4}$. Rewrite the middle terms as a perfect square.

The equation describes a sphere centered at point $(0,\frac{1}{2},0)$ with radius $\frac{1}{2}$.

Exercise 12.7.5

Describe the surfaces defined by the following equations.

a. $\rho = 13$
b. $\theta = \frac{2\pi}{3}$
c. $\varphi = \frac{\pi}{4}$

Hint

Think about what each component represents and what it means to hold that component constant.

Answer a

This is the set of all points 13 units from the origin. This set forms a sphere with radius 13.

Answer b

This set of points forms a half plane. The angle between the half plane and the positive x-axis is $\theta = \dfrac{2\pi}{3}$.

Answer c

Let P be a point on this surface. The position vector of this point forms an angle of $\varphi = \dfrac{\pi}{4}$ with the positive z-axis, which means that points closer to the origin are closer to the axis. These points form a half-cone.

Spherical coordinates are useful in analyzing systems that have some degree of symmetry about a point, such as the volume of the space inside a domed stadium or wind speeds in a planet's atmosphere. A sphere that has Cartesian equation $x^2 + y^2 + z^2 = c^2$ has the simple equation $\rho = c$ in spherical coordinates.

In geography, latitude and longitude are used to describe locations on Earth's surface, as shown in Figure $12.7.16$. Although the shape of Earth is not a perfect sphere, we use spherical coordinates to communicate the locations of points on Earth. Let's assume Earth has the shape of a sphere with radius 4000 mi. We express angle measures in degrees rather than radians because latitude and longitude are measured in degrees.

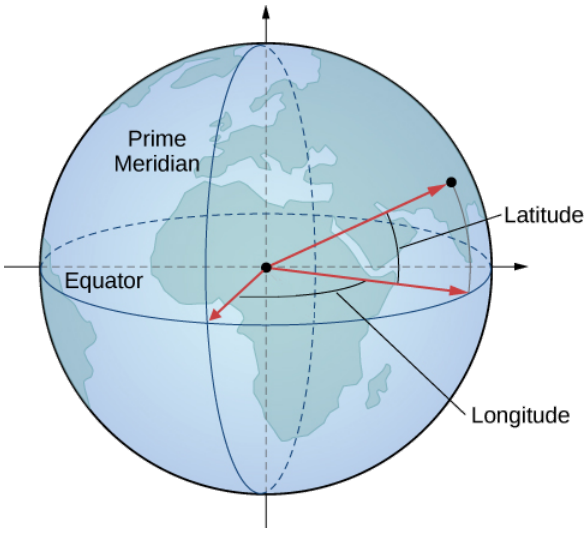


Figure $12.7.16$: In the latitude–longitude system, angles describe the location of a point on Earth relative to the equator and the prime meridian.

Let the center of Earth be the center of the sphere, with the ray from the center through the North Pole representing the positive z-axis. The prime meridian represents the trace of the surface as it intersects the xz-plane. The equator is the trace of the sphere intersecting the xy-plane.

Example 12.7.7: Converting Latitude and Longitude to Spherical Coordinates

The latitude of Columbus, Ohio, is $40°$ N and the longitude is $83°$ W, which means that Columbus is $40°$ north of the equator. Imagine a ray from the center of Earth through Columbus and a ray from the center of Earth through the equator directly south of Columbus. The measure of the angle formed by the rays is $40°$. In the same way, measuring from the prime meridian, Columbus lies $83°$ to the west. Express the location of Columbus in spherical coordinates.

Solution

The radius of Earth is 4000mi, so $\rho = 4000$. The intersection of the prime meridian and the equator lies on the positive x-axis. Movement to the west is then described with negative angle measures, which shows that $\theta = -83°$, Because Columbus lies $40°$ north of the equator, it lies $50°$ south of the North Pole, so $\varphi = 50°$. In spherical coordinates, Columbus lies at point $(4000, -83°, 50°)$.

? Exercise 12.7.6

Sydney, Australia is at $34°S$ and $151°E$. Express Sydney's location in spherical coordinates.

Hint

Because Sydney lies south of the equator, we need to add $90°$ to find the angle measured from the positive z-axis.

Answer

$(4000, 151°, 124°)$

Cylindrical and spherical coordinates give us the flexibility to select a coordinate system appropriate to the problem at hand. A thoughtful choice of coordinate system can make a problem much easier to solve, whereas a poor choice can lead to unnecessarily complex calculations. In the following example, we examine several different problems and discuss how to select the best coordinate system for each one.

✓ Example 12.7.8: Choosing the Best Coordinate System

In each of the following situations, we determine which coordinate system is most appropriate and describe how we would orient the coordinate axes. There could be more than one right answer for how the axes should be oriented, but we select an orientation that makes sense in the context of the problem. Note: There is not enough information to set up or solve these problems; we simply select the coordinate system (Figure 12.7.17).

a. Find the center of gravity of a bowling ball.
b. Determine the velocity of a submarine subjected to an ocean current.
c. Calculate the pressure in a conical water tank.
d. Find the volume of oil flowing through a pipeline.
e. Determine the amount of leather required to make a football.



(a) (b) (c)



(d) (e)

Figure 12.7.17: (credit: (a) modification of work by scl hua, Wikimedia, (b) modification of work by DVIDSHUB, Flickr, (c) modification of work by Michael Malak, Wikimedia, (d) modification of work by Sean Mack, Wikimedia, (e) modification of work by Elvert Barnes, Flickr)

Solution

a. Clearly, a bowling ball is a sphere, so spherical coordinates would probably work best here. The origin should be located at the physical center of the ball. There is no obvious choice for how the x-, y- and z-axes should be oriented. Bowling balls normally have a weight block in the center. One possible choice is to align the z-axis with the axis of symmetry of the weight block.

b. A submarine generally moves in a straight line. There is no rotational or spherical symmetry that applies in this situation, so rectangular coordinates are a good choice. The z-axis should probably point upward. The x- and y-axes could be aligned to point east and north, respectively. The origin should be some convenient physical location, such as the starting position of the submarine or the location of a particular port.

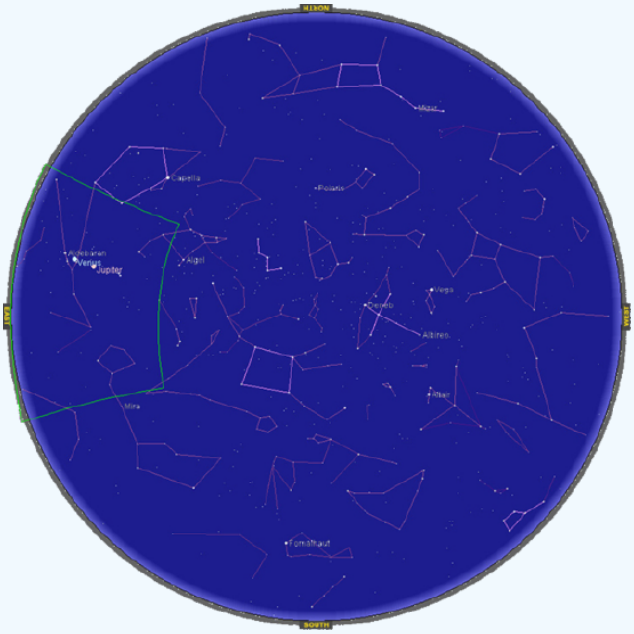
c. A cone has several kinds of symmetry. In cylindrical coordinates, a cone can be represented by equation $z = kr$, where k is a constant. In spherical coordinates, we have seen that surfaces of the form $\varphi = c$ are half-cones. Last, in rectangular coordinates, elliptic cones are quadric surfaces and can be represented by equations of the form $z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$. In this case, we could choose any of the three. However, the equation for the surface is more complicated in rectangular coordinates than in the other two systems, so we might want to avoid that choice. In addition, we are talking about a water tank, and the depth of the water might come into play at some point in our calculations, so it might be nice to have a component that represents height and depth directly. Based on this reasoning, cylindrical coordinates might be the best choice. Choose the z -axis to align with the axis of the cone. The orientation of the other two axes is arbitrary. The origin should be the bottom point of the cone.

d. A pipeline is a cylinder, so cylindrical coordinates would be best the best choice. In this case, however, we would likely choose to orient our z-axis with the center axis of the pipeline. The x-axis could be chosen to point straight downward or to some other logical direction. The origin should be chosen based on the problem statement. Note that this puts the z-axis in a horizontal orientation, which is a little different from what we usually do. It may make sense to choose an unusual orientation for the axes if it makes sense for the problem.

e. A football has rotational symmetry about a central axis, so cylindrical coordinates would work best. The z-axis should align with the axis of the ball. The origin could be the center of the ball or perhaps one of the ends. The position of the x-axis is arbitrary.

Exercise 12.7.7

Which coordinate system is most appropriate for creating a star map, as viewed from Earth (see the following figure)?



How should we orient the coordinate axes?

Hint

What kinds of symmetry are present in this situation?

Answer

Spherical coordinates with the origin located at the center of the earth, the z-axis aligned with the North Pole, and the x-axis aligned with the prime meridian

Key Concepts

- In the cylindrical coordinate system, a point in space is represented by the ordered triple (r, θ, z), where (r, θ) represents the polar coordinates of the point's projection in the xy-plane and z represents the point's projection onto the z-axis.
- To convert a point from cylindrical coordinates to Cartesian coordinates, use equations $x = r\cos\theta, y = r\sin\theta$, and $z = z$.
- To convert a point from Cartesian coordinates to cylindrical coordinates, use equations $r^2 = x^2 + y^2, \tan\theta = \dfrac{y}{x}$, and $z = z$.
- In the spherical coordinate system, a point P in space is represented by the ordered triple (ρ, θ, φ), where ρ is the distance between P and the origin $(\rho \neq 0), \theta$ is the same angle used to describe the location in cylindrical coordinates, and φ is the angle formed by the positive z-axis and line segment $\overline{OP}$, where O is the origin and $0 \leq \varphi \leq \pi$.
- To convert a point from spherical coordinates to Cartesian coordinates, use equations $x = \rho\sin\varphi\cos\theta, y = \rho\sin\varphi\sin\theta$, and $z = \rho\cos\varphi$.
- To convert a point from Cartesian coordinates to spherical coordinates, use equations $\rho^2 = x^2 + y^2 + z^2, \tan\theta = \dfrac{y}{x}$, and $\varphi = \arccos\left(\dfrac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$.
- To convert a point from spherical coordinates to cylindrical coordinates, use equations $r = \rho\sin\varphi, \theta = \theta$, and $z = \rho\cos\varphi$.
- To convert a point from cylindrical coordinates to spherical coordinates, use equations $\rho = \sqrt{r^2 + z^2}, \theta = \theta$, and $\varphi = \arccos\left(\dfrac{z}{\sqrt{r^2 + z^2}}\right)$.

Glossary

cylindrical coordinate system

a way to describe a location in space with an ordered triple (r, θ, z), where (r, θ) represents the polar coordinates of the point's projection in the xy-plane, and z represents the point's projection onto the z-axis

spherical coordinate system

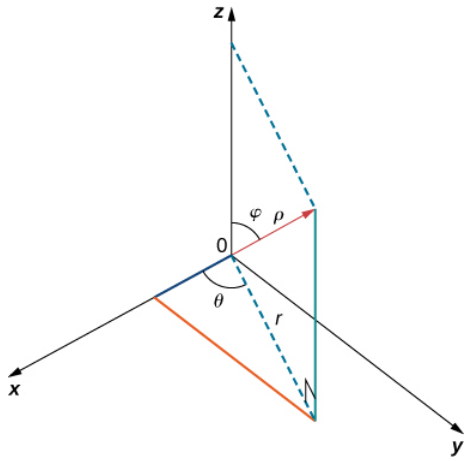
a way to describe a location in space with an ordered triple (ρ, θ, φ), where ρ is the distance between P and the origin $(\rho \neq 0)$, θ is the same angle used to describe the location in cylindrical coordinates, and φ is the angle formed by the positive z-axis and line segment $\bar{OP}$, where O is the origin and $0 \leq \varphi \leq \pi$

- Gilbert Strang (MIT) and Edwin "Jed" Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger edited the LaTeX on the page

This page titled 12.7: Cylindrical and Spherical Coordinates is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.7E: Exercises for Section 12.7

Use the following figure as an aid in identifying the relationship between the rectangular, cylindrical, and spherical coordinate systems.



For exercises 1 - 4, the cylindrical coordinates (r, θ, z) of a point are given. Find the rectangular coordinates (x, y, z) of the point.

1) $\left(4, \frac{\pi}{6}, 3\right)$

Answer

$(2\sqrt{3}, 2, 3)$

2) $\left(3, \frac{\pi}{3}, 5\right)$

3) $\left(4, \frac{7\pi}{6}, 3\right)$

Answer

$(-2\sqrt{3}, -2, 3)$

4) $(2, \pi, -4)$

For exercises 5 - 8, the rectangular coordinates (x, y, z) of a point are given. Find the cylindrical coordinates (r, θ, z) of the point.

5) $(1, \sqrt{3}, 2)$

Answer

$\left(2, \frac{\pi}{3}, 2\right)$

6) $(1, 1, 5)$

7) $(3, -3, 7)$

Answer

$\left(3\sqrt{2}, -\frac{\pi}{4}, 7\right)$

8) $(-2\sqrt{2}, 2\sqrt{2}, 4)$

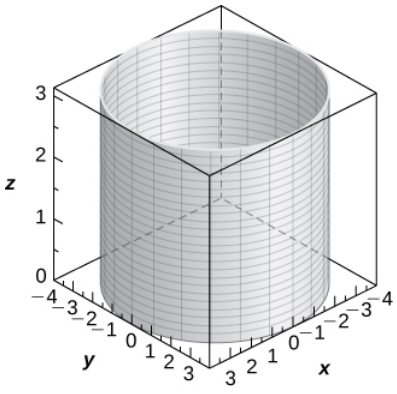
For exercises 9 - 16, the equation of a surface in cylindrical coordinates is given. Find the equation of the surface in rectangular coordinates. Identify and graph the surface.

9) [T] $r = 4$

-

Answer

A cylinder of equation $x^2 + y^2 = 16$, with its center at the origin and rulings parallel to the z-axis,

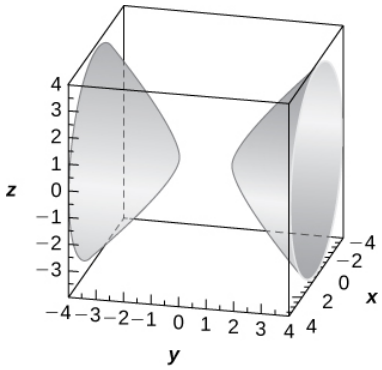


10) [T] $z = r^2 \cos^2 \theta$

11) [T] $r^2 \cos(2\theta) + z^2 + 1 = 0$

Answer

Hyperboloid of two sheets of equation $-x^2 + y^2 - z^2 = 1$, with the y-axis as the axis of symmetry,

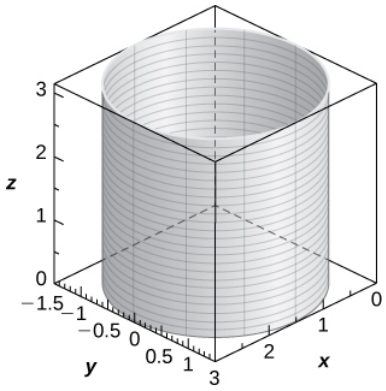


12) [T] $r = 3 \sin \theta$

13) [T] $r = 2 \cos \theta$

Answer

Cylinder of equation $x^2 - 2x + y^2 = 0$, with a center at $(1, 0, 0)$ and radius 1, with rulings parallel to the z-axis,

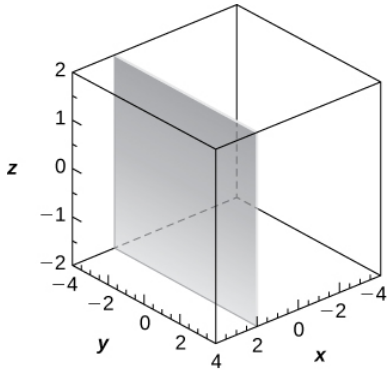


14) [T] $r^2 + z^2 = 5$

15) [T] $r = 2 \sec \theta$

Answer

Plane of equation $x = 2$,



16) [T] $r = 3\csc\theta$

For exercises 17 - 22, the equation of a surface in rectangular coordinates is given. Find the equation of the surface in cylindrical coordinates.

17) $z = 3$

Answer

$z = 3$

18) $x = 6$

19) $x^2 + y^2 + z^2 = 9$

Answer

$r^2 + z^2 = 9$

20) $y = 2x^2$

21) $x^2 + y^2 - 16x = 0$

Answer

$r = 16\cos\theta, \quad r = 0$

22) $x^2 + y^2 - 3\sqrt{x^2 + y^2} + 2 = 0$

For exercises 23 - 26, the spherical coordinates (ρ, θ, φ) of a point are given. Find the rectangular coordinates (x, y, z) of the point.

23) $(3, 0, \pi)$

Answer

$(0, 0, -3)$

24) $\left(1, \frac{\pi}{6}, \frac{\pi}{6}\right)$

25) $\left(12, -\frac{\pi}{4}, \frac{\pi}{4}\right)$

Answer

$(6, -6, \sqrt{2})$

26) $\left(3, \frac{\pi}{4}, \frac{\pi}{6}\right)$

For exercises 27 - 30, the rectangular coordinates (x, y, z) of a point are given. Find the spherical coordinates (ρ, θ, φ) of the point. Express the measure of the angles in degrees rounded to the nearest integer.

27) $(4, 0, 0)$

Answer

$(4, 0, 90°)$

28) $(-1, 2, 1)$

29) $(0, 3, 0)$

Answer

$(3, 90°, 90°)$

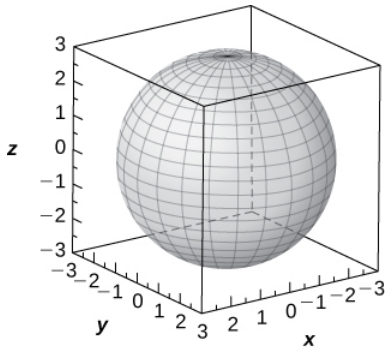
30) $(-2, 2\sqrt{3}, 4)$

For exercises 31 - 36, the equation of a surface in spherical coordinates is given. Find the equation of the surface in rectangular coordinates. Identify and graph the surface.

31) [T] $\rho = 3$

Answer

Sphere of equation $x^2 + y^2 + z^2 = 9$ centered at the origin with radius 3,

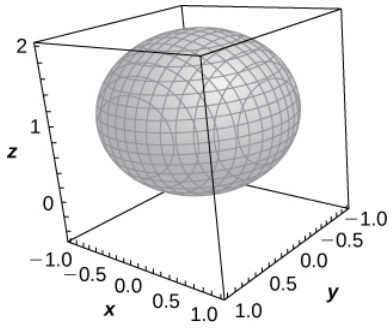


32) [T] $\varphi = \frac{\pi}{3}$

33) [T] $\rho = 2\cos\varphi$

Answer

Sphere of equation $x^2 + y^2 + (z-1)^2 = 1$ centered at $(0, 0, 1)$ with radius 1,

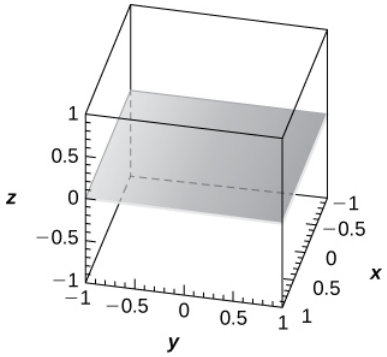


34) [T] $\rho = 4\csc\varphi$

35) [T] $\varphi = \frac{\pi}{2}$

Answer

The xy-plane of equation $z = 0$,



36) [T] $\rho = 6 \csc \varphi \sec \theta$

For exercises 37 - 40, the equation of a surface in rectangular coordinates is given. Find the equation of the surface in spherical coordinates. Identify the surface.

37) $x^2 + y^2 - 3z^2 = 0, \quad z \neq 0$

Answer

$\varphi = \frac{\pi}{3}$ or $\varphi = \frac{2\pi}{3}$; Elliptic cone

38) $x^2 + y^2 + z^2 - 4z = 0$

39) $z = 6$

Answer

$\rho \cos \varphi = 6$; Plane at $z = 6$

40) $x^2 + y^2 = 9$

For exercises 41 - 44, the cylindrical coordinates of a point are given. Find its associated spherical coordinates, with the measure of the angle φ in radians rounded to four decimal places.

41) [T] $(1, \frac{\pi}{4}, 3)$

Answer

$\left(\sqrt{10}, \frac{\pi}{4}, 0.3218\right)$

42) [T] $(5, \pi, 12)$

43) $(3, \frac{\pi}{2}, 3)$

Answer

$(3\sqrt{2}, \frac{\pi}{2}, \frac{\pi}{4})$

44) $\left(3, -\frac{\pi}{6}, 3\right)$

For exercises 45 - 48, the spherical coordinates of a point are given. Find its associated cylindrical coordinates.

45) $(2, -\frac{\pi}{4}, \frac{\pi}{2})$

Answer

$(2, -\frac{\pi}{4}, 0)$

46) $(4, \frac{\pi}{4}, \frac{\pi}{6})$

47) $\left(8, \frac{\pi}{3}, \frac{\pi}{2}\right)$

Answer

$\left(8, \frac{\pi}{3}, 0\right)$

48) $\left(9, -\frac{\pi}{6}, \frac{\pi}{3}\right)$

For exercises 49 - 52, find the most suitable system of coordinates to describe the solids.

49) The solid situated in the first octant with a vertex at the origin and enclosed by a cube of edge length a, where $a > 0$

Answer

Cartesian system, $\{(x, y, z) \mid 0 \leq x \leq a,\ 0 \leq y \leq a,\ 0 \leq z \leq a\}$

50) A spherical shell determined by the region between two concentric spheres centered at the origin, of radii of a and b, respectively, where $b > a > 0$

51) A solid inside sphere $x^2 + y^2 + z^2 = 9$ and outside cylinder $\left(x - \frac{3}{2}\right)^2 + y^2 = \frac{9}{4}$

Answer

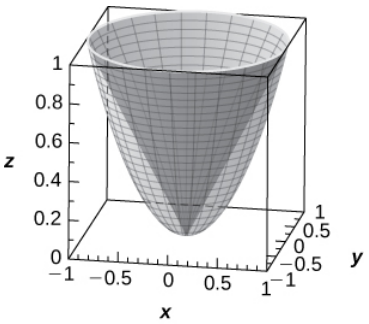
Cylindrical system, $\{(r, \theta, z) \mid r^2 + z^2 \leq 9,\ r \geq 3\cos\theta,\ 0 \leq \theta \leq 2\pi\}$

52) A cylindrical shell of height 10 determined by the region between two cylinders with the same center, parallel rulings, and radii of 2 and 5, respectively

53) [T] Use a CAS or CalcPlot3D to graph in cylindrical coordinates the region between elliptic paraboloid $z = x^2 + y^2$ and cone $x^2 + y^2 - z^2 = 0$.

Answer

The region is described by the set of points $\{(r, \theta, z) \mid 0 \leq r \leq 1,\ 0 \leq \theta \leq 2\pi,\ r^2 \leq z \leq r\}$.



54) [T] Use a CAS or CalcPlot3D to graph in spherical coordinates the "ice cream-cone region" situated above the xy-plane between sphere $x^2 + y^2 + z^2 = 4$ and elliptical cone $x^2 + y^2 - z^2 = 0$.

55) Washington, DC, is located at $39°$ N and $77°$ W (see the following figure). Assume the radius of Earth is 4000 mi. Express the location of Washington, DC, in spherical coordinates.



Answer

$(4000, -77°, 51°)$

56) San Francisco is located at $37.78°N$ and $122.42°W$. Assume the radius of Earth is 4000mi. Express the location of San Francisco in spherical coordinates.

57) Find the latitude and longitude of Rio de Janeiro if its spherical coordinates are $(4000, -43.17°, 102.91°)$.

Answer

$43.17°W, 22.91°S$

58) Find the latitude and longitude of Berlin if its spherical coordinates are $(4000, 13.38°, 37.48°)$.

59) [T] Consider the torus of equation $\left(x^2+y^2+z^2+R^2-r^2\right)^2 = 4R^2(x^2+y^2)$, where $R \geq r > 0$.

a. Write the equation of the torus in spherical coordinates.

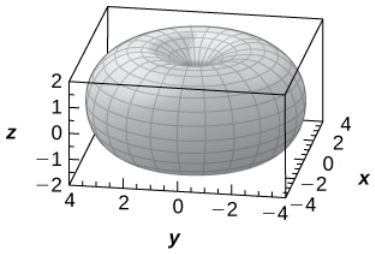
b. If $R = r$, the surface is called a horn torus. Show that the equation of a horn torus in spherical coordinates is $\rho = 2R\sin\varphi$.

c. Use a CAS or CalcPlot3D to graph the horn torus with $R = r = 2$ in spherical coordinates.

Answer

a. $\rho = 0, \quad \rho + R^2 - r^2 - 2R\sin\varphi = 0$

c.



60) [T] The "bumpy sphere" with an equation in spherical coordinates is $\rho = a + b\cos(m\theta)\sin(n\varphi)$, with $\theta \in [0, 2\pi]$ and $\varphi \in [0, \pi]$, where a and b are positive numbers and m and n are positive integers, may be used in applied mathematics to model tumor growth.

a. Show that the "bumpy sphere" is contained inside a sphere of equation $\rho = a + b$. Find the values of θ and φ at which the two surfaces intersect.

b. Use a CAS or CalcPlot3D to graph the surface for $a = 14, b = 2, m = 4$, and $n = 6$ along with sphere $\rho = a + b$.

c. Find the equation of the intersection curve of the surface at b. with the cone $\varphi = \frac{\pi}{12}$. Graph the intersection curve in the plane of intersection.

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

Links to CalcPlot3D added by Paul Seeburger (Monroe Community College).

This page titled 12.7E: Exercises for Section 12.7 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.R: Chapter 12 Review Exercises

For exercises 1 - 4, determine whether the statement is *true* or *false.* Justify the answer with a proof or a counterexample.

1) For vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ and any given scalar c, $c(\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}) = (c\vec{\mathbf{a}}) \cdot \vec{\mathbf{b}}$.

Answer

True; See proof in Section 11.3

2) For vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ and any given scalar c, $c(\vec{\mathbf{a}} \times \vec{\mathbf{b}}) = (c\vec{\mathbf{a}}) \times \vec{\mathbf{b}}$.

3) The symmetric equation for the line of intersection between two planes $x + y + z = 2$ and $x + 2y - 4z = 5$ is given by $-\frac{x-1}{6} = \frac{y-1}{5} = z$.

Answer

False, Converting the symmetric equations above to the parametric equations of the line, we get:

$x = 1 - 6t$

$y = 1 + 5t$

$z = t$

If this line lies on each plane, we should get an identity (like 5 = 5) when we substitute each expression of t in the each plane's equation.

Substituting into the first plane equation, we get: $(1 - 6t) + (1 + 5t) + t = 2$ ✓

Thus we know that this line does lie on the first plane.

But when we substitute into the second plane equation, we obtain: $(1 - 6t) + 2(1 + 5t) - 4(t) = 1 - 6t + 2 + 10t - 4t = 3 \neq 5$

Since we do not get an identity, we know this line is **not** on the second plane and can therefore not be the line of intersection of the two planes.

4) If $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = 0$, then $\vec{\mathbf{a}}$ is perpendicular to $\vec{\mathbf{b}}$.

Answer

False, since $\vec{\mathbf{a}}$ or $\vec{\mathbf{b}}$ could also be the zero vector.

For exercises 5 and 6, use the given vectors to find the quantities.

5) $\vec{\mathbf{a}} = 9\hat{\mathbf{i}} - 2\hat{\mathbf{j}}, \quad \vec{\mathbf{b}} = -3\hat{\mathbf{i}} + \hat{\mathbf{k}}$

a. $3\vec{\mathbf{a}} + \vec{\mathbf{b}}$

b. $\|\vec{\mathbf{a}}\|$

c. $\vec{\mathbf{a}} \times \|\vec{\mathbf{b}} \times \vec{\mathbf{a}}\|$

d. $\|\vec{\mathbf{b}} \times \vec{\mathbf{a}}\|$

Answer

a. $\langle 24, -6, 1 \rangle$

b. $\sqrt{85}$

c. Can't cross a vector with a scalar

d. 11

6) $\vec{\mathbf{a}} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 9\hat{\mathbf{k}}, \quad \vec{\mathbf{b}} = -\hat{\mathbf{i}} + 2\hat{\mathbf{k}}, \quad \vec{\mathbf{c}} = 4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$

a. $2\vec{\mathbf{a}} - \vec{\mathbf{b}}$

b. $\|\vec{\mathbf{b}} \times \vec{\mathbf{c}}\|$

c. $\vec{\mathbf{b}} \times \left(\vec{\mathbf{b}} \times \vec{\mathbf{c}}\right)$

d. $\vec{\mathbf{c}} \times \|\vec{\mathbf{b}} \times \vec{\mathbf{a}}\|$

e. $\text{Proj}_{\vec{\mathbf{a}}}\, \vec{\mathbf{b}}$

7) Find the values of a such that vectors $\langle 2, 4, a\rangle$ and $\langle 0, -1, a\rangle$ are orthogonal.

Answer

$a = \pm 2$

For exercises 8 and 9, find the unit vectors.

8) Find the unit vector that has the same direction as vector $\vec{\mathbf{v}}$ that begins at $(0, -3)$ and ends at $(4, 10)$.

9) Find the unit vector that has the same direction as vector $\vec{\mathbf{v}}$ that begins at $(1, 4, 10)$ and ends at $(3, 0, 4)$.

Answer

$\langle \frac{1}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, -\frac{3}{\sqrt{14}}\rangle = \langle \frac{\sqrt{14}}{14}, -\frac{\sqrt{14}}{7}, -\frac{3\sqrt{14}}{14}\rangle$

For exercises 10 and 11, find the area or volume of the given shapes.

10) The parallelogram spanned by vectors $\vec{\mathbf{a}} = \langle 1, 13\rangle$ and $\vec{\mathbf{b}} = \langle 3, 21\rangle$

11) The parallelepiped formed by $\vec{\mathbf{a}} = \langle 1, 4, 1\rangle$ and $\vec{\mathbf{b}} = \langle 3, 6, 2\rangle$, and $\vec{\mathbf{c}} = \langle -2, 1, -5\rangle$

Answer

27 units^2

For exercises 12 and 13, find the parametric equations and the vector equation of the line with the given properties.

12) The line that passes through point $(2, -3, 7)$ that is parallel to vector $\langle 1, 3, -2\rangle$

13) The line that passes through points $(1, 3, 5)$ and $(-2, 6, -3)$

Answer

$x = 1 - 3t, y = 3 + 3t, z = 5 - 8t, \quad \vec{\mathbf{r}}(t) = (1 - 3t)\hat{\mathbf{i}} + 3(1 + t)\hat{\mathbf{j}} + (5 - 8t)\hat{\mathbf{k}}$

For exercises 14 and 15, find the equation of the plane with the given properties.

14) The plane that passes through point $(4, 7, -1)$ and has normal vector $\vec{\mathbf{n}} = \langle 3, 4, 2\rangle$

15) The plane that passes through points $(0, 1, 5), (2, -1, 6)$, and $(3, 2, 5)$.

Answer

$-x + 3y + 8z = 43$

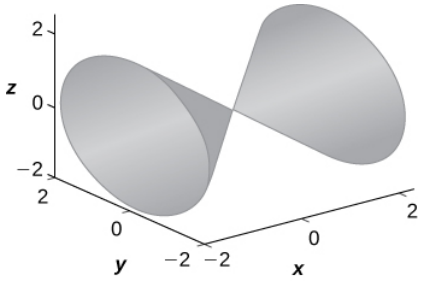
For exercises 16 and 17, find the traces for the surfaces in planes $x = k$, $y = k$, and $z = k$. Then, describe and draw the surfaces.

16) $9x^2 + 4y^2 - 16y + 36z^2 = 20$

17) $x^2 = y^2 + z^2$

Answer

$x = k$ trace: $k^2 = y^2 + z^2$ is a circle, $y = k$ trace: $x^2 - z^2 = k^2$ is a hyperbola (or a pair of lines if $k = 0$), $z = k$ trace: $x^2 - y^2 = k^2$ is a hyperbola (or a pair of lines if $k = 0$). The surface is a cone.



For exercises 18 and 19, write the given equation in cylindrical coordinates and spherical coordinates.

18) $x^2 + y^2 + z^2 = 144$

19) $z = x^2 + y^2 - 1$

Answer

Cylindrical: $z = r^2 - 1$, spherical: $\cos\varphi = \rho \sin^2 \varphi - \frac{1}{\rho}$

For exercises 20 and 21, convert the given equations from cylindrical or spherical coordinates to rectangular coordinates. Identify the given surface.

20) $\rho^2(\sin^2(\varphi) - \cos^2(\varphi)) = 1$

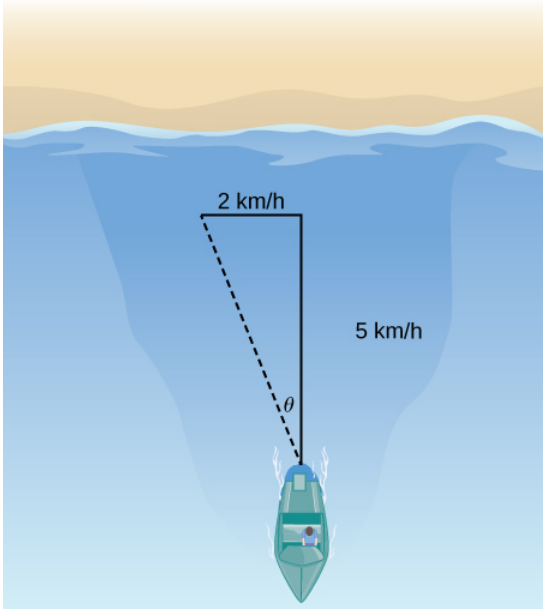
21) $r^2 - 2r\cos(\theta) + z^2 = 1$

Answer

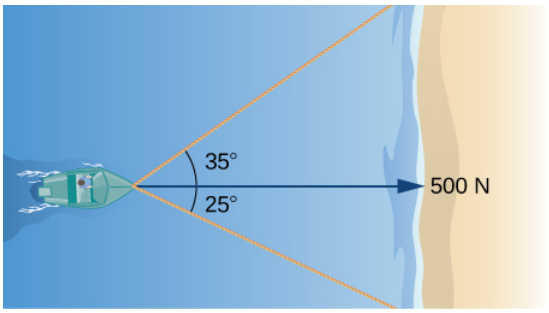
$x^2 - 2x + y^2 + z^2 = 1$, sphere

For exercises 22 and 23, consider a small boat crossing a river.

22) If the boat velocity is 5 km/h due north in still water and the water has a current of 2 km/h due west (see the following figure), what is the velocity of the boat relative to shore? What is the angle θ that the boat is actually traveling?



23) When the boat reaches the shore, two ropes are thrown to people to help pull the boat ashore. One rope is at an angle of $25°$ and the other is at $35°$. If the boat must be pulled straight and at a force of 500 N, find the magnitude of force for each rope (see the following figure).



Answer

331 N and 244 N

24) An airplane is flying in the direction of 52° east of north with a speed of 450 mph. A strong wind has a bearing 33° east of north with a speed of 50 mph. What is the resultant ground speed and bearing of the airplane?

25) Calculate the work done by moving a particle from position $(1, 2, 0)$ to $(8, 4, 5)$ along a straight line with a force $\vec{\mathbf{F}} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$.

Answer

15 J

In problems 26 and 27, consider your unsuccessful attempt to take the tire off your car using a wrench to loosen the bolts. Assume the wrench is 0.3 m long and you are able to apply a 200-N force.

26) Because your tire is flat, you are only able to apply your force at a $60°$ angle. What is the torque at the center of the bolt? Assume this force is not enough to loosen the bolt.

27) Someone lends you a tire jack and you are now able to apply a 200-N force at an $80°$ angle. Is your resulting torque going to be more or less? What is the new resulting torque at the center of the bolt? Assume this force is not enough to loosen the bolt.

Answer

More, 59.09 J

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 12.R: Chapter 12 Review Exercises is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

CHAPTER OVERVIEW

13: Vector-Valued Functions

A vector-valued function, also referred to as a vector function, is a mathematical function of one or more variables whose range is a set of multidimensional vectors or infinite-dimensional vectors. The input of a vector-valued function could be a scalar or a vector. Vector-valued functions provide a useful method for studying various curves both in the plane and in three-dimensional space. We can apply this concept to calculate the velocity, acceleration, arc length, and curvature of an object's trajectory. In this chapter, we examine these methods and show how they are used.

13.0: Prelude to Vector-Valued Functions

13.1: Vector-Valued Functions and Space Curves

13.1E: Exercises for Section 13.1

13.2: Calculus of Vector-Valued Functions

13.2E: Exercises for Section 13.2

13.3: Arc Length and Curvature

13.3E: Exercises for Section 13.3

13.4: Motion in Space

13.4E: Exercises for Section 13.4

13.R: Chapter 13 Review Exercises

This page titled 13: Vector-Valued Functions is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

13.0: Prelude to Vector-Valued Functions

In 1705, using Sir Isaac Newton's new laws of motion, the astronomer Edmond Halley made a prediction. He stated that comets that had appeared in 1531, 1607, and 1682 were actually the same comet and that it would reappear in 1758. Halley was proved to be correct, although he did not live to see it. However, the comet was later named in his honor. Halley's Comet follows an elliptical path through the solar system, with the Sun appearing at one focus of the ellipse. This motion is predicted by Johannes Kepler's first law of planetary motion, which we mentioned briefly previously. Kepler's third law of planetary motion can be used with the calculus of vector-valued functions to find the average distance of Halley's Comet from the Sun.



Figure 13.0.1: Halley's Comet appeared in view of Earth in 1986 and will appear again in 2061.

Vector-valued functions provide a useful method for studying various curves both in the plane and in three-dimensional space. We can apply this concept to calculate the velocity, acceleration, arc length, and curvature of an object's trajectory. In this chapter, we examine these methods and show how they are used.

This page titled 13.0: Prelude to Vector-Valued Functions is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

13.1: Vector-Valued Functions and Space Curves

Learning Objectives

- Write the general equation of a vector-valued function in component form and unit-vector form.
- Recognize parametric equations for a space curve.
- Describe the shape of a helix and write its equation.
- Define the limit of a vector-valued function.

Our study of vector-valued functions combines ideas from our earlier examination of single-variable calculus with our description of vectors in three dimensions from the preceding chapter. In this section, we extend concepts from earlier chapters and also examine new ideas concerning curves in three-dimensional space. These definitions and theorems support the presentation of material in the rest of this chapter and also in the remaining chapters of the text.

Definition of a Vector-Valued Function

Our first step in studying the calculus of vector-valued functions is to define what exactly a vector-valued function is. We can then look at graphs of vector-valued functions and see how they define curves in both two and three dimensions.

Definition: Vector-valued Functions

A vector-valued function is a function of the form

$$\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} \quad \text{or} \quad \vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}},$$

where the component functions f, g, and h, are real-valued functions of the parameter t. Vector-valued functions are also written in the form

$$\vec{\mathbf{r}}(t) = \langle f(t),\, g(t)\rangle \quad \text{or} \quad \vec{\mathbf{r}}(t) = \langle f(t),\, g(t),\, h(t)\rangle.$$

In both cases, the first form of the function defines a two-dimensional vector-valued function; the second form describes a three-dimensional vector-valued function.

The parameter t can lie between two real numbers: $a \le t \le b$. Another possibility is that the value of t might take on all real numbers. Last, the component functions themselves may have domain restrictions that enforce restrictions on the value of t. We often use t as a parameter because t can represent time.

Example 13.1.1: Evaluating Vector-Valued Functions and Determining Domains

For each of the following vector-valued functions, evaluate $\vec{\mathbf{r}}(0)$, $\vec{\mathbf{r}}(\frac{\pi}{2})$, and $\vec{\mathbf{r}}(\frac{2\pi}{3})$. Do any of these functions have domain restrictions?

a. $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}}$
b. $\vec{\mathbf{r}}(t) = 3\tan t\,\hat{\mathbf{i}} + 4\sec t\,\hat{\mathbf{j}} + 5t\,\hat{\mathbf{k}}$

Solution

a. To calculate each of the function values, substitute the appropriate value of t into the function:

$$\begin{aligned}\vec{\mathbf{r}}(0) &= 4\cos(0)\hat{\mathbf{i}} + 3\sin(0)\hat{\mathbf{j}} \\ &= 4\hat{\mathbf{i}} + 0\hat{\mathbf{j}} = 4\hat{\mathbf{i}} \\ \vec{\mathbf{r}}\left(\frac{\pi}{2}\right) &= 4\cos\left(\frac{\pi}{2}\right)\hat{\mathbf{i}} + 3\sin\left(\frac{\pi}{2}\right)\hat{\mathbf{j}} \\ &= 0\hat{\mathbf{i}} + 3\hat{\mathbf{j}} = 3\hat{\mathbf{j}} \\ \vec{\mathbf{r}}\left(\frac{2\pi}{3}\right) &= 4\cos\left(\frac{2\pi}{3}\right)\hat{\mathbf{i}} + 3\sin\left(\frac{2\pi}{3}\right)\hat{\mathbf{j}} \\ &= 4\left(-\tfrac{1}{2}\right)\hat{\mathbf{i}} + 3\left(\tfrac{\sqrt{3}}{2}\right)\hat{\mathbf{j}} = -2\hat{\mathbf{i}} + \tfrac{3\sqrt{3}}{2}\hat{\mathbf{j}}\end{aligned}$$

To determine whether this function has any domain restrictions, consider the component functions separately. The first component function is $f(t) = 4\cos t$ and the second component function is $g(t) = 3\sin t$. Neither of these functions has a domain restriction, so the domain of $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}}$ is all real numbers.

b. To calculate each of the function values, substitute the appropriate value of t into the function:

$$\begin{aligned}\vec{\mathbf{r}}(0) &= 3\tan(0)\hat{\mathbf{i}} + 4\sec(0)\hat{\mathbf{j}} + 5(0)\hat{\mathbf{k}} \\ &= 0\hat{\mathbf{i}} + 4j + 0\hat{\mathbf{k}} = 4\hat{\mathbf{j}} \\ \vec{\mathbf{r}}\left(\frac{\pi}{2}\right) &= 3\tan\left(\frac{\pi}{2}\right)\hat{\mathbf{i}} + 4\sec\left(\frac{\pi}{2}\right)\hat{\mathbf{j}} + 5\left(\frac{\pi}{2}\right)\hat{\mathbf{k}}, \text{ which does not exist} \\ \vec{\mathbf{r}}\left(\frac{2\pi}{3}\right) &= 3\tan\left(\frac{2\pi}{3}\right)\hat{\mathbf{i}} + 4\sec\left(\frac{2\pi}{3}\right)\hat{\mathbf{j}} + 5\left(\frac{2\pi}{3}\right)\hat{\mathbf{k}} \\ &= 3(-\sqrt{3})\hat{\mathbf{i}} + 4(-2)\hat{\mathbf{j}} + \frac{10\pi}{3}\hat{\mathbf{k}} \\ &= (-3\sqrt{3})\hat{\mathbf{i}} - 8\hat{\mathbf{j}} + \frac{10\pi}{3}\hat{\mathbf{k}}\end{aligned}$$

To determine whether this function has any domain restrictions, consider the component functions separately. The first component function is $f(t) = 3\tan t$, the second component function is $g(t) = 4\sec t$, and the third component function is $h(t) = 5t$. The first two functions are not defined for odd multiples of $\frac{\pi}{2}$, so the function is not defined for odd multiples of $\frac{\pi}{2}$. Therefore,

$$\mathrm{D}_{\vec{\mathbf{r}}} = \left\{t \,|\, t \neq \frac{(2n+1)\pi}{2}\right\},$$

where n is any integer.

? Exercise 13.1.1

For the vector-valued function $\vec{\mathbf{r}}(t) = (t^2 - 3t)\,\hat{\mathbf{i}} + (4t + 1)\,\hat{\mathbf{j}}$, evaluate $\vec{\mathbf{r}}(0)$, $\vec{\mathbf{r}}(1)$, and $\vec{\mathbf{r}}(-4)$. Does this function have any domain restrictions?

Hint

Substitute the appropriate values of t into the function.

Answer

$\vec{\mathbf{r}}(0) = \hat{\mathbf{j}}$, $\vec{\mathbf{r}}(1) = -2\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$, $\vec{\mathbf{r}}(-4) = 28\hat{\mathbf{i}} - 15\hat{\mathbf{j}}$

The domain of $\vec{\mathbf{r}}(t) = (t^2 - 3t)\hat{\mathbf{i}} + (4t + 1)\hat{\mathbf{j}}$ is all real numbers.

Example 13.1.1 illustrates an important concept. The domain of a vector-valued function consists of real numbers. The domain can be all real numbers or a subset of the real numbers. The range of a vector-valued function consists of vectors. Each real number in the domain of a vector-valued function is mapped to either a two- or a three-dimensional vector.

Graphing Vector-Valued Functions

Recall that a plane vector consists of two quantities: direction and magnitude. Given any point in the plane (the *initial point*), if we move in a specific direction for a specific distance, we arrive at a second point. This represents the *terminal point* of the vector. We calculate the components of the vector by subtracting the coordinates of the initial point from the coordinates of the terminal point.

A vector is considered to be in *standard position* if the initial point is located at the origin. When graphing a vector-valued function, we typically graph the vectors in the domain of the function in standard position, because doing so guarantees the uniqueness of the graph. This convention applies to the graphs of three-dimensional vector-valued functions as well. The graph of a vector-valued function of the form

$$\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}$$

consists of the set of all points $(f(t),\, g(t))$, and the path it traces is called a **plane curve**. The graph of a vector-valued function of the form

$$\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}$$

consists of the set of all points $(f(t),\, g(t),\, h(t))$, and the path it traces is called a **space curve**. Any representation of a plane curve or space curve using a vector-valued function is called a **vector parameterization** of the curve.

Each plane curve and space curve has an **orientation**, indicated by arrows drawn in on the curve, that shows the direction of motion along the curve as the value of the parameter t increases.

Example 13.1.2 : Graphing a Vector-Valued Function

Create a graph of each of the following vector-valued functions:

a. The plane curve represented by $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}}$, $0 \le t \le 2\pi$
b. The plane curve represented by $\vec{\mathbf{r}}(t) = 4\cos(t^3)\,\hat{\mathbf{i}} + 3\sin(t^3)\,\hat{\mathbf{j}}$, $0 \le t \le \sqrt[3]{2\pi}$
c. The space curve represented by $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$, $0 \le t \le 4\pi$

Solution

a. As with any graph, we start with a table of values. We then graph each of the vectors in the second column of the table in standard position and connect the terminal points of each vector to form a curve (Figure 13.1.1). This curve turns out to be an ellipse centered at the origin.

Table 13.1.1: Table of Values for $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}}$, $0 \le t \le 2\pi$

t	$\vec{\mathbf{r}}(t)$	t	$\vec{\mathbf{r}}(t)$
0	$4\hat{\mathbf{i}}$	π	$-4\hat{\mathbf{i}}$
$\frac{\pi}{4}$	$2\sqrt{2}\hat{\mathbf{i}} + \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$	$\frac{5\pi}{4}$	$-2\sqrt{2}\hat{\mathbf{i}} - \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$
$\frac{\pi}{2}$	$3\hat{\mathbf{j}}$	$\frac{3\pi}{2}$	$-3\hat{\mathbf{j}}$
$\frac{3\pi}{4}$	$-2\sqrt{2}\hat{\mathbf{i}} + \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$	$\frac{7\pi}{4}$	$2\sqrt{2}\hat{\mathbf{i}} - \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$
2π	$4\hat{\mathbf{i}}$		

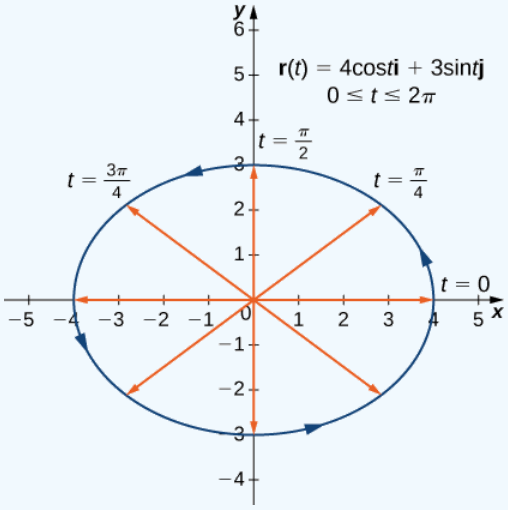


Figure 13.1.1: The graph of the first vector-valued function is an ellipse.

b. The table of values for $\vec{\mathbf{r}}(t) = 4\cos(t^3)\,\hat{\mathbf{i}} + 3\sin(t^3)\,\hat{\mathbf{j}}$, $0 \le t \le \sqrt[3]{2\pi}$ is as follows:

Table of Values for $\vec{\mathbf{r}}(t) = 4\cos(t^3)\,\hat{\mathbf{i}} + 3\sin(t^3)\,\hat{\mathbf{j}}$, $0 \le t \le \sqrt[3]{2\pi}$

t	$\vec{\mathbf{r}}(t)$	t	$\vec{\mathbf{r}}(t)$
0	$4\hat{\mathbf{i}}$	$\sqrt[3]{\pi}$	$-4\hat{\mathbf{i}}$
$\sqrt[3]{\frac{\pi}{4}}$	$2\sqrt{2}\hat{\mathbf{i}} + \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$	$\sqrt[3]{\frac{5\pi}{4}}$	$-2\sqrt{2}\hat{\mathbf{i}} - \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$
$\sqrt[3]{\frac{\pi}{2}}$	$3\hat{\mathbf{j}}$	$\sqrt[3]{\frac{3\pi}{2}}$	$-3\hat{\mathbf{j}}$
$\sqrt[3]{\frac{3\pi}{4}}$	$-2\sqrt{2}\hat{\mathbf{i}} + \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$	$\sqrt[3]{\frac{7\pi}{4}}$	$2\sqrt{2}\hat{\mathbf{i}} - \frac{3\sqrt{2}}{2}\hat{\mathbf{j}}$
$\sqrt[3]{2\pi}$	$4\hat{\mathbf{i}}$		

The graph of this curve is also an ellipse centered at the origin.

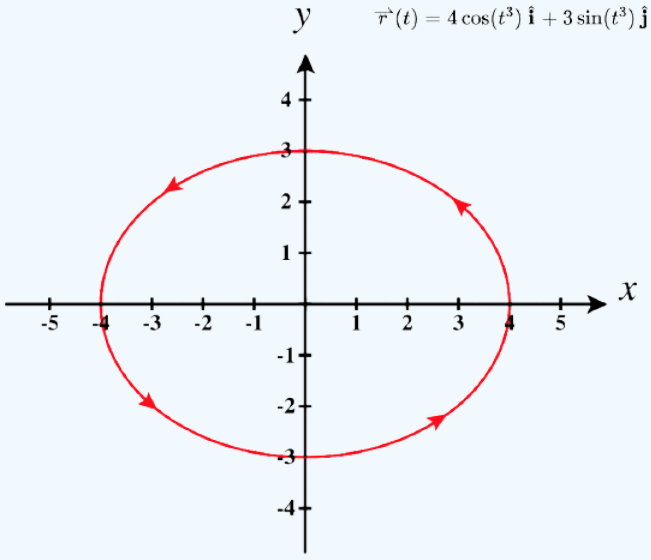


Figure 13.1.2: The graph of the second vector-valued function is also an ellipse.

c. We go through the same procedure for a three-dimensional vector function.

Table of Values for $\mathrm{r}(\mathrm{t}) = 4\cos\mathrm{t}\hat{\mathbf{i}} + 4\sin\mathrm{t}\hat{\mathbf{j}} + \mathrm{t}\hat{\mathbf{k}}$, $0 \le \mathrm{t} \le 4\pi$

t	$\vec{\mathbf{r}}(t)$	t	$\vec{\mathbf{r}}(t)$
0	$4\hat{\mathbf{i}}$	π	$-4\hat{\mathbf{i}}+\pi\hat{\mathbf{k}}$
$\frac{\pi}{4}$	$2\sqrt{2}\hat{\mathbf{i}}+2\sqrt{2}\hat{\mathbf{j}}+\frac{\pi}{4}\hat{\mathbf{k}}$	$\frac{5\pi}{4}$	$-2\sqrt{2}\hat{\mathbf{i}}-2\sqrt{2}\hat{\mathbf{j}}+\frac{5\pi}{4}\hat{\mathbf{k}}$
$\frac{\pi}{2}$	$4\hat{\mathbf{j}}+\frac{\pi}{2}\hat{\mathbf{k}}$	$\frac{3\pi}{2}$	$-4\hat{\mathbf{j}}+\frac{3\pi}{2}\hat{\mathbf{k}}$
$\frac{3\pi}{4}$	$-2\sqrt{2}\hat{\mathbf{i}}+2\sqrt{2}\hat{\mathbf{j}}+\frac{3\pi}{4}\hat{\mathbf{k}}$	$\frac{7\pi}{4}$	$2\sqrt{2}\hat{\mathbf{i}}-2\sqrt{2}\hat{\mathbf{j}}+\frac{7\pi}{4}\hat{\mathbf{k}}$
2π	$4\hat{\mathbf{j}}+2\pi\hat{\mathbf{k}}$		

The values then repeat themselves, except for the fact that the coefficient of $\hat{\mathbf{k}}$ is always increasing (13.1.3). This curve is called a helix. Notice that if the $\hat{\mathbf{k}}$ component is eliminated, then the function becomes $\vec{\mathbf{r}}(t)=4\cos t\hat{\mathbf{i}}+4\sin t\hat{\mathbf{j}}$, which is a circle of radius 4 centered at the origin.

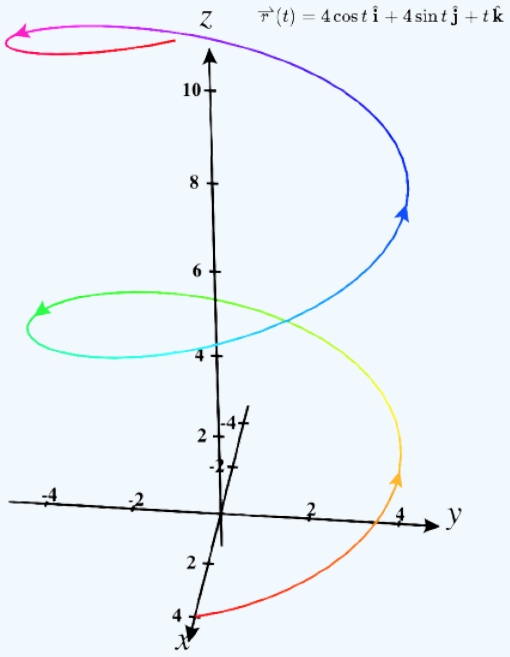


Figure 13.1.3: The graph of the third vector-valued function is a helix.

You may notice that the graphs in parts a. and b. are identical. This happens because the function describing the curve in part b. is a so-called reparameterization of the function describing the curve in part a. In fact, any curve has an infinite number of reparameterizations; for example, we can replace t with $2t$ in any of the three previous curves without changing the shape of the curve. The interval over which t is defined may change, but that is all. We return to this idea later in this chapter when we study arc-length parameterization. As mentioned, the name of the shape of the curve of the graph in 13.1.3 is a *helix*. The curve resembles a spring, with a circular cross-section looking down along the z-axis. It is possible for a helix to be elliptical in cross-section as well. For example, the vector-valued function $\vec{\mathbf{r}}(t)=4\cos t\,\hat{\mathbf{i}}+3\sin t\,\hat{\mathbf{j}}+t\,\hat{\mathbf{k}}$ describes an elliptical helix. The projection of this helix into the xy-plane is an ellipse. Last, the arrows in the graph of this helix indicate the orientation of the curve as t progresses from 0 to 4π.

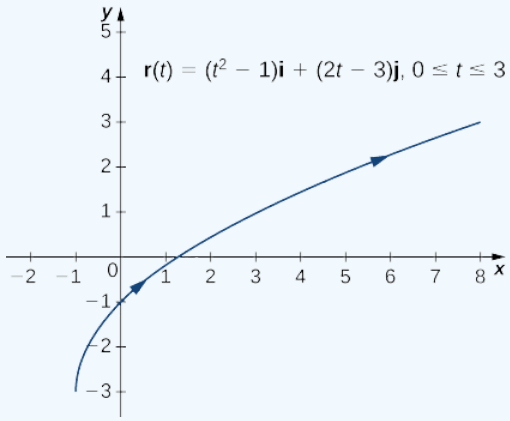
? Exercise 13.1.2

Create a graph of the vector-valued function $\vec{\mathbf{r}}(t)=(t^2-1)\hat{\mathbf{i}}+(2t-3)\hat{\mathbf{j}}$, $0\le t\le 3$.

Hint

Start by making a table of values, then graph the vectors for each value of t.

Answer



At this point, you may notice a similarity between vector-valued functions and parameterized curves. Indeed, given a vector-valued function $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}$ we can define $x = f(t)$ and $y = g(t)$. If a restriction exists on the values of t (for example, t is restricted to the interval $[a, b]$ for some constants $a < b$, then this restriction is enforced on the parameter. The graph of the parameterized function would then agree with the graph of the vector-valued function, except that the vector-valued graph would represent vectors rather than points. Since we can parameterize a curve defined by a function $y = f(x)$, it is also possible to represent an arbitrary plane curve by a vector-valued function.

Limits and Continuity of a Vector-Valued Function

We now take a look at the limit of a vector-valued function. This is important to understand to study the calculus of vector-valued functions.

Definition: limit of a vector-valued function

A vector-valued function $\vec{\mathbf{r}}$ approaches the limit $\vec{\mathbf{L}}$ as t approaches a, written

$$\lim_{t \to a} \vec{\mathbf{r}}(t) = \vec{\mathbf{L}},$$

provided

$$\lim_{t \to a} \left\| \vec{\mathbf{r}}(t) - \vec{\mathbf{L}} \right\| = 0.$$

This is a rigorous definition of the limit of a vector-valued function. In practice, we use the following theorem:

Theorem: Limit of a vector-valued function

Let f, g, and h be functions of t. Then the limit of the vector-valued function $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ as t approaches a is given by

$$\lim_{t \to a} \vec{\mathbf{r}}(t) = [\lim_{t \to a} f(t)]\hat{\mathbf{i}} + [\lim_{t \to a} g(t)]\hat{\mathbf{j}}, \tag{13.1.1}$$

provided the limits $\lim_{t \to a} f(t)$ and $\lim_{t \to a} g(t)$ exist.

Similarly, the limit of the vector-valued function $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$ as t approaches a is given by

$$\lim_{t \to a} \vec{\mathbf{r}}(t) = [\lim_{t \to a} f(t)]\hat{\mathbf{i}} + [\lim_{t \to a} g(t)]\hat{\mathbf{j}} + [\lim_{t \to a} h(t)]\hat{\mathbf{k}}, \tag{13.1.2}$$

provided the limits $\lim_{t \to a} f(t)$, $\lim_{t \to a} g(t)$ and $\lim_{t \to a} h(t)$ exist.

In the following example, we show how to calculate the limit of a vector-valued function.

Example 13.1.3: Evaluating the Limit of a Vector-Valued Function

For each of the following vector-valued functions, calculate $\lim\limits_{t\to 3}\vec{\mathbf{r}}(t)$ for

a. $\vec{\mathbf{r}}(t)=(t^2-3t+4)\hat{\mathbf{i}}+(4t+3)\hat{\mathbf{j}}$
b. $\vec{\mathbf{r}}(t)=\frac{2t-4}{t+1}\hat{\mathbf{i}}+\frac{t}{t^2+1}\hat{\mathbf{j}}+(4t-3)\hat{\mathbf{k}}$

Solution

a. Use Equation 13.1.1 and substitute the value $t=3$ into the two component expressions:

$$\begin{aligned}\lim_{t\to 3}\vec{\mathbf{r}}(t) &= \lim_{t\to 3}\left[(t^2-3t+4)\hat{\mathbf{i}}+(4t+3)\hat{\mathbf{j}}\right]\\ &= \left[\lim_{t\to 3}(t^2-3t+4)\right]\hat{\mathbf{i}}+\left[\lim_{t\to 3}(4t+3)\right]\hat{\mathbf{j}}\\ &= 4\hat{\mathbf{i}}+15\hat{\mathbf{j}}\end{aligned}$$

b. Use Equation 13.1.2 and substitute the value $t=3$ into the three component expressions:

$$\begin{aligned}\lim_{t\to 3}\vec{\mathbf{r}}(t) &= \lim_{t\to 3}\left(\frac{2t-4}{t+1}\hat{\mathbf{i}}+\frac{t}{t^2+1}\hat{\mathbf{j}}+(4t-3)\hat{\mathbf{k}}\right)\\ &= \left[\lim_{t\to 3}\left(\frac{2t-4}{t+1}\right)\right]\hat{\mathbf{i}}+\left[\lim_{t\to 3}\left(\frac{t}{t^2+1}\right)\right]\hat{\mathbf{j}}+\left[\lim_{t\to 3}(4t-3)\right]\hat{\mathbf{k}}\\ &= \tfrac{1}{2}\hat{\mathbf{i}}+\tfrac{3}{10}\hat{\mathbf{j}}+9\hat{\mathbf{k}}\end{aligned}$$

Exercise 13.1.3

Calculate $\lim\limits_{t\to 2}\vec{\mathbf{r}}(t)$ for the function $\vec{\mathbf{r}}(t)=\sqrt{t^2+3t-1}\,\hat{\mathbf{i}}-(4t-3)\hat{\mathbf{j}}-\sin\frac{(t+1)\pi}{2}\hat{\mathbf{k}}$

Hint

Use Equation 13.1.2 from the preceding theorem.

Answer

$$\lim_{t\to 2}\vec{\mathbf{r}}(t)=3\hat{\mathbf{i}}-5\hat{\mathbf{j}}+\hat{\mathbf{k}}$$

Now that we know how to calculate the limit of a vector-valued function, we can define **continuity at a point** for such a function.

Definitions

Let f, g, and h be functions of t. Then, the vector-valued function $\vec{\mathbf{r}}(t)=f(t)\hat{\mathbf{i}}+g(t)\hat{\mathbf{j}}$ is **continuous at point** $t=a$ if the following three conditions hold:

1. $\vec{\mathbf{r}}(a)$ exists
2. $\lim\limits_{t\to a}\vec{\mathbf{r}}(t)$ exists
3. $\lim\limits_{t\to a}\vec{\mathbf{r}}(t)=\vec{\mathbf{r}}(a)$

Similarly, the vector-valued function $\vec{\mathbf{r}}(t)=f(t)\hat{\mathbf{i}}+g(t)\hat{\mathbf{j}}+h(t)\hat{\mathbf{k}}$ is **continuous at point** $t=a$ if the following three conditions hold:

1. $\vec{\mathbf{r}}(a)$ exists
2. $\lim\limits_{t\to a}\vec{\mathbf{r}}(t)$ exists
3. $\lim\limits_{t\to a}\vec{\mathbf{r}}(t)=\vec{\mathbf{r}}(a)$

Summary

- A vector-valued function is a function of the form $\vec{\mathbf{r}}(t)=f(t)\hat{\mathbf{i}}+g(t)\hat{\mathbf{j}}$ or $\vec{\mathbf{r}}(t)=f(t)\hat{\mathbf{i}}+g(t)\hat{\mathbf{j}}+h(t)\hat{\mathbf{k}}$, where the component functions f, g, and h are real-valued functions of the parameter t.

- The graph of a vector-valued function of the form $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ is called a *plane curve*. The graph of a vector-valued function of the form $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$ is called a *space curve*.
- It is possible to represent an arbitrary plane curve by a vector-valued function.
- To calculate the limit of a vector-valued function, calculate the limits of the component functions separately.

Key Equations

- **Vector-valued function**
  $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ or $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$,or $\vec{\mathbf{r}}(t) = \langle f(t), g(t)\rangle$ or $\vec{\mathbf{r}}(t) = \langle f(t), g(t), h(t)\rangle$
- **Limit of a vector-valued function**
  $\lim\limits_{t\to a} \vec{\mathbf{r}}(t) = [\lim\limits_{t\to a} f(t)]\hat{\mathbf{i}} + [\lim\limits_{t\to a} g(t)]\hat{\mathbf{j}}$ or $\lim\limits_{t\to a} \vec{\mathbf{r}}(t) = [\lim\limits_{t\to a} f(t)]\hat{\mathbf{i}} + [\lim\limits_{t\to a} g(t)]\hat{\mathbf{j}} + [\lim\limits_{t\to a} h(t)]\hat{\mathbf{k}}$

Glossary

component functions

the component functions of the vector-valued function $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ are $f(t)$ and $g(t)$, and the component functions of the vector-valued function $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$ are $f(t)$, $g(t)$ and $h(t)$

helix

a three-dimensional curve in the shape of a spiral

limit of a vector-valued function

a vector-valued function $\vec{\mathbf{r}}(t)$ has a limit $\vec{\mathbf{L}}$ as t approaches a if $\lim t \to a \left|\vec{\mathbf{r}}(t) - \vec{\mathbf{L}}\right| = 0$

plane curve

the set of ordered pairs $(f(t), g(t))$ together with their defining parametric equations $x = f(t)$ and $y = g(t)$

reparameterization

an alternative parameterization of a given vector-valued function

space curve

the set of ordered triples $(f(t), g(t), h(t))$ together with their defining parametric equations $x = f(t)$, $y = g(t)$ and $z = h(t)$

vector parameterization

any representation of a plane or space curve using a vector-valued function

vector-valued function

a function of the form $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ or $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$,where the component functions f, g, and h are real-valued functions of the parameter t.

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Edited by Paul Seeburger (Monroe Community College)

This page titled 13.1: Vector-Valued Functions and Space Curves is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

13.1E: EXERCISES FOR SECTION 13.1

INTRODUCTION TO VECTOR-VALUED FUNCTIONS

1) Give the component functions $x = f(t)$ and $y = g(t)$ for the vector-valued function $\vec{\mathbf{r}}(t) = 3\sec t\,\hat{\mathbf{i}} + 2\tan t\,\hat{\mathbf{j}}$.

2) Given $\vec{\mathbf{r}}(t) = 3\sec t\hat{\mathbf{i}} + 2\tan t\hat{\mathbf{j}}$, find the following values (if possible).

a. $\vec{\mathbf{r}}\left(\frac{\pi}{4}\right)$
b. $\vec{\mathbf{r}}(\pi)$
c. $\vec{\mathbf{r}}\left(\frac{\pi}{2}\right)$

3) Sketch the curve of the vector-valued function $\vec{\mathbf{r}}(t) = 3\sec t\hat{\mathbf{i}} + 2\tan t\hat{\mathbf{j}}$ and give the orientation of the curve. Sketch asymptotes as a guide to the graph.

LIMITS OF VECTOR-VALUED FUNCTIONS

4) Evaluate $\lim\limits_{t\to 0}\left(e^t\hat{\mathbf{i}} + \frac{\sin t}{t}\hat{\mathbf{j}} + e^{-t}\hat{\mathbf{k}}\right)$

5) Given the vector-valued function $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t\rangle$ find the following values:

a. $\lim\limits_{t\to\frac{\pi}{4}} \vec{\mathbf{r}}(t)$
b. $\vec{\mathbf{r}}\left(\frac{\pi}{3}\right)$
c. Is $\vec{\mathbf{r}}(t)$ continuous at $t = \frac{\pi}{3}$?
d. Graph $\vec{\mathbf{r}}(t)$.

6) Given the vector-valued function $\vec{\mathbf{r}}(t) = \langle t, t^2 + 1\rangle$, find the following values:

a. $\lim\limits_{t\to -3} \vec{\mathbf{r}}(t)$
b. $\vec{\mathbf{r}}(-3)$
c. Is $\vec{\mathbf{r}}(t)$ continuous at $x = -3$?
d. $\vec{\mathbf{r}}(t+2) - \vec{\mathbf{r}}(t)$

7) Let $\vec{\mathbf{r}}(t) = e^t\hat{\mathbf{i}} + \sin t\hat{\mathbf{j}} + \ln t\hat{\mathbf{k}}$. Find the following values:

a. $\vec{\mathbf{r}}\left(\frac{\pi}{4}\right)$
b. $\lim\limits_{t\to\frac{\pi}{4}} \vec{\mathbf{r}}(t)$
c. Is $\vec{\mathbf{r}}(t)$ continuous at $t = \frac{\pi}{4}$?

For exercises 8 - 13, find the limit of the following vector-valued functions at the indicated value of t.

8) $\lim\limits_{t\to 4}\langle\sqrt{t-3}, \frac{\sqrt{t}-2}{t-4}, \tan\left(\frac{\pi}{t}\right)\rangle$

9) $\lim\limits_{t\to\frac{\pi}{2}} \vec{\mathbf{r}}(t)$ for $\vec{\mathbf{r}}(t) = e^t\hat{\mathbf{i}} + \sin t\hat{\mathbf{j}} + \ln t\hat{\mathbf{k}}$

10) $\lim\limits_{t\to\infty}\langle e^{-2t}, \frac{2t+3}{3t-1}, \arctan(2t)\rangle$

11) $\lim\limits_{t\to e^2}\langle t\ln(t), \frac{\ln t}{t^2}, \sqrt{\ln(t^2)}\rangle$

12) $\lim\limits_{t\to\frac{\pi}{6}}\langle\cos 2t, \sin 2t, 1\rangle$

13) $\lim\limits_{t\to\infty} \vec{\mathbf{r}}(t)$ for $\vec{\mathbf{r}}(t) = 2e^{-t}\mathbf{i} + e^{-t}\hat{\mathbf{j}} + \ln(t-1)\hat{\mathbf{k}}$

DOMAIN OF A VECTOR-VALUED FUNCTION

For problems 14 - 17, find the domain of the vector-valued functions.

14) Domain: $\vec{\mathbf{r}}(t) = \langle t^2, t, \sin t\rangle$

15) Domain: $\vec{\mathbf{r}}(t) = \langle t^2, \tan t, \ln t\rangle$

16) Domain: $\vec{\mathbf{r}}(t) = \langle t^2, \sqrt{t-3}, \frac{3}{2t+1}\rangle$

17) Domain: $\vec{\mathbf{r}}(t) = \langle \csc(t), \frac{1}{\sqrt{t-3}}, \ln(t-2)\rangle$

18) a. Find the domain of $\vec{\mathbf{r}}(t) = 2e^{-t}\hat{\mathbf{i}} + e^{-t}\hat{\mathbf{j}} + \ln(t-1)\hat{\mathbf{k}}$.

b. For what values of t is $\vec{\mathbf{r}}(t) = 2e^{-t}\hat{\mathbf{i}} + e^{-t}\hat{\mathbf{j}} + \ln(t-1)\hat{\mathbf{k}}$ continuous?

19) Domain: $\vec{\mathbf{r}}(t) = (\arccos t)\,\hat{\mathbf{i}} + \sqrt{2t-1}\,\hat{\mathbf{j}} + \ln(t)\,\hat{\mathbf{k}}$

VISUALIZING VECTOR-VALUED FUNCTIONS

20) Describe the curve defined by the vector-valued function $\vec{\mathbf{r}}(t) = (1+t)\hat{\mathbf{i}} + (2+5t)\hat{\mathbf{j}} + (-1+6t)\hat{\mathbf{k}}$.

21) Let $\vec{\mathbf{r}}(t) = \langle \cos t, t, \sin t \rangle$ and use it to answer the following questions.

a. For what values of t is $\vec{\mathbf{r}}(t)$ continuous?

b. Sketch the graph of $\vec{\mathbf{r}}(t)$.

22) Produce a careful sketch of the graph of $\vec{\mathbf{r}}(t) = t^2\hat{\mathbf{i}} + t\hat{\mathbf{j}}$.

In questions 23 - 25, use a graphing utility to sketch each of the vector-valued functions:

23) **[T]** $\vec{\mathbf{r}}(t) = 2\cos^2 t\hat{\mathbf{i}} + (2-\sqrt{t})\hat{\mathbf{j}}$

24) **[T]** $\vec{\mathbf{r}}(t) = \langle e^{\cos(3t)}, e^{-\sin(t)} \rangle$

25) **[T]** $\vec{\mathbf{r}}(t) = \langle 2 - \sin(2t), 3 + 2\cos t \rangle$

FINDING EQUATIONS IN x AND y FOR THE PATH TRACED OUT BY VECTOR-VALUED FUNCTIONS

For questions 26-33, eliminate the parameter t, write the equation in Cartesian coordinates, then sketch the graph of the vector-valued functions.

26) $\vec{\mathbf{r}}(t) = 2t\hat{\mathbf{i}} + t^2\hat{\mathbf{j}}$

(*Hint:* Let $x = 2t$ and $y = t^2$. Solve the first equation for t in terms of x and substitute this result into the second equation.)

27) $\vec{\mathbf{r}}(t) = t^3\hat{\mathbf{i}} + 2t\hat{\mathbf{j}}$

28) $\vec{\mathbf{r}}(t) = \sin t\hat{\mathbf{i}} + \cos t\hat{\mathbf{j}}$

29) $\vec{\mathbf{r}}(t) = 3\cos t\hat{\mathbf{i}} + 3\sin t\hat{\mathbf{j}}$

30) $\vec{\mathbf{r}}(t) = \langle \sin t, 4\cos t \rangle$

31) $\vec{\mathbf{r}}(t) = 2\sin t\hat{\mathbf{i}} - 3\cos t\hat{\mathbf{j}}$

32) $\vec{\mathbf{r}}(t) = \tan t\hat{\mathbf{i}} - 2\sec t\hat{\mathbf{j}}$

33) $\vec{\mathbf{r}}(t) = 3\sec t\hat{\mathbf{i}} + 4\tan t\hat{\mathbf{j}}$

FINDING A VECTOR-VALUED FUNCTION TO TRACE OUT THE GRAPH OF AN EQUATION IN x AND y

For questions 34 - 40, find a vector-valued function that traces out the given curve in the indicated direction.

34) $4x^2 + 9y^2 = 36$; clockwise and counterclockwise

35) $y = x^2$; from left to right

36) The line through P and Q where P is $(1, 4, -2)$ and Q is $(3, 9, 6)$

37) The circle, $x^2 + y^2 = 36$, oriented clockwise, with position $(-6, 0)$ at time $t = 0$.

38) The ellipse, $x^2 + \dfrac{y^2}{36} = 1$, oriented counterclockwise

39) The hyperbola, $\dfrac{y^2}{36} - x^2 = 1$, top piece is oriented from left-to-right

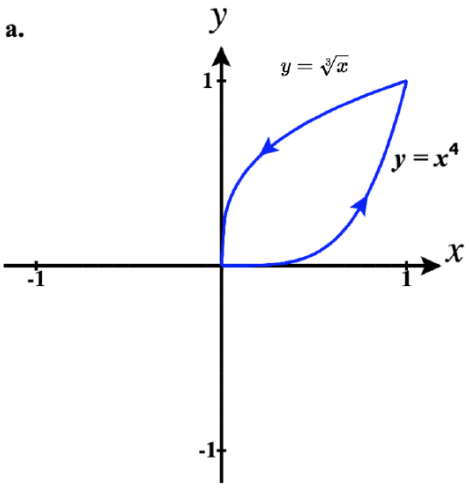
40) The hyperbola, $\dfrac{x^2}{49} - \dfrac{y^2}{64} = 1$, right piece is oriented from bottom-to-top

PARAMETERIZING A PIECEWISE PATH

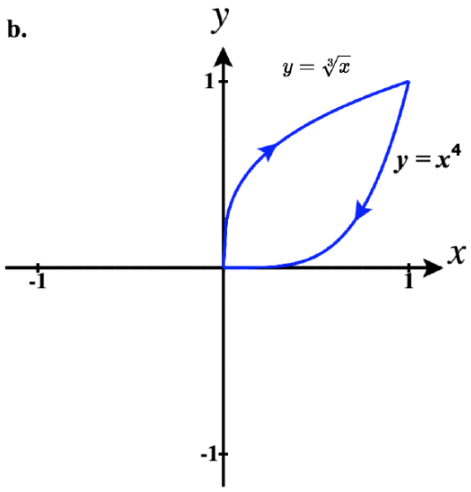
For questions 41 - 44, provide a parameterization for each piecewise path. Try to write a parameterization that starts with $t = 0$ and progresses on through values of t as you move from one piece to another.

41)

a.

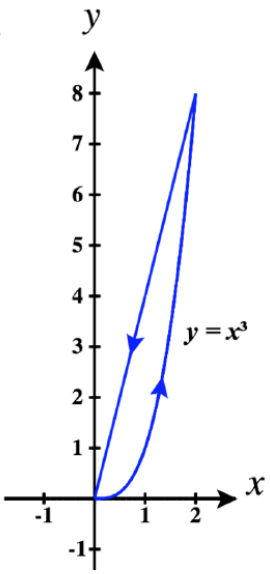


b.

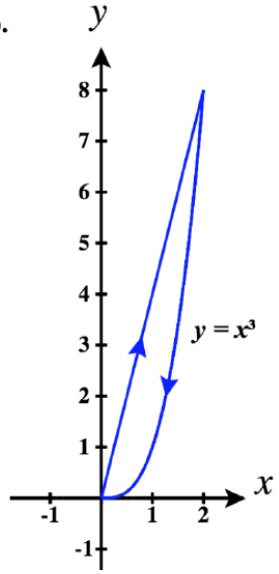


42)

a.

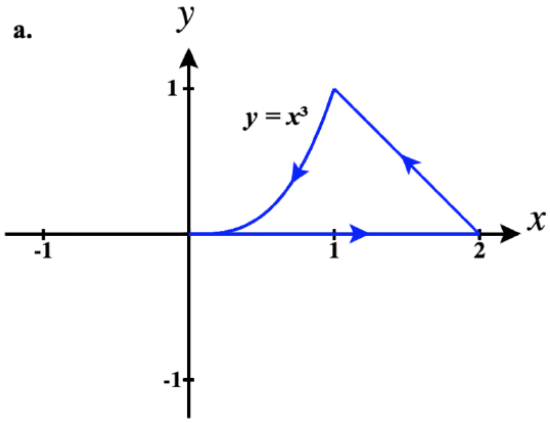


b.

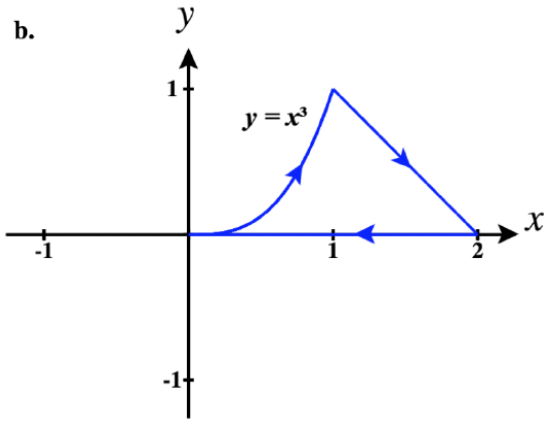


43)

a.

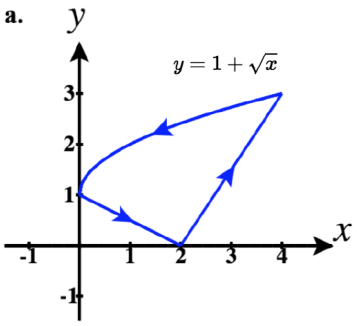


b.

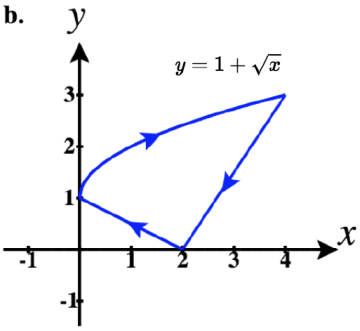


44)

a.



b.



ADDITIONAL VECTOR-VALUED FUNCTION QUESTIONS

For questions 45 - 48, consider the curve described by the vector-valued function $\vec{\mathbf{r}}(t) = (50e^{-t}\cos t)\hat{\mathbf{i}} + (50e^{-t}\sin t)\hat{\mathbf{j}} + (5 - 5e^{-t})\hat{\mathbf{k}}$.

45) What is the initial point of the path corresponding to $\vec{\mathbf{r}}(0)$?

46) What is $\lim_{t\to\infty} \vec{\mathbf{r}}(t)$?

47) **[T]** Use technology to sketch the curve.

48) Eliminate the parameter t to show that $z = 5 - \frac{r}{10}$ where $r^2 = x^2 + y^2$.

49) **[T]** Let $\vec{\mathbf{r}}(t) = \cos t\hat{\mathbf{i}} + \sin t\hat{\mathbf{j}} + 0.3\sin(2t)\hat{\mathbf{k}}$. Use technology to graph the curve (called the *roller-coaster curve*) over the interval $[0, 2\pi)$. Choose at least two views to determine the peaks and valleys.

50) **[T]** Use the result of the preceding problem to construct an equation of a roller coaster with a steep drop from the peak and steep incline from the "valley." Then, use technology to graph the equation.

51) Use the results of the preceding two problems to construct an equation of a path of a roller coaster with more than two turning points (peaks and valleys).

52) Complete the following investigation.

a. Graph the curve $\vec{\mathbf{r}}(t) = (4 + \cos(18t))\cos(t)\hat{\mathbf{i}} + (4 + \cos(18t)\sin(t))\hat{\mathbf{j}} + 0.3\sin(18t)\hat{\mathbf{k}}$ using two viewing angles of your choice to see the overall shape of the curve.
b. Does the curve resemble a "slinky"?
c. What changes to the equation should be made to increase the number of coils of the slinky?

CONTRIBUTORS

Gilbert Strang (MIT) and Edwin "Jed" Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

Paul Seeburger (Monroe Community College) created problems 12, 14, 19, 22, 30-33, 37- 44.

13.1E: Exercises for Section 13.1 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

13.2: Calculus of Vector-Valued Functions

Learning Objectives

- Write an expression for the derivative of a vector-valued function.
- Find the tangent vector at a point for a given position vector.
- Find the unit tangent vector at a point for a given position vector and explain its significance.
- Calculate the definite integral of a vector-valued function.

To study the calculus of vector-valued functions, we follow a similar path to the one we took in studying real-valued functions. First, we define the derivative, then we examine applications of the derivative, then we move on to defining integrals. However, we will find some interesting new ideas along the way as a result of the vector nature of these functions and the properties of space curves.

Derivatives of Vector-Valued Functions

Now that we have seen what a vector-valued function is and how to take its limit, the next step is to learn how to differentiate a vector-valued function. The definition of the derivative of a vector-valued function is nearly identical to the definition of a real-valued function of one variable. However, because the range of a vector-valued function consists of vectors, the same is true for the range of the derivative of a vector-valued function.

Definition: Derivative of Vector-Valued Functions

The derivative of a vector-valued function $\vec{\mathbf{r}}(t)$ is

$$\vec{\mathbf{r}}'(t) = \lim_{\Delta t \to 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t} \tag{13.2.1}$$

provided the limit exists. If $\vec{\mathbf{r}}'(t)$ exists, then $\vec{\mathbf{r}}(t)$ is differentiable at t. If $\vec{\mathbf{r}}'(t)$ exists for all t in an open interval (a, b) then $\vec{\mathbf{r}}(t)$ is differentiable over the interval (a, b). For the function to be differentiable over the closed interval $[a, b]$, the following two limits must exist as well:

$$\vec{\mathbf{r}}'(a) = \lim_{\Delta t \to 0^+} \frac{\vec{\mathbf{r}}(a+\Delta t) - \vec{\mathbf{r}}(a)}{\Delta t}$$

and

$$\vec{\mathbf{r}}'(b) = \lim_{\Delta t \to 0^-} \frac{\vec{\mathbf{r}}(b+\Delta t) - \vec{\mathbf{r}}(b)}{\Delta t}$$

Many of the rules for calculating derivatives of real-valued functions can be applied to calculating the derivatives of vector-valued functions as well. Recall that the derivative of a real-valued function can be interpreted as the slope of a tangent line or the instantaneous rate of change of the function. The derivative of a vector-valued function can be understood to be an instantaneous rate of change as well; for example, when the function represents the position of an object at a given point in time, the derivative represents its velocity at that same point in time.

We now demonstrate taking the derivative of a vector-valued function.

Example 13.2.1: Finding the Derivative of a Vector-Valued Function

Use the definition to calculate the derivative of the function

$$\vec{\mathbf{r}}(t) = (3t+4)\,\hat{\mathbf{i}} + (t^2 - 4t + 3)\,\hat{\mathbf{j}}.$$

Solution

Let's use Equation 13.2.1:

$$
\begin{aligned}
\vec{\mathbf{r}}'(t) &= \lim_{\Delta t \to 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{[(3(t+\Delta t)+4)\,\hat{\mathbf{i}} + ((t+\Delta t)^2 - 4(t+\Delta t)+3)\,\hat{\mathbf{j}}] - [(3t+4)\,\hat{\mathbf{i}} + (t^2-4t+3)\,\hat{\mathbf{j}}]}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{(3t+3\Delta t+4)\,\hat{\mathbf{i}} - (3t+4)\,\hat{\mathbf{i}} + (t^2+2t\Delta t+(\Delta t)^2-4t-4\Delta t+3)\,\hat{\mathbf{j}} - (t^2-4t+3)\,\hat{\mathbf{j}}}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{(3\Delta t)\,\hat{\mathbf{i}} + (2t\Delta t + (\Delta t)^2 - 4\Delta t)\,\hat{\mathbf{j}}}{\Delta t} \\
&= \lim_{\Delta t \to 0} (3\,\hat{\mathbf{i}} + (2t + \Delta t - 4)\,\hat{\mathbf{j}}) \\
&= 3\,\hat{\mathbf{i}} + (2t-4)\,\hat{\mathbf{j}}
\end{aligned}
$$

? Exercise 13.2.1

Use the definition to calculate the derivative of the function $\vec{\mathbf{r}}(t) = (2t^2+3)\,\hat{\mathbf{i}} + (5t-6)\,\hat{\mathbf{j}}$.

Hint

Use Equation 13.2.1.

Answer

$$\vec{\mathbf{r}}'(t) = 4t\,\hat{\mathbf{i}} + 5\,\hat{\mathbf{j}}$$

Notice that in the calculations in Example 13.2.1, we could also obtain the answer by first calculating the derivative of each component function, then putting these derivatives back into the vector-valued function. This is always true for calculating the derivative of a vector-valued function, whether it is in two or three dimensions. We state this in the following theorem. The proof of this theorem follows directly from the definitions of the limit of a vector-valued function and the derivative of a vector-valued function.

Theorem 13.2.1: Differentiation of Vector-Valued Functions

Let f, g, and h be differentiable functions of t.

1. If $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}$ then
$$\vec{\mathbf{r}}'(t) = f'(t)\,\hat{\mathbf{i}} + g'(t)\,\hat{\mathbf{j}}.$$
2. If $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}$ then
$$\vec{\mathbf{r}}'(t) = f'(t)\,\hat{\mathbf{i}} + g'(t)\,\hat{\mathbf{j}} + h'(t)\,\hat{\mathbf{k}}.$$

✓ Example 13.2.2: Calculating the Derivative of Vector-Valued Functions

Use Theorem 13.2.1 to calculate the derivative of each of the following functions.

a. $\vec{\mathbf{r}}(t) = (6t+8)\,\hat{\mathbf{i}} + (4t^2+2t-3)\,\hat{\mathbf{j}}$
b. $\vec{\mathbf{r}}(t) = 3\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}}$
c. $\vec{\mathbf{r}}(t) = e^t \sin t\,\hat{\mathbf{i}} + e^t \cos t\,\hat{\mathbf{j}} - e^{2t}\,\hat{\mathbf{k}}$

Solution

We use Theorem 13.2.1 and what we know about differentiating functions of one variable.

a. The first component of

$$\vec{\mathbf{r}}(t) = (6t+8)\,\hat{\mathbf{i}} + (4t^2+2t-3)\,\hat{\mathbf{j}}$$

is $f(t) = 6t+8$. The second component is $g(t) = 4t^2+2t-3$. We have $f'(t) = 6$ and $g'(t) = 8t+2$, so the Theorem 13.2.1 gives $\vec{\mathbf{r}}'(t) = 6\,\hat{\mathbf{i}} + (8t+2)\,\hat{\mathbf{j}}$.

b. The first component is $f(t) = 3\cos t$ and the second component is $g(t) = 4\sin t$. We have $f'(t) = -3\sin t$ and $g'(t) = 4\cos t$, so we obtain $\vec{\mathbf{r}}'(t) = -3\sin t\,\hat{\mathbf{i}} + 4\cos t\,\hat{\mathbf{j}}$.

c. The first component of $\vec{\mathbf{r}}(t) = e^t \sin t\,\hat{\mathbf{i}} + e^t\cos t\,\hat{\mathbf{j}} - e^{2t}\,\hat{\mathbf{k}}$ is $f(t) = e^t\sin t$, the second component is $g(t) = e^t\cos t$, and the third component is $h(t) = -e^{2t}$. We have $f'(t) = e^t(\sin t + \cos t)$, $g'(t) = e^t(\cos t - \sin t)$, and $h'(t) = -2e^{2t}$, so the theorem gives $\vec{\mathbf{r}}'(t) = e^t(\sin t + \cos t)\,\hat{\mathbf{i}} + e^t(\cos t - \sin t)\,\hat{\mathbf{j}} - 2e^{2t}\,\hat{\mathbf{k}}$.

Exercise 13.2.2

Calculate the derivative of the function

$$\vec{\mathbf{r}}(t) = (t\ln t)\,\hat{\mathbf{i}} + (5e^t)\,\hat{\mathbf{j}} + (\cos t - \sin t)\,\hat{\mathbf{k}}.$$

Hint

Identify the component functions and use Theorem 13.2.1.

Answer

$$\vec{\mathbf{r}}'(t) = (1+\ln t)\,\hat{\mathbf{i}} + 5e^t\,\hat{\mathbf{j}} - (\sin t + \cos t)\,\hat{\mathbf{k}}$$

We can extend to vector-valued functions the properties of the derivative that we presented previously. In particular, the constant multiple rule, the sum and difference rules, the product rule, and the chain rule all extend to vector-valued functions. However, in the case of the product rule, there are actually three extensions:

1. for a real-valued function multiplied by a vector-valued function,
2. for the dot product of two vector-valued functions, and
3. for the cross product of two vector-valued functions.

Theorem: Properties of the Derivative of Vector-Valued Functions

Let $\vec{\mathbf{r}}$ and $\vec{\mathbf{u}}$ be differentiable vector-valued functions of t, let f be a differentiable real-valued function of t, and let c be a scalar.

i.	$\frac{d}{dt}[c\vec{\mathbf{r}}(t)]$	$=$	$c\vec{\mathbf{r}}'(t)$	Scalar multiple
ii.	$\frac{d}{dt}[\vec{\mathbf{r}}(t) \pm \vec{\mathbf{u}}(t)]$	$=$	$\vec{\mathbf{r}}'(t) \pm \vec{\mathbf{u}}'(t)$	Sum and difference
iii.	$\frac{d}{dt}[f(t)\vec{\mathbf{u}}(t)]$	$=$	$f'(t)\vec{\mathbf{u}}(t) + f(t)\vec{\mathbf{u}}'(t)$	Scalar product
iv.	$\frac{d}{dt}[\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}(t)]$	$=$	$\vec{\mathbf{r}}'(t)\cdot\vec{\mathbf{u}}(t) + \vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}'(t)$	Dot product
v.	$\frac{d}{dt}[\vec{\mathbf{r}}(t)\times\vec{\mathbf{u}}(t)]$	$=$	$\vec{\mathbf{r}}'(t)\times\vec{\mathbf{u}}(t) + \vec{\mathbf{r}}(t)\times\vec{\mathbf{u}}'(t)$	Cross product
vi.	$\frac{d}{dt}[\vec{\mathbf{r}}(f(t))]$	$=$	$\vec{\mathbf{r}}'(f(t))\cdot f'(t)$	Chain rule
vii.	If $\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{r}}(t)$	$=$	c, then $\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{r}}'(t) = 0$.	

Proof

The proofs of the first two properties follow directly from the definition of the derivative of a vector-valued function. The third property can be derived from the first two properties, along with the product rule. Let $\vec{\mathbf{u}}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Then

$$\begin{aligned}\frac{d}{dt}[f(t)\vec{\mathbf{u}}(t)] &= \frac{d}{dt}[f(t)(g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}})] \\ &= \frac{d}{dt}[f(t)g(t)\,\hat{\mathbf{i}} + f(t)h(t)\,\hat{\mathbf{j}}] \\ &= \frac{d}{dt}[f(t)g(t)]\,\hat{\mathbf{i}} + \frac{d}{dt}[f(t)h(t)]\,\hat{\mathbf{j}} \\ &= (f'(t)g(t) + f(t)g'(t))\,\hat{\mathbf{i}} + (f'(t)h(t) + f(t)h'(t))\,\hat{\mathbf{j}} \\ &= f'(t)\vec{\mathbf{u}}(t) + f(t)\vec{\mathbf{u}}'(t).\end{aligned}$$

To prove property iv. let $\vec{\mathbf{r}}(t) = f_1(t)\,\hat{\mathbf{i}} + g_1(t)\,\hat{\mathbf{j}}$ and $\vec{\mathbf{u}}(t) = f_2(t)\,\hat{\mathbf{i}} + g_2(t)\,\hat{\mathbf{j}}$. Then

$$\begin{aligned}\frac{d}{dt}[\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}(t)] &= \frac{d}{dt}[f_1(t)f_2(t) + g_1(t)g_2(t)] \\ &= f_1{}'(t)f_2(t) + f_1(t)f_2{}'(t) + g_1{}'(t)g_2(t) + g_1(t)g_2{}'(t) = f_1{}'(t)f_2(t) + g_1 \\ &\quad {}'(t)g_2(t) + f_1(t)f_2{}'(t) + g_1(t)g_2{}'(t) \\ &= (f_1{}'\,\hat{\mathbf{i}} + g_1{}'\,\hat{\mathbf{j}})\cdot(f_2\,\hat{\mathbf{i}} + g_2\,\hat{\mathbf{j}}) + (f_1\,\hat{\mathbf{i}} + g_1\,\hat{\mathbf{j}})\cdot(f_2{}'\,\hat{\mathbf{i}} + g_2{}'\,\hat{\mathbf{j}}) \\ &= \vec{\mathbf{r}}'(t)\cdot\vec{\mathbf{u}}(t) + \vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}'(t).\end{aligned}$$

The proof of property v. is similar to that of property iv. Property vi. can be proved using the chain rule. Last, property vii. follows from property iv:

$$\begin{aligned}\frac{d}{dt}[\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{r}}(t)] &= \frac{d}{dt}[c] \\ \vec{\mathbf{r}}'(t)\cdot\vec{\mathbf{r}}(t) + \vec{\mathbf{r}}(t)\cdot\vec{\mathbf{r}}'(t) &= 0 \\ 2\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{r}}'(t) &= 0 \\ \vec{\mathbf{r}}(t)\cdot\vec{\mathbf{r}}'(t) &= 0\end{aligned}$$

Now for some examples using these properties.

Example 13.2.3: Using the Properties of Derivatives of Vector-Valued Functions

Given the vector-valued functions

$$\vec{\mathbf{r}}(t) = (6t+8)\,\hat{\mathbf{i}} + (4t^2 + 2t - 3)\,\hat{\mathbf{j}} + 5t\,\hat{\mathbf{k}}$$

and

$$\vec{\mathbf{u}}(t) = (t^2 - 3)\,\hat{\mathbf{i}} + (2t+4)\,\hat{\mathbf{j}} + (t^3 - 3t)\,\hat{\mathbf{k}},$$

calculate each of the following derivatives using the properties of the derivative of vector-valued functions.

a. $\frac{d}{dt}[\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}(t)]$
b. $\frac{d}{dt}[\vec{\mathbf{u}}(t)\times\vec{\mathbf{u}}'(t)]$

Solution

We have $\vec{\mathbf{r}}'(t) = 6\,\hat{\mathbf{i}} + (8t+2)\,\hat{\mathbf{j}} + 5\,\hat{\mathbf{k}}$ and $\vec{\mathbf{u}}'(t) = 2t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + (3t^2 - 3)\,\hat{\mathbf{k}}$. Therefore, according to **property iv**:

1.
$$\begin{aligned}\frac{d}{dt}[\vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}(t)] &= \vec{\mathbf{r}}'(t)\cdot\vec{\mathbf{u}}(t) + \vec{\mathbf{r}}(t)\cdot\vec{\mathbf{u}}'(t) \\ &= (6\,\hat{\mathbf{i}} + (8t+2)\,\hat{\mathbf{j}} + 5\,\hat{\mathbf{k}})\cdot((t^2-3)\,\hat{\mathbf{i}} + (2t+4)\,\hat{\mathbf{j}} + (t^3-3t)\,\hat{\mathbf{k}}) \\ &\quad +((6t+8)\,\hat{\mathbf{i}} + (4t^2+2t-3)\,\hat{\mathbf{j}} + 5t\,\hat{\mathbf{k}})\cdot(2t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + (3t^2-3)\,\hat{\mathbf{k}}) \\ &= 6(t^2-3) + (8t+2)(2t+4) + 5(t^3-3t) \\ &\quad +2t(6t+8) + 2(4t^2+2t-3) + 5t(3t^2-3) \\ &= 20t^3 + 42t^2 + 26t - 16.\end{aligned}$$

2. First, we need to adapt **property v** for this problem:

$$\frac{d}{dt}[\vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}'(t)] = \vec{\mathbf{u}}'(t) \times \vec{\mathbf{u}}'(t) + \vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}''(t).$$

Recall that the cross product of any vector with itself is zero. Furthermore, $\vec{\mathbf{u}}''(t)$ represents the second derivative of $\vec{\mathbf{u}}(t)$:

$$\vec{\mathbf{u}}''(t) = \frac{d}{dt}[\vec{\mathbf{u}}'(t)] = \frac{d}{dt}[2t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + (3t^2 - 3)\,\hat{\mathbf{k}}] = 2\,\hat{\mathbf{i}} + 6t\,\hat{\mathbf{k}}.$$

Therefore,

$$\begin{aligned}\frac{d}{dt}[\vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}'(t)] &= 0 + ((t^2 - 3)\,\hat{\mathbf{i}} + (2t + 4)\,\hat{\mathbf{j}} + (t^3 - 3t)\,\hat{\mathbf{k}}) \times (2\,\hat{\mathbf{i}} + 6t\,\hat{\mathbf{k}}) \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ t^2 - 3 & 2t + 4 & t^3 - 3t \\ 2 & 0 & 6t \end{vmatrix} \\ &= 6t(2t + 4)\,\hat{\mathbf{i}} - (6t(t^2 - 3) - 2(t^3 - 3t))\,\hat{\mathbf{j}} - 2(2t + 4)\,\hat{\mathbf{k}} \\ &= (12t^2 + 24t)\,\hat{\mathbf{i}} + (12t - 4t^3)\,\hat{\mathbf{j}} - (4t + 8)\,\hat{\mathbf{k}}.\end{aligned}$$

? Exercise 13.2.3

Calculate $\frac{d}{dt}[\vec{\mathbf{r}}(t) \cdot \vec{\mathbf{r}}'(t)]$ and $\frac{d}{dt}[\vec{\mathbf{u}}(t) \times \vec{\mathbf{r}}(t)]$ for the vector-valued functions:

- $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} - e^{2t}\,\hat{\mathbf{k}}$
- $\vec{\mathbf{u}}(t) = t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} + \cos t\,\hat{\mathbf{k}}$,

Hint

Follow the same steps as in Example 13.2.3.

Answer

$\frac{d}{dt}[\vec{\mathbf{r}}(t) \cdot \vec{\mathbf{r}}'(t)] = 8e^{4t}$

$\frac{d}{dt}[\vec{\mathbf{u}}(t) \times \vec{\mathbf{r}}(t)] = -(e^{2t}(\cos t + 2\sin t) + \cos 2t)\,\hat{\mathbf{i}} + (e^{2t}(2t + 1) - \sin 2t)\,\hat{\mathbf{j}} + (t\cos t + \sin t - \cos 2t)\,\hat{\mathbf{k}}$

Tangent Vectors and Unit Tangent Vectors

Recall that the derivative at a point can be interpreted as the slope of the tangent line to the graph at that point. In the case of a vector-valued function, the derivative provides a tangent vector to the curve represented by the function. Consider the vector-valued function

$$\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} \tag{13.2.2}$$

The derivative of this function is

$$\vec{\mathbf{r}}'(t) = -\sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}}$$

If we substitute the value $t = \pi/6$ into both functions we get

$$\vec{\mathbf{r}}\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\,\hat{\mathbf{i}} + \frac{1}{2}\,\hat{\mathbf{j}}$$

and

$$\vec{\mathbf{r}}'\left(\frac{\pi}{6}\right) = -\frac{1}{2}\,\hat{\mathbf{i}} + \frac{\sqrt{3}}{2}\,\hat{\mathbf{j}}.$$

The graph of this function appears in Figure 13.2.1, along with the vectors $\vec{\mathbf{r}}\left(\frac{\pi}{6}\right)$ and $\vec{\mathbf{r}}'\left(\frac{\pi}{6}\right)$.

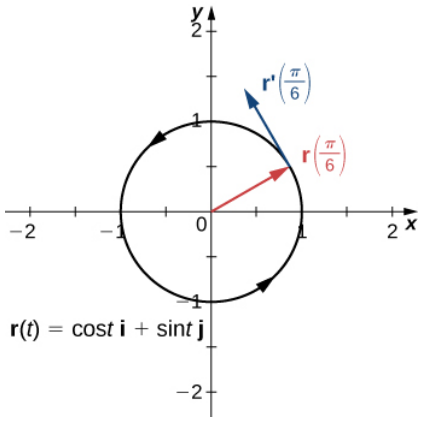


Figure 13.2.1: The tangent line at a point is calculated from the derivative of the vector-valued function $\vec{\mathbf{r}}(t)$.

Notice that the vector $\vec{\mathbf{r}}'\left(\frac{\pi}{6}\right)$ is tangent to the circle at the point corresponding to $t = \frac{\pi}{6}$. This is an example of a tangent vector to the plane curve defined by Equation 13.2.2.

Definition: principal unit tangent vector

Let C be a curve defined by a vector-valued function $\vec{\mathbf{r}}$, and assume that $\vec{\mathbf{r}}'(t)$ exists when $\mathrm{t} = \mathrm{t}_0$ A tangent vector $\vec{\mathbf{r}}$ at $t = t_0$ is any vector such that, when the tail of the vector is placed at point $\vec{\mathbf{r}}(t_0)$ on the graph, vector $\vec{\mathbf{r}}$ is tangent to curve C. Vector $\vec{\mathbf{r}}\,'(t_0)$ is an example of a tangent vector at point $t = t_0$. Furthermore, assume that $\vec{\mathbf{r}}'(t) \neq 0$. The principal unit tangent vector at t is defined to be

$$\vec{\mathbf{T}}(t) = \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|},$$

provided $\|\vec{\mathbf{r}}'(t)\| \neq 0$.

The unit tangent vector is exactly what it sounds like: a unit vector that is tangent to the curve. To calculate a unit tangent vector, first find the derivative $\vec{\mathbf{r}}\,'(t)$. Second, calculate the magnitude of the derivative. The third step is to divide the derivative by its magnitude.

Example 13.2.4: Finding a Unit Tangent Vector

Find the unit tangent vector for each of the following vector-valued functions:

a. $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}}$
b. $\vec{\mathbf{u}}(t) = (3t^2 + 2t)\,\hat{\mathbf{i}} + (2 - 4t^3)\,\hat{\mathbf{j}} + (6t + 5)\,\hat{\mathbf{k}}$

Solution

a.

$$\begin{aligned}
\text{First step:} \quad \vec{\mathbf{r}}'(t) &= -\sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}} \\
\text{Second step:} \quad \|\vec{\mathbf{r}}'(t)\| &= \sqrt{(-\sin t)^2 + (\cos t)^2} = 1 \\
\text{Third step:} \quad \vec{\mathbf{T}}(t) &= \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|} = \frac{-\sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}}}{1} = -\sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}}
\end{aligned}$$

b.

$$\begin{aligned}
\text{First step:} \quad \vec{\mathbf{r}}'(t) &= (6t + 2)\,\hat{\mathbf{i}} - 12t^2\,\hat{\mathbf{j}} + 6\,\hat{\mathbf{k}} \\
\text{Second step:} \quad \|\vec{\mathbf{r}}'(t)\| &= \sqrt{(6t + 2)^2 + (-12t^2)^2 + 6^2} \\
&= \sqrt{144t^4 + 36t^2 + 24t + 40} \\
&= 2\sqrt{36t^4 + 9t^2 + 6t + 10} \\
\text{Third step:} \quad \vec{\mathbf{T}}(t) &= \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|} = \frac{(6t + 2)\,\hat{\mathbf{i}} - 12t^2\,\hat{\mathbf{j}} + 6\,\hat{\mathbf{k}}}{2\sqrt{36t^4 + 9t^2 + 6t + 10}} \\
&= \frac{3t + 1}{\sqrt{36t^4 + 9t^2 + 6t + 10}}\,\hat{\mathbf{i}} - \frac{6t^2}{\sqrt{36t^4 + 9t^2 + 6t + 10}}\,\hat{\mathbf{j}} + \frac{3}{\sqrt{36t^4 + 9t^2 + 6t + 10}}\,\hat{\mathbf{k}}
\end{aligned}$$

Exercise 13.2.4

Find the unit tangent vector for the vector-valued function

$$\vec{\mathbf{r}}(t) = (t^2 - 3)\,\hat{\mathbf{i}} + (2t + 1)\,\hat{\mathbf{j}} + (t - 2)\,\hat{\mathbf{k}}.$$

Hint

Follow the same steps as in Example 13.2.4.

Answer

$$\vec{\mathbf{T}}(t) = \frac{2t}{\sqrt{4t^2 + 5}}\,\hat{\mathbf{i}} + \frac{2}{\sqrt{4t^2 + 5}}\,\hat{\mathbf{j}} + \frac{1}{\sqrt{4t^2 + 5}}\,\hat{\mathbf{k}}$$

Integrals of Vector-Valued Functions

We introduced antiderivatives of real-valued functions in Antiderivatives and definite integrals of real-valued functions in The Definite Integral. Each of these concepts can be extended to vector-valued functions. Also, just as we can calculate the derivative of a vector-valued function by differentiating the component functions separately, we can calculate the antiderivative in the same manner. Furthermore, the Fundamental Theorem of Calculus applies to vector-valued functions as well.

The antiderivative of a vector-valued function appears in applications. For example, if a vector-valued function represents the velocity of an object at time *t*, then its antiderivative represents position. Or, if the function represents the acceleration of the object at a given time, then the antiderivative represents its velocity.

Definition: Definite and Indefinite Integrals of Vector-Valued Functions

Let f, g, and h be integrable real-valued functions over the closed interval $[a, b]$.

1. The indefinite integral of a vector-valued function $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}$ is

$$\int [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}]\,dt = \left[\int f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int g(t)\,dt\right]\hat{\mathbf{j}}.$$

   The definite integral of a vector-valued function is

$$\int_a^b [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}]\,dt = \left[\int_a^b f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int_a^b g(t)\,dt\right]\hat{\mathbf{j}}.$$

2. The indefinite integral of a vector-valued function $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}$ is

$$\int [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}]\,dt = \left[\int f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int g(t)\,dt\right]\hat{\mathbf{j}} + \left[\int h(t)\,dt\right]\hat{\mathbf{k}}.$$

   The definite integral of the vector-valued function is

$$\int_a^b [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}]\,dt = \left[\int_a^b f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int_a^b g(t)\,dt\right]\hat{\mathbf{j}} + \left[\int_a^b h(t)\,dt\right]\hat{\mathbf{k}}.$$

Since the indefinite integral of a vector-valued function involves indefinite integrals of the component functions, each of these component integrals contains an integration constant. They can all be different. For example, in the two-dimensional case, we can have

$$\int f(t)\,dt = F(t) + C_1 \text{ and } \int g(t)\,dt = G(t) + C_2,$$

where F and G are antiderivatives of f and g, respectively. Then

$$\int [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}]\,dt = \left[\int f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int g(t)\,dt\right]\hat{\mathbf{j}}$$
$$= (F(t) + C_1)\,\hat{\mathbf{i}} + (G(t) + C_2)\,\hat{\mathbf{j}}$$
$$= F(t)\,\hat{\mathbf{i}} + G(t)\,\hat{\mathbf{j}} + C_1\,\hat{\mathbf{i}} + C_2\,\hat{\mathbf{j}}$$
$$= F(t)\,\hat{\mathbf{i}} + G(t)\,\hat{\mathbf{j}} + \vec{\mathbf{C}}$$

where $\vec{\mathbf{C}} = C_1\,\hat{\mathbf{i}} + C_2\,\hat{\mathbf{j}}$. Therefore, the *integration constants* becomes a *constant vector*.

Example 13.2.5: Integrating Vector-Valued Functions

Calculate each of the following integrals:

a. $\int [(3t^2 + 2t)\,\hat{\mathbf{i}} + (3t - 6)\,\hat{\mathbf{j}} + (6t^3 + 5t^2 - 4)\,\hat{\mathbf{k}}]\,dt$

b. $\int [\langle t, t^2, t^3\rangle \times \langle t^3, t^2, t\rangle]\,dt$

c. $\int_0^{\frac{\pi}{3}} [\sin 2t\,\hat{\mathbf{i}} + \tan t\,\hat{\mathbf{j}} + e^{-2t}\,\hat{\mathbf{k}}]\,dt$

Solution

a. We use the first part of the definition of the integral of a space curve:

$$\int [(3t^2 + 2t)\,\hat{\mathbf{i}} + (3t - 6)\,\hat{\mathbf{j}} + (6t^3 + 5t^2 - 4)\,\hat{\mathbf{k}}]\,dt = \left[\int 3t^2 + 2t\,dt\right]\hat{\mathbf{i}} + \left[\int 3t - 6\,dt\right]\hat{\mathbf{j}} + \left[\int 6t^3 + 5t^2 - 4\,dt\right]\hat{\mathbf{k}}$$
$$= (t^3 + t^2)\,\hat{\mathbf{i}} + \left(\frac{3}{2}t^2 - 6t\right)\hat{\mathbf{j}} + \left(\frac{3}{2}t^4 + \frac{5}{3}t^3 - 4t\right)\hat{\mathbf{k}} + \vec{\mathbf{C}}.$$

b. First calculate $\langle t, t^2, t^3\rangle \times \langle t^3, t^2, t\rangle$:

$$\langle t, t^2, t^3\rangle \times \langle t^3, t^2, t\rangle = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ t & t^2 & t^3 \\ t^3 & t^2 & t \end{vmatrix}$$
$$= (t^2(t) - t^3(t^2))\,\hat{\mathbf{i}} - (t^2 - t^3(t^3))\,\hat{\mathbf{j}} + (t(t^2) - t^2(t^3))\,\hat{\mathbf{k}}$$
$$= (t^3 - t^5)\,\hat{\mathbf{i}} + (t^6 - t^2)\,\hat{\mathbf{j}} + (t^3 - t^5)\,\hat{\mathbf{k}}.$$

Next, substitute this back into the integral and integrate:

$$\int [\langle t, t^2, t^3\rangle \times \langle t^3, t^2, t\rangle]\,dt = \int (t^3 - t^5)\,\hat{\mathbf{i}} + (t^6 - t^2)\,\hat{\mathbf{j}} + (t^3 - t^5)\,\hat{\mathbf{k}}\,dt$$
$$= \left(\frac{t^4}{4} - \frac{t^6}{6}\right)\hat{\mathbf{i}} + \left(\frac{t^7}{7} - \frac{t^3}{3}\right)\hat{\mathbf{j}} + \left(\frac{t^4}{4} - \frac{t^6}{6}\right)\hat{\mathbf{k}} + \vec{\mathbf{C}}.$$

d. Use the second part of the definition of the integral of a space curve:

$$\int_0^{\frac{\pi}{3}} [\sin 2t\,\hat{\mathbf{i}} + \tan t\,\hat{\mathbf{j}} + e^{-2t}\,\hat{\mathbf{k}}]\,dt = \left[\int_0^{\frac{\pi}{3}} \sin 2t\,dt\right]\hat{\mathbf{i}} + \left[\int_0^{\frac{\pi}{3}} \tan t\,dt\right]\hat{\mathbf{j}} + \left[\int_0^{\frac{\pi}{3}} e^{-2t}\,dt\right]\hat{\mathbf{k}}$$
$$= \left(-\tfrac{1}{2}\cos 2t\right)\Big|_0^{\pi/3}\hat{\mathbf{i}} - \left(\ln|\cos t|\right)\Big|_0^{\pi/3}\hat{\mathbf{j}} - \left(\tfrac{1}{2}e^{-2t}\right)\Big|_0^{\pi/3}\hat{\mathbf{k}}$$
$$= \left(-\tfrac{1}{2}\cos\tfrac{2\pi}{3} + \tfrac{1}{2}\cos 0\right)\hat{\mathbf{i}} - \left(\ln\left(\cos\tfrac{\pi}{3}\right) - \ln(\cos 0)\right)\hat{\mathbf{j}} - \left(\tfrac{1}{2}e^{-2\pi/3} - \tfrac{1}{2}e^{-2(0)}\right)\hat{\mathbf{k}}$$
$$= \left(\tfrac{1}{4} + \tfrac{1}{2}\right)\hat{\mathbf{i}} - (-\ln 2)\,\hat{\mathbf{j}} - \left(\tfrac{1}{2}e^{-2\pi/3} - \tfrac{1}{2}\right)\hat{\mathbf{k}}$$
$$= \tfrac{3}{4}\,\hat{\mathbf{i}} + (\ln 2)\,\hat{\mathbf{j}} + \left(\tfrac{1}{2} - \tfrac{1}{2}e^{-2\pi/3}\right)\hat{\mathbf{k}}.$$

Exercise 13.2.5

Calculate the following integral:

$$\int_1^3 [(2t+4)\,\hat{\mathbf{i}} + (3t^2-4t)\,\hat{\mathbf{j}}]\,dt$$

Hint

Use the definition of the definite integral of a plane curve.

Answer

$$\int_1^3 [(2t+4)\,\hat{\mathbf{i}} + (3t^2-4t)\,\hat{\mathbf{j}}]\,dt = 16\,\hat{\mathbf{i}} + 10\,\hat{\mathbf{j}}$$

Summary

- To calculate the derivative of a vector-valued function, calculate the derivatives of the component functions, then put them back into a new vector-valued function.
- Many of the properties of differentiation of scalar functions also apply to vector-valued functions.
- The derivative of a vector-valued function $\vec{\mathbf{r}}(t)$ is also a tangent vector to the curve. The unit tangent vector $\vec{\mathbf{T}}(t)$ is calculated by dividing the derivative of a vector-valued function by its magnitude.
- The antiderivative of a vector-valued function is found by finding the antiderivatives of the component functions, then putting them back together in a vector-valued function.
- The definite integral of a vector-valued function is found by finding the definite integrals of the component functions, then putting them back together in a vector-valued function.

Key Equations

- **Derivative of a vector-valued function**

$$\vec{\mathbf{r}}'(t) = \lim_{\Delta t \to 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t}$$

- **Principal unit tangent vector**

$$\vec{\mathbf{T}}(t) = \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|}$$

- **Indefinite integral of a vector-valued function**

$$\int [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}]\,dt = \left[\int f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int g(t)\,dt\right]\hat{\mathbf{j}} + \left[\int h(t)\,dt\right]\hat{\mathbf{k}}$$

- **Definite integral of a vector-valued function**

$$\int_a^b [f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}]\,dt = \left[\int_a^b f(t)\,dt\right]\hat{\mathbf{i}} + \left[\int_a^b g(t)\,dt\right]\hat{\mathbf{j}} + \left[\int_a^b h(t)\,dt\right]\hat{\mathbf{k}}$$

Glossary

definite integral of a vector-valued function

the vector obtained by calculating the definite integral of each of the component functions of a given vector-valued function, then using the results as the components of the resulting function

derivative of a vector-valued function

the derivative of a vector-valued function $\vec{\mathbf{r}}(t)$ is $\vec{\mathbf{r}}'(t) = \lim_{\Delta t \to 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t}$, provided the limit exists

indefinite integral of a vector-valued function

a vector-valued function with a derivative that is equal to a given vector-valued function

principal unit tangent vector

a unit vector tangent to a curve C

tangent vector

to $\vec{\mathbf{r}}(t)$ at $t = t_0$ any vector $\vec{\mathbf{v}}$ such that, when the tail of the vector is placed at point $\vec{\mathbf{r}}(t_0)$ on the graph, vector $\vec{\mathbf{v}}$ is tangent to curve C

This page titled 13.2: Calculus of Vector-Valued Functions is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

13.2E: EXERCISES FOR SECTION 13.2

TAKING DERIVATIVES OF VECTOR-VALUED FUNCTIONS

In questions 1 - 10, compute the derivative of each vector-valued function.

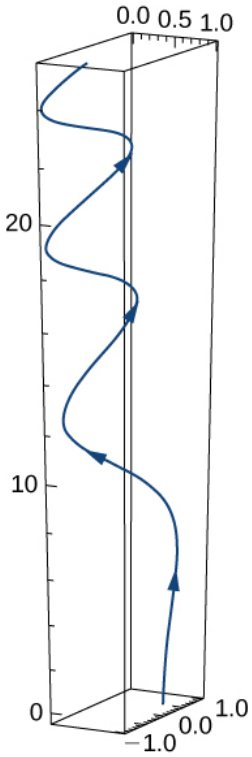
1) $\vec{\mathbf{r}}(t) = t^3\,\hat{\mathbf{i}} + 3t^2\,\hat{\mathbf{j}} + \frac{t^3}{6}\,\hat{\mathbf{k}}$

Answer

$\vec{\mathbf{r}}'(t) = 3t^2\,\hat{\mathbf{i}} + 6t\,\hat{\mathbf{j}} + \frac{1}{2}t^2\,\hat{\mathbf{k}}$

2) $\vec{\mathbf{r}}(t) = \sin(t)\,\hat{\mathbf{i}} + \cos(t)\,\hat{\mathbf{j}} + e^t\,\hat{\mathbf{k}}$

3) $\vec{\mathbf{r}}(t) = e^{-t}\,\hat{\mathbf{i}} + \sin(3t)\,\hat{\mathbf{j}} + 10\sqrt{t}\,\hat{\mathbf{k}}$. A sketch of the graph is shown here. Notice the varying periodic nature of the graph.



Answer

$\vec{\mathbf{r}}'(t) = -e^{-t}\,\hat{\mathbf{i}} + 3\cos(3t)\,\hat{\mathbf{j}} + \frac{5}{\sqrt{t}}\,\hat{\mathbf{k}}$

4) $\vec{\mathbf{r}}(t) = e^t\,\hat{\mathbf{i}} + 2e^t\,\hat{\mathbf{j}} + \hat{\mathbf{k}}$

5) $\vec{\mathbf{r}}(t) = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$

Answer

$\vec{\mathbf{r}}'(t) = \langle 0, 0, 0\rangle = \vec{\mathbf{0}}$

6) $\vec{\mathbf{r}}(t) = te^t\,\hat{\mathbf{i}} + t\ln(t)\,\hat{\mathbf{j}} + \sin(3t)\,\hat{\mathbf{k}}$

7) $\vec{\mathbf{r}}(t) = \left\langle \frac{1}{t+1}, \arctan(t), \ln t^3 \right\rangle$

Answer

$\vec{\mathbf{r}}'(t) = \langle \frac{-1}{(t+1)^2}, \frac{1}{1+t^2}, \frac{3}{t} \rangle$

8) $\vec{\mathbf{r}}(t) = \langle \tan 2t,\ \sec 2t,\ \sin^2 t \rangle$

https://math.libretexts.org/@go/page/69508

9) $\vec{\mathbf{r}}(t) = \langle 3,\ 4\sin(3t),\ t\cos(t)\rangle$

Answer

$\vec{\mathbf{r}}'(t) = \langle 0, 12\cos(3t), \cos t - t\sin t\rangle$

10) $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + te^{-2t}\,\hat{\mathbf{j}} - 5e^{-4t}\,\hat{\mathbf{k}}$

11) a. Describe and sketch the curve represented by the vector-valued function $\vec{\mathbf{r}}(t) = \langle 6t, 6t - t^2\rangle$.

b. Locate the highest point on the curve $\vec{\mathbf{r}}(t) = \langle 6t, 6t - t^2\rangle$ and give the value of the function at this point.

Answer

b. $\vec{\mathbf{r}}(t) = \langle 18, 9\rangle$ at $t = 3$

12) Find the parametric equations of the tangent line to the curve $\vec{\mathbf{r}}(t) = \langle t, t^2, t\rangle$ at $t = 2$.

13) Find the parametric equations of the tangent line to the curve $\vec{\mathbf{r}}(t) = \langle e^t, e^{-t}, 0\rangle$ at $t = 0$.

Answer

$x = 1 + t, \quad y = 1 - t, \quad z = 0$

14) Compute the first, second, and third derivatives of $\vec{\mathbf{r}}(t) = 3t\,\hat{\mathbf{i}} + 6\ln(t)\,\hat{\mathbf{j}} + 5e^{-3t}\,\hat{\mathbf{k}}$.

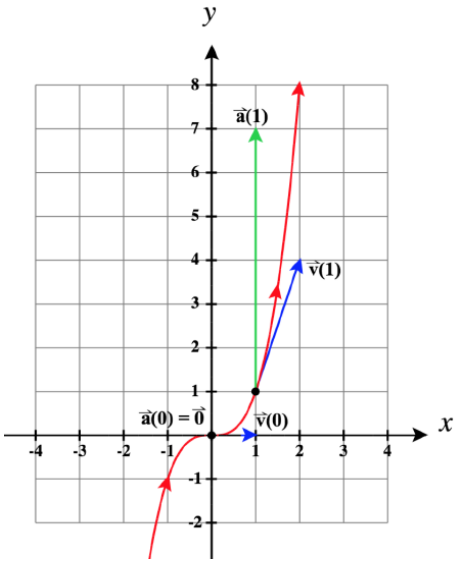
DESCRIBING MOTION WITH VECTOR-VALUED FUNCTIONS

In questions 15 - 17, find the velocity and acceleration at the given times, plot the graph of the position function, and draw in the velocity and acceleration vectors at the corresponding locations on the curve.

15) $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^3\,\hat{\mathbf{j}}$, at $t = 0$ and at $t = 1$

Answer

	$t = 0:$	$t = 1:$
$\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^3\,\hat{\mathbf{j}},$	$\vec{\mathbf{r}}(0) = \vec{\mathbf{0}},$	$\vec{\mathbf{r}}(1) = \hat{\mathbf{i}} + \hat{\mathbf{j}}$
$\vec{\mathbf{v}}(t) = \hat{\mathbf{i}} + 3t^2\,\hat{\mathbf{j}},$	$\vec{\mathbf{v}}(0) = \hat{\mathbf{i}},$	$\vec{\mathbf{v}}(1) = \hat{\mathbf{i}} + 3\,\hat{\mathbf{j}}$
$\vec{\mathbf{a}}(t) = 6t\,\hat{\mathbf{j}},$	$\vec{\mathbf{a}}(0) = \vec{\mathbf{0}},$	$\vec{\mathbf{a}}(1) = 6\,\hat{\mathbf{j}}$

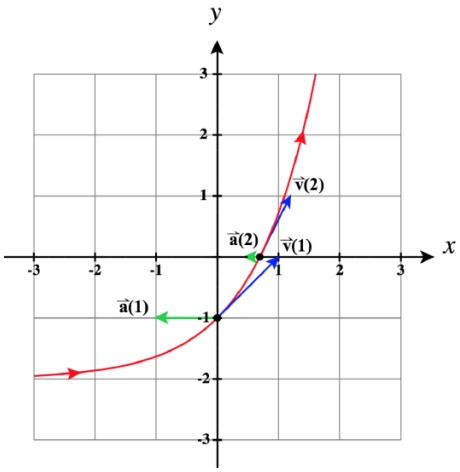


16) $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin 3t\,\hat{\mathbf{j}}$, at $t = 0$, at $t = \frac{\pi}{4}$, and at $t = \frac{\pi}{2}$

17) $\vec{\mathbf{r}}(t) = \ln t\,\hat{\mathbf{i}} + (t-2)\,\hat{\mathbf{j}}$, at $t = 1$ and at $t = 2$

Answer

	$t=1:$	$t=2:$
$\vec{\mathbf{r}}(t)=(\ln t)\,\hat{\mathbf{i}}+(t-2)\,\hat{\mathbf{j}},$	$\vec{\mathbf{r}}(1)=-\hat{\mathbf{j}},$	$\vec{\mathbf{r}}(2)=(\ln 2)\,\hat{\mathbf{i}}$
$\vec{\mathbf{v}}(t)=\frac{1}{t}\,\hat{\mathbf{i}}+\hat{\mathbf{j}},$	$\vec{\mathbf{v}}(1)=\hat{\mathbf{i}}+\hat{\mathbf{j}},$	$\vec{\mathbf{v}}(2)=\frac{1}{2}\hat{\mathbf{i}}+\hat{\mathbf{j}}$
$\vec{\mathbf{a}}(t)=-\frac{1}{t^2}\,\hat{\mathbf{i}},$	$\vec{\mathbf{a}}(1)=-\hat{\mathbf{i}},$	$\vec{\mathbf{a}}(2)=-\frac{1}{4}\,\hat{\mathbf{i}}$



In questions 18 - 24, find the velocity, speed, and acceleration of a particle with the given position function. Remember that the speed is the magnitude of the velocity represented by $\|\vec{\mathbf{v}}(t)\|$ or $\|\vec{\mathbf{r}}'(t)\|$.

18) $\vec{\mathbf{r}}(t)=e^{2t}\,\hat{\mathbf{i}}+\sin t\,\hat{\mathbf{j}}$

19) $\vec{\mathbf{r}}(t)=\cos t^3\,\hat{\mathbf{i}}+\sin t^3\,\hat{\mathbf{j}}$

Answer

$\vec{\mathbf{v}}(t)=-3t^2\sin t^3\,\hat{\mathbf{i}}+3t^2\cos t^3\,\hat{\mathbf{j}},\quad \text{speed}(t)=\|\vec{\mathbf{v}}(t)\|=3t^2,\quad \vec{\mathbf{a}}(t)=\left(-6t\sin t^3-9t^4\cos t^3\right)\hat{\mathbf{i}}+\left(6t\cos t^3-9t^4\sin t^3\right)\hat{\mathbf{j}}$

20) $\vec{\mathbf{r}}(t)=\langle e^t,e^{-t},0\rangle$

21) $\vec{\mathbf{r}}(t)=\langle t+\cos t,t-\sin t\rangle$

Answer

$\vec{\mathbf{v}}(t)=\langle 1-\sin t,1-\cos t\rangle,\quad \text{speed}(t)=\|\vec{\mathbf{v}}(t)\|=\sqrt{3-2(\sin t+\cos t)},\quad \vec{\mathbf{a}}(t)=\langle-\cos t,\sin t\rangle$

22) $\vec{\mathbf{r}}(t)=\dfrac{2t-1}{2t+1}\,\hat{\mathbf{i}}+\ln(1-4t^2)\,\hat{\mathbf{j}}$

23) $\vec{\mathbf{r}}(t)=\cos 3t\,\hat{\mathbf{i}}+\sin 3t\,\hat{\mathbf{j}}+0.5t\,\hat{\mathbf{k}}$

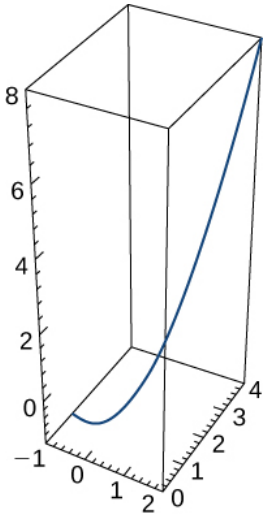
Answer

$\vec{\mathbf{v}}(t)=-3\sin 3t\,\hat{\mathbf{i}}+3\cos 3t\,\hat{\mathbf{j}}+0.5\,\hat{\mathbf{k}},\quad \text{speed}(t)=\|\vec{\mathbf{v}}(t)\|=\sqrt{9.25}\ \text{units/sec},\quad \vec{\mathbf{a}}(t)=-9\cos 3t\,\hat{\mathbf{i}}-9\sin 3t\,\hat{\mathbf{j}}$

24) $\vec{\mathbf{r}}(t)=e^{-t}\,\hat{\mathbf{i}}+(\ln t)\,\hat{\mathbf{j}}+(\sin 7t)\,\hat{\mathbf{k}}$

25) Consider the position vector for a particle to be $\vec{\mathbf{r}}(t)=t\,\hat{\mathbf{i}}+t^2\,\hat{\mathbf{j}}+t^3\,\hat{\mathbf{k}}$. The graph is shown here:

https://math.libretexts.org/@go/page/69508



a. Find the velocity vector at any time.

b. Find the speed of the particle at time $t = 2$ sec.

c. Find the acceleration at time $t = 2$ sec.

Answer

a. $\vec{\mathbf{v}}(t) = \hat{\mathbf{i}} + 2t\hat{\mathbf{j}} + 3t^2\hat{\mathbf{k}}$

b. $\sqrt{161}$ units/sec

Note that $\text{speed}(t) = \|\vec{\mathbf{v}}(t)\| = \sqrt{1^2 + (2t)^2 + (3t^2)^2} = \sqrt{1 + 4t^2 + 9t^4}$.

Hence, $\text{speed}(2) = \sqrt{1 + 16 + 9(16)} = \sqrt{161}$ units/sec.

c. Since $\vec{\mathbf{a}}(t) = 2\hat{\mathbf{j}} + 6t\hat{\mathbf{k}}$, $\vec{\mathbf{a}}(2) = 2\hat{\mathbf{j}} + 12\hat{\mathbf{k}}$

26) A particle travels along the path of an ellipse with the equation $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + 2\sin t\,\hat{\mathbf{j}} + 0\hat{\mathbf{k}}$. Find the following:

a. Velocity of the particle

Answer

$\vec{\mathbf{v}}(t) = \langle -\sin t, 2\cos t, 0\rangle$

b. Speed of the particle at $t = \frac{\pi}{4}$

c. Acceleration of the particle at $t = \frac{\pi}{4}$

Answer

$\vec{\mathbf{a}}(t) = \langle -\frac{\sqrt{2}}{2}, -\sqrt{2}, 0\rangle$

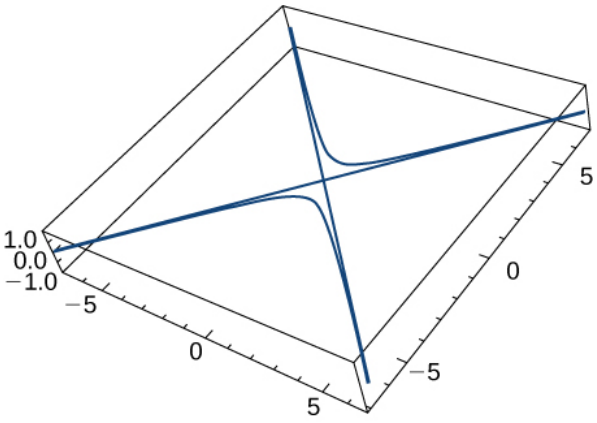
27) Show that if the speed of a particle traveling along a curve represented by a vector-valued function is constant, then the velocity function is always perpendicular to the acceleration function.

Answer

$$\begin{aligned} \|\vec{\mathbf{v}}(t)\| &= k \\ \vec{\mathbf{v}}(t)\cdot\vec{\mathbf{v}}(t) &= k^2 \\ \frac{d}{dt}\left(\vec{\mathbf{v}}(t)\cdot\vec{\mathbf{v}}(t)\right) &= \frac{d}{dt}\left(k^2\right) = 0 \\ \vec{\mathbf{v}}(t)\cdot\vec{\mathbf{v}}'(t) + \vec{\mathbf{v}}'(t)\cdot\vec{\mathbf{v}}(t) &= 0 \\ 2\vec{\mathbf{v}}(t)\cdot\vec{\mathbf{v}}'(t) &= 0 \\ \vec{\mathbf{v}}(t)\cdot\vec{\mathbf{v}}'(t) &= 0 \end{aligned}$$

The last statement implies that the velocity and acceleration are perpendicular or orthogonal.

28) Given the vector-valued function $\vec{\mathbf{r}}(t) = \langle \tan t, \sec t, 0\rangle$ (graph is shown here), find the following:



a. Velocity

Answer

$\vec{\mathbf{v}}(t) = \langle \sec^2 t,\ \sec t \tan t \rangle$

b. Speed

Answer

$\|\vec{\mathbf{v}}(t)\| = \sqrt{\sec^4 t + \sec^2 t \tan^2 t} = \sqrt{(\sec^2 t)(\sec^2 t + \tan^2 t)}$

c. Acceleration

Answer

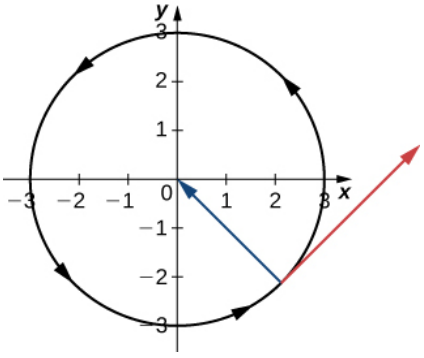
$\vec{\mathbf{a}}(t) = \langle 2\sec^2 t \tan t,\ \sec t \tan^2 t + \sec^3 t \rangle$

29) Find the minimum speed of a particle traveling along the curve $\vec{\mathbf{r}}(t) = \langle t + \cos t, t - \sin t \rangle$, where $t \in [0, 2\pi)$. Then also find its maximum speed on this interval.

Answer

Min. speed is $\sqrt{3 - 2\sqrt{2}} \approx 0.41421$ when $t = \frac{\pi}{4}$.
Max. speed is $\sqrt{3 + 2\sqrt{2}} \approx 2.41421$ when $t = \frac{5\pi}{4}$.

For questions 30 - 31, consider a particle that moves on a circular path of radius b according to the function $\vec{\mathbf{r}}(t) = b\cos(\omega t)\,\hat{\mathbf{i}} + b\sin(\omega t)\,\hat{\mathbf{j}}$, where ω is the angular velocity, $\frac{d\theta}{dt}$.



30) Show that the speed of the particle is proportional to the angular velocity.

31) Find the velocity function and show that $\vec{\mathbf{v}}(t)$ is always orthogonal to $\vec{\mathbf{r}}(t)$.

Answer

$\vec{\mathbf{r}}'(t) = -b\omega\sin(\omega t)\,\hat{\mathbf{i}} + b\omega\cos(\omega t)\,\hat{\mathbf{j}}$. To show orthogonality, note that $\vec{\mathbf{r}}'(t) \cdot \vec{\mathbf{r}}(t) = 0$.

https://math.libretexts.org/@go/page/69508

SMOOTHNESS OF VECTOR-VALUED FUNCTIONS

For questions 32 - 40,

a. Determine any values of t at which $\vec{\mathbf{r}}$ is not smooth.

b. Determine the open intervals on which $\vec{\mathbf{r}}$ is smooth.

c. Graph the vector-valued function and describe its behavior at the points where it is not smooth.

32) $\vec{\mathbf{r}}(t) = \langle 3t, 5t^2 - 1 \rangle$

33) $\vec{\mathbf{r}}(t) = t^3\,\hat{\mathbf{i}} + 5t^2\,\hat{\mathbf{j}}$

Answer

a. $\vec{\mathbf{r}}$ is not smooth at $t = 0$, since $\vec{\mathbf{r}}'(0) = \vec{\mathbf{0}}$.
b. $\vec{\mathbf{r}}$ is smooth on the open intervals $(-\infty, 0)$ and $(0, \infty)$.
c. There is a cusp when $t = 0$.

34) $\vec{\mathbf{r}}(t) = \langle 5,\ 2\sin(t),\ \cos(t) \rangle$

35) $\vec{\mathbf{r}}(t) = \langle t^3 - 3t^2, 7 \rangle$

Answer

a. $\vec{\mathbf{r}}$ is not smooth at $t = 0$ and $t = 2$, since $\vec{\mathbf{r}}'(0) = \vec{\mathbf{0}}$ and $\vec{\mathbf{r}}'(2) = \vec{\mathbf{0}}$.
b. $\vec{\mathbf{r}}$ is smooth on the open intervals $(-\infty, 0)$, $(0, 2)$, and $(2, \infty)$.
c. The motion on the curve reverses along the same path at both $t = 0$ and $t = 2$.

36) $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + t^3\,\hat{\mathbf{j}} - 5e^{-4t}\,\hat{\mathbf{k}}$

37) $\vec{\mathbf{r}}(t) = \left\langle \ln(t^2 + 4t + 5),\ \dfrac{t^3}{3} - 4t,\ 5 \right\rangle$

Answer

a. $\vec{\mathbf{r}}$ is not smooth at $t = -2$, since $\vec{\mathbf{r}}'(-2) = \vec{\mathbf{0}}$.
Since the domain of $\vec{\mathbf{r}}$ is $(-\infty, \infty)$, this is all we have to remove.
b. $\vec{\mathbf{r}}$ is smooth on the open intervals $(-\infty, -2)$ and $(-2, \infty)$.
c. There is a cusp when $t = -2$.

38) $\vec{\mathbf{r}}(t) = (5\cos t - \cos 5t)\,\hat{\mathbf{i}} + (5\sin t - \sin 5t)\,\hat{\mathbf{j}}$, for $0 \le t \le 2\pi$

39) $\vec{\mathbf{r}}(t) = \sqrt{t^3 + 9t^2}\,\hat{\mathbf{i}} + (t^2 + 12t)\,\hat{\mathbf{j}} + 7\hat{\mathbf{k}}$

Answer

a. The domain of $\vec{\mathbf{r}}$ is $[-9, \infty)$.
And $\vec{\mathbf{r}}$ is not smooth at $t = -6$, since $\vec{\mathbf{r}}'(-6) = \vec{\mathbf{0}}$.
The domain of $\vec{\mathbf{r}}'$ is $(-9, \infty)$, since $\vec{\mathbf{r}}'$ is undefined at $t = -9$.
b. $\vec{\mathbf{r}}$ is smooth on the open intervals $(-9, -6)$ and $(-6, \infty)$.
c. There is a cusp when $t = -6$.

40) $\vec{\mathbf{r}}(t) = \cos^3 t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}}$, for $0 \le t \le 2\pi$

Answer

a. The domain of $\vec{\mathbf{r}}$ is $(-\infty, \infty)$.
$\vec{\mathbf{r}}'(t) = -3(\cos^2 t)(\sin t)\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}}$. It's domain is also $(-\infty, \infty)$.
But note that both components have a factor of $\cos t$, so both components will be 0 when $\cos t = 0$.
Therefore, $\vec{\mathbf{r}}$ is not smooth at $t = \frac{\pi}{2}$ and at $t = \frac{3\pi}{2}$, since $\vec{\mathbf{r}}'\left(\frac{\pi}{2}\right) = \vec{\mathbf{0}}$ and $\vec{\mathbf{r}}'\left(\frac{3\pi}{2}\right) = \vec{\mathbf{0}}$. Note then that $\vec{\mathbf{r}}$ is not smooth for any odd multiple of $\frac{\pi}{2}$, that is for $t = \frac{(2n+1)\pi}{2}$, for any integer value n.
b. $\vec{\mathbf{r}}$ is smooth on the open intervals $\left(\dfrac{(2n-1)\pi}{2}, \dfrac{(2n+1)\pi}{2}\right)$, for any integer value n.
c. There is a cusp when $t = \dfrac{(2n+1)\pi}{2}$, for any integer value n.

PROPERTIES OF THE DERIVATIVE

For questions, 41 - 43, evaluate each expression given that $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} - t^4\,\hat{\mathbf{k}}$ and $\vec{\mathbf{s}}(t) = \sin(t)\,\hat{\mathbf{i}} + e^t\,\hat{\mathbf{j}} + \cos(t)\,\hat{\mathbf{k}}$

41) $\frac{d}{dt}\left[\vec{\mathbf{r}}(t^2)\right]$

Answer

$\frac{d}{dt}\left[\vec{\mathbf{r}}(t^2)\right] = \langle 2t, 4t^3, -8t^7\rangle$

42) $\frac{d}{dt}\left[t^2 \cdot \vec{\mathbf{s}}(t)\right]$

43) $\frac{d}{dt}\left[\vec{\mathbf{r}}(t) \cdot \vec{\mathbf{s}}(t)\right]$

Answer

$\frac{d}{dt}\left[\vec{\mathbf{r}}(t) \cdot \vec{\mathbf{s}}(t)\right] = \sin t + 2te^t - 4t^3\cos t + t\cos t + t^2e^t + t^4\sin t$

44) Find $\vec{\mathbf{r}}'(t) \cdot \vec{\mathbf{r}}''(t)$ *for* $\vec{\mathbf{r}}(t) = -3t^5\,\hat{\mathbf{i}} + 5t\,\hat{\mathbf{j}} + 2t^2\,\hat{\mathbf{k}}$.

Answer

$\vec{\mathbf{r}}'(t) \cdot \vec{\mathbf{r}}''(t) = 900t^7 + 16t$

45) Given $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + 3t\,\hat{\mathbf{j}} + t^2\,\hat{\mathbf{k}}$ and $\vec{\mathbf{u}}(t) = 4t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + t^3\,\hat{\mathbf{k}}$, find $\frac{d}{dt}(\vec{\mathbf{r}}(t) \times \vec{\mathbf{u}}(t))$.

46) Evaluate $\frac{d}{dt}\left[\vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}'(t)\right]$ given $\vec{\mathbf{u}}(t) = t^2\,\hat{\mathbf{i}} - 2t\,\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Answer

$\frac{d}{dt}\left[\vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}'(t)\right] = 0\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + 4t\,\hat{\mathbf{k}}$

47) Given $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + 2\sin t\,\hat{\mathbf{j}} + 2\cos t\,\hat{\mathbf{k}}$ and $\vec{\mathbf{u}}(t) = \frac{1}{t}\,\hat{\mathbf{i}} + 2\sin t\,\hat{\mathbf{j}} + 2\cos t\,\hat{\mathbf{k}}$, find the following:

a. $\vec{\mathbf{r}}(t) \times \vec{\mathbf{u}}(t)$

Answer

$\vec{\mathbf{r}}(t) \times \vec{\mathbf{u}}(t) = \left\langle 0,\ 2(\cos t)\left(\frac{1}{t} - t\right),\ 2(\sin t)\left(t - \frac{1}{t}\right)\right\rangle$

b. $\frac{d}{dt}\left(\vec{\mathbf{r}}(t) \times \vec{\mathbf{u}}(t)\right)$

Answer

$\frac{d}{dt}\left(\vec{\mathbf{r}}(t) \times \vec{\mathbf{u}}(t)\right) = \left\langle 0,\ 2(\sin t)\left(t - \frac{1}{t}\right) - 2(\cos t)\left(1 + \frac{1}{t^2}\right),\ 2(\sin t)\left(1 + \frac{1}{t^2}\right) + 2(\cos t)\left(t - \frac{1}{t}\right)\right\rangle$

c. Now, use the product rule for the derivative of the cross product of two vectors and show this result is the same as the answer for the preceding problem.

UNIT TANGENT VECTORS

For questions 48 - 51, find a *unit tangent vector* at the indicated value of t.

48) $\vec{\mathbf{r}}(t) = 3t^3\,\hat{\mathbf{i}} + 2t^2\,\hat{\mathbf{j}} + \frac{1}{t}\,\hat{\mathbf{k}};\quad t = 1$

49) $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + \sin(2t)\,\hat{\mathbf{j}} + \cos(3t)\,\hat{\mathbf{k}};\quad t = \frac{\pi}{3}$

Answer

$\vec{\mathbf{r}}'\left(\frac{\pi}{3}\right) = \langle 1, -1, 0\rangle$ is a tangent vector, so a unit tangent vector would be:
$\frac{1}{\sqrt{2}}\langle 1, -1, 0\rangle \quad = \quad \left\langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0\right\rangle$

50) $\vec{\mathbf{r}}(t) = \cos(2t)\,\hat{\mathbf{i}} + 2\sin t\,\hat{\mathbf{j}} + t^2\,\hat{\mathbf{k}};\quad t = \frac{\pi}{2}$

51) $\vec{\mathbf{r}}(t) = 3e^t\,\hat{\mathbf{i}} + 2e^{-3t}\,\hat{\mathbf{j}} + 4e^{2t}\,\hat{\mathbf{k}};\quad t = \ln(2)$

Answer

$\vec{\mathbf{r}}'(\ln(2)) = \langle 6, -\frac{3}{4}, 32\rangle$ is a tangent vector, so a unit tangent vector would be:

$$\frac{1}{\sqrt{1060.5625}}\langle 6, -\frac{3}{4}, 32\rangle = \left\langle \frac{24\sqrt{16969}}{16969}, -\frac{12\sqrt{16969}}{67876}, \frac{128\sqrt{16969}}{16969}\right\rangle$$

For questions 52 - 58, find the unit tangent vector $\vec{\mathbf{T}}(t)$ for the following parameterized curves.

52) $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + 3t\,\hat{\mathbf{j}} + t^2\,\hat{\mathbf{k}}$

53) $\vec{\mathbf{r}}(t) = 6\,\hat{\mathbf{i}} + \cos(3t)\,\hat{\mathbf{j}} + 3\sin(4t)\,\hat{\mathbf{k}}, \quad 0 \le t < 2\pi$

Answer

$$\vec{\mathbf{T}}(t) = \frac{1}{\sqrt{9\sin^2(3t) + 144\cos^2(4t)}}\langle 0, -3\sin(3t), 12\cos(4t)\rangle$$

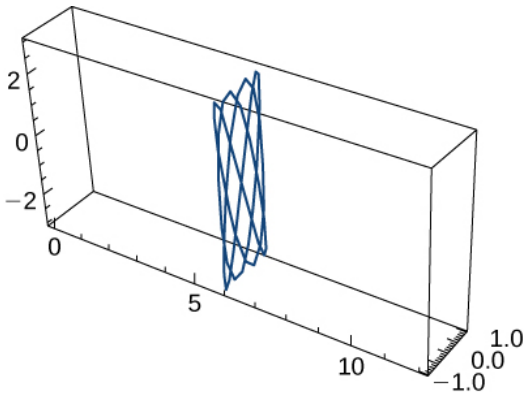
54) $\vec{\mathbf{r}}(t) = \langle t\cos t, t\sin t\rangle$

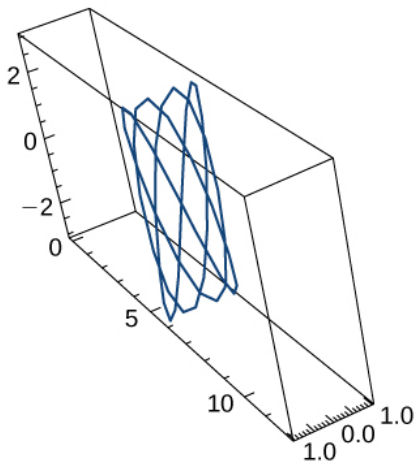
55) $\vec{\mathbf{r}}(t) = \langle t+1, 2t+1, 2t+2\rangle$

Answer

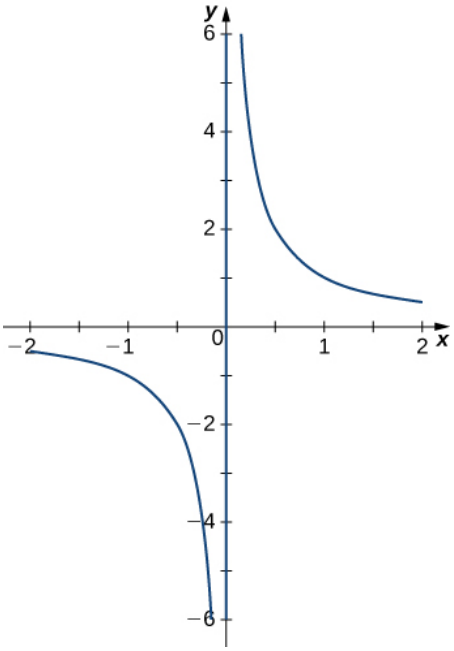
$\vec{\mathbf{T}}(t) = \frac{1}{3}\langle 1, 2, 2\rangle$

56) $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} + \sin t\,\hat{\mathbf{k}}, \quad 0 \le t < 2\pi$. Two views of this curve are presented here:





57) $\vec{\mathbf{r}}(t) = \left\langle t, \frac{1}{t}\right\rangle$. The graph is shown here:



Answer

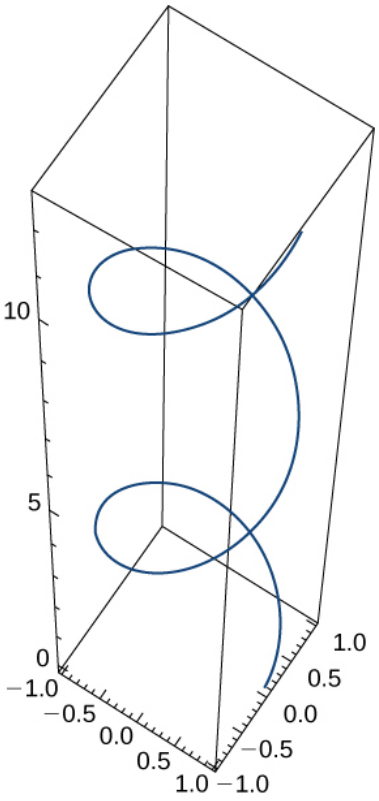
$$\vec{\mathbf{T}}(t) = \left\langle \frac{t^2}{\sqrt{t^4+1}}, \frac{-1}{\sqrt{t^4+1}} \right\rangle$$

58) $\vec{\mathbf{r}}(t) = 3\cos(4t)\,\hat{\mathbf{i}} + 3\sin(4t)\,\hat{\mathbf{j}} + 5t\,\hat{\mathbf{k}}, \quad 1 \le t \le 2$

Answer

$$\vec{\mathbf{T}}(t) = -\tfrac{12}{13}\sin(4t)\,\hat{\mathbf{i}} + \tfrac{12}{13}\cos(4t)\,\hat{\mathbf{j}} + \tfrac{5}{13}\,\hat{\mathbf{k}}$$

59) A particle travels along the path of a helix with the equation $\vec{\mathbf{r}}(t) = \cos(t)\,\hat{\mathbf{i}} + \sin(t)\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$. See the graph presented here:



Find the following:

a. Velocity of the particle at any time

Answer

$\vec{\mathbf{v}}(t) = \langle -\sin t, \cos t, 1 \rangle$

b. Speed of the particle at any time

c. Acceleration of the particle at any time

Answer

$\vec{\mathbf{a}}(t) = -\cos t\,\hat{\mathbf{i}} - \sin t\,\hat{\mathbf{j}} + 0\,\hat{\mathbf{k}}$

d. Find the unit tangent vector for the helix.

INTEGRATION OF VECTOR-VALUED FUNCTIONS

Evaluate the following integrals:

60) $\int \left(e^t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} + \frac{1}{2t-1}\,\hat{\mathbf{k}}\right) dt$

61) $\int_0^1 \vec{\mathbf{r}}(t)\,dt$, where $\vec{\mathbf{r}}(t) = \left\langle \sqrt[3]{t}, \frac{1}{t+1}, e^{-t} \right\rangle$

Answer

$\frac{3}{4}\,\hat{\mathbf{i}} + \ln(2)\,\hat{\mathbf{j}} + (1 - \frac{1}{e})\,\hat{\mathbf{k}}$

62) Evaluate $\int_0^3 \|t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}}\|dt$.

Answer

$\frac{1}{3}(10^{\frac{3}{2}} - 1)$

63) The acceleration function, initial velocity, and initial position of a particle are

$$\vec{\mathbf{a}}(t) = -5\cos t\,\hat{\mathbf{i}} - 5\sin t\,\hat{\mathbf{j}},$$
$$\vec{\mathbf{v}}(0) = 9\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}}, \quad \text{and}$$
$$\vec{\mathbf{r}}(0) = 5\,\hat{\mathbf{i}}$$

Find $\vec{\mathbf{v}}(t)$ and $\vec{\mathbf{r}}(t)$.

Answer

$\vec{\mathbf{v}}(t) = (9 - 5\sin t)\,\hat{\mathbf{i}} + (-3 + 5\cos t)\,\hat{\mathbf{j}}$
$\vec{\mathbf{r}}(t) = (9t + 5\cos t)\,\hat{\mathbf{i}} + (-3t + 5\sin t)\,\hat{\mathbf{j}}$

64) Find the antiderivative of $\vec{\mathbf{r}}'(t) = \cos(2t)\,\hat{\mathbf{i}} - 2\sin t\,\hat{\mathbf{j}} + \frac{1}{1+t^2}\,\hat{\mathbf{k}}$ that satisfies the initial condition $\vec{\mathbf{r}}(0) = 3\,\hat{\mathbf{i}} - 2\,\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

65) An object starts from rest at point $P(1, 2, 0)$ and moves with an acceleration of $\vec{\mathbf{a}}(t) = \hat{\mathbf{j}} + 2\,\hat{\mathbf{k}}$, where $\|\vec{\mathbf{a}}(t)\|$ is measured in feet per second per second. Find the location of the object after $t = 2$ sec.

CONTRIBUTORS:

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) created questions 11 - 19.

13.2E: Exercises for Section 13.2 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

13.3: Arc Length and Curvature

Learning Objectives

- Determine the length of a particle's path in space by using the arc-length function.
- Explain the meaning of the curvature of a curve in space and state its formula.
- Describe the meaning of the normal and binormal vectors of a curve in space.

In this section, we study formulas related to curves in both two and three dimensions, and see how they are related to various properties of the same curve. For example, suppose a vector-valued function describes the motion of a particle in space. We would like to determine how far the particle has traveled over a given time interval, which can be described by the arc length of the path it follows. Or, suppose that the vector-valued function describes a road we are building and we want to determine how sharply the road curves at a given point. This is described by the curvature of the function at that point. We explore each of these concepts in this section.

Arc Length for Vector Functions

We have seen how a vector-valued function describes a curve in either two or three dimensions. Recall that the formula for the arc length of a curve defined by the parametric functions $x = x(t), y = y(t), t_1 \leq t \leq t_2$ is given by

$$s = \int_{t_1}^{t_2} \sqrt{(x'(t))^2 + (y'(t))^2}dt.$$

In a similar fashion, if we define a smooth curve using a vector-valued function $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}$, where $a \leq t \leq b$, the arc length is given by the formula

$$s = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2}dt.$$

In three dimensions, if the vector-valued function is described by $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}$ over the same interval $a \leq t \leq b$, the arc length is given by

$$s = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2}dt.$$

Theorem: Arc-Length Formulas for Plane and Space curves

Plane curve: Given a smooth curve C defined by the function $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}}$, where t lies within the interval $[a, b]$, the arc length of C over the interval is

$$s = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}dt \tag{13.3.1}$$

$$= \int_a^b \|\vec{\mathbf{r}}'(t)\|dt. \tag{13.3.2}$$

Space curve: Given a smooth curve C defined by the function $\vec{\mathbf{r}}(t) = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}$, where t lies within the interval $[a, b]$, the arc length of C over the interval is

$$s = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}dt \tag{13.3.3}$$

$$= \int_a^b \|\vec{\mathbf{r}}'(t)\|dt. \tag{13.3.4}$$

The two formulas are very similar; they differ only in the fact that a space curve has three component functions instead of two. Note that the formulas are defined for smooth curves: curves where the vector-valued function $\vec{\mathbf{r}}(t)$ is differentiable with a non-zero derivative. The smoothness condition guarantees that the curve has no cusps (or corners) that could make the formula problematic.

Example 13.3.1: Finding the Arc Length

Calculate the arc length for each of the following vector-valued functions:

a. $\vec{\mathbf{r}}(t) = (3t-2)\,\hat{\mathbf{i}} + (4t+5)\,\hat{\mathbf{j}}, \quad 1 \le t \le 5$

b. $\vec{\mathbf{r}}(t) = \langle t\cos t, t\sin t, 2t\rangle, 0 \le t \le 2\pi$

Solution

a. Using Equation 13.3.2, $\vec{\mathbf{r}}'(t) = 3\,\hat{\mathbf{i}} + 4\,\hat{\mathbf{j}}$, so

$$\begin{aligned} s &= \int_a^b \|\vec{\mathbf{r}}'(t)\|dt \\ &= \int_1^5 \sqrt{3^2+4^2}dt \\ &= \int_1^5 5dt = 5t\big|_1^5 = 20. \end{aligned}$$

b. Using Equation 13.3.4, $\vec{\mathbf{r}}'(t) = \langle \cos t - t\sin t, \sin t + t\cos t, 2\rangle$, so

$$\begin{aligned} s &= \int_a^b \| \vec{\mathbf{r}}'(t) \| dt \\ &= \int_0^{2\pi} \sqrt{(\cos t - t\sin t)^2 + (\sin t + t\cos t)^2 + 2^2}dt \\ &= \int_0^{2\pi} \sqrt{(\cos^2 t - 2t\sin t\cos t + t^2\sin^2 t) + (\sin^2 t + 2t\sin t\cos t + t^2\cos^2 t) + 4}dt \\ &= \int_0^{2\pi} \sqrt{\cos^2 t + \sin^2 t + t^2(\cos^2 t + \sin^2 t) + 4}dt \\ &= \int_0^{2\pi} \sqrt{t^2+5}dt \end{aligned}$$

Here we can use a table integration formula

$$\int \sqrt{u^2+a^2}du = \frac{u}{2}\sqrt{u^2+a^2} + \frac{a^2}{2}\ln\left|u + \sqrt{u^2+a^2}\right| + C,$$

so we obtain

$$\begin{aligned} \int_0^{2\pi} \sqrt{t^2+5}dt &= \frac{1}{2}\left(t\sqrt{t^2+5} + 5\ln\left|t+\sqrt{t^2+5}\right|\right)_0^{2\pi} \\ &= \frac{1}{2}\left(2\pi\sqrt{4\pi^2+5} + 5\ln\left(2\pi + \sqrt{4\pi^2+5}\right)\right) - \frac{5}{2}\ln\sqrt{5} \\ &\approx 25.343 \text{ units}. \end{aligned}$$

Exercise 13.3.1

Calculate the arc length of the parameterized curve

$$\vec{\mathbf{r}}(t) = \langle 2t^2+1, 2t^2-1, t^3\rangle, \quad 0 \le t \le 3.$$

Hint

Use Equation 13.3.4.

Answer

$\vec{\mathbf{r}}'(t) = \langle 4t, 4t, 3t^2\rangle$, so $s = \frac{1}{27}(113^{3/2} - 32^{3/2}) \approx 37.785$ units

We now return to the helix introduced earlier in this chapter. A vector-valued function that describes a helix can be written in the form

$$\vec{\mathbf{r}}(t) = R\cos\left(\frac{2\pi Nt}{h}\right)\hat{\mathbf{i}} + R\sin\left(\frac{2\pi Nt}{h}\right)\hat{\mathbf{j}} + t\hat{\mathbf{k}}, 0 \le t \le h,$$

where R represents the radius of the helix, h represents the height (distance between two consecutive turns), and the helix completes N turns. Let's derive a formula for the arc length of this helix using Equation 13.3.4. First of all,

$$\vec{\mathbf{r}}\,'(t) = -\frac{2\pi NR}{h}\sin\left(\frac{2\pi Nt}{h}\right)\hat{\mathbf{i}} + \frac{2\pi NR}{h}\cos\left(\frac{2\pi Nt}{h}\right)\hat{\mathbf{j}} + \hat{\mathbf{k}}.$$

Therefore,

$$\begin{aligned}
s &= \int_a^b \|\vec{\mathbf{r}}\,'(t)\|dt \\
&= \int_0^h \sqrt{\left(-\frac{2\pi NR}{h}\sin\left(\frac{2\pi Nt}{h}\right)\right)^2 + \left(\frac{2\pi NR}{h}\cos\left(\frac{2\pi Nt}{h}\right)\right)^2 + 1^2}dt \\
&= \int_0^h \sqrt{\frac{4\pi^2N^2R^2}{h^2}\left(\sin^2\left(\frac{2\pi Nt}{h}\right) + \cos^2\left(\frac{2\pi Nt}{h}\right)\right) + 1}dt \\
&= \int_0^h \sqrt{\frac{4\pi^2N^2R^2}{h^2} + 1}dt \\
&= \left[t\sqrt{\frac{4\pi^2N^2R^2}{h^2} + 1}\right]_0^h \\
&= h\sqrt{\frac{4\pi^2N^2R^2 + h^2}{h^2}} \\
&= \sqrt{4\pi^2N^2R^2 + h^2}.
\end{aligned}$$

This gives a formula for the length of a wire needed to form a helix with N turns that has radius R and height h.

Arc-Length Parameterization

We now have a formula for the arc length of a curve defined by a vector-valued function. Let's take this one step further and examine what an **arc-length function** is.

If a vector-valued function represents the position of a particle in space as a function of time, then the arc-length function measures how far that particle travels as a function of time. The formula for the arc-length function follows directly from the formula for arc length:

$$s = \int_a^t \sqrt{(f'(u))^2 + (g'(u))^2 + (h'(u))^2}du. \tag{13.3.5}$$

If the curve is in two dimensions, then only two terms appear under the square root inside the integral. The reason for using the independent variable u is to distinguish between time and the variable of integration. Since $s(t)$ measures distance traveled as a function of time, $s'(t)$ measures the speed of the particle at any given time. Since we have a formula for $s(t)$ in Equation 13.3.5, we can differentiate both sides of the equation:

$$\begin{aligned}
s'(t) &= \frac{d}{dt}\left[\int_a^t \sqrt{(f'(u))^2 + (g'(u))^2 + (h'(u))^2}du\right] \\
&= \frac{d}{dt}\left[\int_a^t \|\vec{\mathbf{r}}\,'(u)\|du\right] \\
&= \|\vec{\mathbf{r}}\,'(t)\|.
\end{aligned}$$

If we assume that $\vec{\mathbf{r}}(t)$ defines a smooth curve, then the arc length is always increasing, so $s'(t) > 0$ for $t > a$. Last, if $\vec{\mathbf{r}}(t)$ is a curve on which $\|\vec{\mathbf{r}}\,'(t)\| = 1$ for all t, then

$$s(t) = \int_a^t \|\vec{\mathbf{r}}\,'(u)\|\,du = \int_a^t 1\,du = t - a,$$

which means that t represents the arc length as long as $a = 0$.

Theorem: Arc-Length Function

Let $\vec{\mathbf{r}}(t)$ describe a smooth curve for $t \geq a$. Then the arc-length function is given by

$$s(t) = \int_a^t \|\vec{\mathbf{r}}'(u)\| \, du$$

Furthermore,

$$\frac{ds}{dt} = \|\vec{\mathbf{r}}'(t)\| > 0.$$

If $\|\vec{\mathbf{r}}'(t)\| = 1$ for all $t \geq a$, then the parameter t represents the arc length from the starting point at $t = a$.

A useful application of this theorem is to find an alternative parameterization of a given curve, called an **arc-length parameterization**. Recall that any vector-valued function can be reparameterized via a change of variables. For example, if we have a function $\vec{\mathbf{r}}(t) = \langle 3\cos t, 3\sin t \rangle, 0 \leq t \leq 2\pi$ that parameterizes a circle of radius 3, we can change the parameter from t to $4t$, obtaining a new parameterization $\vec{\mathbf{r}}(t) = \langle 3\cos 4t, 3\sin 4t \rangle$. The new parameterization still defines a circle of radius 3, but now we need only use the values $0 \leq t \leq \pi/2$ to traverse the circle once.

Suppose that we find the arc-length function $s(t)$ and are able to solve this function for t as a function of s. We can then reparameterize the original function $\vec{\mathbf{r}}(t)$ by substituting the expression for t back into $\vec{\mathbf{r}}(t)$. The vector-valued function is now written in terms of the parameter s. Since the variable s represents the arc length, we call this an *arc-length parameterization* of the original function $\vec{\mathbf{r}}(t)$. One advantage of finding the arc-length parameterization is that the distance traveled along the curve starting from $s = 0$ is now equal to the parameter s. The arc-length parameterization also appears in the context of curvature (which we examine later in this section) and line integrals.

Example 13.3.2: Finding an Arc-Length Parameterization

Find the arc-length parameterization for each of the following curves:

a. $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}}, \quad t \geq 0$
b. $\vec{\mathbf{r}}(t) = \langle t + 3, 2t - 4, 2t \rangle, \quad t \geq 3$

Solution

a. First we find the arc-length function using Equation 13.3.5:

$$\begin{aligned} s(t) &= \int_a^t \|\vec{\mathbf{r}}'(u)\| \, du \\ &= \int_0^t \|\langle -4\sin u, 4\cos u \rangle\| \, du \\ &= \int_0^t \sqrt{(-4\sin u)^2 + (4\cos u)^2} \, du \\ &= \int_0^t \sqrt{16\sin^2 u + 16\cos^2 u} \, du \\ &= \int_0^t 4 \, du = 4t, \end{aligned}$$

which gives the relationship between the arc length s and the parameter t as $s = 4t$; so, $t = s/4$. Next we replace the variable t in the original function $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}}$ with the expression $s/4$ to obtain

$$\vec{\mathbf{r}}(s) = 4\cos\left(\frac{s}{4}\right)\hat{\mathbf{i}} + 4\sin\left(\frac{s}{4}\right)\hat{\mathbf{j}}.$$

This is the arc-length parameterization of $\vec{\mathbf{r}}(t)$. Since the original restriction on t was given by $t \geq 0$, the restriction on s becomes $s/4 \geq 0$, or $s \geq 0$.

b. The arc-length function is given by Equation 13.3.5:

$$\begin{aligned} s(t) &= \int_a^t \|\vec{\mathbf{r}}'(u)\|\, du \\ &= \int_3^t \|\langle 1,2,2\rangle\|\, du \\ &= \int_3^t \sqrt{1^2+2^2+2^2}\, du \\ &= \int_3^t 3\, du \\ &= 3t-9. \end{aligned}$$

Therefore, the relationship between the arc length s and the parameter t is $s = 3t - 9$, so $t = \frac{s}{3} + 3$. Substituting this into the original function $\vec{\mathbf{r}}(t) = \langle t+3, 2t-4, 2t\rangle$ yields

$$\vec{\mathbf{r}}(s) = \langle\left(\frac{s}{3}+3\right)+3,\, 2\left(\frac{s}{3}+3\right)-4,\, 2\left(\frac{s}{3}+3\right)\rangle = \langle\frac{s}{3}+6, \frac{2s}{3}+2, \frac{2s}{3}+6\rangle.$$

This is an arc-length parameterization of $\vec{\mathbf{r}}(t)$. The original restriction on the parameter t was $t \geq 3$, so the restriction on s is $(s/3)+3 \geq 3$, or $s \geq 0$.

Exercise 13.3.2

Find the arc-length function for the helix

$$\vec{\mathbf{r}}(t) = \langle 3\cos t, 3\sin t, 4t\rangle, \quad t \geq 0.$$

Then, use the relationship between the arc length and the parameter t to find an arc-length parameterization of $\vec{\mathbf{r}}(t)$.

Hint

Start by finding the arc-length function.

Answer

$s = 5t$, or $t = s/5$. Substituting this into $\vec{\mathbf{r}}(t) = \langle 3\cos t, 3\sin t, 4t\rangle$ gives

$$\vec{\mathbf{r}}(s) = \langle 3\cos\left(\frac{s}{5}\right), 3\sin\left(\frac{s}{5}\right), \frac{4s}{5}\rangle, \quad s \geq 0$$

Curvature

An important topic related to arc length is curvature. The concept of curvature provides a way to measure how sharply a smooth curve turns. A circle has constant curvature. The smaller the radius of the circle, the greater the curvature.

Think of driving down a road. Suppose the road lies on an arc of a large circle. In this case you would barely have to turn the wheel to stay on the road. Now suppose the radius is smaller. In this case you would need to turn more sharply to stay on the road. In the case of a curve other than a circle, it is often useful first to inscribe a circle to the curve at a given point so that it is tangent to the curve at that point and "hugs" the curve as closely as possible in a neighborhood of the point (Figure 13.3.1). The curvature of the graph at that point is then defined to be the same as the curvature of the inscribed circle.

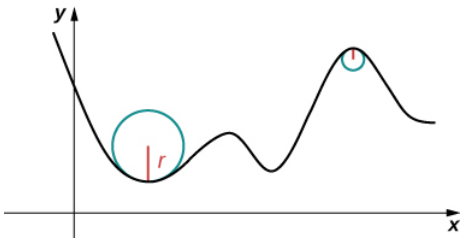


Figure 13.3.1: The graph represents the curvature of a function $y = f(x)$. The sharper the turn in the graph, the greater the curvature, and the smaller the radius of the inscribed circle.

Definition: Curvature

Let C be a smooth curve in the plane or in space given by $\vec{\mathbf{r}}(s)$, where s is the arc-length parameter. The curvature κ at s is

$$\kappa = \left\| \frac{d\vec{\mathbf{T}}}{ds} \right\| = \|\vec{\mathbf{T}}'(s)\|.$$

Visit this video for more information about the curvature of a space curve.

The formula in the definition of curvature is not very useful in terms of calculation. In particular, recall that $\vec{\mathbf{T}}(t)$ represents the unit tangent vector to a given vector-valued function $\vec{\mathbf{r}}(t)$, and the formula for $\vec{\mathbf{T}}(t)$ is

$$\vec{\mathbf{T}}(t) = \frac{\vec{\mathbf{r}}\,'(t)}{\| \vec{\mathbf{r}}\,'(t) \|}.$$

To use the formula for curvature, it is first necessary to express $\vec{\mathbf{r}}(t)$ in terms of the arc-length parameter s, then find the unit tangent vector $\vec{\mathbf{T}}(s)$ for the function $\vec{\mathbf{r}}(s)$, then take the derivative of $\vec{\mathbf{T}}(s)$ with respect to s. This is a tedious process. Fortunately, there are equivalent formulas for curvature.

Theorem: Alternate Formulas of Curvature

If C is a smooth curve given by $\vec{\mathbf{r}}(t)$, then the curvature κ of C at t is given by

$$\kappa = \frac{\|\vec{\mathbf{T}}'(t)\|}{\|\vec{\mathbf{r}}\,'(t)\|}. \tag{13.3.6}$$

If C is a three-dimensional curve, then the curvature can be given by the formula

$$\kappa = \frac{\|\vec{\mathbf{r}}\,'(t) \times \vec{\mathbf{r}}\,''(t)\|}{\|\vec{\mathbf{r}}\,'(t)\|^3}. \tag{13.3.7}$$

If C is the graph of a function $y = f(x)$ and both y' and y'' exist, then the curvature κ at point (x, y) is given by

$$\kappa = \frac{|y''|}{[1 + (y')^2]^{3/2}}. \tag{13.3.8}$$

Proof

The first formula follows directly from the chain rule:

$$\frac{d\vec{\mathbf{T}}}{dt} = \frac{d\vec{\mathbf{T}}}{ds}\frac{ds}{dt},$$

where s is the arc length along the curve C. Dividing both sides by ds/dt, and taking the magnitude of both sides gives

$$\left\| \frac{d\vec{\mathbf{T}}}{ds} \right\| = \left\| \frac{\vec{\mathbf{T}}'(t)}{\frac{ds}{dt}} \right\|.$$

Since $ds/dt = \|\vec{\mathbf{r}}\,'(t)\|$, this gives the formula for the curvature κ of a curve C in terms of any parameterization of C:

$$\kappa = \frac{\|\vec{\mathbf{T}}'(t)\|}{\|\vec{\mathbf{r}}\,'(t)\|}.$$

In the case of a three-dimensional curve, we start with the formulas $\vec{\mathbf{T}}(t) = (\vec{\mathbf{r}}\,'(t))/\|\vec{\mathbf{r}}\,'(t)\|$ and $ds/dt = \|\vec{\mathbf{r}}\,'(t)\|$. Therefore, $\vec{\mathbf{r}}\,'(t) = (ds/dt)\vec{\mathbf{T}}(t)$. We can take the derivative of this function using the scalar product formula:

$$\vec{\mathbf{r}}\,''(t) = \frac{d^2s}{dt^2}\vec{\mathbf{T}}(t) + \frac{ds}{dt}\vec{\mathbf{T}}'(t).$$

Using these last two equations we get

$$\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t) = \frac{ds}{dt}\vec{\mathbf{T}}(t) \times \left(\frac{d^2s}{dt^2}\vec{\mathbf{T}}(t) + \frac{ds}{dt}\vec{\mathbf{T}}'(t)\right)$$
$$= \frac{ds}{dt}\frac{d^2s}{dt^2}\vec{\mathbf{T}}(t) \times \vec{\mathbf{T}}(t) + (\frac{ds}{dt})^2\vec{\mathbf{T}}(t) \times \vec{\mathbf{T}}'(t).$$

Since $\vec{\mathbf{T}}(t) \times \vec{\mathbf{T}}(t) = 0$, this reduces to

$$\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t) = \left(\frac{ds}{dt}\right)^2 \vec{\mathbf{T}}(t) \times \vec{\mathbf{T}}'(t).$$

Since $\vec{\mathbf{T}}'$ is parallel to $\vec{\mathbf{N}}$, and $\vec{\mathbf{T}}$ is orthogonal to $\vec{\mathbf{N}}$, it follows that $\vec{\mathbf{T}}$ and $\vec{\mathbf{T}}'$ are orthogonal. This means that $\|\vec{\mathbf{T}} \times \vec{\mathbf{T}}'\| = \|\vec{\mathbf{T}}\|\|\vec{\mathbf{T}}'\| \sin(\pi/2) = \|\vec{\mathbf{T}}'\|$, so

$$\|\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t)\| = \left(\frac{ds}{dt}\right)^2 \|\vec{\mathbf{T}}'(t)\|.$$

Now we solve this equation for $\|\vec{\mathbf{T}}'(t)\|$ and use the fact that $ds/dt = \|\vec{\mathbf{r}}'(t)\|$:

$$\|\vec{\mathbf{T}}'(t)\| = \frac{\|\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t)\|}{\|\vec{\mathbf{r}}'(t)\|^2}.$$

Then, we divide both sides by $\|\vec{\mathbf{r}}'(t)\|$. This gives

$$\kappa = \frac{\|\vec{\mathbf{T}}'(t)\|}{\|\vec{\mathbf{r}}'(t)\|} = \frac{\|\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t)\|}{\|\vec{\mathbf{r}}'(t)\|^3}.$$

This proves 13.3.7. To prove 13.3.8, we start with the assumption that curve C is defined by the function $y = f(x)$. Then, we can define $\vec{\mathbf{r}}(t) = x\,\hat{\mathbf{i}} + f(x)\,\hat{\mathbf{j}} + 0\,\hat{\mathbf{k}}$. Using the previous formula for curvature:

$$\vec{\mathbf{r}}'(t) = \hat{\mathbf{i}} + f'(x)\,\hat{\mathbf{j}}$$
$$\vec{\mathbf{r}}''(t) = f''(x)\,\hat{\mathbf{j}}$$
$$\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & f'(x) & 0 \\ 0 & f''(x) & 0 \end{vmatrix} = f''(x)\,\hat{\mathbf{k}}.$$

Therefore,

$$\kappa = \frac{\|\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t)\|}{\|\vec{\mathbf{r}}'(t)\|^3} = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{3/2}}$$

Example 13.3.3: Finding Curvature

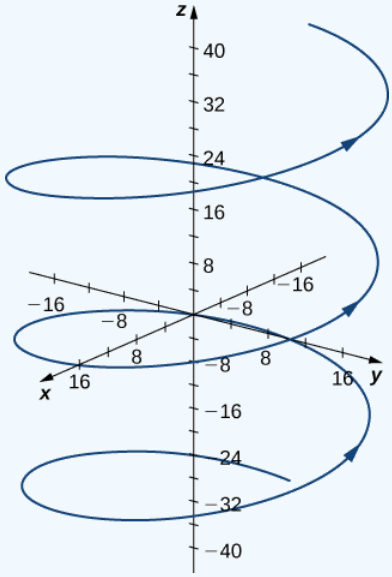
Find the curvature for each of the following curves at the given point:

a. $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}} + 3t\,\hat{\mathbf{k}}, \quad t = \dfrac{4\pi}{3}$

b. $\mathrm{f(x)} = \sqrt{4\mathrm{x} - \mathrm{x}^2}, \mathrm{x} = 2$

Solution

a. This function describes a helix.



The curvature of the helix at $t = (4\pi)/3$ can be found by using 13.3.6. First, calculate $\vec{\mathbf{T}}(t)$:

$$\begin{aligned}\vec{\mathbf{T}}(t) &= \frac{\vec{\mathbf{r}}\,'(t)}{\|\vec{\mathbf{r}}\,'(t)\|} \\ &= \frac{\langle -4\sin t, 4\cos t, 3\rangle}{\sqrt{(-4\sin t)^2 + (4\cos t)^2 + 3^2}} \\ &= \langle -\frac{4}{5}\sin t, \frac{4}{5}\cos t, \frac{3}{5}\rangle.\end{aligned}$$

Next, calculate $\vec{\mathbf{T}}\,'(t)$:

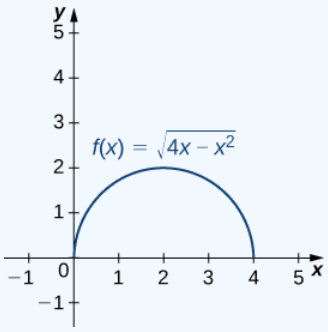
$$\vec{\mathbf{T}}\,'(t) = \langle -\frac{4}{5}\cos t, -\frac{4}{5}\sin t, 0\rangle.$$

Last, apply 13.3.6 :

$$\begin{aligned}\kappa = \frac{\|\vec{\mathbf{T}}\,'(t)\|}{\|\vec{\mathbf{r}}\,'(t)\|} &= \frac{\|\langle -\frac{4}{5}\cos t, -\frac{4}{5}\sin t, 0\rangle\|}{\|\langle -4\sin t, 4\cos t, 3\rangle\|} \\ &= \frac{\sqrt{(-\frac{4}{5}\cos t)^2 + (-\frac{4}{5}\sin t)^2 + 0^2}}{\sqrt{(-4\sin t)^2 + (4\cos t)^2 + 3^2}} \\ &= \frac{4/5}{5} = \frac{4}{25}.\end{aligned}$$

The curvature of this helix is constant at all points on the helix.

2. This function describes a semicircle.



To find the curvature of this graph, we must use 13.3.8. First, we calculate y' and y'':

$$\begin{aligned} y &= \sqrt{4x-x^2} = (4x-x^2)^{1/2} \\ y' &= \frac{1}{2}(4x-x^2)^{-1/2}(4-2x) = (2-x)(4x-x^2)^{-1/2} \\ y'' &= -(4x-x^2)^{-1/2} + (2-x)(-\frac{1}{2})(4x-x^2)^{-3/2}(4-2x) \\ &= -\frac{4x-x^2}{(4x-x^2)^{3/2}} - \frac{(2-x)^2}{(4x-x^2)^{3/2}} \\ &= \frac{x^2-4x-(4-4x+x^2)}{(4x-x^2)^{3/2}} \\ &= -\frac{4}{(4x-x^2)^{3/2}}. \end{aligned}$$

Then, we apply 13.3.8:

$$\begin{aligned} \kappa &= \frac{|y''|}{[1+(y')^2]^{3/2}} \\ &= \frac{\left|-\frac{4}{(4x-x^2)^{3/2}}\right|}{\left[1+((2-x)(4x-x^2)^{-1/2})^2\right]^{3/2}} = \frac{\left|\frac{4}{(4x-x^2)^{3/2}}\right|}{\left[1+\frac{(2-x)^2}{4x-x^2}\right]^{3/2}} \\ &= \frac{\left|\frac{4}{(4x-x^2)^{3/2}}\right|}{\left[\frac{4x-x^2+x^2-4x+4}{4x-x^2}\right]^{3/2}} = \left|\frac{4}{(4x-x^2)^{3/2}}\right| \cdot \frac{(4x-x^2)^{3/2}}{8} \\ &= \frac{1}{2}. \end{aligned}$$

The curvature of this circle is equal to the reciprocal of its radius. There is a minor issue with the absolute value in 13.3.8 ; however, a closer look at the calculation reveals that the denominator is positive for any value of x.

? Exercise 13.3.3

Find the curvature of the curve defined by the function

$$y = 3x^2 - 2x + 4$$

at the point $x = 2$.

Hint

Use 13.3.8.

Answer

$\kappa = \frac{6}{101^{3/2}} \approx 0.0059$

The Normal and Binormal Vectors

We have seen that the derivative $\vec{\mathbf{r}}\,'(t)$ of a vector-valued function is a tangent vector to the curve defined by $\vec{\mathbf{r}}(t)$, and the unit tangent vector $\vec{\mathbf{T}}(t)$ can be calculated by dividing $\vec{\mathbf{r}}\,'(t)$ by its magnitude. When studying motion in three dimensions, two other vectors are useful in describing the motion of a particle along a path in space: the principal unit normal vector and the **binormal vector**.

Definition: Binormal Vectors

Let C be a three-dimensional **smooth** curve represented by $\vec{\mathbf{r}}$ over an open interval I. If $\vec{\mathbf{T}}'(t) \neq \vec{\mathbf{0}}$, then the principal unit normal vector at t is defined to be

$$\vec{\mathbf{N}}(t) = \frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|}. \tag{13.3.9}$$

The binormal vector at t is defined as

$$\vec{\mathbf{B}}(t) = \vec{\mathbf{T}}(t) \times \vec{\mathbf{N}}(t), \tag{13.3.10}$$

where $\vec{\mathbf{T}}(t)$ is the unit tangent vector.

Note that, by definition, the binormal vector is orthogonal to both the unit tangent vector and the normal vector. Furthermore, $\vec{\mathbf{B}}(t)$ is always a unit vector. This can be shown using the formula for the magnitude of a cross product.

$$\|\vec{\mathbf{B}}(t)\| = \|\vec{\mathbf{T}}(t) \times \vec{\mathbf{N}}(t)\| = \|\vec{\mathbf{T}}(t)\|\|\vec{\mathbf{N}}(t)\| \sin\theta,$$

where θ is the angle between $\vec{\mathbf{T}}(t)$ and $\vec{\mathbf{N}}(t)$. Since $\vec{\mathbf{N}}(t)$ is the derivative of a unit vector, property (vii) of the derivative of a vector-valued function tells us that $\vec{\mathbf{T}}(t)$ and $\vec{\mathbf{N}}(t)$ are orthogonal to each other, so $\theta = \pi/2$. Furthermore, they are both unit vectors, so their magnitude is 1. Therefore, $\|\vec{\mathbf{T}}(t)\|\|\vec{\mathbf{N}}(t)\| \sin\theta = (1)(1)\sin(\pi/2) = 1$ and $\vec{\mathbf{B}}(t)$ is a unit vector.

The principal unit normal vector can be challenging to calculate because the unit tangent vector involves a quotient, and this quotient often has a square root in the denominator. In the three-dimensional case, finding the cross product of the unit tangent vector and the unit normal vector can be even more cumbersome. Fortunately, we have alternative formulas for finding these two vectors, and they are presented in Motion in Space.

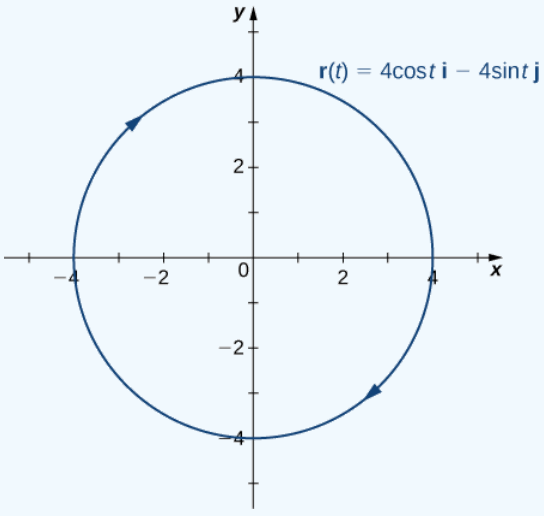
Example 13.3.4: Finding the Principal Unit Normal Vector and Binormal Vector

For each of the following vector-valued functions, find the principal unit normal vector. Then, if possible, find the binormal vector.

1. $\vec{\mathbf{r}}(t) = 4\cos t\,\hat{\mathbf{i}} - 4\sin t\,\hat{\mathbf{j}}$
2. $\vec{\mathbf{r}}(t) = (6t+2)\,\hat{\mathbf{i}} + 5t^2\,\hat{\mathbf{j}} - 8t\,\hat{\mathbf{k}}$

Solution

1. This function describes a circle.



To find the principal unit normal vector, we first must find the unit tangent vector $\vec{\mathbf{T}}(t)$:

$$\begin{aligned}\vec{\mathbf{T}}(t) &= \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|} \\ &= \frac{-4\sin t\,\hat{\mathbf{i}} - 4\cos t\,\hat{\mathbf{j}}}{\sqrt{(-4\sin t)^2 + (-4\cos t)^2}} \\ &= \frac{-4\sin t\,\hat{\mathbf{i}} - 4\cos t\,\hat{\mathbf{j}}}{\sqrt{16\sin^2 t + 16\cos^2 t}} \\ &= \frac{-4\sin t\,\hat{\mathbf{i}} - 4\cos t\,\hat{\mathbf{j}}}{\sqrt{16(\sin^2 t + \cos^2 t)}} \\ &= \frac{-4\sin t\,\hat{\mathbf{i}} - 4\cos t\,\hat{\mathbf{j}}}{4} \\ &= -\sin t\,\hat{\mathbf{i}} - \cos t\,\hat{\mathbf{j}}.\end{aligned}$$

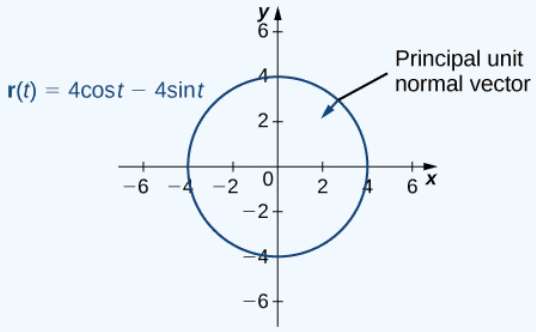
Next, we use 13.3.9 :

$$\begin{aligned}\vec{\mathbf{N}}(t) &= \frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|} \\ &= \frac{-\cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}}}{\sqrt{(-\cos t)^2 + (\sin t)^2}} \\ &= \frac{-\cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}}}{\sqrt{\cos^2 t + \sin^2 t}} \\ &= -\cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}}.\end{aligned}$$

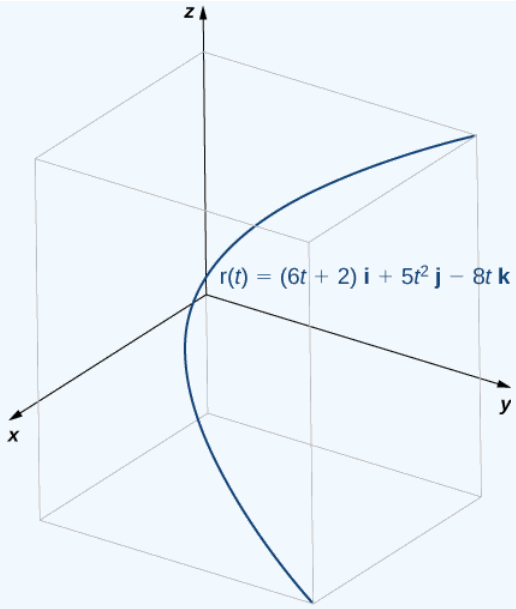
Notice that the unit tangent vector and the principal unit normal vector are orthogonal to each other for all values of t:

$$\begin{aligned}\vec{\mathbf{T}}(t) \cdot \vec{\mathbf{N}}(t) &= \langle -\sin t, -\cos t\rangle \cdot \langle -\cos t, \sin t\rangle \\ &= \sin t\cos t - \cos t\sin t \\ &= 0.\end{aligned}$$

Furthermore, the principal unit normal vector points toward the center of the circle from every point on the circle. Since $\vec{\mathbf{r}}(t)$ defines a curve in two dimensions, we cannot calculate the binormal vector.



2. This function looks like this:



To find the principal unit normal vector, we first find the unit tangent vector $\vec{\mathbf{T}}(t)$:

$$\begin{aligned}\vec{\mathbf{T}}(t) &= \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|} \\ &= \frac{6\,\hat{\mathbf{i}} + 10t\,\hat{\mathbf{j}} - 8\,\hat{\mathbf{k}}}{\sqrt{6^2 + (10t)^2 + (-8)^2}} \\ &= \frac{6\,\hat{\mathbf{i}} + 10t\,\hat{\mathbf{j}} - 8\,\hat{\mathbf{k}}}{\sqrt{36 + 100t^2 + 64}} \\ &= \frac{6\,\hat{\mathbf{i}} + 10t\,\hat{\mathbf{j}} - 8\,\hat{\mathbf{k}}}{\sqrt{100(t^2+1)}} \\ &= \frac{3\,\hat{\mathbf{i}} - 5t\,\hat{\mathbf{j}} - 4\,\hat{\mathbf{k}}}{5\sqrt{t^2+1}} \\ &= \frac{3}{5}(t^2+1)^{-1/2}\,\hat{\mathbf{i}} - t(t^2+1)^{-1/2}\,\hat{\mathbf{j}} - \frac{4}{5}(t^2+1)^{-1/2}\,\hat{\mathbf{k}}.\end{aligned}$$

Next, we calculate $\vec{\mathbf{T}}'(t)$ and $\|\vec{\mathbf{T}}'(t)\|$:

$$\begin{aligned}\vec{\mathbf{T}}'(t) &= \frac{3}{5}(-\frac{1}{2})(t^2+1)^{-3/2}(2t)\,\hat{\mathbf{i}} - ((t^2+1)^{-1/2} - t(\frac{1}{2})(t^2+1)^{-3/2}(2t))\,\hat{\mathbf{j}} \\ &\quad - \frac{4}{5}(-\frac{1}{2})(t^2+1)^{-3/2}(2t)\,\hat{\mathbf{k}} \\ &= -\frac{3t}{5(t^2+1)^{3/2}}\,\hat{\mathbf{i}} - \frac{1}{(t^2+1)^{3/2}}\,\hat{\mathbf{j}} + \frac{4t}{5(t^2+1)^{3/2}}\,\hat{\mathbf{k}} \\ \|\vec{\mathbf{T}}'(t)\| &= \sqrt{\Big(-\frac{3t}{5(t^2+1)^{3/2}}\Big)^2 + \Big(-\frac{1}{(t^2+1)^{3/2}}\Big)^2 + \Big(\frac{4t}{5(t^2+1)^{3/2}}\Big)^2} \\ &= \sqrt{\frac{9t^2}{25(t^2+1)^3} + \frac{1}{(t^2+1)^3} + \frac{16t^2}{25(t^2+1)^3}} \\ &= \sqrt{\frac{25t^2+25}{25(t^2+1)^3}} \\ &= \sqrt{\frac{1}{(t^2+1)^2}} \\ &= \frac{1}{t^2+1}.\end{aligned}$$

Therefore, according to 13.3.9 :

$$\begin{aligned}\vec{\mathbf{N}}(t) &= \frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|} \\ &= \left(-\frac{3t}{5(t^2+1)^{3/2}}\,\hat{\mathbf{i}} - \frac{1}{(t^2+1)^{3/2}}\,\hat{\mathbf{j}} + \frac{4t}{5(t^2+1)^{3/2}}\,\hat{\mathbf{k}}\right)(t^2+1) \\ &= -\frac{3t}{5(t^2+1)^{1/2}}\,\hat{\mathbf{i}} - \frac{5}{5(t^2+1)^{1/2}}\,\hat{\mathbf{j}} + \frac{4t}{5(t^2+1)^{1/2}}\,\hat{\mathbf{k}} \\ &= -\frac{3t\,\hat{\mathbf{i}} + 5\,\hat{\mathbf{j}} - 4t\,\hat{\mathbf{k}}}{5\sqrt{t^2+1}}.\end{aligned}$$

Once again, the unit tangent vector and the principal unit normal vector are orthogonal to each other for all values of t:

$$\begin{aligned}\vec{\mathbf{T}}(t)\cdot\vec{\mathbf{N}}(t) &= \left(\frac{3\,\hat{\mathbf{i}} - 5t\,\hat{\mathbf{j}} - 4\,\hat{\mathbf{k}}}{5\sqrt{t^2+1}}\right)\cdot\left(-\frac{3t\,\hat{\mathbf{i}} + 5\,\hat{\mathbf{j}} - 4t\,\hat{\mathbf{k}}}{5\sqrt{t^2+1}}\right) \\ &= \frac{3(-3t) - 5t(-5) - 4(4t)}{25(t^2+1)} \\ &= \frac{-9t + 25t - 16t}{25(t^2+1)} \\ &= 0.\end{aligned}$$

Last, since $\vec{\mathbf{r}}(t)$ represents a three-dimensional curve, we can calculate the binormal vector using 13.3.10 :

$$\begin{aligned}\vec{\mathbf{B}}(t) &= \vec{\mathbf{T}}(t)\times\vec{\mathbf{N}}(t) \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \dfrac{3}{5\sqrt{t^2+1}} & -\dfrac{5t}{5\sqrt{t^2+1}} & -\dfrac{4}{5\sqrt{t^2+1}} \\ -\dfrac{3t}{5\sqrt{t^2+1}} & -\dfrac{5}{5\sqrt{t^2+1}} & \dfrac{4t}{5\sqrt{t^2+1}} \end{vmatrix} \\ &= \left(\left(-\frac{5t}{5\sqrt{t^2+1}}\right)\left(\frac{4t}{5\sqrt{t^2+1}}\right) - \left(-\frac{4}{5\sqrt{t^2+1}}\right)\left(-\frac{5}{5\sqrt{t^2+1}}\right)\right)\hat{\mathbf{i}} \\ &- \left(\left(\frac{3}{5\sqrt{t^2+1}}\right)\left(\frac{4t}{5\sqrt{t^2+1}}\right) - \left(-\frac{4}{5\sqrt{t^2+1}}\right)\left(-\frac{3t}{5\sqrt{t^2+1}}\right)\right)\hat{\mathbf{j}} \\ &+ \left(\left(\frac{3}{5\sqrt{t^2+1}}\right)\left(-\frac{5}{5\sqrt{t^2+1}}\right) - \left(-\frac{5t}{5\sqrt{t^2+1}}\right)\left(-\frac{3t}{5\sqrt{t^2+1}}\right)\right)\hat{\mathbf{k}} \\ &= \left(\frac{-20t^2-20}{25(t^2+1)}\right)\hat{\mathbf{i}} + \left(\frac{-15-15t^2}{25(t^2+1)}\right)\hat{\mathbf{k}} \\ &= -20\left(\frac{t^2+1}{25(t^2+1)}\right)\hat{\mathbf{i}} - 15\left(\frac{t^2+1}{25(t^2+1)}\right)\hat{\mathbf{k}} \\ &= -\frac{4}{5}\,\hat{\mathbf{i}} - \frac{3}{5}\,\hat{\mathbf{k}}.\end{aligned}$$

Exercise 13.3.4

Find the unit normal vector for the vector-valued function $\vec{\mathbf{r}}(t) = (t^2-3t)\,\hat{\mathbf{i}} + (4t+1)\,\hat{\mathbf{j}}$ and evaluate it at $t=2$.

Hint

First, find $\vec{\mathbf{T}}(t)$, then use 13.3.9.

Answer

$\vec{\mathbf{N}}(2) = \dfrac{\sqrt{2}}{2}(\hat{\mathbf{i}} - \hat{\mathbf{j}})$

For any smooth curve in three dimensions that is defined by a vector-valued function, we now have formulas for the unit tangent vector $\vec{\mathbf{T}}$, the unit normal vector $\vec{\mathbf{N}}$, and the binormal vector $\vec{\mathbf{B}}$. The unit normal vector and the binormal vector form a plane that is perpendicular to the curve at any point on the curve, called the normal plane. In addition, these three vectors form a frame of reference in three-dimensional space called the *Frenet frame of reference* (also called the **TNB** frame) (Figure 13.3.2). Last, the plane determined by the vectors $\vec{\mathbf{T}}$ and $\vec{\mathbf{N}}$ forms the osculating plane of C at any point P on the curve.

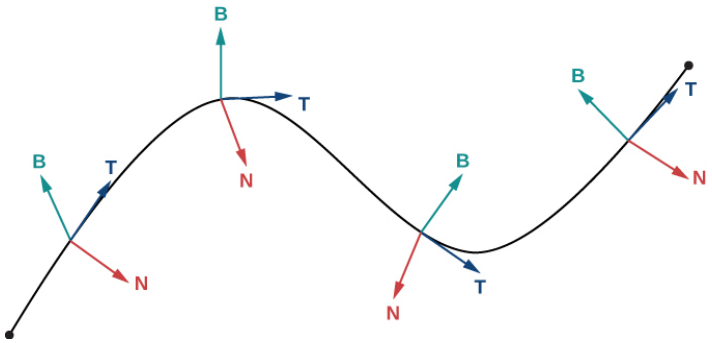


Figure 13.3.2: This figure depicts a Frenet frame of reference. At every point P *on a three-dimensional curve, the unit tangent, unit normal, and binormal vectors form a three-dimensional frame of reference.*

Suppose we form a circle in the osculating plane of C at point P on the curve. Assume that the circle has the same curvature as the curve does at point P and let the circle have radius r. Then, the curvature of the circle is given by $\frac{1}{r}$. We call r the radius of curvature of the curve, and it is equal to the reciprocal of the curvature. If this circle lies on the concave side of the curve and is tangent to the curve at point P, then this circle is called the *osculating circle* of C at P, as shown in Figure 13.3.3.

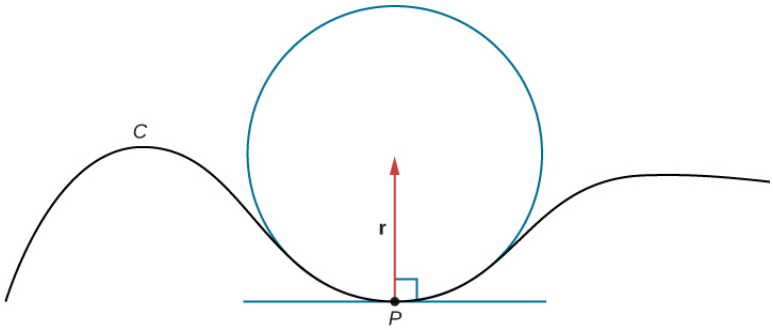


Figure 13.3.3: In this osculating circle, the circle is tangent to curve C at point P *and shares the same curvature.*

For more information on osculating circles, see this demonstration on curvature and torsion, this article on osculating circles, and this discussion of Serret formulas.

To find the equation of an osculating circle in two dimensions, we need find only the center and radius of the circle.

Example 13.3.5: Finding the Equation of an Osculating Circle

Find the equation of the osculating circle of the curve defined by the function $y = x^3 - 3x + 1$ at $x = 1$.

Solution

Figure 13.3.4 shows the graph of $y = x^3 - 3x + 1$.

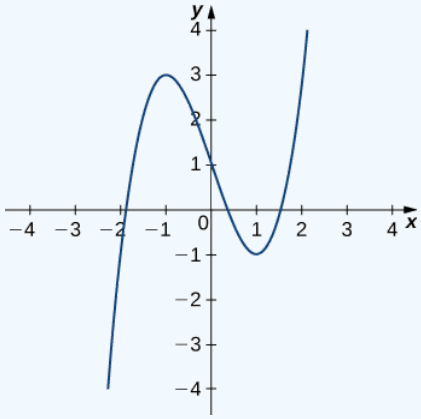


Figure 13.3.4: We want to find the osculating circle of this graph at the point where $x = 1$.

First, let's calculate the curvature at $x = 1$:

$$\kappa = \frac{|f''(x)|}{\left(1 + [f'(x)]^2\right)^{3/2}} = \frac{|6x|}{(1 + [3x^2 - 3]^2)^{3/2}}.$$

This gives $\kappa = 6$. Therefore, the radius of the osculating circle is given by $R = \frac{1}{\kappa} = \frac{1}{6}$. Next, we then calculate the coordinates of the center of the circle. When $x = 1$, the slope of the tangent line is zero. Therefore, the center of the osculating circle is directly above the point on the graph with coordinates $(1, -1)$. The center is located at $(1, -\frac{5}{6})$. The formula for a circle with radius r and center (h, k) is given by $(x - h)^2 + (y - k)^2 = r^2$. Therefore, the equation of the osculating circle is $(x - 1)^2 + (y + \frac{5}{6})^2 = \frac{1}{36}$. The graph and its osculating circle appears in the following graph.

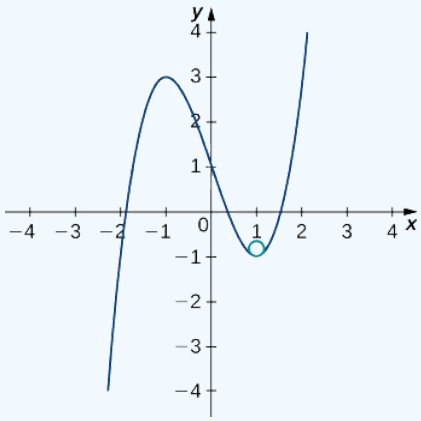


Figure 13.3.5: The osculating circle has radius $R = \frac{1}{6}$.

? Exercise 13.3.5

Find the equation of the osculating circle of the curve defined by the vector-valued function $y = 2x^2 - 4x + 5$ at $x = 1$.

Hint

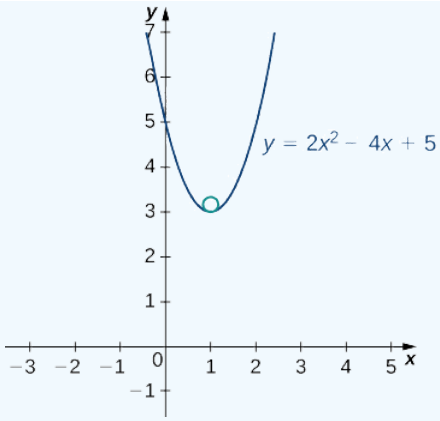
Use 13.3.8 to find the curvature of the graph, then draw a graph of the function around $x = 1$ to help visualize the circle in relation to the graph.

Answer

$\kappa = \frac{4}{[1+(4x-4)^2]^{3/2}}$

At the point $x = 1$, the curvature is equal to 4. Therefore, the radius of the osculating circle is $\frac{1}{4}$.

A graph of this function appears next:



The vertex of this parabola is located at the point $(1, 3)$. Furthermore, the center of the osculating circle is directly above the vertex. Therefore, the coordinates of the center are $(1, \frac{13}{4})$. The equation of the osculating circle is $(x-1)^2 + (y-\frac{13}{4})^2 = \frac{1}{16}$.

Key Concepts

- The arc-length function for a vector-valued function is calculated using the integral formula $s(t) = \int_a^b \|\vec{\mathbf{r}}'(t)\|\, dt$. This formula is valid in both two and three dimensions.
- The curvature of a curve at a point in either two or three dimensions is defined to be the curvature of the inscribed circle at that point. The arc-length parameterization is used in the definition of curvature.
- There are several different formulas for curvature. The curvature of a circle is equal to the reciprocal of its radius.
- The principal unit normal vector at t is defined to be

$$\vec{\mathbf{N}}(t) = \frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|}.$$

- The binormal vector at t is defined as $\vec{\mathbf{B}}(t) = \vec{\mathbf{T}}(t) \times \vec{\mathbf{N}}(t)$, where $\vec{\mathbf{T}}(t)$ is the unit tangent vector.
- The Frenet frame of reference is formed by the unit tangent vector, the principal unit normal vector, and the binormal vector.
- The osculating circle is tangent to a curve at a point and has the same curvature as the tangent curve at that point.

Key Equations

- **Arc length of space curve**
  $s = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}\, dt = \int_a^b \|\vec{\mathbf{r}}'(t)\|\, dt$
- **Arc-length function**
  $s(t) = \int_a^t \sqrt{f'(u))^2 + (g'(u))^2 + (h'(u))^2}\, du \text{ or } s(t) = \int_a^t \|\vec{\mathbf{r}}'(u)\|\, du$
- $\kappa = \frac{\|\vec{\mathbf{T}}'(t)\|}{\|\vec{\mathbf{r}}'(t)\|}$ or $\kappa = \frac{\|\vec{\mathbf{r}}'(t) \times \vec{\mathbf{r}}''(t)\|}{\|\vec{\mathbf{r}}'(t)\|^3}$ or $\kappa = \frac{|y''|}{[1+(y')^2]^{3/2}}$
- **Principal unit normal vector**
  $\vec{\mathbf{N}}(t) = \frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|}$
- **Binormal vector**
  $\vec{\mathbf{B}}(t) = \vec{\mathbf{T}}(t) \times \vec{\mathbf{N}}(t)$

Glossary

arc-length function

a function $s(t)$ that describes the arc length of curve C as a function of t

arc-length parameterization

a reparameterization of a vector-valued function in which the parameter is equal to the arc length

binormal vector

a unit vector orthogonal to the unit tangent vector and the unit normal vector

curvature

the derivative of the unit tangent vector with respect to the arc-length parameter

Frenet frame of reference

(TNB frame) a frame of reference in three-dimensional space formed by the unit tangent vector, the unit normal vector, and the binormal vector

normal plane

a plane that is perpendicular to a curve at any point on the curve

osculating circle

a circle that is tangent to a curve C at a point P and that shares the same curvature

osculating plane

the plane determined by the unit tangent and the unit normal vector

principal unit normal vector

a vector orthogonal to the unit tangent vector, given by the formula $\frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|}$

radius of curvature

the reciprocal of the curvature

smooth

curves where the vector-valued function $\vec{\mathbf{r}}(t)$ is differentiable with a non-zero derivative

---

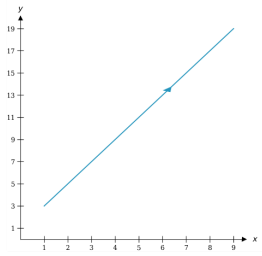
This page titled 13.3: Arc Length and Curvature is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

13.3E: Exercises for Section 13.3

Determining Arc Length

In questions 1 - 5, find the arc length of the curve on the given interval.

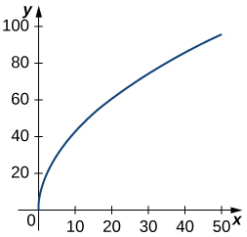
1) $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + (2t^2+1)\,\hat{\mathbf{j}}, \quad 1 \le t \le 3$. This portion of the graph is shown here:



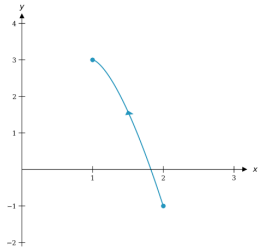
Answer

$8\sqrt{5}$ units ≈ 17.9 units

2) $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + 14t\,\hat{\mathbf{j}}, \quad 0 \le t \le 7$. This portion of the graph is shown here:



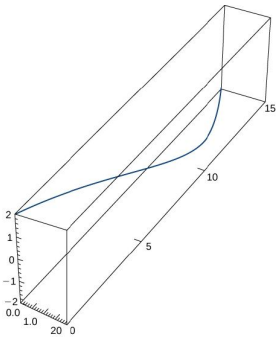
3) $\vec{\mathbf{r}}(t) = \langle t^2+1, 4t^3+3\rangle, \quad -1 \le t \le 0$. This portion of the graph is shown here:



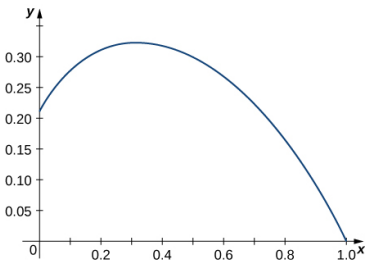
Answer

$\frac{1}{54}(37^{3/2}-1)$ units ≈ 4.15 units

4) $\vec{\mathbf{r}}(t) = \langle 2\sin t, 5t, 2\cos t\rangle, \quad 0 \le t \le \pi$. This portion of the graph is shown here:



5) $\vec{\mathbf{r}}(t) = \langle e^{-t\cos t}, e^{-t\sin t} \rangle$ over the interval $[0, \frac{\pi}{2}]$. Here is the portion of the graph on the indicated interval:



6) Set up an integral to represent the arc length from $t = 0$ to $t = 2$ along the curve traced out by $\vec{\mathbf{r}}(t) = \langle t, t^4 \rangle$. Then use technology to approximate this length to the nearest thousandth of a unit.

7) Find the length of one turn of the helix given by $\vec{\mathbf{r}}(t) = \frac{1}{2}\cos t\,\hat{\mathbf{i}} + \frac{1}{2}\sin t\,\hat{\mathbf{j}} + \frac{\sqrt{3}}{2}\,t\,\hat{\mathbf{k}}$.

Answer

Length $= 2\pi$ units

8) Find the arc length of the vector-valued function $\vec{\mathbf{r}}(t) = -t\,\hat{\mathbf{i}} + 4t\,\hat{\mathbf{j}} + 3t\,\hat{\mathbf{k}}$ over $[0, 1]$.

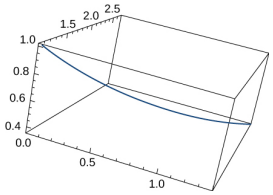
9) A particle travels in a circle with the equation of motion $\vec{\mathbf{r}}(t) = 3\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}} + 0\,\hat{\mathbf{k}}$. Find the distance traveled around the circle by the particle.

Answer

6π units

10) Set up an integral to find the circumference of the ellipse with the equation $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + 2\sin t\,\hat{\mathbf{j}} + 0\,\hat{\mathbf{k}}$.

11) Find the length of the curve $\vec{\mathbf{r}}(t) = \langle \sqrt{2}t, e^t, e^{-t} \rangle$ over the interval $0 \le t \le 1$. The graph is shown here:



Answer

$\left(e - \frac{1}{e}\right)$ units ≈ 2.35 units

12) Find the length of the curve $\vec{\mathbf{r}}(t) = \langle 2\sin t,\ 5t,\ 2\cos t \rangle$ for $t \in [-10, 10]$.

Unit Tangent Vectors and Unit Normal Vectors

13) The position function for a particle is $\vec{\mathbf{r}}(t) = a\cos(\omega t)\,\hat{\mathbf{i}} + b\sin(\omega t)\,\hat{\mathbf{j}}$. Find the unit tangent vector and the unit normal vector at $t = 0$.

Solution:

$$\vec{\mathbf{r}}'(t) = -a\omega\sin(\omega t)\,\hat{\mathbf{i}} + b\omega\cos(\omega t)\,\hat{\mathbf{j}}$$

$$\|\vec{\mathbf{r}}'(t)\| = \sqrt{a^2\omega^2\sin^2(\omega t) + b^2\omega^2\cos^2(\omega t)}$$

$$\vec{\mathbf{T}}(t) = \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|} = \frac{-a\omega\sin(\omega t)\,\hat{\mathbf{i}} + b\omega\cos(\omega t)\,\hat{\mathbf{j}}}{\sqrt{a^2\omega^2\sin^2(\omega t) + b^2\omega^2\cos^2(\omega t)}}$$

$$\vec{\mathbf{T}}(0) = \frac{b\omega\,\hat{\mathbf{j}}}{\sqrt{(b\omega)^2}} = \frac{b\omega\,\hat{\mathbf{j}}}{|b\omega|}$$

If $b\omega > 0,\ \vec{\mathbf{T}}(0) = \hat{\mathbf{j}}$, and if $b\omega < 0,\ T(0) = -\hat{\mathbf{j}}$

Answer

If $b\omega > 0,\ \vec{\mathbf{T}}(0) = \hat{\mathbf{j}}$, and if $b\omega < 0,\ \vec{\mathbf{T}}(0) = -\hat{\mathbf{j}}$

If $a > 0,\ \vec{\mathbf{N}}(0) = -\hat{\mathbf{i}}$, and if $a < 0,\ \vec{\mathbf{N}}(0) = \hat{\mathbf{i}}$

14) Given $\vec{\mathbf{r}}(t) = a\cos(\omega t)\,\hat{\mathbf{i}} + b\sin(\omega t)\,\hat{\mathbf{j}}$, find the binormal vector $\vec{\mathbf{B}}(0)$.

15) Given $\vec{\mathbf{r}}(t) = \langle 2e^t, e^t\cos t, e^t\sin t\rangle$, determine the unit tangent vector $\vec{\mathbf{T}}(t)$.

Answer

$$\vec{\mathbf{T}}(t) = \left\langle \frac{2}{\sqrt{6}}, \frac{\cos t - \sin t}{\sqrt{6}}, \frac{\cos t + \sin t}{\sqrt{6}} \right\rangle$$

$$= \left\langle \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}(\cos t - \sin t), \frac{\sqrt{6}}{6}(\cos t + \sin t) \right\rangle$$

16) Given $\vec{\mathbf{r}}(t) = \langle 2e^t,\ e^t\cos t,\ e^t\sin t\rangle$, find the unit tangent vector $\vec{\mathbf{T}}(t)$ evaluated at $t = 0$, $\vec{\mathbf{T}}(0)$.

17) Given $\vec{\mathbf{r}}(t) = \langle 2e^t,\ e^t\cos t,\ e^t\sin t\rangle$, determine the unit normal vector $\vec{\mathbf{N}}(t)$.

Answer

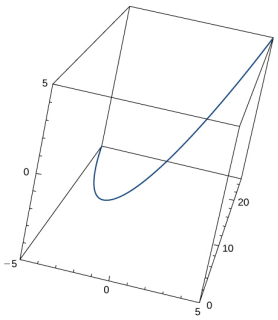
$$\vec{\mathbf{N}}(t) = \langle 0,\ -\tfrac{\sqrt{2}}{2}(\sin t + \cos t),\ \tfrac{\sqrt{2}}{2}(\cos t - \sin t)\rangle$$

18) Given $\vec{\mathbf{r}}(t) = \langle 2e^t,\ e^t\cos t,\ e^t\sin t\rangle$, find the unit normal vector $\vec{\mathbf{N}}(t)$ evaluated at $t = 0$, $\vec{\mathbf{N}}(0)$.

Answer

$$\vec{\mathbf{N}}(0) = \langle 0,\ -\tfrac{\sqrt{2}}{2},\ \tfrac{\sqrt{2}}{2}\rangle$$

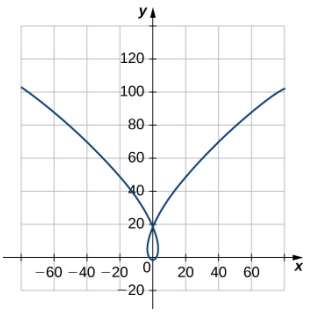
19) Given $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$, find the unit tangent vector $\vec{\mathbf{T}}(t)$. The graph is shown here:



Answer

$$\vec{\mathbf{T}}(t) = \frac{1}{\sqrt{4t^2+2}} < 1, 2t, 1 >$$

20) Find the unit tangent vector $\vec{\mathbf{T}}(t)$ and unit normal vector $\vec{\mathbf{N}}(t)$ at $t = 0$ for the plane curve $\vec{\mathbf{r}}(t) = \langle t^3 - 4t, 5t^2 - 2\rangle$. The graph is shown here:



21) Find the unit tangent vector $\vec{\mathbf{T}}(t)$ for $\vec{\mathbf{r}}(t) = 3t\,\hat{\mathbf{i}} + 5t^2\,\hat{\mathbf{j}} + 2t\,\hat{\mathbf{k}}$.

Answer

$$\vec{\mathbf{T}}(t) = \frac{1}{\sqrt{100t^2 + 13}}(3\,\hat{\mathbf{i}} + 10t\,\hat{\mathbf{j}} + 2\,\hat{\mathbf{k}})$$

22) Find the principal normal vector to the curve $\vec{\mathbf{r}}(t) = \langle 6\cos t, 6\sin t\rangle$ at the point determined by $t = \frac{\pi}{3}$.

23) Find $\vec{\mathbf{T}}(t)$ for the curve $\vec{\mathbf{r}}(t) = (t^3 - 4t)\,\hat{\mathbf{i}} + (5t^2 - 2)\,\hat{\mathbf{j}}$.

Answer

$$\vec{\mathbf{T}}(t) = \frac{1}{\sqrt{9t^4 + 76t^2 + 16}}\left((3t^2 - 4)\,\hat{\mathbf{i}} + 10t\,\hat{\mathbf{j}}\right)$$

24) Find $\vec{\mathbf{N}}(t)$ for the curve $\vec{\mathbf{r}}(t) = (t^3 - 4t)\,\hat{\mathbf{i}} + (5t^2 - 2)\,\hat{\mathbf{j}}$.

25) Find the unit tangent vector $\vec{\mathbf{T}}(t)$ for $\vec{\mathbf{r}}(t) = \langle 2\sin t,\ 5t,\ 2\cos t\rangle$.

Answer

$$\vec{\mathbf{T}}(t) = \langle \tfrac{2\sqrt{29}}{29}\cos t,\ \tfrac{5\sqrt{29}}{29},\ -\tfrac{2\sqrt{29}}{29}\sin t\rangle$$

26) Find the unit normal vector $\vec{\mathbf{N}}(t)$ for $\vec{\mathbf{r}}(t) = \langle 2\sin t,\ 5t,\ 2\cos t\rangle$.

Answer

$$\vec{\mathbf{N}}(t) = \langle -\sin t,\ 0,\ -\cos t\rangle$$

Arc Length Parameterizations

27) Find the arc-length function $\vec{\mathbf{s}}(t)$ for the line segment given by $\vec{\mathbf{r}}(t) = \langle 3 - 3t,\ 4t\rangle$. Then write the arc-length parameterization of r with s as the parameter.

Answer

Arc-length function: $s(t) = 5t$; The arc-length parameterization of $\vec{\mathbf{r}}(t)$: $\vec{\mathbf{r}}(s) = \left(3 - \frac{3s}{5}\right)\,\hat{\mathbf{i}} + \frac{4s}{5}\,\hat{\mathbf{j}}$

28) Parameterize the helix $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$ using the arc-length parameter s, from $t = 0$.

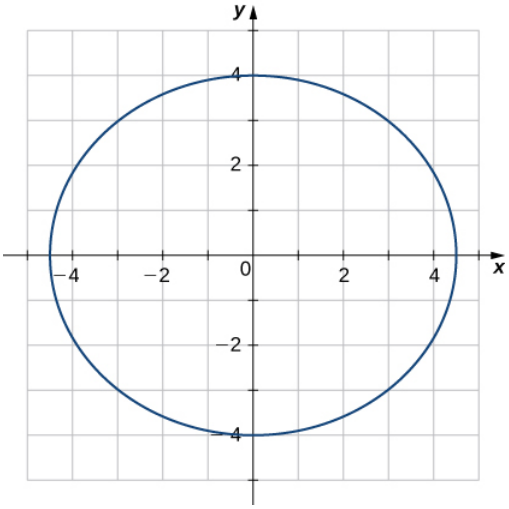
29) Parameterize the curve using the arc-length parameter s, at the point at which $t = 0$ for $\vec{\mathbf{r}}(t) = e^t \sin t\,\hat{\mathbf{i}} + e^t \cos t\,\hat{\mathbf{j}}$

Answer

$$\vec{\mathbf{r}}(s) = \left(1 + \frac{s}{\sqrt{2}}\right)\sin\left(\ln\left(1 + \frac{s}{\sqrt{2}}\right)\right)\,\hat{\mathbf{i}} + \left(1 + \frac{s}{\sqrt{2}}\right)\cos\left(\ln\left(1 + \frac{s}{\sqrt{2}}\right)\right)\,\hat{\mathbf{j}}$$

Curvature and the Osculating Circle

30) Find the curvature of the curve $\vec{\mathbf{r}}(t) = 5\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}}$ at $t = \pi/3$. (*Note:* The graph is an ellipse.)



31) Find the x-coordinate at which the curvature of the curve $y = 1/x$ is a maximum value.

Answer

The maximum value of the curvature occurs at $x = 1$.

32) Find the curvature of the curve $\vec{\mathbf{r}}(t) = 5\cos t\,\hat{\mathbf{i}} + 5\sin t\,\hat{\mathbf{j}}$. Does the curvature depend upon the parameter t?

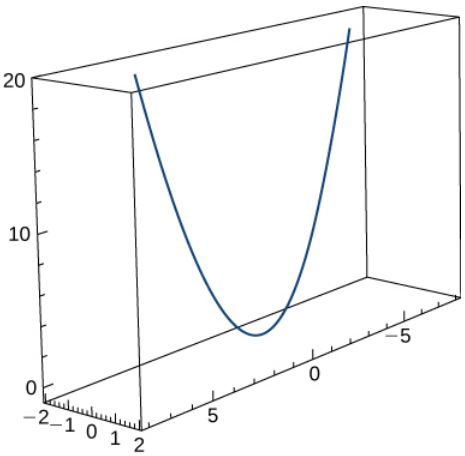
33) Find the curvature κ for the curve $y = x - \frac{1}{4}x^2$ at the point $x = 2$.

Answer

$\frac{1}{2}$

34) Find the curvature κ for the curve $y = \frac{1}{3}x^3$ at the point $x = 1$.

35) Find the curvature κ of the curve $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + 6t^2\,\hat{\mathbf{j}} + 4t\,\hat{\mathbf{k}}$. The graph is shown here:



Answer

$$\kappa \approx \frac{49.477}{(17 + 144t^2)^{3/2}}$$

36) Find the curvature of $\vec{\mathbf{r}}(t) = \langle 2\sin t, 5t, 2\cos t\rangle$.

37) Find the curvature of $\vec{\mathbf{r}}(t) = \sqrt{2}t\,\hat{\mathbf{i}} + e^t\,\hat{\mathbf{j}} + e^{-t}\,\hat{\mathbf{k}}$ at point $P(0, 1, 1)$.

Answer

$\frac{1}{2\sqrt{2}}$

38) At what point does the curve $y = e^x$ have maximum curvature?

39) What happens to the curvature as $x \to \infty$ for the curve $y = e^x$?

Answer

The curvature approaches zero.

40) Find the point of maximum curvature on the curve $y = \ln x$.

41) Find the equations of the normal plane and the osculating plane of the curve $\vec{\mathbf{r}}(t) = \langle 2\sin(3t), t, 2\cos(3t)\rangle$ at point $(0, \pi, -2)$.

Answer

$y = 6x + \pi$ and $x + 6y = 6\pi$

42) Find equations of the osculating circles of the ellipse $4y^2 + 9x^2 = 36$ at the points $(2, 0)$ and $(0, 3)$.

43) Find the equation for the osculating plane at point $t = \pi/4$ on the curve $\vec{\mathbf{r}}(t) = \cos(2t)\,\hat{\mathbf{i}} + \sin(2t)\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$.

Answer

$x + 2z = \frac{\pi}{2}$

44) Find the radius of curvature of $6y = x^3$ at the point $(2, \frac{4}{3})$.

45) Find the curvature at each point (x, y) on the hyperbola $\vec{\mathbf{r}}(t) = \langle a\cosh(t), b\sinh(t)\rangle$.

Answer

$$\frac{a^4b^4}{(b^4x^2 + a^4y^2)^{3/2}}$$

46) Calculate the curvature of the circular helix $\vec{\mathbf{r}}(t) = r\sin(t)\,\hat{\mathbf{i}} + r\cos(t)\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$.

47) Find the radius of curvature of $y = \ln(x + 1)$ at point $(2, \ln 3)$.

Answer

$\frac{10\sqrt{10}}{3}$

48) Find the radius of curvature of the hyperbola $xy = 1$ at point $(1, 1)$.

A particle moves along the plane curve C described by $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}}$. Use this parameterization to answer questions 49 - 51.

49) Find the length of the curve over the interval $[0, 2]$.

Answer

$\frac{1}{4}\left[4\sqrt{17} + \ln\left(4 + \sqrt{17}\right)\right]$ units ≈ 4.64678 units

50) Find the curvature of the plane curve at $t = 0, 1, 2$.

51) Describe the curvature as t increases from $t = 0$ to $t = 2$.

Answer

The curvature is decreasing over this interval.

The surface of a large cup is formed by revolving the graph of the function $y = 0.25x^{1.6}$ from $x = 0$ to $x = 5$ about the y-axis (measured in centimeters).

52) **[T]** Use technology to graph the surface.

53) Find the curvature κ of the generating curve as a function of x.

Answer

$$\kappa = \frac{30}{x^{2/5}\left(25+4x^{6/5}\right)^{3/2}}$$

Note that initially your answer may be:

$$\frac{6}{25x^{2/5}\left(1+\frac{4}{25}x^{6/5}\right)^{3/2}}$$

We can simplify it as follows:

$$\begin{aligned}\frac{6}{25x^{2/5}\left(1+\frac{4}{25}x^{6/5}\right)^{3/2}} &= \frac{6}{25x^{2/5}\left[\frac{1}{25}\left(25+4x^{6/5}\right)\right]^{3/2}} \\ &= \frac{6}{25x^{2/5}\left(\frac{1}{25}\right)^{3/2}\left[25+4x^{6/5}\right]^{3/2}} \\ &= \frac{6}{\frac{25}{125}x^{2/5}\left[25+4x^{6/5}\right]^{3/2}} \\ &= \frac{30}{x^{2/5}\left(25+4x^{6/5}\right)^{3/2}}\end{aligned}$$

54) **[T]** Use technology to graph the curvature function.

Contributors:

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) created question 6.

This page titled 13.3E: Exercises for Section 13.3 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

13.4: Motion in Space

Learning Objectives

- Describe the velocity and acceleration vectors of a particle moving in space.
- Explain the tangential and normal components of acceleration.
- State Kepler's laws of planetary motion.

We have now seen how to describe curves in the plane and in space, and how to determine their properties, such as arc length and curvature. All of this leads to the main goal of this chapter, which is the description of motion along plane curves and space curves. We now have all the tools we need; in this section, we put these ideas together and look at how to use them.

Motion Vectors in the Plane and in Space

Our starting point is using vector-valued functions to represent the position of an object as a function of time. All of the following material can be applied either to curves in the plane or to space curves. For example, when we look at the orbit of the planets, the curves defining these orbits all lie in a plane because they are elliptical. However, a particle traveling along a helix moves on a curve in three dimensions.

Definition: Speed, Velocity, and Acceleration

Let $\vec{\mathbf{r}}(t)$ be a twice-differentiable vector-valued function of the parameter t that represents the position of an object as a function of time.

The velocity vector $\vec{\mathbf{v}}(t)$ of the object is given by

$$\text{Velocity } = \vec{\mathbf{v}}(t) = \vec{\mathbf{r}}\,'(t). \tag{13.4.1}$$

The acceleration vector $\vec{\mathbf{a}}(t)$ is defined to be

$$\text{Acceleration } = \vec{\mathbf{a}}(t) = \vec{\mathbf{v}}\,'(t) = \vec{\mathbf{r}}\,''(t). \tag{13.4.2}$$

The *speed* is defined to be

$$\text{Speed } = v(t) = \|\vec{\mathbf{v}}(t)\| = \|\vec{\mathbf{r}}\,'(t)\| = \frac{ds}{dt}. \tag{13.4.3}$$

Since $\vec{\mathbf{r}}(t)$ can be in either two or three dimensions, these vector-valued functions can have either two or three components. In two dimensions, we define $\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}}$ and in three dimensions $\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$. Then the velocity, acceleration, and speed can be written as shown in the following table.

Table 13.4.1: Formulas for Position, Velocity, Acceleration, and Speed

Quantity	Two Dimensions	Three Dimensions
Position	$\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}}$	$\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$
Velocity	$\vec{\mathbf{v}}(t) = x\,'(t)\hat{\mathbf{i}} + y\,'(t)\hat{\mathbf{j}}$	$\vec{\mathbf{v}}(t) = x\,'(t)\hat{\mathbf{i}} + y\,'(t)\hat{\mathbf{j}} + z\,'(t)\hat{\mathbf{k}}$
Acceleration	$\vec{\mathbf{a}}(t) = x\,''(t)\hat{\mathbf{i}} + y\,''(t)\hat{\mathbf{j}}$	$\vec{\mathbf{a}}(t) = x\,''(t)\hat{\mathbf{i}} + y\,''(t)\hat{\mathbf{j}} + z\,''(t)\hat{\mathbf{k}}$
Speed	$\|\vec{\mathbf{v}}(t)\| = \sqrt{(x\,'(t))^2 + (y\,'(t))^2}$	$\|\vec{\mathbf{v}}(t)\| = \sqrt{(x\,'(t))^2 + (y\,'(t))^2 + (z\,'(t))^2}$

Example 13.4.1: Studying Motion Along a Parabola

A particle moves in a parabolic path defined by the vector-valued function $\vec{\mathbf{r}}(t) = t^2\hat{\mathbf{i}} + \sqrt{5 - t^2}\hat{\mathbf{j}}$, where t measures time in seconds.

1. Find the velocity, acceleration, and speed as functions of time.

2. Sketch the curve along with the velocity vector at time $t = 1$.

Solution

1. We use Equations 13.4.1, 13.4.2, and 13.4.3:

$$\begin{aligned}
\vec{\mathbf{v}}(t) = \vec{\mathbf{r}}\,'(t) &= 2t\hat{\mathbf{i}} - \frac{t}{\sqrt{5-t^2}}\hat{\mathbf{j}} \\
\vec{\mathbf{a}}(t) = \vec{\mathbf{v}}\,'(t) &= 2\hat{\mathbf{i}} - 5(5-t^2)^{-\frac{3}{2}}\hat{\mathbf{j}} \\
||\vec{\mathbf{v}}(t)|| &= ||\vec{\mathbf{r}}\,'(t)|| \\
&= \sqrt{(2t)^2 + \left(-\frac{t}{\sqrt{5-t^2}}\right)^2} \\
&= \sqrt{4t^2 + \frac{t^2}{5-t^2}} \\
&= \sqrt{\frac{21t^2 - 4t^4}{5-t^2}}.
\end{aligned}$$

2. The graph of $\vec{\mathbf{r}}(t) = t^2\hat{\mathbf{i}} + \sqrt{5-t^2}\hat{\mathbf{j}}$ is a portion of a parabola (Figure 13.4.1).

When $t = 1$, $\vec{\mathbf{r}}(1) = (1)^2\hat{\mathbf{i}} + \sqrt{5-(1)^2}\hat{\mathbf{j}} = \hat{\mathbf{i}} + \sqrt{4}\hat{\mathbf{j}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}}$.

Thus the particle would be located at the point $(1, 2)$ when $t = 1$.

The velocity vector at $t = 1$ is

$$\begin{aligned}
\vec{\mathbf{v}}(1) = \vec{\mathbf{r}}\,'(1) &= 2(1)\hat{\mathbf{i}} - \frac{1}{\sqrt{5-1^2}}\hat{\mathbf{j}} \\
&= 2\hat{\mathbf{i}} - \frac{1}{2}\hat{\mathbf{j}}
\end{aligned}$$

and the acceleration vector at $t = 1$ is

$$\vec{\mathbf{a}}(1) = \vec{\mathbf{v}}\,'(1) = 2\hat{\mathbf{i}} - 5(5-1^2)^{-3/2}\hat{\mathbf{j}} = 2\hat{\mathbf{i}} - \frac{5}{8}\hat{\mathbf{j}}.$$

Notice that the velocity vector is tangent to the path, as is always the case.

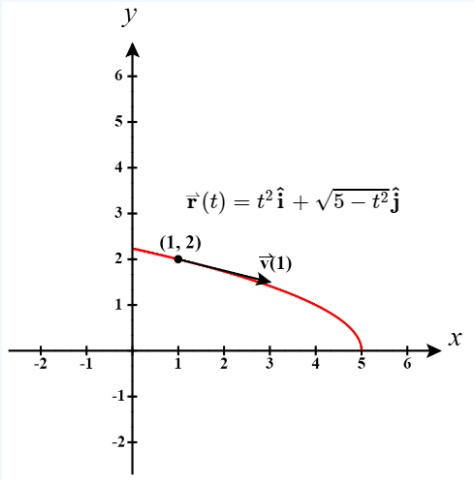


Figure 13.4.1: This graph depicts the velocity vector at time $t = 1$ for a particle moving in a parabolic path.

Exercise 13.4.1

A particle moves in a path defined by the vector-valued function $\vec{\mathbf{r}}(t) = (t^2 - 3t)\,\hat{\mathbf{i}} + (2t - 4)\,\hat{\mathbf{j}} + (t + 2)\,\hat{\mathbf{k}}$, where t measures time in seconds and where distance is measured in feet. Find the velocity, acceleration, and speed as functions of time.

Hint

Use Equations 13.4.1, 13.4.2, and 13.4.3.

Answer

$$\vec{\mathbf{v}}(t) = \vec{\mathbf{r}}\,'(t) = (2t - 3)\,\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\vec{\mathbf{a}}(t) = \vec{\mathbf{v}}\,'(t) = 2\,\hat{\mathbf{i}}$$

$$||\vec{\mathbf{r}}\,'(t)|| = \sqrt{(2t-3)^2 + 2^2 + 1^2} = \sqrt{4t^2 - 12t + 14}$$

The units for velocity and speed are feet per second, and the units for acceleration are feet per second squared.

To gain a better understanding of the velocity and acceleration vectors, imagine you are driving along a curvy road. If you do not turn the steering wheel, you would continue in a straight line and run off the road. The speed at which you are traveling when you run off the road, coupled with the direction, gives a vector representing your velocity, as illustrated in Figure 13.4.2.

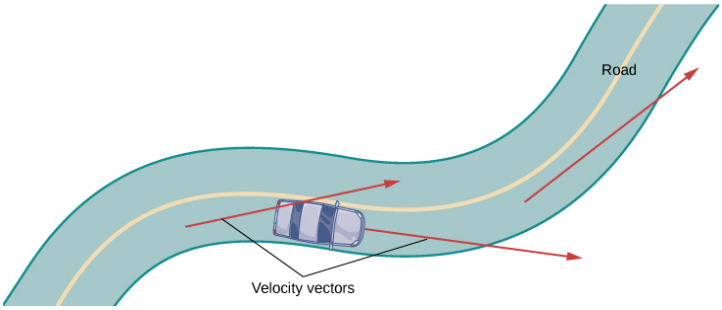


Figure 13.4.2: At each point along a road traveled by a car, the velocity vector of the car is tangent to the path traveled by the car.

However, the fact that you must turn the steering wheel to stay on the road indicates that your velocity is always changing (even if your speed is not) because your *direction* is constantly changing to keep you on the road. As you turn to the right, your acceleration vector also points to the right. As you turn to the left, your acceleration vector points to the left. This indicates that your velocity and acceleration vectors are constantly changing, regardless of whether your actual speed varies (Figure 13.4.3).

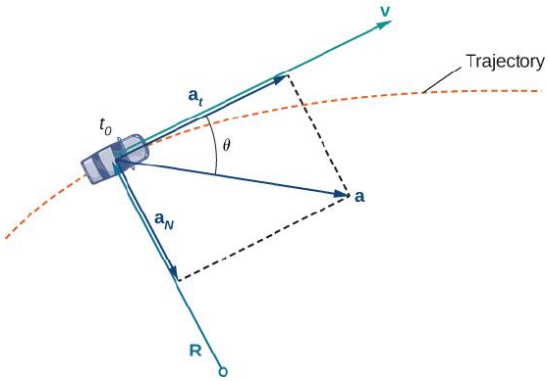


Figure 13.4.3: The dashed line represents the trajectory of an object (a car, for example). The acceleration vector points toward the inside of the turn at all times.

Components of the Acceleration Vector

We can combine some of the concepts discussed in Arc Length and Curvature with the acceleration vector to gain a deeper understanding of how this vector relates to motion in the plane and in space. Recall that the unit tangent vector $\vec{\mathbf{T}}$ and the unit normal vector $\vec{\mathbf{N}}$ form an osculating plane at any point P on the curve defined by a vector-valued function $\vec{\mathbf{r}}(t)$. The following

theorem shows that the acceleration vector $\vec{\mathbf{a}}(t)$ lies in the osculating plane and can be written as a linear combination of the unit tangent and the unit normal vectors.

Theorem 13.4.1: The Plane of the Acceleration Vector

The acceleration vector $\vec{\mathbf{a}}(t)$ of an object moving along a curve traced out by a twice-differentiable function $\vec{\mathbf{r}}(t)$ lies in the plane formed by the unit tangent vector $\vec{\mathbf{T}}(t)$ and the principal unit normal vector $\vec{\mathbf{N}}(t)$ to C. Furthermore,

$$\vec{\mathbf{a}}(t) = v'(t)\vec{\mathbf{T}}(t) + [v(t)]^2\kappa\vec{\mathbf{N}}(t)$$

Here, $v(t) = \|\vec{\mathbf{v}}(t)\|$ is the speed of the object and κ is the curvature of C traced out by $\vec{\mathbf{r}}(t)$.

Proof

Because $\vec{\mathbf{v}}(t) = \vec{\mathbf{r}}\,'(t)$ and $\vec{\mathbf{T}}(t) = \dfrac{\vec{\mathbf{r}}\,'(t)}{||\vec{\mathbf{r}}\,'(t)||}$, we have $\vec{\mathbf{v}}(t) = ||\vec{\mathbf{r}}\,'(t)||\vec{\mathbf{T}}(t) = v(t)\vec{\mathbf{T}}(t)$.

Now we differentiate this equation:

$$\vec{\mathbf{a}}(t) = \vec{\mathbf{v}}\,'(t) = \frac{d}{dt}\left(v(t)\vec{\mathbf{T}}(t)\right) = v'(t)\vec{\mathbf{T}}(t) + v(t)\vec{\mathbf{T}}\,'(t)$$

Since $\vec{\mathbf{N}}(t) = \dfrac{\vec{\mathbf{T}}\,'(t)}{||\vec{\mathbf{T}}\,'(t)||}$, we know $\vec{\mathbf{T}}\,'(t) = ||\vec{\mathbf{T}}\,'(t)||\vec{\mathbf{N}}(t)$, so

$$\vec{\mathbf{a}}(t) = v'(t)\vec{\mathbf{T}}(t) + v(t)||\vec{\mathbf{T}}\,'(t)||\vec{\mathbf{N}}(t).$$

A formula for curvature is $\kappa = \dfrac{||\vec{\mathbf{T}}\,'(t)||}{||\vec{\mathbf{r}}\,'(t)||}$, so $\vec{\mathbf{T}}\,'(t) = \kappa||\vec{\mathbf{r}}\,'(t)|| = \kappa v(t)$.

This gives $\vec{\mathbf{a}}(t) = v'(t)\vec{\mathbf{T}}(t) + \kappa(v(t))^2\vec{\mathbf{N}}(t)$.

□

The coefficients of $\vec{\mathbf{T}}(t)$ and $\vec{\mathbf{N}}(t)$ are referred to as the **tangential component of acceleration** and the **normal component of acceleration**, respectively. We write $a_{\vec{\mathbf{T}}}$ to denote the tangential component and $a_{\vec{\mathbf{N}}}$ to denote the normal component.

Theorem 13.4.2: Tangential and Normal Components of Acceleration

Let $\vec{\mathbf{r}}(t)$ be a vector-valued function that denotes the position of an object as a function of time. Then $\vec{\mathbf{a}}(t) = \vec{\mathbf{r}}\,''(t)$ is the acceleration vector. The tangential and normal components of acceleration $a_{\vec{\mathbf{T}}}$ and $a_{\vec{\mathbf{N}}}$ are given by the formulas

$$a_{\vec{\mathbf{T}}} = \vec{\mathbf{a}} \cdot \vec{\mathbf{T}} = \frac{\vec{\mathbf{v}} \cdot \vec{\mathbf{a}}}{||\vec{\mathbf{v}}||} \tag{13.4.4}$$

and

$$a_{\vec{\mathbf{N}}} = \vec{\mathbf{a}} \cdot \vec{\mathbf{N}} = \frac{||\vec{\mathbf{v}} \times \vec{\mathbf{a}}||}{||\vec{\mathbf{v}}||} = \sqrt{||\vec{\mathbf{a}}||^2 - \left(a_{\vec{\mathbf{T}}}\right)^2}. \tag{13.4.5}$$

These components are related by the formula

$$\vec{\mathbf{a}}(t) = a_{\vec{\mathbf{T}}}\vec{\mathbf{T}}(t) + a_{\vec{\mathbf{N}}}\vec{\mathbf{N}}(t). \tag{13.4.6}$$

Here $\vec{\mathbf{T}}(t)$ is the unit tangent vector to the curve defined by $\vec{\mathbf{r}}(t)$, and $\vec{\mathbf{N}}(t)$ is the unit normal vector to the curve defined by $\vec{\mathbf{r}}(t)$.

The normal component of acceleration is also called the *centripetal component of acceleration* or sometimes the *radial component of acceleration*. To understand centripetal acceleration, suppose you are traveling in a car on a circular track at a constant speed. Then, as we saw earlier, the acceleration vector points toward the center of the track at all times. As a rider in the car, you feel a pull toward the *outside* of the track because you are constantly turning. This sensation acts in the opposite direction of centripetal

acceleration. The same holds true for non-circular paths. The reason is that your body tends to travel in a straight line and resists the force resulting from acceleration that push it toward the side. Note that at point B in Figure 13.4.4 the acceleration vector is pointing backward. This is because the car is decelerating as it goes into the curve.

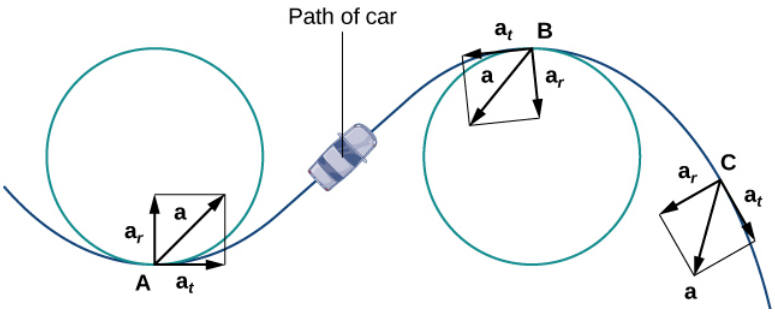


Figure 13.4.4: The tangential and normal components of acceleration can be used to describe the acceleration vector.

The tangential and normal unit vectors at any given point on the curve provide a frame of reference at that point. The tangential and normal components of acceleration are the projections of the acceleration vector onto $\vec{\mathbf{T}}$ and $\vec{\mathbf{N}}$, respectively.

Example 13.4.2: Finding Components of Acceleration

A particle moves in a path defined by the vector-valued function $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + (2t-3)\,\hat{\mathbf{j}} + (3t^2-3t)\,\hat{\mathbf{k}}$, where t measures time in seconds and distance is measured in feet.

a. Find $a_{\vec{\mathbf{T}}}$ and $a_{\vec{\mathbf{N}}}$ as functions of t.
b. Find $a_{\vec{\mathbf{T}}}$ and $a_{\vec{\mathbf{N}}}$ at time $t = 2$.

Solution

a. Let's start deriving the velocity and acceleration functions:

$$\begin{aligned}\vec{\mathbf{v}}(t) &= \vec{\mathbf{r}}\,'(t)\\ &= 2t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + (6t-3)\,\hat{\mathbf{k}}\\ \vec{\mathbf{a}}(t) &= \vec{\mathbf{v}}\,'(t)\\ &= 2\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{k}}\end{aligned}$$

Now we apply Equation 13.4.4:

$$\begin{aligned}a_{\vec{\mathbf{T}}} &= \frac{\vec{\mathbf{v}}\cdot\vec{\mathbf{a}}}{||\vec{\mathbf{v}}||}\\ &= \frac{(2t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + (6t-3)\,\hat{\mathbf{k}})\cdot(2\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{k}})}{||2t\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + (6t-3)\,\hat{\mathbf{k}}||}\\ &= \frac{4t + 6(6t-3)}{\sqrt{(2t)^2 + 2^2 + (6t-3)^2}}\\ &= \frac{40t-18}{40t^2-36t+13}\end{aligned}$$

Now we can apply Equation 13.4.5:

$$
\begin{aligned}
a_{\vec{\mathbf{N}}} &= \sqrt{||\vec{\mathbf{a}}||^2 - \left(a_{\vec{\mathbf{T}}}\right)^2} \\
&= \sqrt{||2\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{k}}||^2 - \left(\frac{40t-18}{\sqrt{40t^2-36t+13}}\right)^2} \\
&= \sqrt{4 + 36 - \frac{(40t-18)^2}{40t^2-36t+13}} \\
&= \sqrt{\frac{40(40t^2-36t+13) - (1600t^2-1440t+324)}{40t^2-36t+13}} \\
&= \sqrt{\frac{196}{40t^2-36t+13}} \\
&= \frac{14}{\sqrt{40t^2-36t+13}}
\end{aligned}
$$

b. We must evaluate each of the answers from part a at $t = 2$:

$$
\begin{aligned}
a_{\vec{\mathbf{T}}}(2) &= \frac{40(2)-18}{\sqrt{40(2)^2-36(2)+13}} \\
&= \frac{80-18}{\sqrt{160-72+13}} \\
&= \frac{62}{\sqrt{101}} \\
a_{\vec{\mathbf{N}}}(2) &= \frac{14}{\sqrt{40(2)^2-36(2)+13}} \\
&= \frac{14}{\sqrt{160-72+13}} = \frac{14}{\sqrt{101}}.
\end{aligned}
$$

The units of acceleration are feet per second squared, as are the units of the normal and tangential components of acceleration.

Exercise 13.4.2

An object moves in a path defined by the vector-valued function $\vec{\mathbf{r}}(t) = 4t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}}$, where t measures time in seconds.

a. Find $a_{\vec{\mathbf{T}}}$ and $a_{\vec{\mathbf{N}}}$ as functions of t.
b. Find $a_{\vec{\mathbf{T}}}$ and $a_{\vec{\mathbf{N}}}$ at time $t = -3$.

Hint

Use Equations 13.4.4 and 13.4.5

Answer

a.

$$
\begin{aligned}
a_{\vec{\mathbf{T}}} = \frac{\vec{\mathbf{v}}(t)\cdot\vec{\mathbf{a}}(t)}{||\vec{\mathbf{v}}(t)||} &= \frac{\vec{\mathbf{r}}\,'(t)\cdot\vec{\mathbf{r}}\,''(t)}{||\vec{\mathbf{r}}\,'(t)||} \\
&= \frac{(4\,\hat{\mathbf{i}} + 2t\,\hat{\mathbf{j}})\cdot(2\,\hat{\mathbf{j}})}{||4\,\hat{\mathbf{i}} + 2t\,\hat{\mathbf{j}}||} \\
&= \frac{4t}{\sqrt{4^2+(2t)^2}} \\
&= \frac{2t}{\sqrt{2+t^2}}
\end{aligned}
$$

https://math.libretexts.org/@go/page/2597

$$a_{\vec{\mathbf{N}}} = \sqrt{||\vec{\mathbf{a}}||^2 - a_{\vec{\mathbf{T}}}^2}$$
$$= \sqrt{||2\hat{\mathbf{j}}||^2 - \left(\frac{2t}{\sqrt{2+t^2}}\right)^2}$$
$$= \sqrt{4 - \frac{4t^2}{2+t^2}}$$

b.

$$a_{\vec{\mathbf{T}}}(-3) = \frac{2(-3)}{\sqrt{2+(-3)^2}}$$
$$= \frac{-6}{\sqrt{11}}$$

$$a_{\vec{\mathbf{N}}}(-3) = \sqrt{4 - \frac{4(-3)^2}{2+(-3)^2}}$$
$$= \sqrt{4 - \frac{36}{11}}$$
$$= \sqrt{\frac{8}{11}}$$
$$= \frac{2\sqrt{2}}{\sqrt{11}}$$

Projectile Motion

Now let's look at an application of vector functions. In particular, let's consider the effect of gravity on the motion of an object as it travels through the air, and how it determines the resulting trajectory of that object. In the following, we ignore the effect of air resistance. This situation, with an object moving with an initial velocity but with no forces acting on it other than gravity, is known as projectile motion. It describes the motion of objects from golf balls to baseballs, and from arrows to cannonballs.

First we need to choose a coordinate system. If we are standing at the origin of this coordinate system, then we choose the positive y-axis to be up, the negative y-axis to be down, and the positive x-axis to be forward (i.e., away from the thrower of the object). The effect of gravity is in a downward direction, so Newton's second law tells us that the force on the object resulting from gravity is equal to the mass of the object times the acceleration resulting from gravity, or $\vec{\mathbf{F}}_g = m\vec{\mathbf{a}}$, where $\vec{\mathbf{F}}_g$ represents the force from gravity and $\vec{\mathbf{a}} = -g\,\hat{\mathbf{j}}$ represents the acceleration resulting from gravity at Earth's surface. The value of g in the English system of measurement is approximately 32 ft/sec^2 and it is approximately 9.8 m/sec^2 in the metric system. This is the only force acting on the object. Since gravity acts in a downward direction, we can write the force resulting from gravity in the form $\vec{\mathbf{F}}_g = -mg\,\hat{\mathbf{j}}$, as shown in Figure 13.4.5.

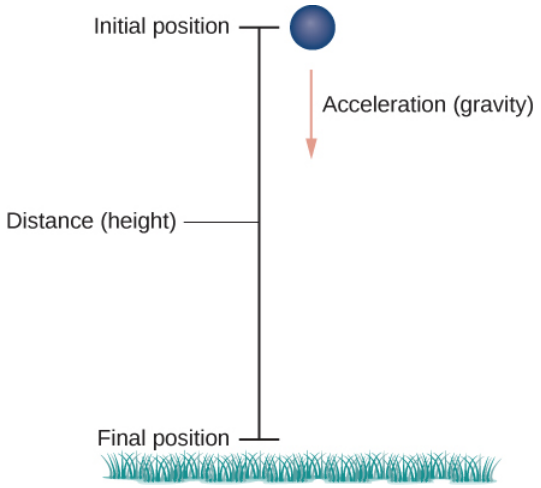


Figure 13.4.5: An object is falling under the influence of gravity.

Newton's second law also tells us that $F = m\vec{\mathbf{a}}$, where $\vec{\mathbf{a}}$ represents the acceleration vector of the object. This force must be equal to the force of gravity at all times, so we therefore know that

$$\begin{aligned} \vec{\mathbf{F}} &= \vec{\mathbf{F}}_g \\ m\vec{\mathbf{a}} &= -mg\hat{\mathbf{j}} \\ \vec{\mathbf{a}} &= -g\hat{\mathbf{j}}. \end{aligned}$$

Now we use the fact that the acceleration vector is the first derivative of the velocity vector. Therefore, we can rewrite the last equation in the form

$$\vec{\mathbf{v}}\,'(t) = -g\hat{\mathbf{j}}$$

By taking the antiderivative of each side of this equation we obtain

$$\vec{\mathbf{v}}(t) = \int -g\hat{\mathbf{j}}\, dt = -gt\hat{\mathbf{j}} + \vec{\mathbf{C}}_1$$

for some constant vector $\vec{\mathbf{C}}_1$. To determine the value of this vector, we can use the velocity of the object at a fixed time, say at time $t = 0$. We call this velocity the *initial velocity*: $\vec{\mathbf{v}}(0) = \vec{\mathbf{v}}_0$. Therefore, $\vec{\mathbf{v}}(0) = -g(0)\,\hat{\mathbf{j}} + \vec{\mathbf{C}}_1 = \vec{\mathbf{v}}_0$ and $\vec{\mathbf{C}}_1 = \vec{\mathbf{v}}_0$. This gives the velocity vector as $\vec{\mathbf{v}}(t) = -gt\hat{\mathbf{j}} + \vec{\mathbf{v}}_0$.

Next we use the fact that velocity $\vec{\mathbf{v}}(t)$ is the derivative of position $\vec{\mathbf{s}}(t)$. This gives the equation

$$\vec{\mathbf{s}}\,'(t) = -gt\hat{\mathbf{j}} + \vec{\mathbf{v}}_0.$$

Taking the antiderivative of both sides of this equation leads to

$$\begin{aligned} \vec{\mathbf{s}}(t) &= \int -gt\hat{\mathbf{j}} + \vec{\mathbf{v}}_0\, dt \\ &= -\frac{1}{2}gt^2\hat{\mathbf{j}} + \vec{\mathbf{v}}_0 t + \vec{\mathbf{C}}_2 \end{aligned}$$

with another unknown constant vector $\vec{\mathbf{C}}_2$. To determine the value of $\vec{\mathbf{C}}_2$, we can use the position of the object at a given time, say at time $t = 0$. We call this position the *initial position*: $\vec{\mathbf{s}}(0) = \vec{\mathbf{s}}_0$. Therefore, $\vec{\mathbf{s}}(0) = -(1/2)g(0)^2\,\hat{\mathbf{j}} + \vec{\mathbf{v}}_0(0) + \vec{\mathbf{C}}_2 = \vec{\mathbf{s}}_0$. This gives the position of the object at any time as

$$\vec{\mathbf{s}}(t) = -\frac{1}{2}gt^2\hat{\mathbf{j}} + \vec{\mathbf{v}}_0 t + \vec{\mathbf{s}}_0.$$

Let's take a closer look at the initial velocity and initial position. In particular, suppose the object is thrown upward from the origin at an angle θ to the horizontal, with initial speed $\vec{\mathbf{v}}_0$. How can we modify the previous result to reflect this scenario? First, we can assume it is thrown from the origin. If not, then we can move the origin to the point from where it is thrown. Therefore, $\vec{\mathbf{s}}_0 = \vec{\mathbf{0}}$, as shown in Figure 13.4.6.

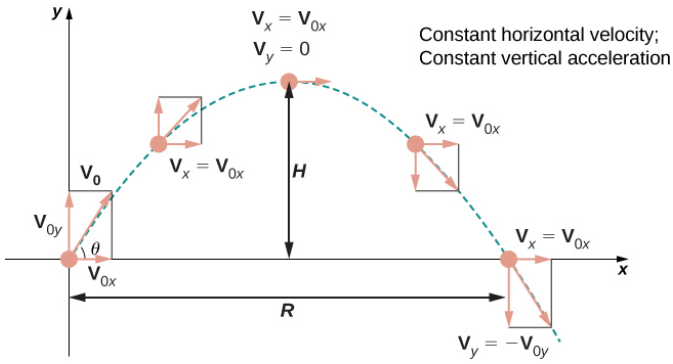


Figure 13.4.6: Projectile motion when the object is thrown upward at an angle θ. The horizontal motion is at constant velocity and the vertical motion is at constant acceleration.

We can rewrite the initial velocity vector in the form $\vec{\mathbf{v}}_0 = v_0\cos\theta\,\hat{\mathbf{i}} + v_0\sin\theta\,\hat{\mathbf{j}}$. Then the equation for the position function $\vec{\mathbf{s}}(t)$ becomes

$$\begin{aligned}\vec{\mathbf{s}}(t) &= -\frac{1}{2}gt^2\,\hat{\mathbf{j}} + v_0 t\cos\theta\,\hat{\mathbf{i}} + v_0 t\sin\theta\,\hat{\mathbf{j}} \\ &= v_0 t\cos\theta\,\hat{\mathbf{i}} + v_0 t\sin\theta\,\hat{\mathbf{j}} - \frac{1}{2}gt^2\,\hat{\mathbf{j}} \\ &= v_0 t\cos\theta\,\hat{\mathbf{i}} + \left(v_0 t\sin\theta - \frac{1}{2}gt^2\right)\hat{\mathbf{j}}.\end{aligned}$$

The coefficient of $\hat{\mathbf{i}}$ represents the horizontal component of $\vec{\mathbf{s}}(t)$ and is the horizontal distance of the object from the origin at time t. The maximum value of the horizontal distance (measured at the same initial and final altitude) is called the range R. The coefficient of $\hat{\mathbf{j}}$ represents the vertical component of $\vec{\mathbf{s}}(t)$ and is the altitude of the object at time t. The maximum value of the vertical distance is the height H.

Example 13.4.3: Motion of a Cannonball

During an Independence Day celebration, a cannonball is fired from a cannon on a cliff toward the water. The cannon is aimed at an angle of 30° above horizontal and the initial speed of the cannonball is 600 ft/sec. The cliff is 100 ft above the water (Figure 13.4.7).

a. Find the maximum height of the cannonball.
b. How long will it take for the cannonball to splash into the sea?
c. How far out to sea will the cannonball hit the water?

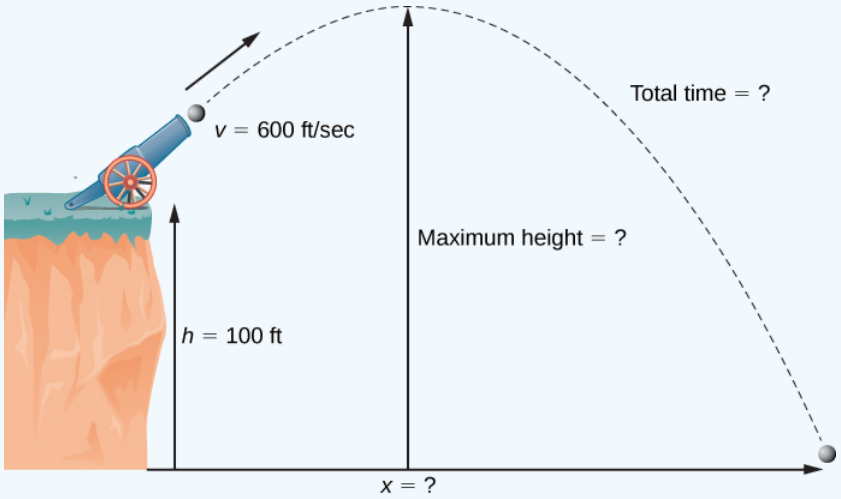


Figure 13.4.7: The flight of a cannonball (ignoring air resistance) is projectile motion.

Solution

We use the equation

$$\vec{\mathbf{s}}(t) = v_0 t\cos\theta\,\hat{\mathbf{i}} + \left(v_0 t\sin\theta - \frac{1}{2}gt^2\right)\hat{\mathbf{j}}$$

with $\theta = 30^\circ$, $g = 32\frac{\text{ft}}{\text{sec}^2}$, and $v_0 = 600\frac{\text{ft}}{\text{sec}}$. Then the position equation becomes

$$\begin{aligned}\vec{\mathbf{s}}(t) &= 600t(\cos 30^\circ)\,\hat{\mathbf{i}} + \left(600t\sin 30^\circ - \frac{1}{2}(32)t^2\right)\hat{\mathbf{j}} \\ &= 300t\sqrt{3}\,\hat{\mathbf{i}} + \left(300t - 16t^2\right)\hat{\mathbf{j}}\end{aligned}$$

a. The cannonball reaches its maximum height when the vertical component of its velocity is zero, because the cannonball is neither rising nor falling at that point. The velocity vector is

$$\begin{aligned}\vec{\mathbf{v}}(t) &= \vec{\mathbf{s}}\,'(t) \\ &= 300\sqrt{3}\,\hat{\mathbf{i}} + (300 - 32t)\,\hat{\mathbf{j}}\end{aligned}$$

Therefore, the vertical component of velocity is given by the expression $300 - 32t$. Setting this expression equal to zero and solving for t gives $t = 9.375$ sec. The height of the cannonball at this time is given by the vertical component of the position vector, evaluated at $t = 9.375$.

$$\begin{aligned}\vec{\mathbf{s}}(9.375) &= 300(9.375)\sqrt{3}\,\hat{\mathbf{i}} + (300(9.375) - 16(9.375)^2)\,\hat{\mathbf{j}} \\ &= 4871.39\,\hat{\mathbf{i}} + 1406.25\,\hat{\mathbf{j}}\end{aligned}$$

Therefore, the maximum height of the cannonball is 1406.39 ft above the cannon, or 1506.39 ft above sea level.

b. When the cannonball lands in the water, it is 100 ft below the cannon. Therefore, the vertical component of the position vector is equal to -100. Setting the vertical component of $\vec{\mathbf{s}}(t)$ equal to -100 and solving, we obtain

$$\begin{aligned}300t - 16t^2 &= -100 \\ 16t^2 - 300t - 100 &= 0 \\ 4t^2 - 75 - 25 &= 0 \\ t &= \frac{75 \pm \sqrt{(-75)^2 - 4(4)(-25)}}{2(4)} \\ &= \frac{75 \pm \sqrt{6025}}{8} \\ &= \frac{75 \pm 5\sqrt{241}}{8}\end{aligned}$$

The positive value of t that solves this equation is approximately 19.08. Therefore, the cannonball hits the water after approximately 19.08 sec.

c. To find the distance out to sea, we simply substitute the answer from part (b) into $\vec{\mathbf{s}}(t)$:

$$\begin{aligned}\vec{\mathbf{s}}(19.08) &= 300(19.08)\sqrt{3}\,\hat{\mathbf{i}} + \left(300(19.08) - 16(19.08)^2\right)\,\hat{\mathbf{j}} \\ &= 9914.26\,\hat{\mathbf{i}} - 100.7424\,\hat{\mathbf{j}}\end{aligned}$$

Therefore, the ball hits the water about 9914.26 ft away from the base of the cliff. Notice that the vertical component of the position vector is very close to −100, which tells us that the ball just hit the water. Note that 9914.26 feet is not the true range of the cannon since the cannonball lands in the ocean at a location below the cannon. The range of the cannon would be determined by finding how far out the cannonball is when its height is 100 ft above the water (the same as the altitude of the cannon).

? Exercise 13.4.3

An archer fires an arrow at an angle of 40° above the horizontal with an initial speed of 98 m/sec. The height of the archer is 171.5 cm. Find the horizontal distance the arrow travels before it hits the ground.

Hint

The equation for the position vector needs to account for the height of the archer in meters.

Answer

967.15 m

One final question remains: In general, what is the maximum distance a projectile can travel, given its initial speed? To determine this distance, we assume the projectile is fired from ground level and we wish it to return to ground level. In other words, we want to determine an equation for the range. In this case, the equation of projectile motion is

$$\vec{\mathbf{s}} = v_0 t\cos\theta\,\hat{\mathbf{i}} + \left(v_0 t\sin\theta - \frac{1}{2}gt^2\right)\,\hat{\mathbf{j}}.$$

Setting the second component equal to zero and solving for t yields

$$v_0 t \sin\theta - \frac{1}{2} g t^2 = 0$$
$$t\left(v_0 \sin\theta - \frac{1}{2} g t\right) = 0$$

Therefore, either $t = 0$ or $t = \dfrac{2v_0 \sin\theta}{g}$. We are interested in the second value of t, so we substitute this into $\vec{\mathbf{s}}(t)$, which gives

$$\begin{aligned}\vec{\mathbf{s}}\left(\frac{2v_0 \sin\theta}{g}\right) &= v_0\left(\frac{2v_0 \sin\theta}{g}\right)\cos\theta\,\hat{\mathbf{i}} + \left(v_0\left(\frac{2v_0 \sin\theta}{g}\right)\sin\theta - \frac{1}{2}g\left(\frac{2v_0 \sin\theta}{g}\right)^2\right)\hat{\mathbf{j}} \\ &= \left(\frac{2v_0^2 \sin\theta\cos\theta}{g}\right)\hat{\mathbf{i}} \\ &= \frac{v_0^2 \sin 2\theta}{g}\,\hat{\mathbf{i}}.\end{aligned}$$

Thus, the expression for the range of a projectile fired at an angle θ is

$$R = \frac{v_0^2 \sin 2\theta}{g}\,\hat{\mathbf{i}}.$$

The only variable in this expression is θ. To maximize the distance traveled, take the derivative of the coefficient of $\hat{\mathbf{i}}$ with respect to θ and set it equal to zero:

$$\begin{aligned}\frac{d}{d\theta}\left(\frac{v_0^2 \sin 2\theta}{g}\right) &= 0 \\ \frac{2v_0^2 \cos 2\theta}{g} &= 0 \\ \theta &= 45^\circ\end{aligned}$$

This value of θ) is the smallest positive value that makes the derivative equal to zero. Therefore, in the absence of air resistance, the best angle to fire a projectile (to maximize the range) is at a 45° angle. The distance it travels is given by

$$\vec{\mathbf{s}}\left(\frac{2v_0 \sin 45^\circ}{g}\right) = \frac{v_0^2 \sin 90^\circ}{g}\,\hat{\mathbf{i}} = \frac{v_0^2}{g}\,\hat{\mathbf{i}}$$

Therefore, the range for an angle of 45° is $\dfrac{v_0^2}{g}$ units.

Kepler's Laws

During the early 1600s, Johannes Kepler was able to use the amazingly accurate data from his mentor Tycho Brahe to formulate his three laws of planetary motion, now known as Kepler's laws of planetary motion. These laws also apply to other objects in the solar system in orbit around the Sun, such as comets (e.g., Halley's comet) and asteroids. Variations of these laws apply to satellites in orbit around Earth.

Theorem 13.4.2: Kepler's Laws of Planetary Motion

1. The path of any planet about the Sun is elliptical in shape, with the center of the Sun located at one focus of the ellipse (the law of ellipses).
2. A line drawn from the center of the Sun to the center of a planet sweeps out equal areas in equal time intervals (the law of equal areas) (Figure 13.4.8).
3. The ratio of the squares of the periods of any two planets is equal to the ratio of the cubes of the lengths of their semimajor orbital axes (the Law of Harmonies).

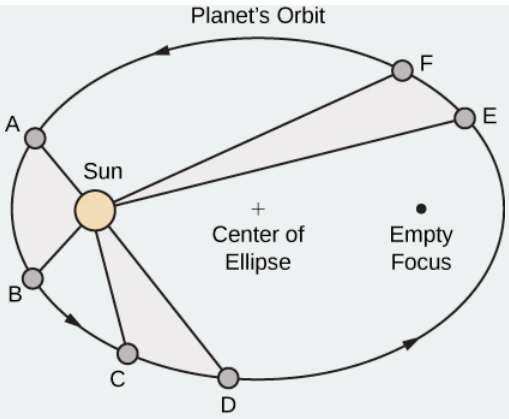


Figure 13.4.8: Kepler's first and second laws are pictured here. The Sun is located at a focus of the elliptical orbit of any planet. Furthermore, the shaded areas are all equal, assuming that the amount of time measured as the planet moves is the same for each region.

Kepler's third law is especially useful when using appropriate units. In particular, *1 astronomical unit* is defined to be the average distance from Earth to the Sun, and is now recognized to be 149,597,870,700 m or, approximately 93,000,000 mi. We therefore write 1 A.U. = 93,000,000 mi. Since the time it takes for Earth to orbit the Sun is 1 year, we use Earth years for units of time. Then, substituting 1 year for the period of Earth and 1 A.U. for the average distance to the Sun, Kepler's third law can be written as

$$T_p^2 = D_p^3$$

for any planet in the solar system, where T_P is the period of that planet measured in Earth years and D_P is the average distance from that planet to the Sun measured in astronomical units. Therefore, if we know the average distance from a planet to the Sun (in astronomical units), we can then calculate the length of its year (in Earth years), and vice versa.

Kepler's laws were formulated based on observations from Brahe; however, they were not proved formally until Sir Isaac Newton was able to apply calculus. Furthermore, Newton was able to generalize Kepler's third law to other orbital systems, such as a moon orbiting around a planet. Kepler's original third law only applies to objects orbiting the Sun.

Proof

Let's now prove Kepler's first law using the calculus of vector-valued functions. First we need a coordinate system. Let's place the Sun at the origin of the coordinate system and let the vector-valued function $\vec{\mathbf{r}}(t)$ represent the location of a planet as a function of time. Newton proved Kepler's law using his second law of motion and his law of universal gravitation. Newton's second law of motion can be written as $\vec{\mathbf{F}} = m\vec{\mathbf{a}}$, where $\vec{\mathbf{F}}$ represents the net force acting on the planet. His law of universal gravitation can be written in the form $\vec{\mathbf{F}} = -\dfrac{GmM}{||\vec{\mathbf{r}}||^2} \cdot \dfrac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||}$, which indicates that the force resulting from the gravitational attraction of the Sun points back toward the Sun, and has magnitude $\dfrac{GmM}{||\vec{\mathbf{r}}||^2}$ (Figure 13.4.9).

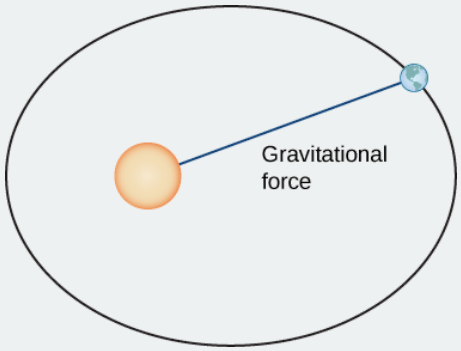


Figure 13.4.9: The gravitational force between Earth and the Sun is equal to the mass of the earth times its acceleration.

Setting these two forces equal to each other, and using the fact that $\vec{\mathbf{a}}(t) = \vec{\mathbf{v}}\,'(t)$, we obtain

$$m\vec{\mathbf{v}}\,'(t) = -\frac{GmM}{\|\vec{\mathbf{r}}\|^2}\cdot\frac{\vec{\mathbf{r}}}{\|\vec{\mathbf{r}}\|},$$

which can be rewritten as

$$\frac{d\vec{\mathbf{v}}}{dt} = -\frac{GM}{||\vec{\mathbf{r}}||^3}\vec{\mathbf{r}}.$$

This equation shows that the vectors $d\vec{\mathbf{v}}/dt$ and $\vec{\mathbf{r}}$ are parallel to each other, so $d\vec{\mathbf{v}}/dt \times \vec{\mathbf{r}} = \vec{\mathbf{0}}$. Next, let's differentiate $\vec{\mathbf{r}} \times \vec{\mathbf{v}}$ with respect to time:

$$\frac{d}{dt}(\vec{\mathbf{r}} \times \vec{\mathbf{v}}) = \frac{d\vec{\mathbf{r}}}{dt} \times \vec{\mathbf{v}} + \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{v}}}{dt} = \vec{\mathbf{v}} \times \vec{\mathbf{v}} + \vec{\mathbf{0}} = \vec{\mathbf{0}}. \tag{13.4.7}$$

This proves that $\vec{\mathbf{r}} \times \vec{\mathbf{v}}$ is a constant vector, which we call $\vec{\mathbf{C}}$. Since $\vec{\mathbf{r}}$ and $\vec{\mathbf{v}}$ are both perpendicular to $\vec{\mathbf{C}}$ for all values of t, they must lie in a plane perpendicular to $\vec{\mathbf{C}}$. Therefore, the motion of the planet lies in a plane.

Next we calculate the expression $d\vec{\mathbf{v}}/dt \times \vec{\mathbf{C}}$:

$$\frac{d\vec{\mathbf{v}}}{dt} \times \vec{\mathbf{C}} = -\frac{GM}{||\vec{\mathbf{r}}||^3}\vec{\mathbf{r}} \times (\vec{\mathbf{r}} \times \vec{\mathbf{v}}) = -\frac{GM}{||\vec{\mathbf{r}}||^3}[(\vec{\mathbf{r}} \cdot \vec{\mathbf{v}})\vec{\mathbf{r}} - (\vec{\mathbf{r}} \cdot \vec{\mathbf{r}})\vec{\mathbf{v}}]. \tag{13.4.8}$$

The last equality in Equation 13.4.8 is from the triple cross product formula (see the cross product section in Introduction to Vectors in Space). We need an expression for $\vec{\mathbf{r}} \cdot \vec{\mathbf{v}}$. To calculate this, we differentiate $\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}$ with respect to time:

$$\frac{d}{dt}(\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) = \frac{d\vec{\mathbf{r}}}{dt} \cdot \vec{\mathbf{r}} + \vec{\mathbf{r}} \cdot \frac{d\vec{\mathbf{r}}}{dt} = 2\vec{\mathbf{r}} \cdot \frac{d\vec{\mathbf{r}}}{dt} = 2\vec{\mathbf{r}} \cdot \vec{\mathbf{v}}. \tag{13.4.9}$$

Since $\vec{\mathbf{r}} \cdot \vec{\mathbf{r}} = ||\vec{\mathbf{r}}||^2$, we also have

$$\frac{d}{dt}(\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) = \frac{d}{dt}||\vec{\mathbf{r}}||^2 = 2||\vec{\mathbf{r}}||\frac{d}{dt}||\vec{\mathbf{r}}||. \tag{13.4.10}$$

Combining Equation 13.4.9 and Equation 13.4.10, we get

$$2\vec{\mathbf{r}} \cdot \vec{\mathbf{v}} = 2||\vec{\mathbf{r}}||\frac{d}{dt}||\vec{\mathbf{r}}||$$
$$\vec{\mathbf{r}} \cdot \vec{\mathbf{v}} = ||\vec{\mathbf{r}}||\frac{d}{dt}||\vec{\mathbf{r}}||.$$

Substituting this into Equation 13.4.8 gives us

$$\begin{aligned}\frac{d\vec{\mathbf{v}}}{dt} \times \vec{\mathbf{C}} &= -\frac{GM}{||\vec{\mathbf{r}}||^3}[(\vec{\mathbf{r}} \cdot \vec{\mathbf{v}})\vec{\mathbf{r}} - (\vec{\mathbf{r}} \cdot \vec{\mathbf{r}})\vec{\mathbf{v}}] \\ &= -\frac{GM}{||\vec{\mathbf{r}}||^3}\left[||\vec{\mathbf{r}}||\left(\frac{d}{dt}||\vec{\mathbf{r}}||\right)\vec{\mathbf{r}} - ||\vec{\mathbf{r}}||^2\vec{\mathbf{v}}\right] \\ &= -GM\left[\frac{1}{||\vec{\mathbf{r}}||^2}\left(\frac{d}{dt}||\vec{\mathbf{r}}||\right)\vec{\mathbf{r}} - \frac{1}{||\vec{\mathbf{r}}||}\vec{\mathbf{v}}\right] \\ &= GM\left[\frac{\vec{\mathbf{v}}}{||\vec{\mathbf{r}}||} - \frac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||^2}\left(\frac{d}{dt}||\vec{\mathbf{r}}||\right)\right].\end{aligned} \tag{13.4.11}$$

However,

$$\begin{aligned}\frac{d}{dt}\frac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||} &= \frac{\frac{d}{dt}(\vec{\mathbf{r}})||\vec{\mathbf{r}}|| - \vec{\mathbf{r}}\frac{d}{dt}||\vec{\mathbf{r}}||}{||\vec{\mathbf{r}}||^2} \\ &= \frac{\frac{d\vec{\mathbf{r}}}{dt}}{||\vec{\mathbf{r}}||} - \frac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||^2}\frac{d}{dt}||\vec{\mathbf{r}}|| \\ &= \frac{\vec{\mathbf{v}}}{||\vec{\mathbf{r}}||} - \frac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||^2}\frac{d}{dt}||\vec{\mathbf{r}}||.\end{aligned}$$

Therefore, Equation 13.4.11 becomes

$$\frac{d\vec{\mathbf{v}}}{dt} \times \vec{\mathbf{C}} = GM\left(\frac{d}{dt}\frac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||}\right).$$

Since $\vec{\mathbf{C}}$ is a constant vector, we can integrate both sides and obtain

$$\vec{\mathbf{v}} \times \vec{\mathbf{C}} = GM\frac{\vec{\mathbf{r}}}{||\vec{\mathbf{r}}||} + \vec{\mathbf{D}},$$

where $\vec{\mathbf{D}}$ is a constant vector. Our goal is to solve for $||\vec{\mathbf{r}}||$. Let's start by calculating $\vec{\mathbf{r}} \cdot \left(\vec{\mathbf{v}} \times \vec{\mathbf{C}}\right)$:

$$\vec{\mathbf{r}} \cdot \left(\vec{\mathbf{v}} \times \vec{\mathbf{C}}\right) = GM\frac{||\vec{\mathbf{r}}||^2}{||\vec{\mathbf{r}}||} + \vec{\mathbf{r}} \cdot \vec{\mathbf{D}} = GM||\vec{\mathbf{r}}|| + \vec{\mathbf{r}} \cdot \vec{\mathbf{D}}.$$

However, $\vec{\mathbf{r}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{C}}) = (\vec{\mathbf{r}} \times \vec{\mathbf{v}}) \cdot \vec{\mathbf{C}}$, so

$$(\vec{\mathbf{r}} \times \vec{\mathbf{v}}) \cdot \vec{\mathbf{C}} = GM||\vec{\mathbf{r}}|| + \vec{\mathbf{r}} \cdot \vec{\mathbf{D}}.$$

Since $\vec{\mathbf{r}} \times \vec{\mathbf{v}} = \vec{\mathbf{C}}$, we have

$$||\vec{\mathbf{C}}||^2 = GM||\vec{\mathbf{r}}|| + \vec{\mathbf{r}} \cdot \vec{\mathbf{D}}.$$

Note that $\vec{\mathbf{r}} \cdot \vec{\mathbf{D}} = ||\vec{\mathbf{r}}||||\vec{\mathbf{D}}|| \cos\theta$, where θ is the angle between $\vec{\mathbf{r}}$ and $\vec{\mathbf{D}}$. Therefore,

$$||\vec{\mathbf{C}}||^2 = GM||\vec{\mathbf{r}}|| + ||\vec{\mathbf{r}}||||\vec{\mathbf{D}}|| \cos\theta$$

Solving for $||\vec{\mathbf{r}}||$,

$$||\vec{\mathbf{r}}|| = \frac{||\vec{\mathbf{C}}||^2}{GM + ||\vec{\mathbf{D}}|| \cos\theta} = \frac{||\vec{\mathbf{C}}||^2}{GM}\left(\frac{1}{1 + e\cos\theta}\right).$$

where $e = ||\vec{\mathbf{D}}||/GM$. This is the polar equation of a conic with a focus at the origin, which we set up to be the Sun. It is a hyperbola if $e > 1$, a parabola if $e = 1$, or an ellipse if $e < 1$. Since planets have closed orbits, the only possibility is an ellipse. However, at this point it should be mentioned that hyperbolic comets do exist. These are objects that are merely passing through the solar system at speeds too great to be trapped into orbit around the Sun. As they pass close enough to the Sun, the gravitational field of the Sun deflects the trajectory enough so the path becomes hyperbolic.

□

Kepler's third law of planetary motion can be modified to the case of one object in orbit around an object other than the Sun, such as the Moon around the Earth. In this case, Kepler's third law becomes

$$P^2 = \frac{4\pi^2 a^3}{G(m+M)}, \tag{13.4.12}$$

where *m* is the mass of the Moon and *M* is the mass of Earth, *a* represents the length of the major axis of the elliptical orbit, and *P* represents the period.

Example 13.4.4: Using Kepler's Third Law for Nonheliocentric Orbits

Given that the mass of the Moon is 7.35×10^{22} kg, the mass of Earth is 5.97×10^{24} kg, $G = 6.67 \times 10^{-11}\text{m/kg} \cdot \text{sec}^2$, and the period of the moon is 27.3 days, let's find the length of the major axis of the orbit of the Moon around Earth.

Solution

It is important to be consistent with units. Since the universal gravitational constant contains seconds in the units, we need to use seconds for the period of the Moon as well:

$$27.3 \text{ days} \times \frac{24 \text{ hr}}{1 \text{ day}} \times \frac{3600 \text{ esc}}{1 \text{ hour}} = 2,358,720 \text{ sec}$$

Substitute all the data into Equation 13.4.12 and solve for a:

$$
\begin{aligned}
(2,358,720sec)^2 &= \frac{4\pi^2 a^3}{\left(6.67\times10^{-11}\frac{m}{\text{kg}\times\text{sec}^2}\right)(7.35\times10^{22}\text{kg}+5.97\times10^{24}\text{kg})} \\
5.563\times10^{12} &= \frac{4\pi^2 a^3}{(6.67\times10^{-11}\text{m}^3)(6.04\times10^{24})} \\
(5.563\times10^{12})(6.67\times10^{-11}\text{m}^3)(6.04\times10^{24}) &= 4\pi^2 a^3 \\
a^3 &= \frac{2.241\times10^{27}}{4\pi^2}\text{m}^3 \\
a &= 3.84\times10^8\text{m} \\
&\approx 384,000\,\text{km}.
\end{aligned}
$$

Analysis

According to solarsystem.nasa.gov, the actual average distance from the Moon to Earth is 384,400 km. This is calculated using reflectors left on the Moon by Apollo astronauts back in the 1960s.

Exercise 13.4.4

Titan is the largest moon of Saturn. The mass of Titan is approximately $1.35\times10^{23}kg$. The mass of Saturn is approximately 5.68×10^{26} kg. Titan takes approximately 16 days to orbit Saturn. Use this information, along with the universal gravitation constant $G=6.67\times10^{-11}\text{m/kg}\cdot\text{sec}^2$ to estimate the distance from Titan to Saturn.

Hint

Make sure your units agree, then use Equation 13.4.12.

Answer

$$a\approx1.224\times10^9\text{m}=1,224,000\text{km}$$

Example 13.4.5: Halley's Comet

We now return to the chapter opener, which discusses the motion of Halley's comet around the Sun. Kepler's first law states that Halley's comet follows an elliptical path around the Sun, with the Sun as one focus of the ellipse. The period of Halley's comet is approximately 76.1 years, depending on how closely it passes by Jupiter and Saturn as it passes through the outer solar system. Let's use $T=76.1$ years. What is the average distance of Halley's comet from the Sun?



Solution

Using the equation $T^2=D^3$ with $T=76.1$, we obtain $D^3=5791.21$, so $D\approx17.96$ A.U. This comes out to approximately 1.67×10^9 mi.

A natural question to ask is: What are the maximum (aphelion) and minimum (perihelion) distances from Halley's Comet to the Sun? The eccentricity of the orbit of Halley's Comet is 0.967 (Source: http://nssdc.gsfc.nasa.gov/planetary...cometfact.html). Recall that the formula for the eccentricity of an ellipse is $e=c/a$, where a is the length of the semimajor axis and c is the distance from the center to either focus. Therefore, $0.967=c/17.96$ and $c\approx17.37$ A.U. Subtracting this from a gives the perihelion distance $p=a-c=17.96-17.37=0.59$ A.U. According to the